CyberSocium: Theoretical Foundations of Cybersocial Economics and the Architecture of a Decentralised Socio-Economic Formation

Foundational Article and Technical Specification of GyberExperiment

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Abstract

This document lays the theoretical foundation of a new interdisciplinary field — **Cybersocial Economics** (CyberSocial Economics) — and presents the technical specification of **GyberExperiment** as its first empirical realisation.

We live in an era in which more than five billion people carry in their pockets computing devices whose aggregate power exceeds by orders of magnitude the entire computational technology of the twentieth century. Information technologies have integrated into the fabric of social relations, dissolving geographical, cultural, and now financial-economic boundaries. Yet the institutional infrastructure governing global resources remains a product of the industrial era — centralised, opaque, oriented toward the concentration of capital and power in the hands of a shrinking circle of beneficiaries. Recurring global crises, mounting inequality, and the systemic fragility of existing economic mechanisms signal that none of the prevailing socio-economic formations is capable of providing sustainable and rational management of the resources of a civilisation whose technological power has long demanded precisely such management.

In this work we: (a) formalise the concept of the **CyberSocial Corporation** (CSC) as the fundamental economic unit of a new type — a decentralised, autonomous, self-governing structure belonging to all its participants; (b) introduce the **Principle of Minimum Individual Participation** (PMIP) and demonstrate that, given a sufficient number of participants, it makes projects of arbitrary scale realisable; (c) describe the **mechanism of Socio-Economic Selection** (SES) — an evolutionary process in which projects compete not for investor capital but for the interest of society; (d) design the **MacroeconomicDAO** system as an instrument of direct public governance of economic processes; (e) present the complete architecture of GyberExperiment — the infrastructure that realises the theoretical constructs described herein through blockchain technologies, distributed computing, cryptographic protocols, and open-source software.

The theory of CyberSocium describes the regularities of the **cybersocialisation of the economy** — the historical process of transition of the global socio-economic mechanism toward distributed peer-to-peer forms controlled directly by society. This process is as objective and irreversible as industrialisation, and is equally in need of new theory, new infrastructure, and new institutional design.

Keywords: CyberSocium, CyberSocial Corporation, decentralised autonomous organisation, cryptoeconomics, Socio-Economic Selection, distributed governance, open-source economics, MacroeconomicDAO, peer-to-peer economy, Principle of Minimum Individual Participation

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1. Introduction: Statement of the Problem

1.1. The Era of Super-Communications: Context and Scale

More than five billion people on planet Earth carry in their pockets computing devices whose aggregate power exceeds by orders of magnitude all the computational technology of the twentieth century. According to the International Telecommunication Union (ITU), by 2025 the number of mobile internet users had surpassed 5.4 billion — more than 67% of the world's population [1]. The aggregate volume of globally generated data reached 120 zettabytes per year (IDC Global DataSphere, 2023), with more than 80% of these data generated directly by users — people, not machines.

This is not the era of supercomputers — it is the **era of super-communications**. Every computing device is, in the first instance, an extension of the human being — a communication instrument that expands, accelerates, and transforms relations among members of society. Manuel Castells, in his foundational work The Information Age: Economy, Society and Culture (2000), defined this process as the formation of the "network society" — a social structure in which key economic, political, and cultural functions are organised around networks of information processing [2]. Two decades later, his thesis has not merely been confirmed — it proved to be conservative.

Information technologies have integrated into the very fabric of social relations. They dissolve geographical, cultural, and now financial-economic boundaries. The process that anthropologists call "time-space compression" (Harvey, 1989 [3]) has reached a qualitatively new level: today a developer in Nairobi and a researcher in Helsinki can work in real time on the same software product, with their collaboration coordinated automatically through distributed version-control systems, requiring not a single bureaucratic intermediary. Yochai Benkler, in The Wealth of Networks (2006), described this phenomenon as **commons-based peer production** — a model of value creation that fits neither the market nor the hierarchical paradigm [4].

The scale of this transformation is remarkable. According to GitHub Octoverse (2024), more than 100 million developers are registered on the largest open-source development platform; the aggregate volume of open code created by volunteers without traditional economic incentives exceeds the code produced by all the commercial corporations in the world. The volume of on-chain transactions in decentralised networks reached $15 trillion per year (Chainalysis, 2024). The number of decentralised autonomous organisations (DAOs) has surpassed 13,000, and the total assets under DAO management have reached $30 billion (DeepDAO, 2024).

Yet the institutional infrastructure governing global resources remains a product of the industrial era — centralised, opaque, oriented toward the concentration of capital and power in the hands of a shrinking circle of beneficiaries. This gap between the **technological capabilities** of society and the **institutional reality** of resource governance is the central problem that this document addresses.

The exclusive goal and exclusive opportunity of society is the most effective unification of its participants in the interests of society itself. The process of development of the human social mechanism is defined as the continuous development of relations among people — from primitive exchange to complex financial instruments, from local markets to global platforms. Each new technological leap expanded the circle of participants in economic interaction and increased the complexity of the forms of coordination among them. Today we stand on the threshold of the next leap — the transition from platform capitalism [5], in which intermediaries extract rent from coordination, to **Cybersocial Economics**, in which coordination is performed by protocols belonging to the participants themselves.

1.2. The Crisis of Existing Socio-Economic Institutions

If the technological dimension of the problem — the era of super-communications — represents a story of expanding possibilities, the institutional dimension is a story of mounting crisis. The gap between the technological potential of society and the institutional capacity to realise that potential manifests across several dimensions.

Concentration of economic power. According to Oxfam (2024), 1% of the world's wealthiest people own more than 40% of global wealth, and the 26 wealthiest individuals own as much as the poorest 50% of the world's population (3.8 billion people). Thomas Piketty, in his foundational work Capital in the Twenty-First Century (2013), demonstrated that inequality is not an anomaly but a structural property of the capitalist system: when the rate of return on capital exceeds the rate of economic growth ($r > g$), inequality inevitably intensifies [30]. Daron Acemoglu and James Robinson, in Why Nations Fail (2012), showed that extractive institutions — those that concentrate power and wealth in the hands of a narrow elite — are the primary cause of economic underdevelopment [31].

CyberSocium offers not a redistribution of existing wealth (a political solution) but the creation of institutional infrastructure in which concentration is impossible by design: Hybrid Voting Power limits the influence of large token-holders, while PMIP creates a mechanism for financing projects without the requirement of large capital.

Opacity of governance. The global financial crisis of 2008 demonstrated that even the largest financial institutions (Lehman Brothers, AIG, Bear Stearns) could operate with critically dangerous risks invisible to regulators and market participants. Nassim Nicholas Taleb, in The Black Swan (2007), theorised this as the problem of "hidden risks" — the

opacity of systems makes them fragile to unforeseen events [32]. Axiom A2 (transparency) is a direct response: in the CSC, all financial flows are recorded on-chain and available for verification by any participant. Opacity is physically impossible — in contrast to traditional institutions, where it is secured by corporate rights to secrecy.

The crisis of democratic governance. Traditional representative democracy faces a crisis of legitimacy: according to the Edelman Trust Barometer (2024), trust in governments in developed countries stands at less than 50%, and in major corporations at less than 40%. Colin Crouch introduced the term “post-democracy” (2004) to describe a situation in which formal democratic institutions persist, but real power is concentrated among economic elites [33]. Helen Langman and Mansour Moya, in Surveillance Capitalism, note that digital platforms do not merely extract value from user data but also shape users’ preferences — which calls into question the very possibility of informed democratic choice.

The four-class DAO taxonomy of CyberSocium proposes a governance model in which: (a) decisions are taken directly by the interested participants, not their representatives; (b) competence is accounted for through differentiated DAO classes and Hybrid Voting Power; (c) transparency is ensured by on-chain voting; (d) AI tools compensate for the cognitive overload of direct democracy (Axiom A8).

Data appropriation as a form of economic extraction. Shoshana Zuboff, in The Age of Surveillance Capitalism (2019), showed that the business model of the largest digital platforms is based on the appropriation of “behavioral surplus” — data generated by users beyond what is necessary for the provision of the service [10]. These data are monetised through advertising, predictive models, and behavioural manipulation. The user is not a customer (they do not pay for the service) but a product (their data are sold to advertisers).

Axiom A3 (data sovereignty) eliminates this model at the architectural level: in DSP, data are stored on users’ devices and encrypted; no server in the ecosystem has access to raw user data (section 6.1). This is not a political solution (regulation such as GDPR) but a technical one — the data-extraction model is physically impossible within the architecture of CyberSocium.

Inefficiency of resource allocation. According to ILO (International Labour Organization) estimates, approximately 60% of the world’s working-age population works in the informal economy, lacking access to institutional financing, social protection, or participation in economic governance. Venture capital, which is the primary mechanism for financing innovation, is geographically concentrated (>50% of global VC investments in Silicon Valley) and thematically biased (preference for software projects over hardware or social innovations). This means that a vast portion of human potential and creative ideas systematically fails to receive the resources for realisation — not because they are unwanted by society, but because they do not fit the selection criteria of centralised financing institutions.

PMIP addresses this problem through a mechanism of decentralised resource allocation: a project receives financing not from VC or a grant committee, but directly from the interested community. The sole criterion is social relevance SR(p): if a sufficient number of people are interested in the outcome of a project (|SIC(p)| ≥ threshold), the project receives resources. This eliminates both the geographical and the thematic bias of traditional financing.

1.2a. Structure of the Document

This document is organised as follows.

Section 1 (Introduction) formulates the problem: the gap between the technological capabilities of society and the outdated institutions of resource governance. We examine the problem of data ownership, the historical tendency toward decentralisation, and the evolution of economic governance systems.

Section 2 (Literature Review) maps the intellectual traditions at whose intersection the theory of CyberSocium stands: the political economy of the digital age, institutional economics and the theory of common-pool resources, mechanism theory and mechanism design, cybernetics and the theory of complex adaptive systems, and cryptoeconomics. We identify the limits of each of these traditions and establish the necessity of a new synthesis.

Section 3 (The Theory of CyberSocium) presents the original theoretical contribution: the axiomatic foundation (A1–A8), the formal definition of the CyberSocial Corporation (CSC), the formalisation of the evolution of governance forms, the Principle of Minimum Individual Participation (PMIP), the mechanism of Socio-Economic Selection (SES), the IPI model, and the formal model of CyberSocium as a complex adaptive system.

Section 4 (Architecture of GyberExperiment) translates theory into technical specification: the tokenomics of GBR, the architecture of DSP, GyberNet, GyberComputer, MacroeconomicDAO, the legal wrapper DUNA, and the AI department (AiC).

Section 5 (DAO Taxonomy) details the four-class model of decentralised decision-making: Social DAO, Code DAO, Commerce DAO, and Economic DAO.

Section 6 (The Applied Ecosystem) describes the specific implementation components: DSP, GyberNet, GyberComputer, G-Plan, and auxiliary projects.

Section 7 (Comparative Analysis) compares GyberExperiment with existing theoretical frameworks and practical projects, identifying its unique contribution.

Section 8 (Discussion) analyses the theoretical and practical implications, limitations, threat model, and open research questions.

Section 9 (Roadmap) presents the four-phase implementation plan: from foundation to maturity.

Section 10 (Conclusion) summarises the findings and articulates a vision of the future.

Sections 11–12 contain the bibliography and appendices (glossary, smart contract addresses, summary of axioms).

1.3. Information Security, Ethics, and Data Ownership

Even those of us who have no direct connection to computer science and electronics are personally involved in the process of generating enormous quantities of data — all manner of valuable information that is readily exploited for all manner of manipulation and is equally readily monetised.

The scale of this process is striking. Each smartphone user generates on average 1.5 GB of data daily [16]. Cumulatively, this forms a flow of data that traditional economic theory is unable to describe adequately: data are neither a classical commodity (they are non-rival and reproducible at zero marginal cost) nor a public good (they are exclusive — controlled by those who own the servers).

The majority of large IT companies treat user accounts as corporate property, and the enormous quantities of data continuously generated by users as a fully-fledged product of the company. From a technical standpoint, this is correct: the accounts created by users are formed and stored on the companies' own servers. Approximately 99% of the internet content known to the general public is stored on closed proprietary servers that are physically secured no less well — and more likely considerably better — than the most heavily guarded military facilities, and all of this is, of course, maintained at the expense of the network's users themselves.

The fundamental technical concept of the internet is simple: it is a collection of computers connected by a network. In reality, the internet consists of people sending one another physical signals through a computer network, with the computer transforming those signals into information of the required type for us. Without people the internet would not be alive; it is obviously inconceivable without society, because it is the living result of society's activity at a real moment in time, and cannot exist without it.

From an ethical standpoint, data created by users constitute at the very least their personal intellectual property and personal data, and the appropriation of these data by obsolete economic institutions by virtue of temporary technical circumstances is a glaring fact that signals the lag of the humanistic development of modern society behind the development

of its technology. When one adds the fact that 99% of the internet runs on open (free) software products and fundamentally on open Linux servers, the absurdity of the capture of internet space by obsolete economic institutions becomes evident.

The trading of user data is, in our time, an already established, commonplace, and entirely legal phenomenon. Companies develop lengthy user agreements that almost nobody still reads, and in which they obtain from users various kinds of consent — such as consent to the processing of personal data — which gives them the legal right to appropriate the data of millions of users worldwide, to analyse them, and to trade in them. Meanwhile, the multimillion-strong community of actual users remains excluded from the management and control of their own data, which in the long term leads to the erosion of the fundamental principles of the freedom of personal information, the security and control of personal data, and the freedom of society, and in every way retards and suppresses the process of the historical socio-economic development of society.

The GDPR (General Data Protection Regulation), adopted by the European Union in 2018, was the first serious attempt at a legal solution to this problem [17]. However, GDPR remains a palliative solution: it regulates the use of data but does not eliminate the structural cause of the problem — the centralised storage and processing of data. As Aaron Swartz noted in the “Guerrilla Open Access Manifesto”: the problem lies not in the policies of specific companies, but in the architecture of systems that makes the concentration of data technically inevitable and economically advantageous [18].

It is evident that, from an ethical perspective, the user is the sole owner of all rights to the content they generate, including metadata and all other data types, to say nothing of the inviolability of personal correspondence, interests expressed online, and so forth. Of course, many technical products that use user data provide users with convenient, useful, and interesting features, but it is clear that all such features must be enabled and disabled at will, that the mechanism of their operation must be transparent, must ensure the security and control of user data, and that this must be confirmed by the open source code of the product.

Equally critically important is individual flexibility and the modifiability of services. The user must have full control over the portion of the overall system that they personally use, and the possibilities for modifications introduced by the user must be unlimited within their personal portion of the system. Individual possibilities for extending the system must also be unlimited — the system must be extensible in all directions.

1.4. Decentralisation and Open Source as a Historical Tendency

Computer science began its development as a closed field of technology, accessible only to corporations and governments. Computers were expensive, cumbersome machines available primarily to large technical institutions and corporations. We may roughly define this as the initial stage. It is followed by a stage of accelerated development of information technologies — which may be defined as the period in which the computer becomes accessible to a larger number of developers. At this time, competition in software development expanded and two principal competing currents emerged in the world of software development: proprietary and open.

Richard Stallman founded the free software movement in 1983, formulating four fundamental freedoms: the freedom to use a program, to study and modify it, to distribute copies of it, and to distribute modified versions [19]. Linus Torvalds, in 1991, created the Linux kernel, demonstrating that a distributed community of volunteers is capable of producing a software product that surpasses in reliability and scalability the products of the largest corporations. Eric Raymond, in his essay “The Cathedral and the Bazaar” (1999), theorised this phenomenon, contrasting two models of development: the “cathedral” (centralised) and the “bazaar” (distributed) [20].

As a result of surmounting these abstract stages of development, we observe an unprecedented surge of activity in OPEN SOURCE development, an improvement in the quality and usability of open-source software products, an expansion of their scope of application, and a significant growth in the number of users. OPEN SOURCE increasingly contributes to the emergence of revolutionary technologies. We can observe an entire evolutionary cycle of global socio-economic relations, directly connected with and nourished by the global OPEN SOURCE community and its ideas, whose general vector is the open and comprehensive development of software and of society as a whole.

Parallel to the evolution of open-source software, the decentralisation of information technologies developed — a process that evolved harmoniously in the depths of the internet community, periodically announcing itself through the emergence of powerful decentralised technological mechanisms resolving particular problems of society:

- **BitTorrent** (2001) — decentralised data distribution, demonstrating that a peer-to-peer protocol can be more efficient than centralised infrastructure [21]
- **Bitcoin** (2009) — a decentralised electronic monetary system, solving the double-spending problem without a trusted intermediary [22]
- **Ethereum** (2015) — a programmable blockchain, transforming decentralisation from a protocol for transactions into a platform for arbitrary applications [23]
- **IPFS** (2015) — the InterPlanetary File System, proposing content addressing in place of location addressing [24]

- **FileCoin** (2020) — an economic layer on top of IPFS, incentivising data storage [25]

Combating such decentralised technologies is practically futile and at best results in a temporary impediment to the operation of services, which subsequently leads to their modernisation and stabilisation — that is, to their development. This empirical fact has deep theoretical significance: it demonstrates that decentralisation is not a design decision but a **historical attractor** — a direction toward which the system tends regardless of attempts at external opposition.

Another of the principal contemporary problems looming over society is the growth of centralised control over the dissemination of ideas that are relevant to society. Obsolete economic institutions, by virtue of their hopeless but still sufficiently powerful position, strive rapidly to refine their own mechanisms for the control and monitoring of society, in order to identify relevant, advanced, socially significant ideas and projects and to counter their activity before they spread through society. Together with the fact that the overwhelming majority of large resource-intensive projects are initiated and financed by those same obsolete socio-economic institutions, we have, for the most part, a world that reflects exclusively the interests of these obsolete institutions and is extremely reactionary toward any kind of revolutionary, fundamental change, and consequently — toward the development of society as a whole.

At the same time, and in part as a result of this, we observe the vigorous development of decentralised technologies, which has already taken shape as a powerful, global, distributed socio-economic movement that has already united fairly broad and educated masses of people. In essence, we are witnessing the process of formation of a new type of global creative intelligentsia, capable of uniting its ideas and interests with other social forces into a single ideology wholly directed toward the common good and development — to become the basis for the final formation of a new, advanced creative class, one capable at last of making the transition to the long-overdue global socio-economic changes adequate to the contemporary demands of global Civilisation.

This emerging class differs fundamentally from the traditional intelligentsia in the historical sense — one enclosed within academic or cultural institutions and dependent on the patronage of the state or capital. What is at issue here is a distributed global network of researchers, developers, engineers, economists, and thinkers, united not by geography, corporate affiliation, or national identity, but by three structural characteristics: first, a shared technological literacy that enables them not only to understand but also to create complex information systems; second, a commitment to the principles of openness, decentralisation, and transparency as fundamental ethical and practical orientations; third — and this is critically important — possession of practical tools for the direct realisation of

their ideas without the mediation of obsolete institutions, ranging from cryptographic protocols and smart contracts to distributed computing networks, open-source governance systems, and mechanisms of decentralised financing.

Richard Florida described the emergence of the “creative class” as an economic force defining the development of post-industrial societies [47], yet his analysis remained within the framework of existing institutions — corporations, cities, universities. Florida’s creative class creates value within the system and depends upon it. What we observe now is qualitatively different: a class is being formed that does not integrate itself into existing institutions but generates its own — decentralised autonomous organisations, open protocols, distributed networks of storage and computation, CyberSocial Corporations. For the first time in history, a significant mass of people simultaneously possesses the intellectual potential to design alternative forms of socio-economic organisation and the technical capacity to realise them — without permission, without mediation, and without control on the part of obsolete structures.

Today, a new global social consciousness is being formed — the principal organ of humanity’s social self-governance. Although the era of “kings”, with all its characteristic rigidly centralised mechanisms of control that restrained the education and development of society, science, and technology, has long since passed, and today we observe a vigorous flourishing of culture, education, science, and technology in the historical sense, and their penetration — as a consequence of the historically successive social transformations that have already been accomplished — into virtually all strata of society, the process of decentralisation of global mechanisms of economic interaction is still far from its final stage. However, the direction of this process is already determined — not by the will of individual leaders or organisations, but by the very logic of technological and social development, which cannot be stopped, but can only be temporarily impeded, which, as the history of decentralised technologies shows, invariably leads to their strengthening and perfection.

There is an evident tendency of the global socio-economic mechanism in its historical development to arrive at the most distributed peer-to-peer form, governed through cybersocial financial mechanisms acting upon the global socio-economic resource — that is, a mechanism in the hands of the public and controlled directly by the public.

As information technologies develop and as they penetrate the real sector of the financial-economic space, society acquires an ever-growing number of levers of substantial influence over the direction of global economic development. Technologically, the global economy stands on the threshold of large-scale transformations associated with society’s transition to a new level of socio-economic interaction, with the direct use of innovative decentralised cybersocial economic mechanisms.

With the emergence of Blockchain technology, the possibility of constructing decentralised financial systems has opened up, and their development genuinely contributes to the emergence of opportunities for the construction of global self-governing cybersocial economic structures, expanding the possibilities of society's intellectual influence on the global economic process.

Whatever happens in the world, and whatever powerful forces influence the course of history, the organism of society invariably produces a cycle of evolutionary transformation — adaptive change. It becomes more perfect, and this process occurs, above all, not on the platforms of debates, financial-economic congresses, and other hypocritical venues, but in the minds of people — initially in isolated cases, then, gaining momentum, it strengthens within society and develops in a spiral. And, of course, all this development is accompanied and ratified by the appearance of revolutionary, fundamental, philosophical, legal, and technical documents and products, which are various moments of the single dialectic of the development of global socio-economic relations, systematising and advancing that development to the degree necessary for the transition to a new stage of development.

This document is an attempt at such a systematisation.

1.5. The Evolution of Economic Governance Systems

The history of economic governance represents a sequence of expanding circles of participation. Each stage of transformation is characterised by three parameters: **who** makes economic decisions, **what volume of resources** is coordinated, and **what mechanism** of coordination is employed.

Stage 1 — Local subsistence economy (before the 18th century). Economic decisions are made by the head of a household or community. Coordination is limited to physical space: the market fair, the guild, the manor estate. Mechanism: tradition, personal relations, direct coercion. Circle of coordination participants: tens to hundreds.

Stage 2 — National market economy (18th–19th centuries). Adam Smith described the market as a coordination mechanism in which the "invisible hand" replaces centralised planning [6]. Stock exchanges, joint-stock companies, and central banks emerge. The circle of participants expands to thousands and tens of thousands. Coordination mechanism: the price signal. However, Marx showed that this mechanism systematically generates inequality and the concentration of capital [7].

Stage 3 — Global corporate economy (20th century). Transnational corporations coordinate resources on a scale exceeding the budgets of many states. Ronald Coase explained the existence of the firm through the minimisation of transaction costs [8]: intra-corporate coordination is cheaper than market coordination under certain conditions. Oliver

Williamson developed this theory, demonstrating the role of opportunism and asset specificity [9]. Circle of decision-makers: thousands of managers, governing the activity of millions. Mechanism: hierarchy + market.

Stage 4 — Platform economy (early 21st century). Digital platforms (Google, Amazon, Facebook, Apple, Microsoft) achieve scales of coordination comparable to those of states. Nick Srnicek, in Platform Capitalism (2017), showed that platforms extract rent from the role of intermediary without creating proportionate value [5]. Shoshana Zuboff, in The Age of Surveillance Capitalism (2019), described the extraction of “behavioral surplus” as a new form of exploitation [10]. Circle of participants: billions of users, but decisions are made by boards of directors. Mechanism: algorithmic coordination controlled by a corporate hierarchy.

Stage 5 — Decentralised cybersocial economy (emergent). The CSC as the fundamental economic unit eliminates the gap between the scale of participation and the scale of governance. For the first time in history, millions of participants can directly coordinate economic resources without hierarchical intermediaries — through formalised protocols executed by smart contracts. Mechanism: algorithmic coordination controlled by participants through DAOs. The PMIP principle ensures the scale of financing, while SES ensures the quality of project selection.

This sequence is not coincidental. It reflects a fundamental law: as the technological capabilities of coordination grow, **the circle of participants in economic decisions steadily expands**, and **the role of hierarchical intermediaries contracts**. Each stage is characterised by a contradiction between expanded coordination capabilities and outdated governance institutions. The resolution of this contradiction requires a new institutional design — and it is precisely this that the theory of CyberSocium proposes.

Stage	Decision participants	Resource scale	Coordination mechanism
1. Local	10^1–10^2	Local	Tradition, personal ties
2. National	10^3–10^4	National	Price signal (market)
3. Corporate	10^3 (managers)	Global	Hierarchy + market
4. Platform	10^1 (directors)	Global	Algorithm (corporate)
5. Cybersocial	10^6–10^9	Global	Protocol (DAO + PMIP)

The formalisation of this evolutionary sequence and the demonstration of the inevitability of the fifth stage constitute one of the central tasks of section 3.

1.6. Methodology

This work employs a combination of several methodological approaches, determined by the interdisciplinary nature of the subject under investigation.

The axiomatic method. The theoretical foundation of CyberSocium is built on a system of eight axioms (A1–A8), formulated as normative design principles of the system. The axiomatic approach is not chosen arbitrarily: it ensures the logical coherence of the architecture and allows formal verification of the correspondence between technical decisions and theoretical principles. Each axiom is formulated, justified through references to existing theoretical traditions, and linked to specific technical implementation mechanisms.

Formal modelling. The key mechanisms (PMIP, SES, IPI, FRP) are formalised as mathematical models with clearly defined parameters, conditions of operability, and demonstrable properties. For the PMIP model, a number of theorems are proved (universal realisability, scale inversion, anti-fragility). CyberSocium is modelled as a complex adaptive system (CAS) in the terminology of Holland [11] and Kauffman [12], with an explicit definition of agents, resources, interactions, feedback loops, and emergent properties.

Systems analysis. The interactions among the components of the GyberExperiment ecosystem are analysed using methods of system dynamics (Forrester [13], Meadows [14]). Positive and negative feedback loops, leverage points, and potential regimes of instability are identified. This approach allows the prediction of system behaviour under various initial conditions and external perturbations.

Comparative institutional analysis. The architecture of GyberExperiment is compared with existing projects (MakerDAO, Aragon, Colony, Gitcoin, Nouns DAO) and theoretical frameworks (mechanism design, Ostrom's theory, cryptoeconomics). This analysis makes it possible to identify the unique contribution and to reveal borrowings and parallels.

The constructive approach. In contrast to purely descriptive studies, the present work not only analyses existing systems but also proposes a specific architecture for a new system. Each theoretical construct receives a technical implementation: axioms are implemented in smart contracts, mechanisms in protocols, models in algorithms. This approach follows the tradition of institutional design (institutional design) established by Hurwicz [15] and developed in mechanism design.

Epistemological position. We acknowledge that the theory of CyberSocium is at an early stage of development. Many of its propositions are hypotheses requiring empirical verification. GyberExperiment is regarded precisely as an experiment — a controlled intervention on a real system with the purpose of testing theoretical predictions. In this sense our approach is closer to the methodology of Lakatos [42] (a research programme with a "hard core" of axioms and a "protective belt" of empirically testable consequences) than to Popperian falsificationism.

1.7. Scope and Boundaries of This Work

To prevent misplaced expectations and to ensure the correct interpretation of results, it is necessary to explicitly define the boundaries of this document — what it contains and what it does not contain.

What this work is:

1. A theoretical foundation. The document defines an axiomatic system (A1–A8), formal mechanisms (PMIP, SES, IPI, FRP), and mathematical models (CAS, Hybrid VP, replicator dynamics) that constitute the theoretical foundation of a new class of socio-economic systems — CyberSocial Corporations (CSC).

2. An architectural specification. For each theoretical construct, a specific technical implementation is defined: smart contracts (UnitManager, Governor, GBRVotingWrapper, LPBurnInitiator), protocols (DSP, GPROD, DDP), infrastructure (GyberNet, GyberComputer), AI components (AiC), and legal wrapper (DUNA). The specification defines what is to be implemented and why, leaving the specific implementation details to technical documentation.

3. An experimental protocol. GyberExperiment is formalised as a controlled socio-economic experiment with explicit hypotheses (H1–H5), observable variables, counterfactuals, and falsification criteria (section 8.6).

4. A positioning document. Through the literature review (section 2) and comparative analysis (section 7), the document positions CyberSocium relative to existing theoretical traditions (institutional economics, mechanism design, cryptoeconomics, CAS theory) and practical projects (MakerDAO, Aragon, Colony, Gitcoin, Nouns DAO).

What this work is NOT:

1. Not a token white paper. Although the document describes the tokenomics of GBR (section 4.2), its goal is not the attraction of investment but the formalisation of economic mechanisms. The document contains no financial projections, promises of returns, or investment recommendations. GBR is treated as a utility token — an instrument of coordination within the ecosystem, not an object of investment (see the Howey analysis, section 8.4.2).

2. Not technical documentation. The document describes the architecture at the level of abstraction necessary for understanding the connection between theory and implementation. Detailed API specifications, deployment guides, Solidity NatSpec, and UI mockups are contained in the technical documentation: `docs/system-`

`architecture/` (specifications 00–06), `integration-docs/` (integration specifications), and inline documentation in the CONTRACTS, GPROD, and DSP Client repositories.

3. Not a legal opinion. The analysis of the legal form of DUNA (section 4.7), the Howey test (section 8.4.2), and the compliance analysis (section 8.4) are informational in nature and do not substitute for qualified legal counsel. Participants in the ecosystem are obliged independently to assess the legal consequences of their actions in the jurisdiction of their residence.

4. Not a completed theory. CyberSocium is at the stage of formulation — analogously to the early works of Nakamoto (2008) on Bitcoin or Buterin (2014) on Ethereum. Many propositions are hypotheses subject to empirical verification. The document consciously adopts this position: an honest recording of the current state of knowledge is more valuable than premature claims to completeness.

Target audience:

- Researchers — economists, computer scientists, legal scholars, sociologists — interested in the formal analysis of decentralised socio-economic systems;
- Developers — blockchain engineers, full-stack developers, AI/ML engineers — implementing components of the ecosystem;
- Participants in GyberExperiment — all levels of MemberStatus — for an understanding of the theoretical foundations of the system in which they operate;
- Regulators and legal professionals — for the assessment of the legal architecture and compliance model;
- Critics and opponents — for constructive critique; the existence of formalised hypotheses and falsification criteria makes criticism productive.

Versioning. This document follows the principle of a living document: it will be updated as empirical data from GyberExperiment are received. Each version is recorded in Git with a complete history of changes. Current version: 2.0 (March 2026). Next planned version: 2.1 (following the completion of Phase I and the receipt of initial empirical data, estimated Q4 2026).

2. Literature Review and Interdisciplinary Context
3. Theory of CyberSocium
3.1. Axiomatic Foundations
The theory of CyberSocium is built upon a system of eight axioms — normative principles that define the fu CyberSocial Corporation must possess. The axioms serve a triple function: (1) they fix the value priorities of the s evaluation criteria for technical decisions; (3) they define the boundaries of permissible behaviour for participants
Each axiom is formulated, justified through references to existing theoretical traditions, and linked to specific impl in the GyberExperiment architecture.
Axiom 1 (Decentralization). No single node, agent, group of agents, or algorithm may exercise control over t Governance functions are distributed among all participants and implemented through formalized protocols.
Justification: Ashby's law of requisite variety [36]: centralized management is incapable of processing the varie system. Empirically: the resilience of Bitcoin and Ethereum, in which the absence of a central point of failure ensu under any external perturbation.
Implementation: P2P architecture of GyberNet, multisig contracts with distributed control, distributed storage (IP single point of failure in every component of the ecosystem.
Axiom 2 (Transparency). All economic transactions, governance decisions, and system parameters are recorde are available for verification by any participant in real time.
Justification: Ostrom's principle of user-side monitoring [30]. The condition of informational symmetry required fo of voting mechanisms (Gibbard–Satterthwaite [31]).
Implementation: On-chain governance for all DAO decisions, AGPL-3.0 licensing of all code, public blockchain smart contracts.
Axiom 3 (Data Sovereignty). The user is the sole owner of all rights to the content, metadata, and any other typ them. No component of the system may use a user's data without their explicit informed consent.
Justification: Zuboff's ethical argument [10] on the inadmissibility of "surveillance capitalism." The legal argument Swartz's technical argument: the problem lies not in policy but in architecture [18].
Implementation: DID (Decentralized Identifiers), IPFS for user-controlled data storage, E2E encryption, zero verification without disclosure of data.
Axiom 4 (Extensibility). The system is designed for unlimited expansion without the permission of a central au may create new modules, applications, and integrations that extend the functionality of the system.
Justification: The principle of permissionless innovation, realized in Ethereum and demonstrated in practice (tho applications on a single platform). Modular architecture as a property of complex adaptive systems [11].
Implementation: Modular DSP architecture, open APIs, SAPP (extension standard), permissionless participation in
Axiom 5 (Meritocratic Fairness). A participant's reward is determined by their contribution to the system, veri mechanisms, rather than by hierarchical position or the volume of capital.

2. Literature Review and Interdisciplinary Context
Justification: Ostrom's principle of user-side monitoring [30]. Smart contracts and activity-tracking systems (G-P verification automatable and manipulation-resistant. The VCG mechanism [32] demonstrates the possibility o which honest behaviour is a rational strategy.
Implementation: UnitManager smart contract (six-level status system: None, Unit, Dev, LeadDev, ArchDev, verification), reputation system with on-chain recording, Activity Scoring (score 0–100, reward multiplier 50–150%
Axiom 6 (Inclusiveness). The entry barrier to the system must be minimal, and every participant must have resources regardless of their personal involvement in specific projects.
Justification: PMIP (Section 3.4) demonstrates mathematically that inclusiveness does not contradict scalability Olson [29] showed that a high entry barrier leads to the collective action problem. PMIP resolves this pro contributes a minimum share, yet the aggregate effect scales without limit.
Implementation: GBR liquidity pools with all projects ensure every community member participates in the eco project. Multilingualism, cross-chain compatibility, free basic access to DSP.
Axiom 7 (Self-Governance). The system is governed exclusively by its participants through formalized me decision-making, without external management.
Justification: Ostrom's principle of the community's right to self-organization [30]. DAO contracts provide the tech self-governance. Hayek's argument [37] on the impossibility of effective centralized planning due to the knowledge.
Implementation: Four-class DAO taxonomy (Social, Code, Commerce, Economic), Governor contracts (OpenZeppe on-chain voting with a quadratic mechanism and reputational multipliers.
Axiom 8 (Cognitive Augmentation). The system uses artificial intelligence as a tool for augmenting the collective participants, but not as a substitute for human judgment in matters affecting values, ethics, and strategic direction
Justification: Direct democracy has historically been considered impractical at scale due to two constraints: cog deeply evaluate every proposal) and coordinational (millions of people cannot deliberate simultaneously). AI re making large-scale direct democracy practically feasible for the first time. However, without an explicit limitation form of centralization — an "AI-oligarchy" in which an algorithm, rather than the community, determines direct limits AI to the role of a cognitive augmenter.
Principle of demarcation: - **AI decides autonomously:** technical optimizations (compute distribution, routing, m and structuring of information, anomaly and pattern detection, generation of recommendations and forecasts - **decides:** strategic decisions about the direction of development, ethical questions, allocation of large reso conflicts of values (FRP at the level of principles) - **AI has no authority to:** independently move treasury funds parameters without a DAO vote, block or censor participant proposals, make irreversible decisions without a huma
Implementation: all AI agents are registered in GyberNet (soulbound tokens), their decisions are recorded on-c parameters of AI agents are established by the vote of the corresponding DAO, any participant may contest an A the standard proposal mechanism. AiC (three contours: internal, research, commercial) is the ecosystem's infrastr

2. Literature Review and Interdisciplinary Context

Hypothesis on Consistency. Axioms A1–A8 are consistent and mutually reinforcing. Decentralization (A1) en transparency (A2); transparency (A2) ensures the verifiability of meritocracy (A5); data sovereignty (A3) and exte conditions for inclusiveness (A6); inclusiveness (A6) provides the scale necessary for self-governance (A7) protects all other axioms from violation by external forces; cognitive augmentation (A8) ensures the quality of any scale, strengthens meritocracy (A5) through objective analysis of contributions, and respects data so federated learning without transfer of raw data.

```
Axiom mutual-reinforcement graph:

A1 (Decentralization) ──→ A2 (Transparency) ──→ A5 (Meritocracy) │ │ └──→ A7 (Self-Governance) ←-
←──┘ │ ↑ └──→ A8 (Cognitive Augmentation) ↑ A3 (Data Sovereignty) ──→ A4 (Extensibility) ```
```

It should be noted that there is a constructive tension between A5 (meritocracy: voting weight proportional (inclusiveness: minimal barrier for all). New participants have access, but little governance weight. This tens balancing mechanisms: quadratic voting (reduces the influence of large holders), reputational multipliers (rew alone), and a temporal horizon (a participant's weight grows with activity, not with investment).

3.2. CyberSocial Corporation: Definition and Properties

Definition 3.2.1. A CyberSocial Corporation (CSC) is an organizational form satisfying the following conditions:

1. **Distributed ownership:** A CSC belongs to all of its participants; no single participant or group of participants h
2. **Algorithmic governance:** key decisions are made through formalized protocols (DAO), executed by smart contr all software code of the CSC is available for audit and modification by any participant 4. **Peer-to-peer structu** without hierarchical intermediaries; statuses define the level of responsibility and reward, but not of power 5. **E** CSC is capable of generating resources for its own operation and development without dependence on external **Social orientation:** the criterion for selecting projects for implementation is Social Relevance, not profit maximizati

Comparative analysis of organizational forms:

```
Characteristic      | Traditional     | DAO             | CSC | Corporation     | (existing)
────────────────────┼─────────────────┼─────────────────┼────────────────── Ownership
Shareholders     | Tokenholders     | All participants |                   |
────────────────────┼─────────────────┼─────────────────┼────────────────── Governance
of          | Token-weighted   | Verified | directors        | voting            | activity + |
|                     | DAO ────────────────────┼─────────────────┼─────────────────┼──────────
criterion     | Shareholder      | TVL / token      | Social | profit            | price
────────────────────┼─────────────────┼─────────────────┼────────────────── Code
Proprietary         |  Partially           |  Fully  |                          |  open
────────────────────┼─────────────────┼─────────────────┼────────────────── Entry  barrier
Hiring  /           | Token                | Participation  |  share  purchase   |  purchase
────────────────────┼─────────────────┼─────────────────┼────────────────── Scalability
Hierarchical     | Limited          | PMIP- |                   | (plutocracy of   | scaling |
tokenholders)       |  ────────────────────┼─────────────────┼─────────────────┼──────────
to          | Depends on       | Depends on       | Antifragility crises                | manag
market    | (Theorem 3)
```

2. Literature Review and Interdisciplinary Context

> 📊 **Professional diagram:** Open SVG

Table 3.1: Comparative analysis of three organizational forms across seven key characteristics. The CSC (C
differs qualitatively from traditional corporations and existing DAOs along all dimensions, particularly in owner
shareholders/tokenholders), governance (verified activity vs capital), success criteria (Social Relevance vs profi
crises (Antifragility vs dependence on management/market).

Formal Definition 3.2.2. A CSC is formally defined as a sextuple:

```
CSC = ⟨N, R, P, G, M, E⟩
```

where: N = {n_1, n_2, ..., n_k} — set of participants (NFC — Network Financial Community) Each n_i is characterized
$status_i$, $reputation_i$, $activity_i$, $staking_i$⟩

R — aggregate resource of the system: R = R_fin ⊕ R_int ⊕ R_rep ⊕ R_inf (financial ⊕ intellectual ⊕ reputatio

P = {p_1, ..., p_m} — set of projects (IPI model) Each p_j passes through the lifecycle: Idea → Discussion → Forma
Implementation → Operation

G = {G_social, G_code, G_commerce, G_economic, G_macro} — governance system (four-class DAO taxonomy +
implements the functions: — Collective decision: G × Proposal → {Accept, Reject, Defer} — Parameter adjustme
— Conflict resolution: G × Conflict → FRP(resolution)

M — set of coordination mechanisms: M = {PMIP, SES, FRP, IPI, HybridVP, ActivityScoring} Each mechanism m_k
properties (incentive compatibility, Pareto efficiency, convergence)

E — ecosystem infrastructure: E = {DSP, GyberNet, GyberComputer, GBR_token, DUNA, AiC} Physical realization
```

**Properties of the CSC.** Definition 3.2.1 together with formal structure 3.2.2 yields a set of theoretically sig
distinguish the CSC from all existing organizational forms.

**Property 3.2.1 (Self-similarity).** The CSC possesses a fractal structure: each project within the CSC may be regar
its own set of participants (SIC(p)), resources, governance (project-level DAO), and coordination mechanisms
scalability: a CSC may contain an arbitrary number of nested CSC-like structures, analogously to Ostrom's "nest
principle 8).

**Property 3.2.2 (Openness of composition).** The set N is not fixed: any agent may join the NFC (A6, inclusiven
moment. Formally: ∀t, N(t) ⊆ N(t+1) ∪ N′(t+1), where N′(t+1) is the set of agents that have left the system. T
distinguishes the CSC from traditional corporations (closed employee roster) and even from most DAOs (token-pu

**Property 3.2.3 (Antifragility, proved in Section 3.4).** The CSC possesses the property of Antifragility in Talel
stresses (participant outflows, token price drops, attacks) not only fail to destroy the system, but strengthen it t
adaptation mechanisms. FRP transforms conflicts into diversity; SES selects for resilient projects; Activity Scorin
favour of active participants.

**Property 3.2.4 (Formal verifiability).** Each axiom A1–A8 translates into specific smart contracts with formally ve
means that conformity of the system to its axioms can be proved mathematically, and not merely declared. Forr
Axiom → Specification → Solidity code → Slither/Mythril/Certora → Proven invariants.
```

2. Literature Review and Interdisciplinary Context

A substantive distinction between the CSC and existing DAOs requires separate clarification. Most contempora Compound, Uniswap — employ token-weighted voting, in which voting weight is proportional to the number of to a plutocracy: large holders determine decisions, while small holders are effectively disenfranchised. Vitalik pointed to this problem [46]. In the CSC, voting weight is determined not by the number of tokens, but by **verified** in projects confirmed by the G-Plan system and by participants of higher status. This implements the principle (A5) and prevents plutocratic degeneration.

To illustrate the fundamental difference between a CSC and a traditional corporation, consider the response to t — the loss of 30% of active participants:

``` Response to the loss of 30% of participants:

Traditional corporation: — Mass layoffs → loss of knowledge → decline in productivity — Share price falls → i weakening — Management reacts in panic → erroneous strategic decisions — Possible outcome: bankruptcy o feedback (crisis amplification)

Existing DAO: — Tokenholder outflow → token price falls — Treasury loses value → funding becomes im participants lose motivation → further outflow — Possible outcome: dead protocol → Positive feedback (death spi

CSC (CyberSocium): — 30% left → Activity Score of the remaining rises (less competition for rewards) — VP of → stronger governance — SES selects the most viable projects (resources concentrate on the best) — PMIP req to maintain financing (cost → 0 as |SIC| → ∞, but it works at smaller |SIC| as well) — FRP allows the remaining Antifragility: the system emerges from the crisis more compact, but more efficient → Negative feedback (stabiliza

This example demonstrates that the CSC is not merely a "better version of a DAO" — it is a qualitatively differen which multiple feedback mechanisms (Activity Score, SES, PMIP, FRP) provide resilience that is impossible withir token-weighted models.

### 3.3. Evolution of Forms of Governance over Economic Processes: A Formalization

The historical trajectory described in Section 1.5 can be formalized as a sequence of phase transitions in the forms.

**Definition 3.3.1.** Let the space of organizational forms be described by a vector of characteristics:

``` Ω = ⟨D, T, A, S, C⟩

where: D ∈ [0, 1] — degree of decentralization of decision-making T ∈ [0, 1] — degree of operational transparenc accessibility of participation (inclusiveness) S ∈ [0, 1] — degree of participants' sovereignty over their data and as of algorithmic coordination (displacement of human hierarchy by formalized protocols) ```

**Definition 3.3.2.** A phase transition in the space Ω is a historically observable qualitative change in the dominan which at least one component of the vector Ω undergoes a jump exceeding a threshold δ > 0.

**Historical sequence of phase transitions:**

``` Phase 0 — Monarchical Centralization (before the 18th century): $\Omega_0$ ≈ ⟨0.05, 0.05, 0.02, 0.10, 0.01⟩
```

2. Literature Review and Interdisciplinary Context
Characteristics: Governance of economic processes concentrated in the hands of the sovereign. Trade guilds operate under royal privilege. Transparency is absent. Participation is determined by social estate. Coordination and tradition.
Dominant form of economic unit: manufactory, trading house, colonial company.
Phase 1 — Parliamentary Capitalism (18th–19th centuries): $\Omega_1 \approx \langle 0.15, 0.15, 0.10, 0.20, 0.05 \rangle$
Transition: δ(D) ≈ 0.10 — governance functions passed from the monarch to parliament, representing the interes forces.
Characteristics: The emergence of the joint-stock company as a means of distributing risk and ownership. The The stock exchange as a mechanism for coordinating capital. However, participation is limited by property qu class is excluded from governance.
Dominant form: joint-stock company, bank.
Key theorists: Adam Smith [26], David Ricardo.
Phase 2 — Corporate Capitalism (late 19th – mid-20th century): $\Omega_2 \approx \langle 0.15, 0.20, 0.15, 0.15, 0.10 \rangle$
Transition: δ(C) ≈ 0.05 — emergence of management as a formalized discipline of coordination (Taylor, Ford). δ(A participation through mass public securities markets.
Characteristics: Transnational corporations. Separation of ownership from management (Berle and Means, revolution. Keynesian state regulation as compensation for market failures.
Paradox: D does not grow, and in some respects declines — concentration of control in the hands of managemen dispersed.
Dominant form: public corporation, state enterprise.
Phase 3 — Financial Capitalism / Neoliberalism (1970s – 2008): $\Omega_3 \approx \langle 0.12, 0.18, 0.20, 0.12, 0.15 \rangle$
Transition: δ(C) ≈ 0.05 — algorithmic trading, derivatives, securitization. δ(A) ≈ 0.05 — expansion of retail investr — effective decline in decentralization: financial conglomerates, concentration of banking capital, too big to fai participant sovereignty: growing dependence on financial intermediaries.
Characteristics: Economic hegemony of the financial sector. Globalization as intensification of financial flows globalization of governance. Result — the 2008 crisis as a systemic failure of an opaque, hyper-concentrated sys
Dominant form: investment bank, hedge fund, financial conglomerate.
Critics: Stiglitz [9], Piketty [8], Minsky.
Phase 4 — Platform Capitalism (2000s – present): $\Omega_4 \approx \langle 0.10, 0.15, 0.35, 0.08, 0.30 \rangle$

2. Literature Review and Interdisciplinary Context
Transition: δ(A) ≈ 0.15 — radical reduction of participation barriers (any person with a phone is a user and a value — algorithmic coordination through platforms (Uber, Airbnb, Amazon Marketplace). However: δ(D) < 0 — decentralization: GAFAM controls ~$10T+ in market capitalization, the data of billions of people. δ(S) < 0 — minir user owns neither the account, nor the data, nor the algorithm. δ(T) < 0 — decline in transparency: closed recommendations, and pricing.
Phase 4 Paradox: maximum accessibility of participation combined with minimum sovereignty and minimum dec people are involved in creating value, but are deprived of control over the results of their labour. This structu driving force behind the transition to Phase 5.
Critics: Zuboff [6], Srnicek [7].
Dominant form: platform (Google, Meta, Amazon).
Phase 5 — Declaration: Cybersocial Economics (emerging): $\Omega_5^* \approx \langle 0.85, 0.90, 0.90, 0.95, 0.85 \rangle$ (target state)
Transition: Simultaneous jump of all components. This is not an incremental improvement of one parameter (as in phase transition in the full sense — a qualitative change in the nature of the organizational form. AI (A8) is a nece transition: D=0.85 requires decentralized monitoring, T=0.90 — processing petabytes of data, C=0.85 — algo millions of participants. Without AI these target values are unattainable.
Characteristics: CSC as the dominant form of economic unit. PMIP as a financing mechanism. SES as a projec MacroeconomicDAO as a coordination tool.
Marker: GyberExperiment as the first empirical realization. ```
Theorem 3.3.1 (On the directionality of evolution). The historical sequence $\Omega_0 \to \Omega_1 \to \Omega_2 \to \Omega_3 \to \Omega_4$ demonstra toward growth in algorithmic coordination (C) and accessibility of participation (A), but a non-monotone dynamic transparency (T), and sovereignty (S). Phases 3 and 4 are characterized by regression of D, T, and S alongside g creates a structural contradiction — mass participation in the absence of control. This contradiction is the d transition to Phase 5, in which all five components simultaneously attain high values.
Justification: The Phase 4 contradiction is not a stable equilibrium. Mass involvement (high A) combined with low low decentralization (low D) means that enormous human potential is directed by governing algorithms (high narrow group of beneficiaries. As participants' awareness grows (the Zuboff effect, the effect of data leaks, the alternatives), the costs of maintaining this disequilibrium increase, while the attractiveness of alternatives rises. for a phase transition analogous to the way in which the contradiction between the social character of production appropriation created conditions for social transformations in the industrial era [26]. A formal proof of this theo and a subject for further research.
Hypothesis 3.3.2 (On the irreversibility of the transition). The transition from Phase 4 to Phase 5 poss irreversibility, conditioned by the following factors:
1. **Technological ratchet:** Cryptographic protocols, once widely disseminated, cannot be "recalled." A user who g encrypts their data acquires a sovereignty that cannot be taken away without physical coercion — and physical to billions of people.

2. Literature Review and Interdisciplinary Context

2. **Knowledge effect:** Understanding of the principles of decentralization, open code, and self-governance, c society, cannot be “forgotten.” This is analogous to the ratchet of literacy: a society that has learned to read does

3. **Economic ratchet:** Every successful decentralized project creates infrastructure (code, protocols, standards) projects are built. The cumulative effect makes a return economically irrational.

4. **Network effect:** The value of a decentralized network grows in proportion to the square of the number of parti After reaching critical mass, the network becomes self-sustaining.

Counter-argument: historical examples (the Great Firewall, the crypto winter of 2022, regulatory bans) demons possible under certain conditions — large-scale state coercion, loss of trust in the technology, or a catastrophi hypothesis of irreversibility requires refinement: the transition is irreversible provided the network effect is mair crisis of trust occurs.

Remark 3.3.3 (On heterogeneity and coexistence of forms). The assertion of a transition to Phase 5 does not disappearance of organizational forms from preceding phases. Just as industrialization did not destroy artisa created a new dominant form coexisting with its predecessors, so cybersocialization creates a new dominant fo will coexist with traditional forms, gradually transforming the economic landscape without a radical, instantaneous

3.4. The PMIP Concept and the Mechanism of Socio-Economic Selection

The Principle of Minimum Individual Participation (PMIP) is the central economic mechanism of the CSC. It formali given a sufficient number of participants, **any** project can be realized at **negligibly small** individual costs for each

3.4.1. Definitions

Definition 3.4.1 (New Financial Community, NFC). NFC = ⟨A, Gbr, Protocols⟩ — a triple consisting of: a set of ag the community token Gbr as a unit of value and coordination; and a set of formalized protocols (DAO, IPI, FRP, SI interaction.

Definition 3.4.2 (Social Investment Circle, SIC). For a project p, SIC(p) ⊆ A is the set of NFC members intereste and prepared to participate in its financing through PMIP. |SIC(p)| is unbounded; its size is determined by the project.

Definition 3.4.3 (Active Group, AG). AG(p) ⊆ SIC(p) is the subset of participants wishing to actively participat (development, testing, management, support). |AG(p)| is determined by the nature and scale of the project. AG is among SIC participants.

Definition 3.4.4 (Coordinating Group, CG). CG(p) is a professional auxiliary resource for managing the realization both members and non-members of the NFC. CG is accountable to SIC(p) through DAO mechanisms.

Definition 3.4.5 (Collective Capital). CC(p) = Σ_i investment_capacity(a_i) for all $a_i \in$ SIC(p) — the aggregate inve participants.

Definition 3.4.6 (PMIP). The cost of project C(p) is distributed across all funding participants:

```
individual_cost(p) = C(p) / |SIC(p)|
```

Fundamental property: as |SIC(p)| → ∞, individual_cost(p) → 0 ```

2. Literature Review and Interdisciplinary Context

Definition 3.4.7 (Social Relevance, SR). SR(p) = 1, if CC(p) ≥ C(p), i.e., the aggregate capital of the SIC covers th
0, otherwise.

Definition 3.4.8 (Threshold of Social Relevance, TSR). TSR(p) = C(p) / min_capacity — the minimum number of S
to realize the project at minimum individual contribution.

3.4.2. The IPI Model: Project Lifecycle

The lifecycle of a project in the CSC is formalized as a finite automaton **Idea → Project → Implementation** (IPI):

```` States: {Idea, Discussion, Formation, Accumulation, Implementation, Operation, Completion}

Transitions: Idea → Discussion: The author publishes the idea in the DSP project space. Any NFC participant r
entry barrier is zero.

Discussion → Formation: A sufficient number of participants have expressed interest (|interested| ≥ θ_formatio
and AG(p) are formed. An AI analyst generates a structured feasibility assessment.

Formation → Accumulation: SR(p) = 1 (Social Relevance confirmed). Burning of 0.1% GBR of C(p) as a com
parameters are fixed: budget, timeline, AG.

Accumulation → Implementation: CC(p) ≥ C(p) (necessary funds have been raised). Internal project tokens are iss
realization under oversight of SIC(p).

Implementation → Operation: The project is operational and generating value. Internal_token/GBR liquidity pool is c

Operation → Completion: Optional, for finite projects. Final report, distribution of outcomes. ```

#### 3.4.3. Economic Dynamics of PMIP

PMIP creates a unique economic dynamic, fundamentally different from all existing financing models:

```
┌──────────────────┬──────────────────┬──────────────┬──────────────┬──────────────┐
│ Characteristic   │ Venture Capital  │ Crowdfunding │ Stock Market │ State Budget │
├──────────────────┼──────────────────┼──────────────┼──────────────┼──────────────┤
│ Who finances     │ Few (funds)      │ Many (crowd) │ Shareholders │ All (taxes)  │
│ Selection criterion│ Profitability   │ Product      │ Market       │ Policy       │
│                  │ for investor     │ attractiveness│ valuation   │              │
│ Entry barrier    │ $100K+           │ $1+          │ Medium       │ None (taxes) │
│ Maximum          │ $1-10B           │ $1-50M       │ $1-100B+     │ National budget │
│ project scale    │                  │              │              │              │
│ Transparency     │ Minimal          │ Medium       │ Regulatory   │ Bureaucratic │
│ Resilience to    │ Low              │ Medium       │ Market       │ Depends on   │
│ manipulation     │ (1 investor)     │              │ volatility   │ sy
(decentralization)│
└──────────────────┴──────────────────┴──────────────┴──────────────┴──────────────┘
```
```

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Key inversion: in traditional models, large-scale projects require concentration of capital; in PMIP, large-scale pr through distribution across many. The more socially significant a project, the larger the SIC → the smaller the indiv the probability of realization.

#### 3.4.4. PMIP Theorems

**Theorem 1 (Universal Realizability).** For any project p with cost C(p) > 0 and any min_capacity > 0, there exis min_capacity⌉ such that for |SIC(p)| ≥ N* the project p is socially relevant (SR(p) = 1).

Proof: For N ≥ N: individual_cost = C(p)/N ≤ C(p)/N ≤ min_capacity. Therefore CC(p) = N × min_capacity ≥ N* × m

**Theorem 2 (Scale Inversion).** In the PMIP model, with a fixed size of the global NFC (|A| = const), the probabil correlates positively with its social significance.

Justification: Let the social significance of project p be defined as the fraction of NFC agents interested in its rea = |SIC(p)| / |A|. Then:

```
individual_cost(p) = C(p) / |SIC(p)| = C(p) / (significance(p) × |A|)
```

With fixed C(p) and |A|: the higher significance(p), the lower individual_cost(p), the higher the probability t min_capacity, and hence the higher the probability of SR(p) = 1.

This inverts the traditional logic, in which large-scale projects are accessible only to concentrated capital. In PM become possible precisely through distribution across many — provided the project is sufficiently significant to at

A formal proof of this theorem is an open problem and a subject for further research.

**Theorem 3 (Antifragility).** A system based on PMIP possesses the property of Antifragility in Taleb's sense number of participants does not merely make the system resilient to shocks, but strengthens its capacity to complexity.

Proof: Let δ be an economic shock reducing min_capacity by ε > 0. Then the new threshold:

```
TSR'(p) = C(p) / (min_capacity - ε)  >  TSR(p)
```

However, if simultaneously the number of participants grows: N′ = N + Δ, then for sufficient Δ:

```
N' ≥ TSR'(p) ⟺ N + Δ ≥ C(p) / (min_capacity - ε) ⟺ Δ ≥ C(p) / (min_capacity - ε) - N
```

For large N, even significant ε is offset by a small relative increment Δ/N. Moreover, economic shocks, as a rule, alternative economic models, which increases the inflow of participants into the NFC. Thus, the system not me but uses them as a source of growth — which is the defining property of Antifragility. ∎

Limitation: the Antifragility of the system depends on the type of shock. The crypto winter of 2022 demonstrate within the crypto industry may cause participant outflow rather than inflow. The theorem holds for shocks in the stimulate a search for alternatives, but requires empirical verification for endogenous crypto-economic crises theorem is an open problem and a subject for further research.

**Theorem 4 ("Millions are Billionaires").** With a sufficient number of NFC participants, the collective capital reache the budgets of the largest institutions and state programs, at individual costs that are negligibly small for each par

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Demonstration:

```
Scenario A: N = 10⁶ (one million participants) min_capacity = $10/month CC = $10⁷/month = $120M/year

Scenario B: N = 10⁷ (ten million participants) min_capacity = $5/month CC = $5 × 10⁷/month = $600M/year

Scenario C: N = 10⁸ (one hundred million participants) min_capacity = $3/month CC = $3 × 10⁸/month = $3.6B/yea

Scenario D: N = 10⁹ (one billion participants) min_capacity = $1/month CC = $10⁹/month = $12B/year

For comparison: — Annual CERN budget: ~$1.2B — Annual NASA budget: ~$25B — Annual National Institute
budget: ~$47B — Total global venture capital market volume (2023): ~$345B — Scenario D at $1/month covers C
— Scenario C at $3/month exceeds ESA's budget (European Space Agency, ~$7.7B) in 2.1 years
```

Corollary: the statement "Millions are Billionaires" is not a slogan but a mathematical consequence of PMIP. This
The question lies not in the mathematics — it lies in creating a coordination infrastructure that ensures: (a) forma
scale; (b) transparent and verifiable fund collection; (c) accountable project implementation; (d) fair distribu
precisely the subject of GyberExperiment. ∎

A formal proof of this theorem is an open problem and a subject for further research.

**Corollary 4.1 (AI Augmentation of PMIP).** Artificial intelligence accelerates the convergence of PMIP toward a zero
threshold simultaneously from two directions, introducing a qualitatively new dynamic:

First direction — reduction of C(p):

AI-assisted development (code generation, automated testing, AI security auditing of smart contracts) reduc
realization. AI coordination within AG(p) reduces management overhead. AI analytics prevent costly design
Therefore:

```
C(p | AI) < C(p)

individual_cost(p | AI) = C(p | AI) / |SIC(p)| < C(p) / |SIC(p)|
```

Second direction — increase of effective |SIC|:

Coordination costs at large |SIC| are a fundamental limitation of PMIP (see Loop 5, Section 3.6). Brooks sh
organizations coordination costs grow as O(n²). In the CSC, formalized protocols reduce this growth to O(n
mediator — aggregating the positions of thousands of participants, clustering opinions, formulating decision optio
the growth of coordination costs to O(n):

```
Without AI: effective_capacity(SIC) = |SIC| / coordination_overhead(|SIC|) coordination_overhead ∝ log(|SIC|)

With AI: effective_capacity(SIC | AI) ≈ |SIC| coordination_overhead ≈ const (AI scales trivially)
```

Combined effect:

```
individual_cost(p | AI) = C(p | AI) / effective_SIC(p | AI)
```

With C(p | AI) < C(p) and effective_SIC(p | AI) > effective_SIC(p): individual_cost(p | AI) << individual_cost(p)

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AI accelerates PMIP's convergence to zero EXPONENTIALLY, not linearly — which in practice means that Relevance TSR(p) is crossed for projects of any cost with significantly fewer participants than predicted by the mo

This corollary has fundamental significance: AI is not an external supplement to PMIP, but a structural factor that limitations of a theoretically sound mechanism. AI coordination substantially reduces the communication com ($O(n^2) \rightarrow O(n)$ for routine coordination tasks), although the core of the law — the complexity of judgment coordin strategic decisions (Axiom A8). ∎

#### 3.4.5. Mechanism for Resolving Internal Contradictions: Fork Resolution Protocol

Any system of collective decision-making confronts the problem of internal disagreements. Traditional hierarchi through the mechanism of managerial authority — the decision is made by whoever is higher in the hierarch mechanism exists by definition (Axiom A1). A formalized protocol is required that preserves decentralization.

**The cost estimation problem.**

Precise determination of C(p) is difficult in the early stages of a project. An underestimate leads to underfunding to inefficient resource allocation and an artificially inflated TSR.

Solution: iterative cost estimation model — the initial estimate is formulated by the idea's author at the Discussion stage the estimate is revised with the participation of AG and CG (if engaged); the final fixation of C(p) oc Accumulation, when the project's internal tokens are issued. In addition, the modular structure of the project allo decomposed into independent subprojects, which has demonstrated both potential and risks; MolochD Governance (2020), Gitcoin (2019), Nouns DAO (2021), ConstitutionDAO (2021).

Significance: Demonstrated the ability of distributed groups to coordinate significant resources without ce ConstitutionDAO raised $47M in 72 hours from >17,000 participants — an empirical demonstration of PMIP- like a formal theory.

At the same time, key problems were identified: tokenholder plutocracy, low voting turnout, vulnerability to govern

Theorists: Buterin [46], Weyl and Posner [44], Zargent.

Stage 4 — Cybersocial Economics (emerging): CSC as the fundamental economic unit. PMIP as a financing mech selection mechanism. MacroeconomicDAO as a tool for macroeconomic governance.

Marker: GyberExperiment.

Significance: Transition from isolated experiments with DAO and DeFi to a coherent system integrating implementation, and distribution of results into a single theoretically grounded framework. ```

### 3.5. Socio-Economic Selection and Project Evolution

The PMIP mechanism resolves the financing problem: how to provide resources for projects without concent equally important is the question of **selection**: which projects to realize? In the traditional economy, selection is market (projects that maximize profit) or through the state (projects determined by political will). The CSC offers a mechanism.

#### 3.5.1. Definition of SES

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**Definition 3.5.1 (Socio-Economic Selection, SES).** SES is an evolutionary mechanism in which projects compete but for **societal interest**. The winner is not the project that is most profitable for the investor, but the one that is r that is, interesting to the greatest number of people.

Formally: let there be a set of projects $P = \{p_1, p_2, ..., p_m\}$ competing for NFC resources. For each project $p_i$ a SIC is realized if and only if $SR(p_i) = 1$. Thus, SES is a **decentralized filter**: no central authority makes the decision decision is made by the aggregate of individual decisions by agents to join the SIC.

This mechanism is analogous to natural selection in biology, but operates in the space of ideas and projects. selects organisms by their fitness to the environment, SES selects projects by their fitness to the needs of society.

#### 3.5.2. Stages of Cybersocialization of the Economy

**Definition 3.5.2 (Cybersocialization of the Economy).** The Cybersocialization of the Economy is a historical proc global socio-economic relations, in which: information technologies and cryptographic protocols become not to the **habitat** of economic relations; governance of economic processes passes from hierarchical institutions to mechanisms; the economic unit transforms from a closed corporation into an open CSC; a collective economic the capacity of society to make rational collective decisions about resource allocation through formalized mechan

``` Stages of cybersocialization:

Stage 0 — Prerequisites (1970–2000): Development of the internet, free software, e-commerce. Markers: ARPANE Amazon, eBay

Stage 1 — Digitization of exchange (2000–2010): Social networks, platforms, Web 2.0. Markers: Google, Facebook

Stage 2 — Decentralization of finance (2009–2020): Blockchain, cryptocurrencies, DeFi. Markers: Bitcoi MakerDAO

Stage 3 — Decentralization of governance (2016–present): DAO, on-chain governance, token-weighted voting. M MolochDAO, Gitcoin, Nouns DAO

Stage 4 — Cybersocial Economics (emerging): CSC as the fundamental economic unit. PMIP as a MacroeconomicDAO. Marker: GyberExperiment ```

#### 3.5.3. Historical Analogy

For a deeper understanding of the scale and character of the transformation underway, it is useful to draw an an Revolution — the last phase transition of comparable scale.

Carlota Perez, in her foundational work “Technological Revolutions and Financial Capital” (2002), showed t revolution passes through a cycle: installation of a new technological paradigm → financial bubble → crisis → which the institutional structure finally adapts to the new technology [42]. We suggest that the digital revolution bubble/crisis phase (the dot-com crash of 2000, the 2008 crisis, the crypto winter of 2022) to the deployme institutional innovation — the CSC — becomes the dominant form.

``` Phase 0 — Monarchical Centralization (before the 18th century): $\Omega_0 \approx \langle 0.05, 0.05, 0.02, 0.10, 0.01 \rangle$

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| --- |
| Characteristics: Governance of economic processes concentrated in the hands of the sovereign. Trade guilds operate under royal privilege. Transparency is absent. Participation is determined by social estate. Coordination and tradition. |
| Dominant form of economic unit: manufactory, trading house, colonial company. |
| Phase 1 — Parliamentary Capitalism (18th–19th centuries): $\Omega_1 \approx \langle 0.15, 0.15, 0.10, 0.20, 0.05 \rangle$ |
| Transition: $\delta(D) \approx 0.10$ — governance functions passed from the monarch to parliament, representing the interes forces. |
| Characteristics: The emergence of the joint-stock company as a means of distributing risk and ownership. The The stock exchange as a mechanism for coordinating capital. However, participation is limited by property qu class is excluded from governance. |
| Dominant form: joint-stock company, bank. |
| Key theorists: Adam Smith [26], David Ricardo. |
| Phase 2 — Corporate Capitalism (late 19th – mid-20th century): $\Omega_2 \approx \langle 0.15, 0.20, 0.15, 0.15, 0.10 \rangle$ |
| Transition: $\delta(C) \approx 0.05$ — emergence of management as a formalized discipline of coordination (Taylor, Ford). δ(A participation through mass public securities markets. |
| Characteristics: Transnational corporations. Separation of ownership from management (Berle and Means, revolution. Keynesian state regulation as compensation for market failures. |
| Paradox: D does not grow, and in some respects declines — concentration of control in the hands of managemen dispersed. |
| Dominant form: public corporation, state enterprise. |
| Phase 3 — Financial Capitalism / Neoliberalism (1970s – 2008): $\Omega_3 \approx \langle 0.12, 0.18, 0.20, 0.12, 0.15 \rangle$ |
| Transition: $\delta(C) \approx 0.05$ — algorithmic trading, derivatives, securitization. $\delta(A) \approx 0.05$ — expansion of retail investr — effective decline in decentralization: financial conglomerates, concentration of banking capital, too big to fai participant sovereignty: growing dependence on financial intermediaries. |
| Characteristics: Economic hegemony of the financial sector. Globalization as intensification of financial flows globalization of governance. Result — the 2008 crisis as a systemic failure of an opaque, hyper-concentrated sys |
| Dominant form: investment bank, hedge fund, financial conglomerate. |
| Critics: Stiglitz [9], Piketty [8], Minsky. |
| Phase 4 — Platform Capitalism (2000s – present): $\Omega_4 \approx \langle 0.10, 0.15, 0.35, 0.08, 0.30 \rangle$ |

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Transition: δ(A) ≈ 0.15 — radical reduction of participation barriers (any person with a phone is a user and a value — algorithmic coordination through platforms (Uber, Airbnb, Amazon Marketplace). However: δ(D) < 0 — decentralization: GAFAM controls ~$10T+ in market capitalization, the data of billions of people. δ(S) < 0 — minir user owns neither the account, nor the data, nor the algorithm. δ(T) < 0 — decline in transparency: closed recommendations, and pricing.

Phase 4 Paradox: maximum accessibility of participation combined with minimum sovereignty and minimum dec people are involved in creating value, but are deprived of control over the results of their labour. This structu driving force behind the transition to Phase 5.

Critics: Zuboff [6], Srnicek [7].

Dominant form: platform (Google, Meta, Amazon).

Phase 5 — Declaration: Cybersocial Economics (emerging): $\Omega_5^* \approx \langle 0.85, 0.90, 0.90, 0.95, 0.85 \rangle$ (target state)

Transition: Simultaneous jump of all components. This is not an incremental improvement of one parameter (as in phase transition in the full sense — a qualitative change in the nature of the organizational form. AI (A8) is a nece transition: D=0.85 requires decentralized monitoring, T=0.90 — processing petabytes of data, C=0.85 — algo millions of participants. Without AI these target values are unattainable.

Characteristics: CSC as the dominant form of economic unit. PMIP as a financing mechanism. SES as a projec MacroeconomicDAO as a coordination tool.

Marker: GyberExperiment as the first empirical realization. ```

> 📊 **Professional diagram:** Open SVG

Figure 3.1: Structural analogy between the Industrial Revolution (18th–19th centuries) and Cybersocializatio technological revolution creates new organizational forms (factory/CSC), economic units (joint-stock company/Cy coordination mechanisms (stock exchange/MacroeconomicDAO), social forces (working class/digital intelligen institutional infrastructure (legislation, theory, philosophy).

The analogy is not superficial. Each element has a deep structural correspondence. Just as the joint-stock comp large enterprises (transoceanic trade, railway construction) to be distributed across many investors, so the CSC t costs of large-scale projects to be distributed across an arbitrarily large number of participants. Just as the sto mechanism for price discovery and aggregation of dispersed capital, so MacroeconomicDAO creates a mechanism Relevance and aggregating dispersed interest. Just as the industrial working class became conscious of itsel formed institutions to protect its interests (trade unions, workers' parties), so the digital intelligentsia becomes co the Open Source movement, crypto-communities, and DAO experiments, and forms institutions of a new type global.

Of course, the analogy has its limits. The Industrial Revolution took place in a world of nation-states an infrastructure (factories, railways). Cybersocialization takes place in a global digital space and is largely ab geography. This means that the pace of transformation may be substantially higher, and its scope — globa bypassing the stage of national-scale development.

#### 3.5.4. AI-Directed Evolution: from Blind Selection to Designed Selection

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The SES model described above represents an analogue of biological evolution (formalized in detail in Secti
However, biological evolution has a fundamental limitation — it is **blind**: mutations are random, selection is
organisms perish in order to consolidate one successful adaptation.

In the CSC a Lamarckian component is already present — acquired traits (experience, code, infrastructure) are i
projects. Artificial intelligence introduces a qualitatively new element — **foresight** — transforming SES from an
process into a designed one:

```` Biological evolution: Mutation (random) → Selection (blind) → Inheritance

SES without AI: Idea (from human) → SES (SR as criterion) → Inheritance of code/experience

SES with AI (directed evolution): AI environment analysis → Generation of directed ideas → AI forecast of SR
Inheritance + AI optimization of the inherited ```

**AI-generated ideas based on ecosystem gap analysis.** An AI agent with access to the full map of projects, t
outcomes is capable of identifying: unfilled niches ("the ecosystem has X and Y, but lacks Z to connect them"
projects are blocked by the absence of component W"), external trends ("demand for category K is growi
competencies to realize it"). Ideas prompted or generated by AI have a higher probability of being socially rele
from individual agents. At the same time, AI-generated proposals are obligatorily marked as such — the decision
with the community (Axiom A7).

**AI predictor of Social Relevance.** Before the author invests significant effort in a full project specification, AI is ca
potential |SIC(p)| based on precedent data, modelling TSR(p) for various budget parameters, and recommen
increase SR. This reduces "evolutionary waste" — efforts expended on projects doomed to SR = 0.

**AI augmentation of FRP.** Fork Resolution Protocol (Section 3.4.5) describes a fork as a last resort; however, a fo
duplication of effort, community split. AI optimizes each stage of FRP: - Deliberation stage: AI clusters particip
points of agreement and real (rather than apparent) points of divergence, visualizes the argument space - Synth
compromise solutions v, maximizing |support(v)| / |SIC(p)|, based on analysis of each group's preferences - Forl
AI models the viability of each branch — forecasts |SIC| for each option, assesses resource sufficiency, warns of

Formally, AI-enhanced SES raises the success rate of the system through three channels:

```` Success_rate_AI = f(directed_diversity, enhanced_selection, accelerated_inheritance)

directed_diversity > random_diversity (AI generates ideas in promising directions)

enhanced_selection > blind_selection (AI reduces noise in SR evaluation — see property E1 below)

accelerated_inheritance > passive_inheritance (AI systematizes and makes instantly accessible the experience of
property E6 below) ```

Analogy with biology: AI transforms SES from a process of natural selection into a process resembling genetic
modifications with predictable outcomes, while maintaining evolutionary diversity through FRP.

**AI transformation of the IPI lifecycle.** At each phase of the Idea-Project-Implementation model, AI introd
capabilities:

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- **Idea phase:** three parallel streams — (1) a human proposes an idea, AI validates it; (2) AI identifies ecosyst proposals; (3) AI forms an "opportunity map" based on environmental analysis - **Formation phase:** AI automat specification based on discussions, identifies unaddressed risks, proposes an optimal composition of AG(p) base participant workloads - **Accumulation phase:** AI forecasts the rate of accumulation, optimizes parameters of inte stability of the future LP - **Implementation phase:** AI as co-developer — code generation, automated testi auditing, automated documentation. AI Project Manager — task decomposition, optimal assignment, earl **Functioning phase:** AI monitoring of metrics, anomaly detection, parameter optimization, revenue and load foreca

Combined effect: AI reduces the IPI cycle time (from idea to functioning) by a factor of 3–5, which implies a prop Loop 1 (Growth Loop, Section 3.6.3) and, consequently, exponential acceleration of the entire ecosystem's develo

#### 3.5.5. Formal Properties of SES: Game-Theoretic Analysis

Socio-Economic Selection can be formalized in terms of evolutionary game theory (Maynard Smith, 1982 [81]), its stability and efficiency.

**Definition 3.5.3 (SES as an evolutionary game).** Let the strategy of agent $a \in A$ be described by a resource allocat

```
σ(a) = (w₁, w₂, ..., wₘ), Σwᵢ = 1, wᵢ ≥ 0
```

where $w_i$ is the share of resources allocated by agent a to project $p_i \in P$.

Payoff function of the agent: $u(a) = \Sigma_i w_i \times R(p_i) \times q(a, p_i)$

where: $R(p_i)$ — total value of the project (0 if SR=0) $q(a, p_i)$ — share of the project's value accruing to agent a (d status, size of SIC) ```

**Theorem 3.5.1 (Pareto Optimality of SES).** In the stationary state, under full information and rational agents, the re is Pareto-optimal: it is impossible to improve the position of one agent without worsening the position of another.

Justification: In the stationary state each agent allocates resources to projects with maximum expected payof reallocation existed, rational agents would already have moved to it. The condition SR(p) = 1 ensures that only societal interest receive resources, which corresponds to the social optimum. Limitation: full information is not agents partially compensate for this by aggregating and analysing information (A8). A formal proof under c information is an open problem. ∎

**Theorem 3.5.2 (Evolutionary Stability of Cooperation in SES).** The strategy "participation in SIC" is an Evolut (ESS) provided $|NFC| \geq N_critical$, where N_critical depends on the reward parameters.

Justification: Consider two strategies: C (cooperator — participates in SIC and contributes) and D (defector participate, but uses the publicly available results). In traditional models D dominates C (the free-rider problem, this dynamic is inverted:

```
Payoff of cooperator: u(C) = R(p) × q(C, p) − cost(C, p) where q(C, p) > 0 (reward through UnitManager, re
access to services)
```

Payoff of defector: $u(D) = external_benefit(p) - 0$ where $external_benefit(p) << R(p) \times q(C, p)$ (the defector externalities, but not internal rewards)

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ESS condition: $u(C) > u(D) \iff R(p) \times q(C, p) - cost(C, p) > external_benefit(p)$ ```

UnitManager ensures $q(C, p) > 0$ through direct GBR rewards. PMIP ensures $cost(C, p) \to 0$ at large $|SIC|$. Act additional gap: cooperators receive a high Activity Score (→ greater voting weight, greater rewards), defector Score (→ status degradation, loss of access). Thus, with a sufficiently large $|NFC|$ and a functioning Activity Scor of cooperation dominates free-riding. ∎

**Theorem 3.5.3 (Diversity as an Evolutionary Advantage).** A system with several competing projects in one adaptability than a system with a single project for each function.

Justification: By analogy with Page's diversity theorem [41], the aggregate prediction accuracy of a group inc approaches, not with individual accuracy of each predictor. In the context of SES: several competing implemen (made possible by FRP and the low IPI entry barrier) provide evolutionary pressure that raises quality. This form intuition: "given enough eyeballs, all bugs are shallow" (Linus's Law).

FRP is the key mechanism for maintaining diversity: rather than forcing a choice between competing approache the system allows forks — parallel development of alternatives with subsequent evaluation of results through SI not destroyed, but archived, preserving accumulated experience for future iterations. ∎

#### 3.5.6. Economic Dynamics of SES: A Formal Model

For a quantitative description of SES dynamics, we employ the Replicator Dynamics model, standard in evolutiona

Let $x(p_i, t)$ be the share of NFC resources allocated to project $p_i$ at time t. The dynamics are described by equations:

```
dx(p_i)/dt = x(p_i) × [f(p_i) − f̄]
```

where: $f(p_i) = SR(p_i) \times performance(p_i)$ — "fitness" of the project $\bar{f} = \Sigma_j x(p_j) \times f(p_j)$ — average fitness of the sy function(revenue, SIC growth, product quality, participant satisfaction) ```

**Properties of this dynamic:**

1. Positive feedback: projects with $f(p_i) > \bar{f}$ grow, attracting more resources. Projects with $f(p_i) < \bar{f}$ shrink, losing pa

2. Stationary states: the system tends toward a state in which projects with the highest fitness survive. monoculture: different niches (Social DAO, Code DAO, Commerce DAO, Economic DAO) have independent fitness

3. Role of noise: random fluctuations (new ideas, external shocks, changes in preferences) prevent prema suboptimal state (lock-in). FRP is the institutional mechanism for maintaining "noise" — diversity of approaches.

4. AI modification: the AI predictor of SR (Section 3.5.4) modifies the dynamic, accelerating reallocation of resour to strong ones. Formally: AI increases the coefficient multiplying $[f(p_i) - \bar{f}]$, accelerating convergence witho recommends, not decides; FRP preserves alternatives).

5. Meta-evolution: SES applies not only to projects but to the DAO rules themselves. Proposals to change syster thresholds, VP formula) compete through the same evolutionary mechanism: successful institutional innovatio DAOs, unsuccessful ones are rejected. This ensures adaptability of the system at the meta-level.

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**Empirical verification.** The formal SES model requires empirical calibration. The necessary data wil GyberExperiment: distribution of resources among projects, SIC dynamics, project success/failure rates, IPI c simulation (Section 6.3) allows testing of model parameters before their application in the real system.

#### 3.5.7. Mechanism Design of SES: Incentive Compatibility and Strategic Robustness

The key question for any mechanism of resource allocation is whether it is **incentive compatible** — that is, whet the optimal strategy for each participant. This question is formalized in terms of mechanism theory (Hurwicz, 19 allows formal properties of SES as an economic mechanism to be proved.

**Definition (SES mechanism).** The SES mechanism is a tuple M_SES = ⟨A, Θ, S, g, u⟩, where:

`A = {a₁, ..., aₙ} — set of participants (NFC) θ = {θ₁, ..., θₙ} — participant types (private i preferences, time resources) S = S₁ × ... × Sₙ — strategy space (for each participant: which in, what contribution to make, whom to vote for) g: S → X — outcome function (distribution projects, accrual of GBR rewards) u: X × θ → ℝ — utility function (depending on outcome and par`

**Theorem 5 (Incentive Compatibility of SES in Dominant Strategies).** Under the following conditions, the SES compatible in dominant strategies (DSIC):

(C1) Activity Score is computed from verifiable on-chain actions (not manipulable without cost); (C2) Hybrid VP i of the stake (√GBR); (C3) MemberStatus is assigned through verifiable thresholds (Activity Score ≥ θ_status); (C4 to f(Activity Score, MemberStatus).

Proof (sketch). Consider the strategy of participant $a_i$ with type $\theta_i$. The participant may deviate from honest behav

**Deviation 1: manipulation of Activity Score.** To increase Activity Score, on-chain activity is required (commits votes, DSP interactions). Falsification of on-chain data is impossible (guaranteed by blockchain cryptography). C activity (spam transactions, fictitious tasks) is detected by the AI Activity Verifier and Sybil detector (Appe undetected falsification grows with the quality of AI models (GNN precision ≥ 0.95), exceeding the potential gain. agent, the optimal strategy for increasing Activity Score = genuine activity.

**Deviation 2: strategic voting.** In Hybrid VP, voting power VP = √(GBR_staked) × AM × SB. The square root c increasing VP through GBR purchase costs quadratically more: doubling VP requires quadrupling the stake. Wit cost of manipulating a voting outcome exceeds the expected benefit (proof analogous to the analysis of quadra Weyl [29]). AM and SB depend on Activity Score and MemberStatus respectively — both based on verifiable metr

**Deviation 3: coalition attack.** A group of participants may coordinate voting to direct resources to a project co Defences: (a) LP Burn — the project initiator irreversibly burns LP tokens, making "shell projects" economically cycle requires passing through stages (Discussion → Formation → Implementation), at each of which unrelated p suspicious project; (c) SES dynamics ensure that projects without real output lose resources ($f(p_i) < \bar{f} \to dx(p_i)/dt$ of maintaining a coalition grows linearly with the number of participants and time (constant LP Burn, constant a benefit is limited to the fund of a single project. ∎

**Corollary.** SES approximates the property of Incentive Compatibility characteristic of Vickrey–Clarke–Groves (V realized through behavioural incentives (Activity Score, MemberStatus) rather than through monetary transfers. mechanism theory to systems in which a participant's utility depends not only on financial outcome, but also on s and a sense of belonging — factors that Ostrom [30] identified as key to sustainable institutions of collective actio

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#### 3.5.8. Multi-population Dynamics: Four DAOs as Ecological Niches

Classical Replicator Dynamics (Section 3.5.6) describes competition among projects within a single DAO class. architecture of CyberSocium creates multi-population dynamics, in which projects of different classes interact y different resources — analogously to ecological niches in evolutionary biology.

Let us formalize this dynamic. Let $x_k(p_i, t)$ be the share of the k-th DAO class's resources ($k \in$ {Social, Code, allocated to project $p_i$. Each DAO class has its own fitness landscape:

```` Multi-population SES:

Social DAO: f_S(p) = engagement(p) × community_growth(p) Projects compete for: attention, moderation, comm GBR from Social DAO treasury

Code DAO: f_C(p) = code_quality(p) × technical_impact(p) Projects compete for: developers, review capacity, me from Code DAO treasury + developer hours

Commerce DAO: f_M(p) = revenue(p) × scalability(p) × SR(p) Projects compete for: capital (LP Burn), users, marke from Commerce DAO treasury + LP

Economic DAO: f_E(p) = ecosystem_impact(p) × sustainability(p) Projects compete for: system parameters, strate MacroeconomicDAO treasury

Cross-class interactions: ———————————————————————————— Commerce → Code: cor demand for technical infrastructure Code → Social: technical tools improve user experience Social → Commerce users for commercial projects Economic → All: meta-rules determine the fitness landscape of all classes ```

**Theorem 6 (Stability of Multi-population SES).** In a four-class system with cross-class interactions, a stable dyna the following conditions are satisfied:

(i) fitness functions f_k are sublinear (diminishing returns to scale within each class); (ii) cross-class interactio benefit between classes); (iii) MacroeconomicDAO adjusts parameters upon deviation from equilibrium.

Sketch proof. Under condition (i), no project can capture the entire resource of a class (diminishing returns). Co the growth of one class stimulates the growth of others (positive feedback between classes). Condition (iii) creat the meta-level (MacroeconomicDAO as stabilizer). The combination of positive feedback within classes and neg levels is a classical recipe for stable complex order (Kauffman, "edge of chaos" [12]). A formal proof via dyn (existence of an Attractor) requires numerical verification through cadCAD and remains an open problem (RQ5, Se

This multi-population model explains why the four-class architecture outperforms a monolithic DAO: instead of c which projects of different types compete by incommensurable criteria, the system creates four specialized land for its type of decisions. This is analogous to biological specialization of ecological niches, which increases the ov ecosystem.
````

3.6. Formal Model of CyberSocium as a Complex Adaptive System

The theoretical constructs introduced in the preceding sections — CSC, PMIP, SES, FRP, IPI — describe individual components and mechanisms. However, CyberSocium as a coherent system possesses properties that cannot be reduced to the sum of its components' properties. To describe these **Emergent Properties**, the apparatus of the theory of Complex Adaptive Systems (CAS) is required [36, 37].

3.6.1. System Components

```
CyberSocium CS = ⟨A, P, R, T, G, Φ⟩

where:
  A = {a₁, ..., aₙ} — set of agents (NFC participants)
    Each agent aᵢ is characterized by a vector:
        aᵢ = ⟨walletᵢ, reputationᵢ, statusᵢ, skillsᵢ, capacityᵢ⟩

        where:
          walletᵢ — cryptographic address identifying the agent in the system
          reputationᵢ ∈ ℝ≥₀ — accumulated reputation, determined by the history
                               of verified activity
          statusᵢ ∈ {Unit, Dev, LeadDev, ArchDev, Core} — status in the hierarchy
                               of responsibility (not authority), determining
                               the level of reward and confirmation powers
          skillsᵢ ⊆ S — subset of the set of competencies S,
                        relevant to the ecosystem's projects
          capacityᵢ ∈ ℝ>₀ — investment capacity of the agent
                             (the resource they are prepared to direct to projects)

      P = {p₁, ..., pₘ} — set of projects (including potential ones)
        Each project pⱼ is characterized by:
        pⱼ = ⟨descriptionⱼ, C(pⱼ), stateⱼ, SIC(pⱼ), AG(pⱼ), tokensⱼ, revenueⱼ⟩

        where:
          descriptionⱼ — formalized description of the project
          C(pⱼ) ∈ ℝ>₀ — estimated cost of realization
          stateⱼ ∈ {Idea, Discussion, Formation, Accumulation,
                    Implementation, Operation, Completion} — current
                    state in the IPI model
          SIC(pⱼ) ⊆ A — Social Investment Circle
          AG(pⱼ) ⊆ SIC(pⱼ) — Active Group
          tokensⱼ — internal project tokens (issued upon
                    transition to Accumulation)
          revenueⱼ(t) — project revenue function of time
                        (defined for state = Operation)

      R = ⟨R_fin, R_int, R_rep, R_inf⟩ — aggregate resource of the system

        where:
          R_fin — financial resource: aggregate Gbr liquidity,
                  funds in pools, accumulated project revenues
          R_int — intellectual resource: aggregate competencies
                  of participants, Σᵢ |skillsᵢ|, accumulated codebase,
                  documentation, research
          R_rep — reputational resource: aggregate reputation
                  of the ecosystem as a whole (distinct from the sum
                  of participants' reputations — an Emergent Property)
          R_inf — infrastructural resource: GyberNet, GyberComputer,
                  DSP, IPFS nodes, deployed smart contracts,
                  liquidity pools

      T = {Gbr} U {token(pⱼ) | pⱼ ∈ P, stateⱼ ∈ {Accumulation,
           Implementation, Operation}} — set of tokens

        Gbr — primary utility token of the ecosystem
        token(pⱼ) — internal token of project pⱼ

        Relation between tokens: each token(pⱼ) is linked to Gbr
```

```
        through a liquidity pool created before opening
        trading on DEX. This ensures Axiom 6 (inclusiveness):
        active participants receive tokens of any project
        through Gbr, regardless of personal participation in that project.

      G = ⟨G_social, G_code, G_commerce, G_economic⟩ — governance structure

        G_social — Social DAO: decision-making within the community,
                   organization of social actions
        G_code — Code DAO: management of the platform's codebase
                 through voting on changes
                 to the main repository branch
        G_commerce — Commerce DAO: crowd-investing in business ideas,
                     proposing and realizing commercial
                     projects at the community's expense
        G_economic — Economic DAO: accumulation of social, financial,
                     and economic resources for the realization
                     of the most large-scale projects

        Each G_x is implemented as a smart contract based on Solidity
        and the OpenZeppelin library, with a proposal
        and voting mechanism.

      Φ = {PMIP, FRP, IPI, RepProtocol, StakingProtocol,
           RewardProtocol} — set of interaction protocols

        PMIP — Principle of Minimum Individual Participation
               (Section 3.4.1)
        FRP — Fork Resolution Protocol (Section 3.4.5)
        IPI — "Idea → Project → Implementation" model (Section 3.4.2)
        RepProtocol — Reputation protocol: set of rules determining
                      how an agent's activity is converted into reputation
        StakingProtocol — Staking protocol:
          Gbr staking: staking percentage determined by vote of
                       Economic DAO and adapted to the economic
                       parameters of the ecosystem
          Project token staking: 10 / [total number of issued
                       tokens] % of the specific project's revenue
        RewardProtocol — Reward protocol through UnitManager:
          Frequency: once per month
          Conditions: status verification + activity confirmation
                   through G-Plan
          Amounts by status:
            Unit:    10,000,000 Gbr
            Dev:     100,000,000 Gbr
            LeadDev: 1,000,000,000 Gbr
            ArchDev: 10,000,000,000 Gbr
            Core:    determined by the community
    ```

3.6.2. Emergent Properties

Define the properties that arise from the interaction of CS components but are not inherent to any of them individually. It is precisely these properties that make CyberSocium a qualitatively new system, rather than a simple aggregate of existing technologies.

E1 — Collective Economic Intelligence.
```
```

```
The capacity of the system to identify and realize socially significant projects
through the SES mechanism. No individual agent possesses sufficient information or
resources to determine which projects are most significant for society as a whole.
However, the aggregate behaviour of agents — SIC formation, participation voting, FRP
— produces an information signal (Social Relevance) that is a collective assessment
of the significance of a project. This is a direct analogue of "the wisdom of crowds"
in Surowiecki's formulation [40], but realized not through surveys or prediction
markets, but through the mechanism of economic participation.

Formally: let utility(p) be the true social utility of project p (unobservable). Then
SR(p), determined through |SIC(p)|, is an unbiased estimator of utility(p) provided
that: (a) agents hold independent utility assessments; (b) the cost of joining SIC(p)
is sufficiently small (ensured by PMIP) so as not to create a barrier to expressing
preferences; (c) agents act in their own interests, which on average correlate with
social utility. Condition (c) is the strongest assumption; however, Page [41] showed
that diversity of assessments in large groups compensates for individual errors and
biases, ensuring the quality of a collective decision that exceeds the quality of any
individual expert's decision.

**E2 — Antifragility.**

The system is strengthened by stresses. This property is formally proved in Theorem 3
for PMIP, but at the level of the entire CS system it manifests more broadly:

- Economic crises → growing disillusionment with traditional institutions → inflow of
participants into NFC → growth of |A| → reduction of TSR → expansion of the range of
realizable projects.
- Failure of individual project pᵢ → accumulation of experience (what does not work)
→ improvement of quality of subsequent projects. Code written for pᵢ remains open and
can be reused.
- Attack on an individual node → redistribution of load to remaining nodes +
incentive to improve security → more resilient network.

An individual agent is fragile — they may lose motivation, resources, or
competencies. An individual project is fragile — it may fail. But the aggregate of
agents and projects, connected by CS mechanisms, possesses Antifragility: stresses at
the component level strengthen the system at the level of the whole.

**E3 — Self-Scaling.**

Successful projects attract new participants → R increases → TSR decreases for future
projects → more large-scale projects are realized → even more participants are
attracted. This is a positive feedback loop without central management — a property
characteristic of complex adaptive systems [36].

Formally: let N(t) be the number of participants at time t, and let a successful
project pⱼ with revenue rⱼ attract ΔN(rⱼ) new participants (where ΔN is an increasing
function of r). Then:
```

$N(t+1) = N(t) + \Sigma_j \Delta N(r_j(t)) — attrition(t)$

where attrition(t) — participant outflow.

When $\Sigma_j \Delta N(r_j(t)) > attrition(t)$, the system grows.

With growth of N(t): — the number of proposed ideas grows (more agents → more ideas) — TSR decreases for each project (Theorem 1) — CC grows for each project — the number of realizable projects grows — total revenue $\Sigma_j r_j$ grows — ΔN grows

This is a self-sustaining growth cycle, limited only by: (a) the finite number of people on the planet; (b) the finite capacity of coordination mechanisms (this constraint is substantially mitigated by the AI coordinator, reducing costs from O(n log n) to O(n) — see Loop 5); (c) competition with alternative systems.

```
**E4 — Economic Evolution.**

Through the SES mechanism the system continuously generates, tests, and selects
projects, adapting to changes in the external environment without centralized
planning. Analogue of biological evolution:
```

Biological	evolution	CyberSocium

Mutations → New ideas proposed by agents Genetic diversity → Diversity of projects and their versions (FRP) Natural selection → Socio-Economic Selection (SES): SR as survival criterion Fitness → Social Relevance Heredity → Reuse of code, knowledge, infrastructure, experience Habitat → Aggregate of interests, needs, and resources of NFC participants Speciation → Branching of projects through FRP Co-evolution → Mutual influence of projects: the infrastructure of one creates opportunities for another Extinction → Completion of projects with SR = 0 (without loss of accumulated experience and code)

```
Unlike biological evolution, economic evolution in CS has a Lamarckian component:
acquired "traits" (experience, code, infrastructure) are inherited by subsequent
projects. This makes evolution in CS substantially faster than biological evolution,
and more directed — accumulation of knowledge ensures not random, but cumulative
progress.

**E5 — Distributed Resilience.**

Failure of any individual project, node, agent, or even group of agents does not lead
to failure of the system as a whole. This property follows from Axiom 1
(decentralization) applied to the scale of the entire system.

Formally: for any subset S' ⊂ A, |S'| < |A|/2, removal of S' does not halt the
functioning of CS. This is true because: (a) the infrastructure (GyberNet, DSP) is
distributed across nodes controlled by different agents; (b) the code is open — any
agent can bring up a new node; (c) data is replicated in the blockchain and IPFS; (d)
governance is distributed through DAO — there is no single decision-making point
whose removal would paralyze the system.

This is qualitatively different from the resilience of a traditional corporation
(dependent on management) or of existing DAOs (dependent on large tokenholders). In
CS, resilience does not depend on specific agents — it is a structural property of
the architecture.

**E6 — Knowledge Compounding.**

Each realized project pᵢ produces:

(a) Open source code, available to future projects. The code of project pᵢ may be a
library, framework, or module that reduces the cost of realizing pᵢ₊₁, ..., pₖ.

(b) Experience and competencies of AG(pᵢ) participants. A developer who participated
in realizing pᵢ possesses skills applicable in pᵢ₊₁.

(c) Infrastructure reusable in subsequent projects: deployed nodes, liquidity pools,
integrations with external services.

(d) Economic resource through Gbr liquidity: each project that enters the market
creates a liquidity pool with Gbr, increasing its aggregate liquidity and utility.

Formally:
```

$R(t+1) = R(t) + \Delta R(p_i)$

where $\Delta R(p_i) = \langle \Delta R_fin(p_i), \Delta R_int(p_i), \Delta R_rep(p_i), \Delta R_inf(p_i) \rangle$

Upon successful realization: $||\Delta R(p_i)|| > cost(p_i)$

That is: the system creates more aggregate resource than it consumes for project realization — owing to the intangible cumulative effect of knowledge.

Moreover, the cumulative effect is superlinear: $\Delta R(p_i \mid p_1, ..., p_{i-1}$ realized$) > \Delta R(p_i \mid$ only p_i realized$)$

That is: each project creates more value in the context of previously realized projects than in isolation. This is a direct consequence of the reuse of code, infrastructure, and experience.

```
This effect is one of the most powerful advantages of the CSC over traditional
organizational forms. In a traditional corporation, knowledge is proprietary and does
not spread beyond its boundaries. In the CSC, all knowledge is open by definition
(Axiom 4), allowing the entire ecosystem to benefit from every project. Raymond [20]
described this effect for individual software projects; in the CSC it scales to the
entire economic system.

#### 3.6.3. System Dynamics — Formalization of Feedback Loops

The behaviour of CyberSocium as a CAS is determined by the interaction of positive
(reinforcing) and negative (stabilizing) feedback loops. Donella Meadows in "Thinking
in Systems" [39] showed that it is the structure of feedback loops, not the
components in isolation, that determines the long-term behaviour of a system.

**Positive feedback loops (reinforcing):**
```

Loop 1 — Growth Loop:

More participants (↑A) → more ideas proposed to the project space → more projects proceed through IPI → more successful realizations → more value created in the ecosystem → higher attractiveness of NFC → more participants (↑A)

Speed of loop: determined by the IPI cycle time (from idea to first results). Optimizing this time is one of the key infrastructure tasks.

Loop 2 — Liquidity Loop:

More projects enter the market → more liquidity pools created with Gbr → higher aggregate Gbr liquidity → higher value of Gbr as a medium of exchange → greater reward for active participants (in real terms) → higher motivation for active participation → more and better quality projects → more projects enter the market

Key role: the mechanism described in the original documentation — “all projects create a liquidity pool with Gbr before opening trading on DEX, allowing active community members to receive tokens of any project, regardless of personal participation in that project” — is the implementation of Axiom 6 (inclusiveness) and the key mechanism of this loop.

Loop 3 — Competence Loop:

More realized projects → more experience accumulated in the community (↑R_int) → higher quality of new projects → higher percentage of successful realizations → higher ecosystem reputation (↑R_rep) → attraction of more qualified participants → even more experience

This loop explains why the CSC becomes more efficient over time, not less — unlike bureaucratic organizations, where growth in size is usually accompanied by a decline in efficiency (Parkinson's law).

```
**Negative feedback loops (stabilizing):**
```

Loop 4 — Quality Filter:

Growth in the number of proposed projects → decline in average quality of ideas (more participants → more inexperienced → more weak ideas) → SES mechanism filters out irrelevant projects (low SIC → SR = 0 → project does not proceed to Accumulation) → only projects with high Social Relevance are realized → increase in average quality of realized projects

This is a stabilizing loop: it prevents ecosystem degradation during rapid growth of participant numbers. SES acts as an immune system, filtering out non-viable projects without a centralized "selection committee."

Loop 5 — Coordination Cost Loop:

Growth in participant numbers → growth in coordination costs (more people → harder to reach agreement → longer deliberations → more conflicts) → incentive to improve coordination tools (G-Plan, DAO, automation, AI-assisted governance) → reduction of coordination costs per participant → possibility of further growth

This loop describes the key challenge of scaling. Brooks in "The Mythical Man-Month" [50] showed that in traditional projects, coordination costs grow as $O(n^2)$, where n is the number of participants. In the CSC modular architecture and formalized protocols (PMIP, FRP) allow keeping cost growth closer to $O(n \log n)$, but this estimate requires empirical verification.

Loop 6 — Accountability Loop:

Growth of aggregate ecosystem resource (↑R) → growth of attractiveness for unscrupulous agents (fraudsters, free-riders, manipulators) → activation of the reputational mechanism: unscrupulous behaviour is detected through G-Plan and blockchain transparency → decline in reputation of the unscrupulous agent → exclusion from active roles (AG, implementor) → growth of trust in the community → growth of willingness to participate in projects → growth of resource (↑R)

Critical: the reputational mechanism works only with transparency (A2) and verifiability of activity (G-Plan). Without these conditions the loop breaks, and the system becomes vulnerable.

```
#### 3.6.4. System Attractor

**Hypothesis 3.6.1 (On the Attractor of CyberSocium).** With sufficient initial
conditions (N₀ > N_critical, R₀ > R_critical), the CS system evolves toward a state
characterized by:
```

1. |A| → A_max — coverage of a significant portion of the global population possessing access to digital technologies (as of 2025: ~5.4 billion people)
2. TSR(p) → accessibility for planetary-scale projects at individual_cost negligibly small for each participant
3. Decentralization index → max (no agent or coalition < |A|/3 controls the system)
4. Gini(R_fin within CS) → min (a more equal distribution of resources than in any existing economic system of comparable scale)
5. Rate of project realization → optimal (not maximum, but determined by quality: loop 4 ensures balance of speed and quality)
6. Diversity of projects → maximum (FRP ensures coexistence of different approaches, not convergence to a single one)

```
This state is an Attractor because the positive feedback loops (Loops 1-3) dominate
the negative ones (Loops 4-6) provided the axioms A1-A8 are maintained. Dominance
does not mean that the negative loops are suppressed — they are active and perform a
critically important stabilizing function. But the net effect of the feedback loops
is directed toward the described state.

**Theorem 3.6.1 (On the Axiomatic Stability of the Attractor).** Loss of any of the
axioms A1-A8 transfers the system to a trajectory toward a different Attractor that
does not possess properties (a)-(f). In particular, loss of A8 (cognitive
augmentation) leads to degradation of E1 (collective intelligence) during scaling and
restoration of the coordination bottleneck (Loop 5). Analysis of degradation is
provided in Section 3.6.2.

**Corollary 3.6.1.** Axioms A1-A8 are not idealistic desiderata, but **structural
conditions for system stability**. Their violation is not an ethical but a systemic
problem, leading to degradation of Emergent Properties E1-E6 and transition to
degenerate attractors that reproduce the pathologies of existing organizational
forms. Protection of the axioms is not a question of ideology, but a question of
engineering: the axioms must be embedded in the system's architecture in such a way
that their violation is technically difficult or economically disadvantageous.

**Assessment of critical initial conditions.** Hypothesis 3.6.1 is conditional on the
sufficiency of initial conditions. Let us estimate N_critical and R_critical:
```

N_critical — minimum number of participants at which Loop 1 (growth) begins to dominate outflow:

Condition: $\Sigma_j \Delta N(r_j(t)) > attrition(t)$

Empirical estimate: based on data from existing DAO ecosystems (MakerDAO: ~70K active addresses; Gitcoin: ~300K unique participants; Ethereum governance: ~50K active voters), N_critical is estimated in the range 10^3–10^4 active participants.

At N < N_critical the system is in “manual launch” mode — growth requires targeted efforts to attract and retain participants.

At N > N_critical the system transitions to self-sustaining growth mode.

R_critical — minimum aggregate resource at which realization of the first projects is possible, creating positive feedback:

R_critical = R_inf_min + R_fin_min

R_inf_min — minimum infrastructure: deployed smart contracts, functioning DSP prototype, GyberComputer nodes, Gbr liquidity pools

R_fin_min — minimum financial resource for realization of the first cascade projects (Section 5, Roadmap): estimated at $50K–$500K (depending on the complexity of the first projects)

```
---

### 3.7. Cybersocial Industry: From Digital Governance to Real Production

#### 3.7.1. Problem Statement: The Individual Barrier to Industrial Production

Sections 3.1-3.6 described CyberSocium as a system of digital governance,
coordination, and financing. However, the theory of CyberSocium is not limited to the
digital sphere. This section formulates the next evolutionary step — the
**Cybersocial Industry** (CSI): the integration of cybersocial mechanisms into
processes of real industrial production.

The central problem addressed by CSI may be called the **individual barrier to
industrial production** — the structural impossibility for an individual or small
group to realize an idea at industrial scale. This barrier has several dimensions:
```

Individual barrier to industrial production:

1. Capital barrier: Industrial production requires significant investment: factory, equipment, raw materials, logistics. Entry threshold: $500K — $50M+ (depending on sector). Accessible to: large corporations, venture funds. Inaccessible to: individuals, small groups, communities.

2. Competence barrier: An industrial product requires competencies that no single person can cover: engineering, design, materials science, electronics, certification, logistics, marketing, legal support, scaling. Result: specialization → dependence on the corporation.

3. Coordination barrier: Even if capital and competencies are available, coordinating hundreds of participants, suppliers, and contractors requires organizational infrastructure, which is traditionally provided by corporate hierarchy.

4. Distribution barrier: Producing a product is insufficient: distribution channels, logistics networks, and after-sales service are needed. These are controlled by retail giants.

5. Certification barrier: Regulatory requirements (CE, FCC, ISO, GOST) presuppose an institutional subject: a legal entity, financial statements, liability insurance.

```
The aggregate of these barriers creates a fundamental asymmetry: **ideas are born
with individuals, but realized only by corporations**. This means that a vast layer
of potentially valuable products is never realized — not because they are technically
impossible or economically unviable, but because their authors lack access to the
institutional infrastructure of production. According to WIPO (2024), fewer than 5%
of patents granted to individual inventors reach the stage of industrial production,
versus ~35% of corporate patents.

CyberSocium proposes a systemic solution to this problem — and this is precisely what
constitutes its most ambitious long-term implication.

#### 3.7.2. The "Prompt → Product" Model: The Cybersocial Production Pipeline

The central idea of CSI is formalized as the **"prompt → product" model** — a
pipeline in which the initial impulse (an idea described by a participant) passes
through a series of cybersocial mechanisms and is transformed into a real industrial
product. Each stage of the pipeline removes one of the barriers described in 3.7.1.
```

Cybersocial production pipeline:

Prompt CSC Mechanism Result (initial impulse) (barrier removal) (intermediate product) ——————————————————— ——

“Product idea” ——————————→ Social DAO ——————————→ Validated concept (text description, (discussion, (community feedback, sketch, AI prompt) voting, demand assessment, SIC formation) AG identification)

Validated ——————————————→ AiC + Code DAO ——————→ Engineering specification concept (AI-generated (3D models, BOM, CAD models, circuit diagrams, simulation, material specification, engineer review) cost estimation)

Engineering ——————————→ Commerce DAO ——————→ Prototype specification (LP Burn = commitment, (3D printing, CNC, financing, lab samples, milestone voting, functional testing) contracting)

Prototype ——————————→ Economic DAO ——————→ Industrial product (large-scale (serial production, financing certification, through PMIP, distribution, partnership with after-sales manufacturers, service) DUNA = legal entity)

Industrial ——————————→ SES ——————————→ Evolution product (feedback, (iterations, Social new versions, Relevance, derivative products, adaptation) product ecosystem)

```
Key observation: **none of the stages requires the idea's author to independently
overcome all five barriers**. Each barrier is removed by the corresponding
cybersocial mechanism:

- **Capital barrier** is removed by PMIP: individual_cost(product) = C(product) / |
SIC(product)| → 0. If an industrial product costing $5M attracts a SIC of 50,000
participants, individual_cost = $100 — accessible to everyone.
- **Competence barrier** is removed by AiC + NFC: AI models generate engineering
specifications from text prompts; the global NFC provides access to expertise in any
domain (engineers, designers, materials scientists — all are CSC participants).
- **Coordination barrier** is removed by the four-class DAO taxonomy: Social DAO
validates the idea, Code DAO manages technical development, Commerce DAO coordinates
production, Economic DAO provides financing.
- **Distribution barrier** is removed by the scale of NFC: 50,000 SIC participants
are simultaneously the first 50,000 buyers. Decentralized distribution through the
participant network (each DSP node = a potential distribution point).
- **Certification barrier** is removed by DUNA: a Wyoming DUNA legal entity may
obtain certificates, conclude contracts with factories, and bear responsibility for
the product.

#### 3.7.3. Formalization: from PMIP to PMIP-Industrial

Let us extend the formal PMIP model for industrial production. We introduce **PMIP-
Industrial** — an extension of the Principle of Minimum Individual Participation,
accounting for the specifics of physical production.

In classical PMIP, the project cost C(p) is predominantly the cost of labour
(developer person-hours). In PMIP-Industrial, the project cost is decomposed into
components:
```

C_industrial(p) = C_design(p) + C_prototype(p) + C_tooling(p) + C_production(p) + C_certification(p) + C_distribution(p) + C_support(p)

where: C_design — cost of engineering design (reduced by AI: CAD generation, simulation) C_prototype — cost of prototyping (reduced: 3D printing, open-source hardware) C_tooling — cost of tooling for serial production (injection moulds, dies, PCB production) C_production — cost of serial production (raw materials, energy, labour, depreciation)

C_certification — cost of certification (CE, FCC, ISO; DUNA bears legal costs) C_distribution — cost of distribution (logistics, marketing, retail) C_support — cost of after-sales service (warranty, repair, updates)

Application of PMIP:

individual_cost_industrial(p) = C_industrial(p) / |SIC(p)|

Example: Product: open-source laptop C_design = $200K (AI-generated schematics + engineer review) C_prototype = $50K (10 prototypes, testing) C_tooling = $500K (injection moulds for case, PCB) C_production = $2M (first batch: 10,000 units) C_certification = $100K (FCC, CE, RoHS) C_distribution = $300K (logistics, warehouses) C_support = $100K (warranty fund, documentation) ——————————————————————————— C_industrial = $3.25M

If |SIC| = 50,000 participants: individual_cost = $3.25M / 50,000 = $65

If |SIC| = 200,000 participants: individual_cost = $3.25M / 200,000 = $16.25

Conclusion: an industrial product costing $3.25M becomes accessible to each participant for $16–$65 — without bank loans, venture capital, or corporate hierarchy.

```
**Theorem 3.7.1 (Universal Industrial Realizability).** For any industrial product p
with finite cost C_industrial(p) < ∞, there exists a threshold SIC* such that for |
SIC(p)| ≥ SIC*, individual_cost_industrial(p) ≤ ε for any given ε > 0.

*Proof* is analogous to the Theorem of Universal Realizability of PMIP (Section
3.4.3): SIC* = ⌈C_industrial(p) / ε⌉. For |SIC| ≥ SIC* we have individual_cost =
C_industrial / |SIC| ≤ C_industrial / SIC* = ε. ∎

Practical corollary: **"Millions are Factories."** If PMIP in the digital sphere
yields "millions are billionaires" (any digital project is realizable with a
sufficient NFC), then PMIP-Industrial yields: any industrial product is realizable
with a sufficient NFC. The difference lies only in the absolute value of SIC*: for a
digital product SIC* ~ 100-10,000; for an industrial product SIC* ~ 10,000-1,000,000.
But with a global NFC ≥ 10M, both ranges are achievable.

#### 3.7.4. The Role of AI in the Cybersocial Production Pipeline

Axiom A8 (Cognitive Augmentation) acquires particular significance in the context of
industrial production. If in the digital sphere AI accelerates development and
reduces cognitive load, then in the industrial sphere AI **fundamentally changes the
architecture of the production process**:

**AI as engineering translator (prompt → specification).** Modern AI models (GPT-4,
Claude, Gemini) are capable of generating structured engineering descriptions from
text prompts. Specialized models (AutoCAD AI, Fusion 360 Generative Design,
SolidWorks xDesign) generate 3D models from text and graphic specifications. In the
context of CSI this means: a participant without engineering education can describe a
product idea as a text prompt, and the AI system AiC (Contour 2) translates this
prompt into an engineering specification, ready for review by professional engineers
(the project's AG).
```

Example of “prompt → specification” translation:

Participant’s prompt: “A portable device for purifying water from contaminants, powered by solar energy, the size of a thermos, costing up to $50 in serial production. Target audience: regions without access to clean water.”

AI generation (AiC Contour 2): → Requirements analysis: portability, autonomy, cost, target purification characteristics → Review of existing solutions: UV purification, filtration, reverse osmosis, electrochemical treatment → Technology recommendation: UV-C LED + activated carbon filter (optimal efficiency/cost ratio) → Generation of 3D housing model (parametric) → BOM (Bill of Materials): ~15 components, $22 cost price → Electrical schematic: solar panel → Li-ion → UV-C LED → Performance calculation: 2L/hour at irradiance >500 W/m² → Preliminary certification: NSF/ANSI 55 (UV), IP67 → Serial cost estimate: $38 at a run of 10,000

Result: complete engineering specification, ready for review by AG (engineers, materials scientists, certification specialists)

```
**AI as production process optimizer.** At the scaling stage (Commerce DAO → Economic
DAO), AI models optimize: selection of production partners (matching factories by
competencies, capacities, geography); logistics routes; production parameters
(tolerances, yield rates, energy efficiency); pricing accounting for production,
distribution, and margin costs.

**AI as generator of derivative products.** After successful production of the first
product, AI models analyse feedback (SES) and generate proposals for: modifications
(size, colour, material); accessories; compatible products; products for adjacent
markets. This launches a SES cycle at the level of a product line, not an individual
product.

#### 3.7.5. Cybersocial Industry and the Global Economy

CSI represents not merely a technical possibility, but a potential structural
transformation of the global economy. If CyberSocium in the digital sphere competes
with technology corporations (Google, Meta, Apple), then CSI competes with industrial
giants (Foxconn, Samsung, Siemens) — not by creating an alternative industrial
infrastructure, but by **democratizing access to the existing one**.
```

Traditional model Cybersocial model of industrial production of industrial production (CSI)

Idea → Corporation: Idea → Prompt: Inventor sells Participant describes idea patent to corporation in DSP; AI translates (or founds startup into specification with venture financing)

Development → R&D department: Development → AiC + AG: Company engineers AI generates models; design product engineers from NFC review (closed process) (open process)

Production → Factory: Production → Contract Manufacturing: Corporation builds/rents Commerce DAO contracts production facilities existing factories (capex = $10M+) (DUNA = legal entity; payment from PMIP fund)

Distribution → Retail: Distribution → NFC: Corporation concludes SIC(p) = first buyers; agreements with retailers decentralized logistics (Walmart, Amazon, etc.) through DUNA partners

Profit → Shareholders: Revenue → CSC: Profit distributed Revenue distributed: among shareholders 10% → CSC (MacroeconomicDAO) proportional to capital AG Reward → AG participants LP Burn → permanent liquidity Buyback → GBR ecosystem

Control → Board of directors: Control → DAO: Decisions made by Decisions made by 10–15 people all SIC participants through on behalf of shareholders four-class DAO system

Scale → Limited by Scale → Limited only by corporation capital: NFC size: $10M → 1 factory
PMIP: $10M / 100K = $100/person

Three fundamental advantages of CSI over the traditional model:

1. Democratization of initiative. In the traditional model, the right to initiate an industrial product belongs to those who control capital (corporations, VC). In CSI, the right of initiative belongs to every NFC participant — it suffices to describe an idea in a prompt and attract a SIC. This removes the filter "the idea must be approved by an investor" and replaces it with the filter "the idea must be socially relevant" (SR(p) = 1).

2. Alignment of incentives. In the traditional model, there is a conflict between: shareholders (maximizing profit), workers (maximizing wages), consumers (minimizing price), society (minimizing externalities). In CSI, these roles are combined: a SIC participant is simultaneously an investor (contributes individual_cost), a developer/worker (participates in AG), a consumer (receives the product), and a manager (votes in DAO). The conflict of interest is structurally eliminated.

3. Adaptability through SES. A traditional corporation is committed to its products (sunk cost fallacy, organizational inertia). CSI through the SES mechanism ensures evolutionary selection: products with high SR(p) receive more resources; products with low SR(p) receive less. There is no corporate policy protecting unprofitable product lines. There are no "sacred cows." Only Social Relevance determines a product's survival.

3.7.6. Precedents and Analogues

CSI does not arise in a vacuum. Precedents exist that demonstrate individual elements of the model:

Crowdfunding (Kickstarter, Indiegogo). Removes the capital barrier, but does not remove the competence, coordination, or certification barriers. The project author must independently organize production. According to Kickstarter data (2024), only 35% of successfully funded hardware projects deliver the product on time; 9% never deliver. CSI differs: the DAO structure provides coordination; AiC provides competencies; DUNA provides certification.

Open-Source Hardware (Arduino, RISC-V, Open Source Ecology). Removes the competence barrier through open specifications, but does not remove the capital or coordination barriers. Production remains the responsibility of the individual or a commercial company. CSI differs: PMIP provides financing; Commerce DAO coordinates production.

Platform Cooperativism (Mondragon, Fairphone). Removes the incentive alignment problem through cooperative ownership, but is limited geographically and in scale. Fairphone — perhaps the closest analogue: an ethical smartphone produced with regard for the interests of all stakeholders. But Fairphone remains a traditional company (BV in the Netherlands), managed by managers rather than by a community. CSI differs: global scale; DAO governance; AI augmentation.

Contract Manufacturing (Foxconn, Jabil). The contract manufacturing industry demonstrates that physical ownership of a factory is not a necessary condition for production: Apple does not own a single factory. CSI uses the same model: DUNA contracts production through Commerce DAO, without owning production facilities. The difference: in the traditional model, contract manufacturing is accessible only to large corporations (MOQ = 100K+ units); CSI through PMIP aggregates demand and reaches minimum order volumes through the scale of NFC.

3.7.7. Formal Conditions of CSI Realizability

Cybersocial Industry is realizable upon satisfaction of four conditions:

Condition 1: Sufficient NFC. |NFC| ≥ 10^5 (one hundred thousand active participants) ensures: sufficient SIC size for any product with C_industrial ≤ $10M; sufficient diversity of competencies (engineers, designers, lawyers, marketers); sufficient market for the first batch. At |NFC| ≥ 10^6 (one million), products with C_industrial ≤ $100M — the scale of the automotive industry — become realizable.

Condition 2: Maturity of AiC. AI models of Contour 2 must ensure: generation of engineering specifications from text prompts with accuracy sufficient for review (rather than complete design from scratch); optimization of production parameters; market analysis and pricing. Current state of AI (2026): partially achievable for simple products; full realization — a horizon of 3-5 years.

Condition 3: Functioning DAO infrastructure. The four-class DAO system must ensure: coordination of dozens of parallel production projects; milestone voting with integration of production metrics; management of financial flows (PMIP → contract manufacturing → distribution → revenue → CSC). Current state: prototype (Phase I); full realization — Phase III.

Condition 4: Legal infrastructure. DUNA must be capable of: concluding contracts with manufacturing factories; obtaining industrial certificates; bearing responsibility for the product before consumers; providing warranty and after-sales service. Current state: DUNA registered; expansion of functions — Phase II-III.

3.7.8. Vision: from the Digital Economy to the Cybersocial Industry

If CyberSocium in the digital sphere answers the question "how to organize collective governance?", then CSI answers the question "how to organize collective production?" Together they form a complete model of the cybersocial economy — from idea to physical product, from prompt to industrial batch, from individual to global ecosystem.

A prompt is not merely a text input. A prompt is a **formalized intention** which, passing through cybersocial mechanisms (SIC formation, DAO voting, AI translation, PMIP financing, contract manufacturing), acquires material form. In this lies the fundamental difference of CSI from all preceding models: not capital produces a product, but Social Relevance. Capital (PMIP) is merely a consequence of relevance (SR(p) = 1), not its cause.

This is an inversion of the traditional economic logic: not "there is capital → we produce a product → we seek a buyer," but "there is relevance → SIC forms → capital = consequence → product = result of collective action." Industrial production ceases to be a privilege of those who control capital, and becomes a possibility for anyone whose idea is socially relevant.

---

4. Architecture of GyberExperiment: Technical Specification

4.1. From Theory to Implementation: A Correspondence Cartography

Sections 1-3 established the theoretical foundation: they defined the problem, formulated the axioms, introduced the formal models CSC, PMIP, SES, FRP, described CyberSocium as a complex adaptive system, and articulated the concept of cyber-social industry. The present section translates theory into architecture — describing how each theoretical construct is realised technically within the GyberExperiment

```
framework.

![Correspondence of theory and implementation](diagrams/architecture/
theory_implementation_map.svg)

*Figure 4.2: Correspondence cartography between the theoretical concepts of
CyberSocium (left column) and their technical realisation in GyberExperiment (right
column). Each axiom (A1-A7), mechanism (PMIP, SES, IPI), and system component (GBR,
Agents, Resources) has an explicit technical embodiment in the architecture.*

**High-level diagram:**
```

```
┌──────────────────────────────┐
│      MacroeconomicDAO        │
│  ┌─────────┐  ┌─────────┐    │
│  │Social   │  │ Code    │    │
│  │  DAO    │  │  DAO    │    │
│  └─────────┘  └─────────┘    │
│                              │
│  ┌─────────┐  ┌─────────┐    │
│  │Commerce │  │Economic │    │
│  │  DAO    │  │  DAO    │    │
│  └─────────┘  └─────────┘    │
└──────────────────────────────┘
                │
                │ governance
                ▼
┌──────────────────────────────┐
│     GyberNet Blockchain      │
│  (Security & Transparency)   │
└──────────────────────────────┘
                │
                │ trust layer
```

```
┌──────────────────────────────────────────────────────────────────────────────────────────────┐
│        │        │        ┌──────────────────────▼──────────────────────┐
┌──────────────────────▼──────────────────────┐
┌──────────────────────────────────────▼──────────────────────────────────────┐ │ Decentralised │ │ GyberComputer
│ │ Gbr Token Economy │ │ Social Platform │ │ (Distributed │ │ │ │ │ │
Computing) │ │ ┌──────────────────────────────────────────────┐ │ │
┌──────────────────────────────────────────┐ │ │ │ │ │ UnitManager │ │ │ │ Project
Spaces │ │ │ ┌──────────────────────────────┐ │ │ │ (BSC, Solidity) │ │ │ │ Chat /
Voice │ │ │ │ AI Models │ │ │ ├──────────────────────────────────────────┤ │ │ │ Video
Hosting │ │ │ │ (AiC Project) │ │ │ │ Project Liquidity │ │ │ │ File Sharing │ │ │
│ Compute │ │ │ │ Pools │ │ │ │ G-Plan (Tasks) │ │ │ │ Services │ │ │ │
Staking Contracts │ │ │ │ Reputation │ │ │ └──────────────────────────────────────┘ │ │ │
Reputation Oracle │ │ │ │ Shared Spaces │ │ │ │ │
└──────────────────────────────────────────┘ │ │ └──────────────────────────────────────────┘
│ │ │ │ │ └──────────────────────────────────────────────────────────────────────────────────────┘
└──────────────────────────────────────────┘
└──────────────────────────────────────────────────────────────────────────────────────────────┘ │ │ │
└──────────────────────────────────────────────────────────────────────────────────────────────┘
```

```
| data layer ┌──────────────────────────────▼──────────────────────────────┐ | IPFS |
|   (Distributed   File   Storage)   |   |   +   Community   Pinning   Nodes   |
└──────────────────────────────────────────────────────────────────────────┘
```

```
> 📊 **Professional version:** [Open SVG](diagrams/architecture/
system_architecture.svg)

*Figure 4.1: Complete architecture of the GyberExperiment ecosystem, showing four
layers: (1) Governance Layer (MacroeconomicDAO with four DAO classes), (2) Trust
Layer (GyberNet Blockchain), (3) Application Layer (DSP, GyberComputer, GBR
tokenomics), and (4) Data Layer (IPFS with community pinning nodes).*

#### 4.1.1. Formal Correspondence Matrix: Theory → Implementation

Every theoretical construct introduced in Sections 1–3 must have an explicit,
verifiable technical embodiment. The complete correspondence matrix below links
abstract concepts to concrete contracts, protocols, and system components.

**Axioms → technical mechanisms:**

| Axiom | Formulation | Technical realisation | Contract / Module | Verification |
|---|---|---|---|---|
| A1 (Decentralisation) | Absence of a single point of control | UUPS proxy + Gnosis
Safe (M-of-N) + Timelock | `UnitManager.sol`, `Governor.sol` | Gnosis Safe: ≥3-of-5
signers |
| A2 (Transparency) | All economic decisions on-chain | BSC → GyberNet: all votes,
transactions, statuses recorded on blockchain | `Governor.sol`, BSC explorer |
Verification via Etherscan/BscScan |
| A3 (Data sovereignty) | User = sole data owner | 3-layer encryption in DSP; DID
standard; Verifiable Credentials | DSP encryption layer, DDP (IPFS) | Proof: server
has no decryption keys |
| A4 (Extensibility) | Unlimited modularity | SAPP standard (Socio-App Protocol
Package); Plugin API in DSP | DSP Module Registry | Any participant may publish a
module |
| A5 (Meritocratic fairness) | Weight proportional to contribution | Hybrid Voting
Power: VP = √(GBR_staked) × ActivityMultiplier × StatusBonus | `GBRVotingWrapper.sol`
| Activity Score 0–100, transparent formula |
| A6 (Inclusivity) | Minimal entry threshold | UnitManager: free registration; PMIP:
individual_cost → 0 | `UnitManager.sol` | MemberStatus.Unit = base level |
| A7 (Self-governance) | All rules modifiable through DAO | OpenZeppelin Governor +
TimelockController | `Governor.sol`, `TimelockController.sol` | 4 DAO classes with
parameters |
| A8 (Cognitive augmentation) | AI augments but does not replace | AiC: three
circuits (internal, research, commercial); soulbound audit tokens | AiC Agent
Framework | Division principle: AI ⊂ {recommends, decides} |

**Mechanisms → technical components:**

| Mechanism | Formal model | Technical realisation | Parameters |
|---|---|---|---|
| PMIP | individual_cost(p) = C(p) / \|SIC(p)\| → 0 | LP Burn: participant burns LP
tokens to initiate a project; UnitManager distributes rewards | burn_threshold =
$1,000 in GBR/USDT LP |
| SES | Fitness f(p) = w₁·SR(p) + w₂·EV(p) - w₃·C(p) | Commerce DAO + Economic DAO
voting; cadCAD parameter simulation | w₁ = 0.4, w₂ = 0.4, w₃ = 0.2 (calibrable) |
| IPI (Idea→Project→Implementation) | Finite automaton: Idea → Project →
Implementation | Social DAO voting (Idea→Project); Commerce DAO management
(Project→Impl) | Idea: quorum 5%, 72h; Project: quorum 15% |
| FRP (Fork Resolution) | Protocol for resolving conflicting bifurcations | On-chain
Dispute Resolution: escrow, arbitration, fork-voting | Cooling period: 7 days;
arbitration quorum: 30% |
```

```
| CAS (Complex Adaptive System) | Agents + environment + rules → emergence | All
components together: DSP (agents), GyberNet (environment), MacroDAO (rules) |
Monitoring via cadCAD + Dune Analytics |

**System properties → observable metrics:**

| Property (Section 3.6) | Formal definition | Metric | Measurement instrument |
|---|---|---|---|
| Self-organisation | Spontaneous formation of SIC without external coordination |
Number of SIC formed organically (without promotion) | G-Plan analytics |
| Emergent governance | Quality of DAO decisions > average individual participant
decision | Condorcet indicator: P(DAO_correct) > P(individual_correct) |
Retrospective decision analysis |
| Adaptability | Response time to external shocks | Time from problem detection to
adoption of a corrective DAO proposal | Timestamped on-chain events |
| Antifragility | System improves under stress | Growth in active participants and
TVL following crisis events | Dune Analytics dashboards |
| Scalable coordination | O(n) → O(1) via AI | Cognitive load per participant does
not grow with increasing N | Decision time × quality |

#### 4.1.2. Architectural Principles of Implementation

The translation of theory into code is governed by several fundamental architectural
principles that ensure systemic integrity.

**Trustless First principle.** Any component in which trust in a third party can be
eliminated through cryptographic verification is implemented as trustless. Smart
contracts execute automatically; votes are recorded on-chain; financial flows pass
through auditable contracts. Trust is required only at system boundaries (DUNA legal
obligations, interactions with the banking system) and in the bootstrap phase
(Founder as initiator).

**Progressive Decentralisation principle.** The architecture is designed so that each
component can be decentralised independently of the others. This allows commencement
with a partially centralised implementation (BSC instead of GyberNet, GitHub instead
of on-chain governance, manual moderation instead of AI) with a sequential transition
to full decentralisation as the community grows and technologies mature.

Concrete plan of progressive decentralisation:
```

Component	Phase I (Bootstrap)	Phase II (Growth)	Phase III (Maturity)
Blockchain	BSC (centralised)	BSC + GyberNet testnet	GyberNet mainnet
Governance	Founder + Safe	MacroDAO (delegated)	Full DAO autonomy
Data storage	Supabase (self-hosted)	Supabase + IPFS	IPFS + DDP
Frontend	DSP Client (hosted)	DSP Client + community	Federated nodes
AI	External APIs (OpenAI)	Fine-tuned + local	GyberComputer FL
Moderation	Manual + AI flags	AI + Social DAO appeal	AI + DAO (autonomous)
Finance	80% Governance Pool	50% → Smart contracts	100% DAO-controlled

```
**Compositionality principle.** Each contract implements exactly one function and
interacts with others through standardised interfaces (ERC-20, OpenZeppelin Governor,
UUPS Proxy). This ensures: (a) auditability — each contract can be verified in
isolation; (b) replaceability — any contract can be upgraded through a UUPS proxy
without disrupting system operation; (c) testability — 100% coverage of invariants by
unit tests and property-based tests (Foundry, Echidna).

**Observability principle.** All significant system states must be observable
externally without privileged access. On-chain data is read through public view
functions; off-chain metrics are exposed through open APIs and Dune Analytics
dashboards. This not only ensures transparency (A2) but also enables the community
and AI systems to conduct continuous monitoring of ecosystem health.

#### 4.1.3. Component Interaction: Information Flows

The GyberExperiment architecture is organised as a system of coupled layers, each
providing services to the layer above. Information flows between components are
formalised as follows:

**Flow 1: Registration and identity (User → DSP → UnitManager)**
```

1. User registers in DSP (email/password → Session)
2. User connects wallet (MetaMask/WalletConnect)
3. DSP sends EIP-712 signature to GPROD
4. GPROD calls UnitManager.registerMember(wallet)
5. On-chain: status = MemberStatus.Unit
6. Supabase: wallet_address recorded as primary key
7. User receives access to full functionality

```
**Flow 2: Project lifecycle (IPI: Idea → Project → Implementation)**
```

1. Participant publishes an Idea in DSP (off-chain, stored in Supabase + IPFS hash)
2. SIC forms organically: other participants join
3. Initiator calls LPBurnInitiator.proposeProject()
4. Social DAO voting: Idea → Project (quorum 5%, 72h)
5. Upon approval: Commerce DAO takes the project into operation
6. G-Plan: tasks, milestone tracking, activity proofs
7. UnitManager.claimReward() — reward for verified activity
8. Upon completion: liquidity pool token/GBR, project token
9. Buyback & burn: 2% of project revenue → GBR burn

```
**Flow 3: Governance (Participant → DAO → Smart Contracts)**
```

1. Participant creates a proposal in DSP
2. AI analyst generates impact analysis
3. Governor.propose() — on-chain record
4. Voting delay: 24–48h (for informing)
5. Voting period: 72h (Social), 120h (Economic)
6. Upon adoption: TimelockController.schedule() — delay 24–48h
7. After timelock: TimelockController.execute() — automatic execution
8. Result recorded on-chain, notifications via DSP

```
**Flow 4: Monitoring and feedback (On-chain → AI → DAO → Correction)**
```

1. On-chain events indexed (The Graph / custom indexer)
2. AI monitoring: analysis of voting patterns, tokenomics, activity
3. Upon anomaly detection: AI generates alert
4. Class “AI recommends”: notification to Social DAO
5. Class “AI decides”: automatic contract pause (emergency pause)
6. DAO voting: confirmation or cancellation of AI action
7. Result: parameter adaptation (quorum, thresholds, rewards)

```
These four flows form a closed feedback loop: users interact via DSP, their actions
are recorded on-chain, AI systems monitor and optimise parameters, DAO adopts
corrective decisions that alter the rules of interaction. This is a direct
realisation of the CAS model from Section 3.6: agents (users + AI), environment
(GyberNet + DSP), rules (smart contracts), feedback loops (monitoring → DAO →
correction).

#### 4.1.4. Security Architecture: Defense-in-Depth

The security of CSC is not reducible to the security of individual smart contracts —
it is a *systemic property* emerging from the interaction of multiple protective
mechanisms at different levels. The present section describes the Defense-in-Depth
strategy applied to the GyberExperiment architecture.

**Level 1: Cryptographic.** The foundational security level is provided by the
blockchain's cryptographic primitives: ECDSA signatures for transaction
authentication, Keccak-256 hashing for integrity, Merkle trees for state
verification. At this level, security is determined by mathematical guarantees:
breaking ECDSA requires solving the discrete logarithm problem on the elliptic curve
secp256k1, which is computationally infeasible at current technological levels
(estimate: 2^128 operations).

**Level 2: Smart Contract Security.** The smart contracts UnitManager, Governor,
Timelock, GBRVotingWrapper, and LPBurnInitiator are developed using security best
practices:
```

Smart Contract Security Stack:

Patterns: — Checks-Effects-Interactions: prevention of reentrancy — UUPS Proxy: upgradeability without state loss — ReentrancyGuard (OpenZeppelin): explicit protection against reentrant calls — Pausable: emergency halt capability — AccessControl: role-based access model — EIP-712: typed signatures for activity proofs

Audit pipeline (CI/CD): 1. Slither (static analysis) — detection of known vulnerability patterns 2. Mythril (symbolic execution) — vulnerability discovery through formal verification of execution paths 3. Echidna (fuzzing) — property-based testing with randomised inputs 4. Manual review (peer review via Code DAO process) 5. External audit (every 6 months or upon major upgrade)

Post-deployment monitoring: — Forta Network: real-time anomaly detection — Custom watchers: monitoring of Fund balance, unusual voting patterns, mass operations — Alert → Emergency multisig → Pause → DAO review

```
**Level 3: Governance Security.** Protection against governance attacks — attacks in
which an adversary exploits legitimate DAO mechanisms to cause harm:

- **Flash loan governance attack**: an adversary takes a flash loan to temporarily
increase VP. Defence: GBRVotingWrapper requires staking for a minimum period (≥7
days) — VP is available only for locked tokens, rendering flash loans inapplicable.
- **Vote buying**: an adversary purchases votes off-chain. Defence: the square root
in Hybrid VP makes buying 51% of VP economically impractical (cost grows
exponentially with the target share); voter anonymity at the on-chain level (no on-
chain record of individual votes until voting concludes — snapshot voting).
- **Proposal spam**: an adversary clutters governance with meaningless proposals.
Defence: minimum threshold for creating a proposal (MemberStatus ≥ Dev(2) + minimum
GBR staked); LP Burn for Commerce DAO proposals — an irreversible financial
commitment.
- **Governance capture**: gradual accumulation of VP up to DAO control. Defence:
Activity Score decay (inactive participants lose VP); qualified majority (≥67%) for
critical decisions; Emergency pause for Core (Kill Switch).

**Level 4: Operational security.** Protection of infrastructure (DSP servers, GPROD
databases, DDP IPFS nodes):

- TLS 1.3 for all communications;
- Container isolation (Docker) for each module;
- Secret management via HashiCorp Vault or sealed secrets;
- Regular security updates: automated dependency updates via Dependabot/Renovate;
- Incident response plan: severity levels (P0–P3), escalation matrix, post-mortem
process.

**Level 5: Economic security.** Protection against economic attacks that exploit
market mechanisms to extract value:

- **Oracle manipulation**: protection via Chainlink price feeds (decentralised
oracles with economic guarantees);
- **Token price manipulation**: LP Burn creates a "floor price" through irreversibly
burned liquidity;
- **Fund exhaustion**: Fund Monitoring + Halving mechanism — automatic reward
reduction upon reaching the threshold;
- **Sybil attack**: GNN detector (Appendix F.3) + MemberStatus hierarchy + AI
Activity Verifier;
- **MEV (Miner Extractable Value)**: on GyberNet (Phase III+) — private mempool +
subsidised gas for DAO operations.

The multi-level security model of CyberSocium differs fundamentally from the approach
of most DAOs, which confine themselves to Levels 1–2 (cryptography + contract
audits). Levels 3–5 (governance, operational, economic security) are specific to
fully-fledged socio-economic systems and reflect the architectural maturity
characteristic of CyberSocium.

#### 4.1.5. The Principle of Modular Degradation

A key architectural decision of GyberExperiment is the principle of **modular
degradation** (*graceful degradation*): the failure of any individual component does
not lead to failure of the system as a whole. The system continues to function with
degraded but acceptable performance.
```

Degradation matrix:

Component failed	Functions that remain	Degradation
DSP Client	On-chain: voting, staking, interaction (Etherscan/BSCScan)	No UI; use CLI/BSCScan	rewards via direct contract	participants
GPROD	DSP Client operates in read-only mode (data from cache/IPFS); off-chain operations available	No new on-chain operations	off-chain operations
Supabase (PostgreSQL) read-through	On-chain data available; profiles;	No off-chain from blockchain	GPROD transitions to slowdown
AI (all AiC agents) processing;	Manual Activity Score; without summarisation	Slower manual moderation; proposals increased	participant load
BSC (blockchain) failure; plan: GyberNet	STOP: core functions (token, voting, rewards) unavailable GyberNet as alternative	Critical Mitigation: migration to backup transition to RPC endpoint; Phase III+:	
DDP (IPFS) A1, but availability	Data accessible via temporary preserved	Central centralised fallback (S3); fallback;	violation of centralisation

Single point of failure: the blockchain itself (BSC in Phases I–II). Mitigation: BSC health monitoring + plan B (GyberNet, Phase III).

```
This principle follows from Axiom A4 (extensibility): the system must be capable of
adapting to environmental changes, including component failures. The modular
architecture (DSP Client ↔ GPROD ↔ CONTRACTS ↔ Supabase ↔ DDP) ensures that each
module can be replaced, upgraded, or deactivated without cascading failure. This is a
direct realisation of Ashby's Law (Section 2.4.1): the variety of protective
mechanisms must match the variety of threats.

### 4.2. GyberCommunityToken (GBR): Tokenomics

#### 4.2.1. Token Specification
```

Name: GyberCommunityToken Ticker: GBR Standard: BEP-20 (Binance Smart Chain) Contract address: 0xa970cae9fa1d7cca913b7c19df45bf33d55384a9 Type: Utility Token Total supply: Fixed (determined at contract deployment) Additional issuance: Impossible (contract contains no mint function after initialisation)

```
**Token functions within the ecosystem:**
```

1. Governance: — Voting in DAO (Social, Code, Commerce, Economic) — Signing proposals — Condition: wallet must be active (verified by a dedicated algorithm at voting time)
2. Work evaluation: — Means of evaluating participant contributions to projects — Reward via UnitManager (proportional to status and verified activity)
3. Infrastructure interaction: — Interaction with GyberNet Blockchain — Interaction with GyberComputer — Confirmation of seriousness of intent when initiating projects (burning 0.1% of C(p) upon transition to Accumulation)
4. Staking: — GBR staking: staking rate determined by Economic DAO vote and adapted to the economic parameters of the ecosystem — Enhancement of participant reputation — Passive income
5. Medium of exchange: — Trading on DEX (PancakeSwap) — Interaction with project liquidity pools — Discounts for special project clients

```
#### 4.2.2. Emission Distribution and Governance
```

Distribution (fixed supply, deflationary model):

Governance Pool (~80%+): — Controlled by initiators and key community participants at the deployment stage of the experiment — Used for on-chain voting, UnitManager financing, and reserve operations — As the system becomes automated, funds are transferred to a stabilisation fund — a smart contract governed exclusively through DAO voting — Managed via the BEP-20 delegation mechanism

Trading Float (~20% or less): — Listed on DEX (PancakeSwap) for liquidity provision and price discovery — Creates a market mechanism for determining the value of GBR

```
#### 4.2.3. Reward Mechanism: UnitManager

UnitManager is a smart contract implementing the RewardProtocol and constituting the
technical realisation of Axiom 5 (meritocratic fairness).

**Implementation status:** the current version of UnitManager (GybernatyUnitManager)
is a working prototype demonstrating the principle of automatic reward distribution
based on participant status. The prototype implements four levels (Unit, Dev,
LeadDev, ArchDev), a two-level approval system (confirmation by participants one and
two levels higher), and restrictions on withdrawal frequency (up to 2 times per
month). Integration with G-Plan for activity verification is the next development
stage.

**Reward payment algorithm:**
```

1. Fixed reward per Unit Type: Each participant receives a fixed amount determined by their status (Unit Type):

   — Unit (level 1): 10,000,000 GBR / period — Dev (level 2): 100,000,000 GBR / period — LeadDev (level 3): 1,000,000,000 GBR / period — ArchDev (level 4): 10,000,000,000 GBR / period

   The reward is paid for verified activity in the current period.

2. Project completion bonus (×5 multiplier): Upon verified completion of a project in the IPI model (transition to the Operation phase), all members of the project team receive a bonus:

   $reward_completion(a_i) = reward_base(status_i) \times 5$

   The ×5 multiplier creates a colossal economic incentive to complete what was started: a team that brings a project to Operation earns in a single period as much as in five ordinary periods. This resolves the systemic problem of incomplete projects — a blight on both open-source communities and corporate innovation programmes.

3. Activity verification: UnitManager queries G-Plan — an innovative task manager developed by the community.

   G-Plan verifies activity a_i by checking: — Participation in tasks during the reporting period — Task completion confirmed by participants of higher status — Overall level of engagement in ecosystem projects

4. Payment: If activity confirmed: UnitManager.transfer($wallet_i$, reward_amount) If project completed: UnitManager.transfer($wallet_i$, reward_amount × 5) If activity not confirmed: Transaction rejected. Participant may retry after accumulating activity.

5. Record: All operations are recorded on the BSC blockchain, ensuring full transparency and auditability.

```
**Connection to theory.** UnitManager realises several axioms simultaneously:
- A2 (transparency): all payments are on-chain, verifiable by anyone
- A5 (meritocracy): reward is determined by status and activity, not capital
- A6 (inclusivity): minimum status (Unit) is accessible to every active participant

**Free-riding countermeasure:** reward is paid not for "presence in the system" but
for verifiable completion of specific tasks in G-Plan. The ×5 multiplier for project
completion creates an additional incentive for genuine contribution rather than
minimal simulation of activity. AI verification of task quality (Section 6.4)
provides a primary filter against formal completion without substantive output.

Critical distinction from existing DAOs: in most DAOs, rewards are determined by
token-holder or administrator voting — creating subjectivity and politicisation. In
UnitManager, rewards are determined by **objectively verifiable activity** through G-
Plan, and the amount is tied to status determined by the community. This minimises
the space for manipulation.

#### 4.2.4. Liquidity Pools and Connection to Projects

The liquidity pool mechanism is a key element in the realisation of Axiom 6
(inclusivity) and Loop 2 (liquidity):
```

For each project p_j transitioning to the Operation state:

1. Internal project tokens token(p_j) are issued (at the Accumulation stage, for sale at 1 BUSD)

2. Before opening trading on DEX, a liquidity pool is created: Pool(p_j) = {token(p_j), GBR}

3. This means token(p_j) trades in a pair with GBR

4. Consequences for participants: — Any GBR holder may acquire token(p_j) via DEX, regardless of participation in SIC(p_j) — Active participants receiving GBR through UnitManager automatically have access to the tokens of any project — This realises the principle: “active community members receive tokens of any project, regardless of personal participation in that specific project”

5. Staking of internal tokens: token(p_j) staking: 10 / [total_supply(token(p_j))] % of the revenue of project p_j

   This creates a direct economic link between the success of a project and the reward of its token holders.

6. Project client tiers: — External clients: use fiat currency, pay full price — Special clients: use GBR, receive a discount — Super-clients: use token(p_j), receive maximum discount

   This creates an economic incentive for acquiring and holding GBR and token(p_j), supporting token liquidity and value. ```

To prevent liquidity fragmentation at ecosystem scale, minimum liquidity thresholds for each pool are prescribed, determined by Commerce DAO voting, as well as an aggregation mechanism through PowerSwapMeta (Section 6.5).

4.2.5. Liquidity Pool: Architecture and Management

GBR liquidity is provided through DEX pools on PancakeSwap with a dual trading pair. The liquidity architecture is designed to ensure stable price discovery and market depth.

```
Liquidity parameters:

  DEX:              PancakeSwap v2 (migration to v3 via
                    governance vote upon reaching maturity)
  Trading pairs:    GBR/BNB (50% of liquidity) +
                    GBR/USDT (50% of liquidity)
  LP provider:      Exclusively DUNA — the sole
                    provider of base liquidity
  Locked Liquidity: Team.Finance, locked for 12 months
                    with extension via governance vote
  Initial capital:  $50,000-100,000 from DUNA treasury
                    (approved by governance vote)

Rationale for 50/50 split:
  GBR/BNB (50%):   Primary trading pair for
                   BSC users, high volumes
  GBR/USDT (50%):  Stable USD price discovery,
                   simplifies buyback & burn,
                   lower Impermanent Loss

"DUNA only as LP" model:
  — Full community control over liquidity
    through governance
  — All trading fees (0.25% on v2) go to
    DUNA treasury
  — Guaranteed market depth, independent
    of the discretion of individual LP providers
  — Transparency: all liquidity visible on-chain,
    managed through Timelock
  — LP tokens locked via Team.Finance for
    12 months → public page with countdown timer
    to confirm rug pull protection
```

4.2.6. LP Burn Mechanism: Proof-of-Commitment for Project Initiation

LP Burn is a unique economic mechanism with no direct analogues in the existing DAO ecosystem. It solves the problem of filtering serious proposals from spam, while simultaneously deepening ecosystem liquidity.

Operating principle. A participant wishing to initiate a new project, research effort, or critical vote creates their own LP position (GBR + BNB/USDT) on PancakeSwap and burns the resulting LP tokens through a dedicated Gybernaty smart contract. Burning LP tokens is an irreversible financial commitment (proof-of-commitment), after which the participant receives the right to initiate: - A new research project via Commerce DAO or Economic DAO - A product development proposal - A critical governance vote (when GBR is insufficient for the Proposal Threshold)

```
LP Burn economics:

  Procedure:
    1. Participant creates an LP position (GBR + BNB/USDT)
       on PancakeSwap
    2. Receives LP tokens (CAKE-LP)
    3. Burns LP tokens via the LPBurnInitiator contract
    4. Contract verifies minimum threshold
       ($1,000–$5,000, determined by governance)
    5. Participant receives the right to initiate

  Why burn, not stake:
    — LP burn = irreversible commitment
      (cannot be recovered → not speculation, but conviction)
    — Double stake: participant deposits both GBR
      and BNB/USDT simultaneously
    — Upon burn, liquidity remains in the pool permanently →
      market depth increase → benefit for the entire ecosystem
    — If proposal rejected: the burner lost funds,
      but deepened liquidity for all → no
      "free spam"

  Example:
    Participant wishes to launch a research project
    on swarm intelligence algorithms:
    1. Creates an LP position for $5,000 (GBR + BNB)
    2. Burns LP tokens via the contract
    3. Submits a proposal for a $30,000 grant from DUNA
    4. Community votes:
       Approved → participant receives the grant
       Rejected → participant lost $5,000,
       but ecosystem liquidity increased

  Protection against abuse:
    — Minimum LP burn threshold ($1,000–$5,000)
    — Limit on the number of LP burn proposals per month
      (determined by governance)
    — Whitelist for participants with a history of successful
      projects (exemption from LP burn)

  Implementation status: contracts implemented
  and in the testing phase
```

Theoretical justification. LP Burn realises the principle of costly signalling from evolutionary biology (Zahavi, 1975): a costly signal (irreversible burning of funds) is a reliable indicator of seriousness of intent, since it cannot be falsified. In mechanism design theory, this corresponds to the screening principle (Rothschild & Stiglitz, 1976): the mechanism separates agents with high and low motivation through self-selection, without requiring centralised evaluation.

4.2.7. Deflationary Model and Buyback & Burn

GBR tokenomics is built on a deflationary model: the total supply decreases monotonically through several burning mechanisms.

```
GBR burning mechanisms:

  1. Buyback & Burn (operational):
     — DUNA directs a portion of ecosystem revenues
       (LP trading fees, project revenues)
       toward buyback of GBR from the DEX
     — Purchased tokens are burned
       (transfer to 0x000...dead)
     — Legal characterisation: "protocol liquidity
       management operation"
       (asset management operation),
       NOT profit distribution
     — Effect: at stable demand — growth in
       GBR value → growth in real value of
       UnitManager rewards

  2. LP Burn (initiatory):
     — Participants burn LP tokens upon initiating
       projects (Section 4.2.6)
     — Effect: liquidity remains in the pool permanently,
       but the burner's LP share disappears

  3. Project Initiation Burn:
     — Burning of 0.1% of C(p) upon project
       transition to the Accumulation phase
     — Confirmation of seriousness of intent +
       deflationary pressure

  4. Activity Scoring Penalties:
     — Penalties for inactivity (burning of a portion
       of staked GBR when Activity Score falls
       below the threshold)

  Aggregate deflationary effect:
    total_supply(t+1) = total_supply(t)
      — buyback_burn(t)
      — lp_burn(t)
      — initiation_burn(t)
      — penalty_burn(t)

  As the ecosystem grows, all four components grow →
  accelerating deflation → increase in GBR value
  with growing demand (new participants, new projects)
```

Protection against classification as a security (Howey Test). Buyback & Burn does not constitute profit distribution among participants. The DUNA Association Agreement explicitly states: “GBR tokens do NOT entitle holders to dividends or profit distributions”.

Buyback & Burn is formalised as an “asset management operation to maintain protocol liquidity” — analogous to a corporation buying back its own shares from the market without any obligation to pay dividends. A detailed Howey Test analysis is provided in Section 8.4.2.

4.2.8. Economic Equilibrium and Tokenomics Sustainability

GBR tokenomics is designed to achieve long-term equilibrium between emission (UnitManager rewards) and deflation (four burning mechanisms). We formalise the conditions of this equilibrium.

Flow balance model:

Let S(t) denote the circulating supply of GBR at time t. The dynamics of S(t) are determined by:

```
dS/dt = emission(t) - burn(t)

where:
  emission(t) = Σ_status reward(status) × N_active(status, t)
  — aggregate reward to all active participants by status

  burn(t) = buyback_burn(t) + lp_burn(t) + initiation_burn(t) + penalty_burn(t)
  — aggregate burning from all sources
```

Three regimes:

1. **Inflationary regime (dS/dt > 0):** emission > burn. Characteristic of early phases, when the ecosystem distributes rewards to attract participants but has not yet generated sufficient revenue for buyback & burn. In this regime, the GBR price experiences downward pressure, offset by growing utility (voting, staking, access to services).

2. **Equilibrium regime (dS/dt ≈ 0):** emission ≈ burn. The target state of a mature ecosystem. Circulating supply stabilises; the GBR price is determined by the balance of supply and demand at the utility value level.

3. **Deflationary regime (dS/dt < 0):** burn > emission. Characteristic of a mature ecosystem with multiple revenue-generating projects. Buyback & burn from project revenues exceeds the emission of new tokens. In this regime, circulating supply contracts, creating upward price pressure at stable or growing demand.

Parameter estimates for three phases:

Parameter	Phase I (Bootstrap)	Phase II (Growth)	Phase III+ (Maturity)
N_active	50–500	500–5,000	5,000–100,000+
Monthly emission	~10B GBR	~50B GBR	~100B GBR
Project revenues (monthly)	$0	$1K–$50K	$50K–$5M
Buyback & burn (monthly)	0 GBR	~1B GBR	~50B–200B GBR
LP burn (monthly)	~0.1B GBR	~1B GBR	~5B GBR
Regime	Inflationary	Transitional	Deflationary

Halving formula:

To prevent exhaustion of the reward fund (60% of total supply = 6T GBR), a halving mechanism is provided — an automatic reduction of rewards upon reaching threshold levels:

```
reward_multiplier(t) = {
  1.0,   if remaining_fund > 50% of initial_fund
  0.75,  if remaining_fund ∈ (25%, 50%] of initial_fund
  0.50,  if remaining_fund ∈ (10%, 25%] of initial_fund
  0.25,  if remaining_fund ≤ 10% of initial_fund
}

actual_reward(status, t) = base_reward(status) × reward_multiplier(t) ×
activity_multiplier(i)
```

Halving ensures that the reward fund is never fully exhausted (asymptotic approach to zero), while rewards remain meaningful over decades (provided GBR price growth compensates for halving in real terms).

FundMonitor contract:

Automated monitoring of the reward fund state is implemented through the FundMonitor smart contract:

```
// FundMonitor pseudocode
contract FundMonitor {
    uint256 public constant INITIAL_FUND = 6_000_000_000_000; // 6T GBR
    uint256 public remainingFund;

    function getMultiplier() public view returns (uint256) {
        uint256 ratio = (remainingFund * 100) / INITIAL_FUND;
        if (ratio > 50) return 100; // 1.0x
        if (ratio > 25) return 75;  // 0.75x
        if (ratio > 10) return 50;  // 0.50x
        return 25;                   // 0.25x
    }

    function monthlyReport() external returns (
        uint256 remaining,
        uint256 monthlyEmission,
        uint256 projectedRunway
    ) {
        // Publishes an on-chain report on fund status
        // projectedRunway = remaining / monthlyEmission in months
    }
}
```

Stress-testing (cadCAD scenarios):

Prior to launch, Economic DAO will conduct cadCAD modelling of the following stress scenarios:

1. **Rapid growth:** 10× participants in 6 months. Test: does reward fund sustain emission? Expectation: halving activates, but real value of rewards grows through increased GBR utility.

2. **Price crash:** GBR loses 90% of value in one week. Test: death spiral or recovery? Expectation: fixed supply + buyback from stabilisation fund + preservation of utility → recovery.

3. **Mass sell-off:** 30% of stakers withdraw funds. Test: does LP liquidity hold? Expectation: LP burn ensures permanent liquidity; sell-off creates price pressure but does not destroy liquidity.

4. **Regulatory shock:** GBR is delisted from PancakeSwap. Test: does the ecosystem continue to function? Expectation: migration to alternative DEXs (SushiSwap, 1inch); GBR utility within the ecosystem is independent of external DEXs.

Target equilibrium parameters (Phase III):

```
  Circulating supply:       ~7-8T GBR (accounting for emission + burn)
  Velocity:                 ~4-6 turnovers per year (comparable to ETH)
  Staked ratio:             30-50% of circulating supply
  TVL (all pools):          ≥$10M
  Gini (GBR distribution):  < 0.5
  Active participation rate: ≥30% of registered participants
  Revenue/market_cap ratio: ≥5% (indicates genuine utility value rather than
speculation)
```

4.3. Decentralised Social Platform (DSP): Architecture

4.3.1. Overview and Principles

DSP (Decentralised Social Platform) is a social platform owned by all users and governed by them through DAO. DSP is not merely a communication tool but the **operating system of CSC** — the environment in which the entire project lifecycle (IPI) takes place, SICs form, AGs operate, G-Plan functions, and the day-to-day interaction of participants occurs.

Architectural principles (derived from axioms):

```
From A1 (decentralisation):
  — DSP is a network of nodes, not
    a client-server application
  — Each node contains a set of microservice containers
  — No single point of failure
  — Each participant may deploy their own node

From A3 (data sovereignty):
  — Three levels of data protection:
    Level 1 (basic): server-side encryption
      using the client's public key
    Level 2 (enhanced): client-side encryption
      prior to transmission to the server
    Level 3 (maximum): full end-to-end
      encryption; the server stores only encrypted
      data; keys held exclusively by the user
  — Even the basic level provides optimal protection
  — The user has 100% control over their data

From A4 (extensibility):
  — Modular architecture: each user may write
    their own code module
  — A module can be included in a node and offered
    to other participants
  — Each node can be supplemented with any
    existing or new modules
  — Extensibility via GitHub repository
  — The system expands in all directions

From A7 (self-governance):
  — The state of the platform's codebase is determined
    by the main branch of the GitHub repository
  — Changes to the main branch are made
    only through Code DAO voting
  — Any participant may propose a change
    (pull request), but adoption is through DAO

From A8 (cognitive augmentation):
  — AI content moderation: a language model
    trained on community rules classifies
    content in real time. The model's decisions
    are advisory — the final decision
    is made by Social DAO participants.
    Formally: flag(m) = LLM(m, Rules_DAO) ∈ {ok, review, reject},
    where review → queue for human moderators
  — AI recommendations: a system for recommending projects,
    tasks, and SICs based on the participant's
    competency profile. The model runs locally (on-device);
    raw data does not leave the device (A3)
  — AI participant assistant: a contextual assistant
    for navigating the ecosystem — explaining
    DAO mechanics, assisting in drafting proposals,
    summarising discussions. Operates as an interface
    to documentation and on-chain data, without access
    to users' private data
```

4.3.2. DSP Technology Stack

The choice of technology stack for DSP is determined by a trade-off between technological maturity (minimising bootstrap-phase risks), compatibility with blockchain infrastructure, and readiness for progressive decentralisation.

Frontend (DSP Client — client-side component of DSP):

- **Framework:** Next.js 15 (React 19, App Router, Server Components)
- **Rationale:** server components provide SEO and initial-load performance; the React ecosystem offers the largest developer pool for open-source contribution; App Router provides type-safe routing
- **Web3 integration:** wagmi + viem for interaction with BSC/GyberNet; RainbowKit for wallet connection (MetaMask, WalletConnect, Coinbase Wallet)
- **Real-time:** WebSocket (Socket.IO) for chat and notifications; Server-Sent Events for DAO voting updates
- **Localisation:** i18next with support for Russian and English as a minimum; the architecture allows addition of arbitrary locales

Backend (GPROD — Gyber Production Backend):

- **Framework:** NestJS 11 (TypeScript, ESM modules)
- **Role:** blockchain middleware — not a full-featured backend, but a linking layer between DSP and on-chain infrastructure
- **ORM:** Prisma 6 with PostgreSQL (Supabase self-hosted)
- **Key functions:** EIP-712 signature verification; on-chain event indexing; blockchain data caching; API for DSP
- **Authentication:** two-level model: (1) password session → restricted access; (2) wallet connection → full access to Web3 functionality

Database:

- **Supabase (self-hosted PostgreSQL):** primary off-chain data store (profiles, projects, G-Plan tasks, metadata)
- **Wallet Address = primary key:** all tables are linked via wallet_address, providing unified identity across modules
- **Row Level Security (RLS):** native Supabase support ensures A3 (data sovereignty) at the database level
- **Realtime:** built-in Supabase Realtime for push notifications on data changes

Distributed storage (DDP — Distributed Data Protocol, Phase 2):

- **IPFS:** content-addressed storage for documents, media files, voting hashes

- **Content Registry (smart contract):** on-chain registry of CIDs (Content Identifiers), ensuring immutability of references
- **ACL (Access Control List):** cryptographic access control: public, SIC-only, personal-only
- **Community Pinning Nodes:** community participants provide storage in exchange for GBR rewards

4.3.3. Data Model and Identity

The central concept of DSP is the **wallet as a unified identity**. Instead of the traditional model (email/password → internal ID → relations), DSP uses the cryptographic wallet address as the root identifier:

```
┌──────────────────────────────────────────┐
│              Wallet Address (PK)          │
│    0xa1b2c3d4e5f6...                      │
├──────────────────────────────────────────┤
│ On-chain:                                 │
│   MemberStatus (UnitManager)              │
│   GBR Balance (ERC-20)                    │
│   Voting Power (GBRVotingWrapper)         │
│   Activity Proofs (EIP-712 signed)        │
│   Staking Position                        │
│   Soulbound Tokens (audit, reputation)    │
├──────────────────────────────────────────┤
│ Off-chain (Supabase):                     │
│   Profile (name, avatar, bio, skills)     │
│   Projects (created, participated)        │
│   G-Plan Tasks (assigned, completed)      │
│   Social Graph (connections, SIC membership)│
│   Preferences (language, notifications)   │
├──────────────────────────────────────────┤
│ Distributed (IPFS/DDP):                   │
│   Personal Documents                      │
│   Published Content                       │
│   Encrypted Messages                      │
│   Portfolio & Credentials                 │
└──────────────────────────────────────────┘
```

DID (Decentralized Identifier): in Phase II, each participant receives a DID in the format `did:gybernaty:<wallet_address>`, linked to an on-chain record in UnitManager. A DID enables: (a) cross-platform identification; (b) issuance and verification of Verifiable Credentials (VC); (c) identity portability — a participant may transfer their identity to any DSP-compatible node.

Verifiable Credentials (VC): documents signed with the digital signature of the issuing party (DAO, another participant, AI auditor), confirming specific attributes: status in UnitManager, project participation, AI audit results, qualifications. VCs are stored by the participant (self-sovereign) and presented on demand without reference to a central registry.

4.4. GyberNet: Community Blockchain

GyberNet is a specialised community blockchain serving as the trust layer for the entire GyberExperiment ecosystem. GyberNet is a direct realisation of Axioms A1 (decentralisation) and A2 (transparency) at the infrastructure level.

4.4.1. Purpose and Functional Profile

Unlike general-purpose blockchains (Ethereum, BSC), GyberNet is designed as a **specialised blockchain for CSC** — optimised for a specific set of operations:

- **Recording DAO decisions:** every vote, every proposal, every decision execution is recorded immutably. This ensures full auditability of governance — any participant may at any time verify the entire history of decisions
- **Registering statuses and activity:** changes to MemberStatus (Unit → Dev → LeadDev → ArchDev → Core), activity proofs (EIP-712 signatures), G-Plan records — all significant ecosystem state changes are recorded on-chain
- **Document hashing:** all critical documents (proposals, reports, audit results) are hashed and recorded on-chain, providing proof of existence and immutability without storing the documents themselves in the blockchain
- **Financial flows:** all GBR transactions, LP operations, staking, buyback & burn — transparent and verifiable
- **Identity:** DID registry, Verifiable Credentials anchor, soulbound tokens

4.4.2. Current Implementation: BSC as a Transitional Platform

In the initial stage (Phases I–II), GyberNet functions are fulfilled through BNB Smart Chain (BSC). The choice of BSC is motivated by:

Parameter	BSC	Ethereum L1	Polygon	Solana
Fee	~$0.01–0.03	~$5–50	~$0.01	~$0.001
Block time	~3 sec	~12 sec	~2 sec	~0.4 sec
EVM compatibility	Full	Native	Full	None
DeFi infrastructure	Mature (PancakeSwap)	Maximum	Mature	Mature
Stability	High (>3 years without downtime)	Maximum	High	Medium (historical outages)
Validator centralisation	Medium (21 validators)	Low (>500k)	Medium (100+)	Low (1500+)

BSC provides an optimal balance between cost, speed, and infrastructure for the bootstrap phase. The centralisation of BSC validators (21 nodes) is a deliberate compromise: in the initial stage, the priority is functionality and accessibility, not maximum decentralisation. Migration to GyberNet mainnet (Phase III) will eliminate this limitation.

4.4.3. Prospective Architecture of GyberNet

GyberNet is designed as an **application-specific blockchain** with a five-layer architecture:

Layer 1 — Consensus (Reputational Proof-of-Stake, RPoS):

A modified PoS in which validator weight is determined not only by the quantity staked but also by reputation within the ecosystem:

```
validator_weight(v) = √(staked_GBR(v)) × (1 + ActivityScore(v) / 100)
```

Where ActivityScore is the aggregated activity score of the validator, computed from G-Plan data, voting participation, and Code DAO contributions. The square root of staking reduces the weight of large holders (anti-plutocratic mechanism), while the reputational multiplier incentivises genuine ecosystem participation.

Consensus parameters: - Minimum validator stake: 100,000 GBR (calibrable through Economic DAO) - Validator set size: 50–200 (grows with the ecosystem) - Epoch length: 100 blocks (~5 minutes) - Slashing conditions: double signing (100% slash), offline >24h (1% slash per day), malicious proposals (50% slash + status reduction) - Block leader rotation: weighted random selection proportional to validator_weight

Layer 2 — Execution (EVM-compatible):

Full compatibility with the Ethereum Virtual Machine for reuse of all existing contracts without modification. Additional precompiled contracts for CSC-specific operations: - `0x0100` — ActivityProof verification (native EIP-712 activity proof verification) - `0x0101` — ReputationQuery (reading Activity Score without external call) - `0x0102` — DAOStatus (fast query of voting status)

Layer 3 — Storage (State + History):

Separation into “hot” (active state: balances, statuses, current votes) and “cold” (archival: completed votes, historical activity proofs) storage. Hot storage is optimised for fast access; cold storage for long-term retention with Merkle proofs of integrity.

Layer 4 — Network (P2P + Cross-chain):

LibP2P for peer discovery and transaction propagation. Cross-chain bridge with BSC and Ethereum: lock-mint scheme with M-of-N threshold signatures (not a centralised relayer). The bridge is governed by Economic DAO — adding new chains requires a vote.

Layer 5 — API and indexing:

Built-in JSON-RPC API (Ethereum-compatible), The Graph protocol for event indexing, WebSocket API for real-time subscriptions. All APIs are public and free of charge (A2).

4.5. GyberComputer: Distributed Computing

GyberComputer is a private distributed computing network of the community, providing computational resources for the GyberExperiment ecosystem without dependence on centralised cloud providers (AWS, GCP, Azure).

4.5.1. Purpose and Economic Model

GyberComputer realises Axiom A1 (decentralisation) in the domain of computational infrastructure. Its purposes are:

- **DSP node hosting:** ensuring fault tolerance of the social platform without dependence on a single provider
- **AI model training and inference (AiC project):** distributed training through Federated Learning, local inference to ensure A3 (data sovereignty)
- **IPFS content pinning:** ensuring availability of ecosystem files through a distributed storage network
- **Computationally intensive tasks:** cadCAD simulations, cryptographic proofs, off-chain batch data processing

Economic model:

Participants providing computational resources (CPU, GPU, RAM, storage) receive GBR rewards according to the formula:

```
reward(node, epoch) = base_reward × resource_utilization(node) ×
uptime_multiplier(node) × quality_score(node)
```

Where: - `base_reward` — base rate per epoch, determined by Economic DAO - `resource_utilization` — fraction of resources utilised (0..1) - `uptime_multiplier` — multiplier for availability time (0.5 at uptime <90%, 1.0 at >99%) - `quality_score` — quality rating: response speed, computation correctness, absence of downtime (0.5..1.5)

4.5.2. Computing Network Architecture

```
┌──────────────────────────────────────────────────────────┐
│                    GyberComputer Orchestrator            │
│    (smart contract + off-chain coordinator)              │
├──────────────────────────────────────────────────────────┤
│                                                          │
│    ┌──────────────┐  ┌──────────────┐  ┌──────────────┐  │
│    │   Node A     │  │   Node B     │  │   Node C     │  │
│    │ GPU: 2×A100  │  │ CPU: 32-core │  │ Storage: 10TB│  │
│    │ Role: AI     │  │ Role: DSP    │  │ Role: IPFS   │  │
│    │ Stake: 50K   │  │ Stake: 20K   │  │ Stake: 10K   │  │
│    └──────────────┘  └──────────────┘  └──────────────┘  │
│                                                          │
│    ┌──────────────────────────────────────────────┐      │
│    │              Resource Registry               │      │
│    │  — Catalogue of available resources (CPU/GPU/RAM/SSD)│
│    │  — Matching: tasks ↔ nodes (by capabilities)  │      │
│    │  — Scheduling: task prioritisation            │      │
│    │  — Billing: accounting of consumed resources  │      │
│    └──────────────────────────────────────────────┘      │
└──────────────────────────────────────────────────────────┘
```

Resource classes:

Class	Requirements	Stake	Application	Reward
Compute-Light	8+ CPU cores, 16GB RAM	10,000 GBR	DSP hosting, IPFS pinning, indexing	1× base_reward
Compute-Heavy	32+ CPU cores, 64GB RAM	25,000 GBR	cadCAD simulations, batch processing	2× base_reward
GPU-Inference	1+ GPU (RTX 3090+), 24GB VRAM	50,000 GBR	AI model inference (LLM, GNN)	3× base_reward
GPU-Training	2+ GPU (A100+), 80GB+ VRAM	100,000 GBR	Federated Learning, model training	5× base_reward
Storage	5TB+ SSD, 99.9% uptime	10,000 GBR	IPFS pinning, archival storage	1× base_reward × TB

Quality assurance mechanism:

- **Proof of Computation:** for computational tasks — verifiable computation (optimistic verification: the result is accepted if none of k random validators challenged it within the challenge period)
- **Proof of Storage:** for storage nodes — periodic challenges: the orchestrator requests a Merkle proof for a random data block
- **Slashing:** upon failure to fulfil obligations (downtime, incorrect computations, data loss) — stake confiscation proportional to the severity of the violation

4.5.3. Connection to the AiC Project

GyberComputer serves as the computational foundation for AiC (Artificial Intelligence Community, Section 4.8). Key integration points:

1. **Federated Learning:** AI models are trained distributedly on GyberComputer GPU nodes. Each node trains the model on local data and sends only gradient updates (not raw data) to an aggregating server. This ensures A3 (data sovereignty) while building high-quality AI models

2. **Model Registry:** trained models are stored in IPFS with on-chain metadata (version, quality metrics, authorship, AGPL licence). DAO manages access to models: internal models (free for participants), research models (open access), commercial models (licensable)

3. **Inference API:** API for AI model inference, distributed across GPU nodes with load balancing. Ecosystem participants invoke AI through a single endpoint; the orchestrator routes requests to the nearest available node

4.6. MacroeconomicDAO: Governance System

MacroeconomicDAO is the central mechanism for realising Axiom 7 (self-governance). It is a transparent system of interaction and decision-making, based on blockchain and verified Solidity smart contracts using the OpenZeppelin library.

4.6.1. Governance Architecture

```
Technical implementation:

  Governor Contract (OpenZeppelin Governor):
- Deployed on BNB Smart Chain (BSC)
- Accepts proposals from GBR holders
- Manages voting: vote counting,
  quorum determination, result recording
- Transfers approved proposals to Timelock

  Timelock Contract:
    - Delay in the execution of approved decisions
      (24-48 hours — parameter determined by the community)
    - Provides the possibility of emergency response
      in the event of a malicious proposal being detected
    - Automatic execution after the delay expires

  Gnosis Safe (emergency multisig):
    - Emergency pause in the event of
      detection of a critical vulnerability
    - Keys held by trusted community members
      (determined by vote)
    - Cannot initiate expenditure — only pause
```

4.6.2. Decision-Making Process

```
Standard governance cycle:

  1. Proposal:
     Any participant with an active wallet and sufficient
     GBR (Proposal Threshold) submits
     a proposal to the relevant DAO.

     Requirement: the wallet must be classified
     as "active" by the verification algorithm — this
     prevents governance capture by large but
     inactive holders.

  2. Discussion:
     The proposal is published in the project space
     of DSP associated with the relevant DAO.
     Discussion period: 1–2 days (Voting Delay).

  3. Voting:
     Upon expiry of the discussion period, voting opens.
     Voting period: 3–7 days.
     Quorum: 10–20% of total supply
     (specific parameters determined by the community).

     The voting mechanism may vary
     by DAO class and proposal type:
     simple majority, qualified
     majority, quadratic voting.

  4. Execution:
     Approved proposals are automatically queued
     in Timelock. Upon expiry of the delay —
     automatic execution of on-chain operations.

     For off-chain operations — authorised
     implementors begin work.

  5. Accountability:
     Execution results are recorded on-chain.
     Implementors report via DSP.
     The reputation system records the quality
     of execution.
```

4.6.3. Four DAO Classes

MacroeconomicDAO unites four classes of decentralised organisations, each responsible for a specific category of decisions:

```
┌──────────────────────────────────────────────────────────────────┐
│                      MacroeconomicDAO                            │
│                (meta-level of coordination)                      │
│                                                                  │
│  ┌──────────────────┐          ┌──────────────────┐             │
│  │   Social DAO     │          │    Code DAO      │             │
│  │                  │          │                  │             │
│  │ Community social │          │ Technical        │             │
│  │ life:            │          │ infrastructure:  │             │
│  │ norms, events,   │          │ codebase,        │             │
│  │ membership,      │          │ architecture,    │             │
│  │ disputes         │          │ standards        │             │
│  └──────────────────┘          └──────────────────┘             │
│                                                                  │
│  ┌──────────────────┐          ┌──────────────────┐             │
│  │  Commerce DAO    │          │  Economic DAO    │             │
│  │                  │          │                  │             │
│  │ Commercial       │          │ Systemic         │             │
│  │ projects:        │          │ economic         │             │
│  │ crowd-investing, │          │ decisions:       │             │
│  │ business plans,  │          │ treasury,        │             │
│  │ profitability    │          │ strategy,        │             │
│  │                  │          │ large-scale      │             │
│  │                  │          │ projects         │             │
│  └──────────────────┘          └──────────────────┘             │
└──────────────────────────────────────────────────────────────────┘
```

📊 **Professional version:** Open SVG

Figure 4.3: Structure of MacroeconomicDAO, uniting four classes of decentralised autonomous organisations: Social DAO (community social life), Code DAO (technical infrastructure and codebase), Commerce DAO (commercial projects and crowd-investing), Economic DAO (systemic economic decisions and treasury management). MacroeconomicDAO functions as the meta-level of coordination among the classes.

A detailed specification of each class is presented in Section 5. #### 4.6.4. Voting Parameters by DAO Class

Each DAO class has its own set of voting parameters, optimised for the type of decisions being made. Parameters are determined by a vote of Economic DAO and are adapted as the ecosystem matures.

```
Voting Parameters:

  Social DAO (social decisions):
    Voting Delay:       1 day
    Voting Period:      3 days
    Quorum:             10% of total supply
    Proposal Threshold: 0.5% of total supply
    Timelock Delay:     24 hours
    Mechanism:          simple majority (>50%)
    Examples:           codes of conduct, event
                        organisation, dispute resolution,
                        appointment/replacement of Administrator

  Code DAO (technical decisions):
    Voting Delay:       2 days
    Voting Period:      5 days
    Quorum:             15% of total supply
    Proposal Threshold: 1% of total supply
    Timelock Delay:     48 hours
    Mechanism:          qualified majority
                        (>66%) for critical changes
    Examples:           merge into main branch, smart
                        contract upgrades, architectural
                        decisions, development standards
    Additional:         AI Code Reviewer must approve
                        the proposal before voting

  Commerce DAO (commercial decisions):
    Voting Delay:       2 days
    Voting Period:      7 days
    Quorum:             15% of total supply
    Proposal Threshold: 1% of total supply
    Timelock Delay:     48 hours
    Mechanism:          simple majority + mandatory
                        due diligence (AI analyst
                        generates report)
    Examples:           launch of a new project,
                        approval of a business plan,
                        determination of LP parameters

  Economic DAO (economic decisions):
    Voting Delay:       3 days
    Voting Period:      7 days
    Quorum:             20% of total supply
    Proposal Threshold: 2% of total supply
    Timelock Delay:     72 hours (or 7 days for
                        critical changes)
    Mechanism:          qualified majority
                        (>66%)
    Examples:           changes to staking parameters,
                        major treasury expenditures
                        (>$50,000), amendment of Governor
                        contract, blockchain migration

Proposal types by priority:
  Routine    (<$10K):  standard class parameters
  Significant ($10K-$50K): elevated quorum (+5%)
  Critical   (>$50K):  maximum quorum + 7-day
```

```
                          Timelock + mandatory AI audit
  Emergency  (security): Gnosis Safe → pause → then
                          standard governance for resolution
```

4.6.5. Hybrid Voting Power: A Combined Voting Model

Standard token-weighted voting produces plutocracy: one token = one vote means that large holders dominate. GyberExperiment implements the Hybrid Voting Power model, combining three components:

```
Hybrid Voting Power Formula:

  VP(aᵢ) = √(GBR_staked(aᵢ)) × AM(aᵢ) × SB(statusᵢ)

  where:
    GBR_staked(aᵢ)  — number of GBR staked
                      by participant aᵢ in the Governor contract
    √()             — square root ensures
                      sub-quadratic scaling:
                      doubling the stake yields only √2 ≈ 1.41×
                      increase in voting power
    AM(aᵢ)          — Activity Multiplier, determined
                      by Activity Score (0-100):
                        score ∈ [0, 30)  → AM = 0.5
                        score ∈ [30, 60) → AM = 0.75
                        score ∈ [60, 80) → AM = 1.0
                        score ∈ [80, 100] → AM = 1.5
    SB(statusᵢ)     — Status Bonus:
                        Unit:    1.0
                        Dev:     1.1
                        LeadDev: 1.2
                        ArchDev: 1.3
                        Core:    1.5

Example:
  Participant A: 1,000,000 GBR staked, score=90, Dev
    VP(A) = √(1,000,000) × 1.5 × 1.1 = 1000 × 1.65 = 1,650

  Participant B: 10,000,000 GBR staked, score=20, Unit
    VP(B) = √(10,000,000) × 0.5 × 1.0 = 3162 × 0.5 = 1,581

  Result: participant A, with a 10× smaller stake, possesses
  MORE voting power than B, owing to high
  activity and Dev status. This realises Axiom A5
  (meritocratic fairness).

Model properties:
  — Completely eliminates linear plutocracy
  — Rewards active participants
  — Preserves economic incentives (staking)
  — Resistant to Sybil attacks: Activity Score
    requires genuine verified activity
  — Implemented via GBRVotingWrapper —
    a wrapper around OpenZeppelin Governor
```

4.6.6. Cross-DAO Coordination

Decisions affecting multiple DAO classes simultaneously — for instance, the launch of a major project requiring commercial approval from Commerce DAO, technical approval from Code DAO, and economic approval from Economic DAO — pass through the Cross-DAO coordination mechanism.

```
Cross-DAO Mechanism:

  1. The initiator submits a multi-DAO proposal
     specifying the affected classes
  2. AI Cross-DAO Coordinator automatically
     routes the proposal to the relevant
     DAO contracts
  3. Each DAO votes independently according to its
     own parameters
  4. The proposal is considered approved only upon
     approval by ALL affected DAOs
  5. If any one DAO rejects → the entire proposal
     is returned for revision
  6. In the event of divergence → FRP (Conflict
     Fork Protocol, section 3.6.3)

Examples of Cross-DAO decisions:
  — Launch of a new project (Commerce + Code + Economic)
  — Upgrade of UnitManager (Code + Economic)
  — Amendment of membership rules (Social + Economic)
  — Integration of a new AI model (Code + Social)
```

4.6.7. Wallet Activity Verification

A distinguishing feature of MacroeconomicDAO is the requirement for **wallet activity verification**. To participate in governance (voting and signing proposals), a tokenholder's wallet must be classified as "active" by a dedicated verification algorithm.

```
The verification algorithm checks:
  — Participation in ecosystem tasks (G-Plan data)
  — Engagement with projects (commits, discussions,
    funding)
  — Regularity of activity (not one-off actions,
    but a stable pattern of participation)

Objectives of the mechanism:
  — Preventing governance capture by inactive
    large holders
  — Ensuring that governance power is exercised
    by participants engaged in the current
    activities of the experiment
  — Aligning governance influence with active
    contribution rather than passive wealth
```

4.7. Legal Wrapper: Gybernaty DUNA

4.7.1. Rationale for a Legal Form

The cyber-social corporation operates in the domain of blockchain and smart contracts; however, its interaction with the traditional economy — banking operations, contracts with counterparties, tax obligations, intellectual property protection — requires a legal entity. The choice of legal form is a critical decision: the wrong form could undermine Axioms A1 (decentralisation) and A7 (self-governance).

The fundamental distinction: Gybernaty DUNA ≠ MacroeconomicDAO. Gybernaty DUNA is a centralised legal gateway for the bootstrap phase of the experiment. It serves as an administrative shell enabling the organisers to conduct the experiment within the legal framework of the traditional economy: to open bank accounts, conclude contracts, and pay taxes. DUNA is deliberately centralised at this stage — the organisers retain control to ensure operational efficiency and the project's survivability in its early phase. MacroeconomicDAO, described in section 5, is the target state of full decentralisation towards which the experiment evolves as the community grows and the protocol matures. The transition from DUNA to MacroeconomicDAO is a planned trajectory, not a one-time event.

```
Selected form: DUNA (Decentralized Unincorporated
Nonprofit Association) — State of Wyoming, USA

Rationale:

  1. Specialised legal form for DAOs:
     DUNA was established in 2024 specifically for
     the recognition of on-chain governance within
     the legal framework. This means that decisions
     of the Governor contract carry legal force —
     a direct link between the smart contract and
     the legal entity.

  2. Elimination of fiduciary duties:
     DUNA permits the exclusion of fiduciary duties
     of members, enabling full reliance on code
     and voting. This is critically important for
     realising Axiom A7 (self-governance):
     decisions are made by algorithm, not by
     trusted individuals.

  3. Limitation of liability:
     Limits the liability of participants (community
     members) for the DAO's obligations —
     risk mitigation for participants.

  4. Nonprofit status:
     Revenue is reinvested rather than distributed
     as dividends. This is consistent with the CSC
     model, where the criterion of success is social
     relevance, not profit maximisation
     (Definition 3.2.1, point 6).

  5. Public identifier of the smart contract:
     The address of the Governor contract is
     specified upon registration — this creates a
     direct legal link between the legal entity and
     the on-chain governance mechanism.
```

4.7.2. Control Architecture: Community → DUNA → Real World

```
    ```
    Principle of Absolute Control:

      Gybernaty Community (GBR holders)
        │
        │ on-chain voting
        │ via Governor + Timelock
        ▼
      Gybernaty DUNA (legal entity)
        │
        │ legally binding
        │ voting decisions
        ▼
      Administrator (Ministerial Agent)
        │
        │ technical execution:
        │ fiat payments, taxes, compliance
        ▼
      Traditional Economy
        (banks, counterparties, regulators)

    Key restrictions on Administrator:
      — Has no right to initiate expenditures
without on-chain voting
```

— Has no access to the crypto-treasury (no keys held) — Cannot block approved decisions — May be instantly replaced via a vote in Social DAO — Their role is exclusively technical execution: ministerial agent, not a manager

Two-tier treasury:

Crypto-treasury (DAO governance): All crypto assets → Timelock contract Any movement → Governor vote Direct community control

Fiat treasury: Treasury Fiat Account → governance via DAO (large payments, capital assets) Operating Fiat Account → Administrator (small operational expenditures within the approved budget, max $2,000 per transaction)

```
Replenishment of Operating Account → only
via governance vote (e.g., $10K/month)

Surpluses → automatic return
to Treasury Account
```
```

```
#### 4.7.3. Mechanisms for Legal Value Extraction by Participants

Participants of CSC make real contributions — intellectual labour, code, research,
coordination — and must receive remuneration for these. DUNA provides legal
mechanisms for this purpose:
```

1. Compensation for services: Participants conclude a service contract with DUNA (strategic guidance, technical development, research, coordination). The amount of remuneration is approved by vote. Legal qualification: “reasonable compensation”, permitted by law.

2. Research grants: DUNA issues grants for the development of new products, the conduct of research, and educational programmes. Grants are approved via Economic DAO or Social DAO.

3. Buyback & Burn: DUNA directs a portion of revenues towards the repurchase of GBR from a DEX and its burning. Legal qualification: “asset management operation to maintain protocol liquidity”, and NOT a distribution of profits. Effect: deflationary pressure on supply → given stable demand → growth in GBR value → growth in the real value of remuneration through UnitManager.

All operations pass through a vote (the crypto portion is executed automatically via Timelock; the fiat portion — via Administrator).

```
#### 4.7.4. Compliance with Axioms
```

A1 (decentralisation): — DUNA has no single controlling party — Governance pool (~80%+) is distributed across dozens of wallets — No single participant holds <25% control

A2 (transparency): — All votes are on-chain on BSC — All crypto-treasury operations via Timelock — Monthly Administrator reports — Public dashboard (Dune Analytics) — Mandatory smart contract audit

A3 (sovereignty): — Every participant controls their own keys — No one has access to the wallets of others

A5 (meritocracy): — Remuneration is determined by activity (G-Plan) and status, not by token holdings

A6 (inclusivity): — Minimum barrier to entry — Limitation of liability via DUNA — Liquidity pools ensure participation in the success of any project

A7 (self-governance): — Governor contract = legally recognised governance mechanism — Administrator is subordinate to a vote — All critical decisions — via DAO

A8 (cognitive augmentation): — AI Administrator monitor: an autonomous agent continuously analysing Administrator actions for compliance with DAO decisions: ∀ action(Adm) : Verify(action, DAO_decisions) ∈ {compliant, alert} — Upon detection of a divergence (alert) — automatic notification to Social DAO with a detailed description of the violation — The AI monitor has no executive authority: observation and notification only (watchdog principle) — Monitoring logs are recorded on-chain, ensuring auditability of control

```
#### 4.7.5. Legal Constraints and Priorities

**Nonprofit constraint upon dissolution:** in accordance with Wyoming legislation,
upon dissolution of DUNA, the association's assets may not be distributed among
participants for their private benefit. Assets must be transferred to another
nonprofit organisation with a similar mission. GBR token holders have no right to
receive assets in exchange for tokens upon dissolution. This is a fundamental
property of the nonprofit form, ensuring that the value of the ecosystem cannot be
"privatised."

**Priority of the agreement over code:** in the event of a discrepancy between the
text of the Association Agreement and the behaviour of the smart contracts, the
Agreement takes legal precedence. Smart contracts are instruments for implementing
the Agreement, not a substitute for it. Upon discovering a discrepancy, the community
is obliged to bring the code into conformity with the Agreement through a vote. This
is a legal reality that does not undermine the principle of algorithmic governance:
code implements the law, it does not create it.

#### 4.7.6. Compliance, Tax, and Regulatory Obligations

The operation of DUNA as the legal shell of CSC gives rise to a set of regulatory
obligations, the disregard of which creates legal risks for participants.

**Beneficial owners (FinCEN BOI Report).** Any participant with >25% control over
DUNA is required to submit a Beneficial Ownership Information Report (FinCEN). The
distribution of the Governance Pool across dozens of wallets ensures that no single
participant exceeds the 25% threshold. However, FinCEN also requires disclosure of
persons with "substantial control" (senior officers, important decision-makers).
Penalties for non-disclosure are criminal.

**Tax obligations:**
```

DUNA as a legal entity: — Files Form 8832 to elect tax status — Then files Form 990-N (for nonprofits with income <$50,000) or Form 990/990-EZ — DUNA's income (LP trading fees, buyback) is subject to corporate income tax — Expenditures (salaries, grants, operational) reduce the taxable base

Participants (personal taxes): — Salaries and grants from DUNA are subject to income tax based on the participant's residence — Contractor participants receive Form 1099-NEC (for US residents) or W-8BEN (for non-residents) — KYC is mandatory for all payment recipients >$600 (IRS requirement)

Administrator: — Responsible for timely filing of returns — All expenditures on tax consultants are approved by vote

```
**AML/KYC obligations:**
- KYC of grant recipients: Administrator conducts KYC of all payment recipients >$600
- Wallet screening: Chainalysis or Elliptic for checking incoming/outgoing
transactions (required for banking services)
- Travel Rule: for transfers >$3,000 — disclosure of recipient information (FATF)

**Insurance:**
- D&O Insurance: liability insurance for Administrator
- Smart Contract Insurance: Nexus Mutual for coverage of vulnerability risks
- General Liability: general DUNA insurance to protect against third-party claims

**Audit and reporting:**
- Smart contracts: mandatory audit of Governor, Timelock, UnitManager,
LPBurnInitiator prior to production deployment (CertiK, OpenZeppelin, or Trail of
Bits)
- Financial reporting: quarterly publication via DSP + annual report
- Public dashboard: on-chain treasury data via Dune Analytics
- Annual report: Form 990 (IRS) + public report for the community

#### 4.7.7. Risk Matrix and Mitigation

| Risk | Probability | Impact | Mitigation |
|---|---|---|---|
| Administrator refuses to execute a decision | Low | High | Immediate dismissal via
Social DAO, contractual liability, financial sanctions, AI monitoring |
| Loss of access to wallets | Low | Critical | Distribution across dozens of wallets,
backup seed phrases, Gnosis Safe multisig, documented recovery procedure |
| SEC classifies GBR as a security | Medium | Critical | Legal opinion (Howey Test),
absence of dividends, buyback as asset management, wallet distribution, utility
functions |
| Bank refuses service | Medium | Medium | Mercury, Relay, Wyoming Community Bank;
preparation of compliance package; minimisation of fiat operations (Phase 1) |
| Smart contract errors | Medium | Critical | Audit (CertiK/OZ/ToB), standard
OpenZeppelin contracts, bug bounty, Timelock delay, Gnosis Safe pause |
| Change in Wyoming legislation | Low | High | Monitoring, migration plan to LLC or
alternative form |
| Conflict of interest (self-dealing) | Medium | Medium | Public disclosure,
limitation to "reasonable compensation", AI audit, transparent reporting |
| IRS challenges nonprofit status | Low | High | Strict compliance with nonprofit
rules, legal structuring of buyback, tax attorney consultations |
| BSC ceases operation / 51% attack | Very Low | Critical | Migration plan to an EVM-
compatible network, off-chain backup copies of governance data |

#### 4.7.8. DUNA Roadmap
```

Stage 1 — Legal structuring: — Engagement of a law firm (Wyoming DUNA specialisation) — Drafting of Association Agreement — Contract with Administrator (Ministerial Agent Agreement) — Template contracts with participants

Stage 2 — Registration: — Filing Articles of Organization with Secretary of State — Obtaining EIN from IRS — Filing BOI Report (FinCEN) — Appointment of registered agent

Stage 3 — Technical implementation: — Deployment of Governor contract (BSC) — Deployment of Timelock contract (BSC) — Configuration of Gnosis Safe (emergency multisig) — Contract audit — Deployment of LPBurnInitiator

Stage 4 — Launch: — Vote to approve Association Agreement — Vote to appoint Administrator — First grants and compensation (governance vote) — Opening of fiat bank account — Transfer of assets to Timelock — Creation of LP positions on PancakeSwap — Locking LP via Team.Finance

Stage 5 — Automation (6–12 months): — Opening of Bridge Business Account (Treasury Fiat) — Development of Automation Contract with Chainlink Functions — Automated crypto→USD conversion — Programmatic wire transfers pursuant to governance decisions

Stage 6 — Full automation (12+ months): — 90%+ of operations without human involvement: Governance vote → Timelock → Chainlink Oracle → Bridge API → wire transfer — Administrator only for: Operating Account, compliance, emergency cases

```
### 4.8. AiC — Gybernaty Artificial Intelligence Department

#### 4.8.1. Status and Role in the Ecosystem

AiC (*Artificial Intelligence Community*) is not a standalone project among others —
it is an **internal department** of Gybernaty Community, responsible for one of the
key domains of computer science. Artificial intelligence permeates every aspect of
the community's activities: from the automation of governance and verification of
participant activity to the creation of fundamentally new products entering the open
market. The integration of AI into the course of the experiment and into everything
Gybernaty does is one of the key aspects of community development.

This status fundamentally distinguishes AiC from external AI startups or isolated
research laboratories. AiC is embedded in the architecture of CSC as its intellectual
layer — in the same way that the nervous system is embedded in an organism, rather
than being a separate organ. Every component of the GyberExperiment ecosystem — DSP,
G-Plan, MacroeconomicDAO, GyberComputer, GyberNet — is simultaneously a consumer and
a supplier of AI solutions.
```

Position of AiC in the CyberSocium architecture:

```
┌──────────────────────────────────┐
│        AiC Department            │
│   (Artificial Intelligence       │
│    Community)                    │
│                                  │
│   ┌──────────┐  ┌──────────────┐ │
│   │ Research │  │  Applied AI  │ │
│   │   Lab    │  │ Engineering  │ │
│   └──────────┘  └──────────────┘ │
└──────────────────────────────────┘
          │              │
          │              │
   ┌──────┴──────┬───────┴──────┬──────┐
   │             │              │      │
   ▼             ▼              ▼      ▼
┌──────────┐  ┌──────────┐  ┌──────────┐  ┌──────────┐
│   DSP    │  │  G-Plan  │  │ Macro-   │  │ Gyber-   │
│          │  │          │  │ economic │  │ Computer │
│ • Recom- │  │ • Auto-  │  │ DAO      │  │          │
│   mend-  │  │   matic  │  │          │  │ • Model  │
│   ations │  │   activ- │  │ • Proposal│  │   training│
│ • Moder- │  │   ity    │  │   analysis│  │ • Infer- │
│   ation  │  │   verifi-│  │ • Fore-  │  │   ence   │
│ • NLP for│  │   cation │  │   casting│  │ • Distrib-│
│   chats  │  │ • Task   │  │          │  │   uted   │
│ • Search │  │   eval-  │  │          │  │   comput-│
│ • Transl.│  │   uation │  │          │  │   ing    │
│          │  │ • Distrib-│  │          │  │          │
└──────────┘  └──────────┘  └──────────┘  └──────────┘
   │             │              │      │
   └──────┬──────┴───────┬──────┴──────┘
          │              │
          ▼              ▼
┌──────────────────────────────────┐
│      Open Market                 │
│  AI products and services based  │
│  on AiC models                   │
└──────────────────────────────────┘
```

```
> 📊 **Professional version:** [Open SVG](diagrams/architecture/aic_integration.svg)

*Figure 4.4: Position of AiC Department in the CyberSocium architecture. AiC
functions as a horizontal AI infrastructure layer permeating all components of the
ecosystem: DSP (recommendations, moderation, NLP, search, translation), G-Plan
(automatic activity verification, task evaluation, distribution), MacroeconomicDAO
(proposal analysis, forecasting), GyberComputer (model training, inference,
distributed computing). AI products reach the open market via Commerce DAO.*

#### 4.8.2. The Problem AiC Addresses

The current state of the artificial intelligence industry is characterised by a
paradox structurally identical to the Phase 4 paradox (platform capitalism) described
in Section 3.3. A technology that is by its very nature the product of decades of
open academic research — from Rosenblatt's perceptron (1958) to the Transformer by
Vaswani et al. (2017) [51] — is being monopolised by a narrow circle of corporations
that transform public knowledge into a proprietary product.
```

Structure of the problem:

1. Concentration of resources: Training frontier models (GPT-4, Gemini, Claude) requires $50M–$500M+ in computational resources. This creates an oligopoly: only 5–7 organisations worldwide are capable of training models at this scale.

2. Closedness: Despite the fact that fundamental architectures (Transformer, LSTM, CNN) are published in open papers and implemented in open libraries (TensorFlow, PyTorch), the final models — trained weights, training data, fine-tuning methods — are proprietary.

   The underlying paradox: open science → closed product.

3. Control over data: AI models are trained on data generated by billions of users (text, images, interactions). Users receive neither control nor compensation for the contribution of their data to the creation of models worth billions of dollars.

   This is a direct reproduction of the problem described in Section 1.3: the appropriation of user data by obsolete economic institutions.

4. Opacity: Closed models are black boxes. It is impossible to verify their safety, absence of bias, or conformity with stated properties. This is a direct violation of the transparency principle (Axiom A2).

5. Economic inaccessibility: APIs of frontier models cost $0.01–$0.06 per 1K tokens. For large-scale usage (millions of requests) this amounts to hundreds of thousands of dollars per year — unaffordable for the majority of developers, researchers, and organisations in developing countries.

6. Risk of dependency: Dependence on closed AI providers creates a strategic risk: the provider may change terms, revoke access, or impose censorship. This violates Axiom A3 (sovereignty) and Axiom A7 (self-governance).

```
The open AI movement (Meta's LLaMA, Mistral, Stability AI, EleutherAI) partially
addresses problems 2 and 4, but does not resolve problems 1, 3, 5, and 6 — the models
remain products of individual corporations or collectives not embedded within a
sustainable economic system that provides long-term financing and democratic
governance.

AiC proposes a systemic solution grounded in the theory of CyberSocium: **the
development and use of AI models as a public good, financed through PMIP, governed
through DAO, and deployed on the decentralised GyberComputer infrastructure**.

#### 4.8.3. AiC Architecture
```

Three functional circuits:

Circuit 1 — Internal (AI for the ecosystem):

Integration of AI into all components of GyberExperiment. Goal: increase efficiency, automate processes, improve user experience.

Applications:

DSP: — Recommendation systems: personalisation of the news feed, project suggestions, matching of like-minded participants — Content moderation: automatic detection of spam, violations, and bad-faith behaviour without centralised censorship (federated moderation models) — NLP functions: automatic translation (critically important for global NFC), summarisation of discussions, extraction of key decisions from debates — Semantic search across projects, documentation, and codebase — AI assistants for new participants: onboarding, navigation of the ecosystem

G-Plan: — Automatic activity verification: analysis of commits in repositories, participation in discussions, quality of completed tasks — to confirm activity when requesting rewards from UnitManager — Task complexity evaluation: automatic preliminary estimation of effort based on historical data — Optimal task distribution: matching of participants and tasks based on competencies, availability, and workload — Deadline forecasting: ML models for estimating implementation time based on project characteristics and historical data — Risk identification: early detection of problematic projects (slowdown in activity, decline in SIC engagement, conflicts in AG)

MacroeconomicDAO: — Proposal analysis: automatic checking of completeness, consistency, and realism of proposals — Social relevance forecasting: ML models predicting |SIC(p)| based on project characteristics — Simulation of economic effects: modelling of the impact of proposed decisions on tokenomics, liquidity, and ecosystem

growth — Anomaly detection: identification of suspicious voting patterns (Sybil attacks, coordinated manipulation) — Information aggregation for voters: automatic summaries of discussions, visualisation of arguments for/against

GyberNet: — Network monitoring: anomaly detection in blockchain transactions — Optimisation of consensus parameters — Predictive load analytics

GyberComputer: — Optimal distribution of computational tasks across nodes — Predictive resource scaling — Monitoring of node performance

Circuit 2 — Research (AI science):

Organisation and conduct of fundamental and applied research in the field of AI. Goal: advancing science and building the intellectual capacity of the community.

Directions: — Distributed model training (federated learning, distributed training) — Private machine learning (differential privacy, homomorphic encryption for training on confidential data) — AI alignment and safety (open research, critically important for society) — Efficient architectures for edge devices (models running on participants' devices, without cloud infrastructure) — Multimodal models (text, images, audio, video, code) — AI for scientific discovery (drug discovery, materials science, climate modeling — via Economic DAO)

Organisation: — Research groups are formed as SIC/AG within the IPI model — Financing via Economic DAO or DUNA grants — Results are published as open access (AGPL for code, CC-BY for papers) — Peer review within the community and in international journals

Circuit 3 — Commercial (AI products):

Creation of AI products and services for the open market. Goal: generation of revenue for the ecosystem (Loop 2 — liquidity).

Model: — AI products are developed via IPI (as any ecosystem project) — Products utilise models developed in Circuit 2 and the infrastructure of Circuit 1 — Market entry: via Commerce DAO — Tokenomics: project internal tokens → liquidity pool with GBR → DEX — Client tiers: external (fiat), special (GBR), super-clients (project tokens)

Examples of potential products: — AI as a Service (AIaaS) platform on GyberComputer — Specialised models for verticals: medicine, finance, education, law — Automation tools for developers — AI analytics for blockchain projects — Private AI assistants (running on the user's device, without sending data to servers)

```
#### 4.8.4. AiC Technical Stack
```

Programming languages: — Python: primary language for ML/AI (TensorFlow, PyTorch, Keras, Scikit-learn, OpenCV, Hugging Face Transformers, LangChain, JAX) — C++: high-performance components, inference engines (ONNX Runtime, TensorRT) — Rust: safe system components, WASM modules for edge inference, integration with blockchain infrastructure — TypeScript: AI integrations in DSP, web interfaces for AI tools — Solidity: smart contracts for DAO governance of AI resources

Frameworks and libraries: — Training: PyTorch (primary), TensorFlow, JAX — Distributed training: DeepSpeed, Megatron-LM, Hivemind, PyTorch Distributed — Inference: vLLM, TGI, ONNX Runtime, llama.cpp, ExLlamaV2 — Computer Vision: OpenCV, Detectron2, Segment Anything — NLP: Hugging Face ecosystem, spaCy, NLTK — MLOps: MLflow, Weights & Biases, DVC — Privacy: PySyft, OpenDP, Microsoft SEAL (homomorphic encryption)

Blockchain integration: — Ethereum/BSC: DAO contracts for governance of AI resources and models — TON: potential platform for AI micropayments (high throughput, low fees) — Solana: high-performance AI applications with on-chain components — IPFS/Filecoin: decentralised storage of training data and model weights

Licensing: — All AiC code: AGPL-3.0 (guarantees that any modifications and derivative works remain open) — Trained models: open weights (licence determined for each model by vote in Code DAO) — Research publications: CC-BY-4.0 — Training data: open datasets or data collected with the explicit consent of participants (Axiom A3 — data sovereignty)

```
#### 4.8.5. DAO Governance of AI Resources

One of the fundamental innovations of AiC is the governance of AI models and
resources through DAO mechanisms. This provides democratic control over a technology
that, in the traditional model, is controlled exclusively by corporate management.
```

Decisions made through DAO:

Code DAO: — Adoption of architectural decisions (selection of base models, frameworks) — Merge of pull requests into primary AI code repositories — Approval of code quality and documentation standards — Decisions on model licensing

Social DAO: — Prioritisation of research directions — Ethical issues: permissible and impermissible applications of AI models — Data policy: what data is used for training, how privacy is ensured — Organisation of AI conferences, hackathons, and educational programmes within the community

Commerce DAO: — Approval of AI products for launch on the commercial market — Pricing of AI services — Partnerships with external organisations

Economic DAO: — Financing of major research programmes (training of large models, GPU procurement, data collection) — Distribution of GyberComputer computational resources among AiC tasks — Investments in AI infrastructure

```
#### 4.8.6. AiC Economic Model
```

Funding sources:

1. Gybernaty internal budget — Allocation from DUNA treasury via Economic DAO vote — Targeted financing of infrastructure (GPU clusters, data storage)
2. Grants and partnerships — Decentralised grant programmes (Ethereum Foundation, Protocol Labs, etc.) — Academic partnerships with research institutions — Corporate partnerships for applied AI solutions
3. Commercial revenues (Circuit 3) — AIaaS (AI as a Service) on GyberComputer — Licensing of specialised models — Consulting and AI solution integration — Private AI assistants for enterprises
4. AiC tokenomics — Issuance of AiC internal token via the IPI model (Section 3.4) — AiC/GBR liquidity pool on DEX — Staking: 10 / [total number of AiC tokens] % of project revenue

```
**Revenue distribution:**
```

┌───┐ | AiC PROJECT REVENUES |
├───┤ | | | | 40% → AiC token holders | | (SIC participants) | | | | 25% → Operational expenditures | | (infrastructure, GPU, | | data storage) | | | | 20% → Reinvestment in research | | (Circuit 2 — AI science) | | | | 10% → Gybernaty treasury | | (via GBR LP) | | | | 5% → Staking rewards | | (for AiC token stakers) | | |
└───┘

```
| Direction | Share | Purpose |
|------------|------|------------|
| **AiC token holders** | 40% | Remuneration of SIC participants |
| **Operational expenditures** | 25% | Infrastructure, GPU, data storage |
| **Reinvestment in research** | 20% | Circuit 2 — AI science |
| **Gybernaty treasury** | 10% | Via GBR/AiC liquidity pool |
| **Staking rewards** | 5% | For AiC token stakers |

*Table 4.1: Distribution of revenues from AiC projects. All revenues pass through the
AiC/GBR liquidity pool, providing a direct link between AiC's success and the value
of the GBR ecosystem — realisation of Loop 2 (liquidity).*

  Key mechanism: all revenues pass
  through the AiC/GBR liquidity pool, which
  provides a direct link between AiC's success
  and the value of the GBR ecosystem —
  realisation of Loop 2 (liquidity)
  from the project lifecycle model.
```

4.8.7. The Cyber-Social Significance of AiC

AiC represents the application of cyber-social principles to the technology that is arguably the most significant for the current era. By demonstrating that frontier AI development can be organised through decentralised, community-governed structures, AiC challenges the assumption that only centralised corporations with massive capital reserves are capable of participating in the advancement of the AI frontier.

AiC also provides a model for the democratic governance of AI systems — a critically important need in an era when artificial intelligence is increasingly shaping economic, social, and political outcomes. Within AiC:

- **Decisions on research priorities** are made through community voting, not by a corporate board of directors
- **Training data** is collected with explicit consent and under the control of participants (Axiom A3)
- **Trained models** are available to all participants and licensed as open (AGPL for code, open weights for models)
- **Revenues from commercialisation** are distributed among participants through transparent smart contracts, rather than being appropriated by shareholders

4.8.8. Applied AI Engineering: Concrete Implementations

Based on an analysis of the current state of the AI industry (2026) and the GyberExperiment architecture, the following describes concrete AI integration points — from the most mature (production-ready) to experimental.

1. AI Governance Agents (maturity level: high).

The problem of voter fatigue in DAOs is addressed by three types of AI agents:

- AI proposal summariser — an agent that analyses each proposal and generates a structured summary: essence, risks, economic consequences, arguments for/against. Integrated with Governor contracts via API.
- AI delegate with limited mandate — an agent that votes on routine matters in Social DAO and Code DAO according to pre-established rules (constraint-first governance), with a mandatory human override right. Identified by a soulbound token.
- Consequences simulator — before a vote, models: "if this proposal passes, here is what will happen to the treasury / liquidity / GyberComputer load". Uses cadCAD models (section 6.3).

Technology stack: CrewAI (multi-agent orchestration), open-source LLM (Llama, Mixtral) on GyberComputer for sovereignty, ETHOS-like system (soulbound tokens for agent identification, ZK-proofs for decision auditing).

2. Continuous AI Audit Pipeline (maturity level: production-ready).

Every pull request with Solidity code changes automatically passes through a multi-level AI audit:

```
CI/CD AI Audit Architecture:

  Git Push → GitHub Actions
    │
    ├─ Level 1: Static analysis
    │  (Slither, Mythril — deterministic)
    │
    ├─ Level 2: AI audit
    │  (Octane Security, Sherlock AI, AuditAgent)
    │  — analysis of logical vulnerabilities
    │  — simulation of attack scenarios
    │  — comparison with the database of past exploits
    │
    ├─ Level 3: Formal verification
    │  (critical invariants of UnitManager,
    │   Governor, Timelock)
    │
    └─ Result:
       Critical vulnerability → block merge
         + alert to Core/ArchDev
       Warning → comment in PR
         + review in Code DAO
       Clean → admission to Code DAO vote

  Deployed contracts:
    Chainalysis AI monitoring → alerts
    to Gnosis Safe signers upon anomalies
```

Specific tools: Octane Security (CI/CD-native, helped protect >$8M in a real DeFi protocol), Sherlock AI (analysis during code writing), AuditAgent (Nethermind, trained on real audits), ChainGPT Auditor (trained on historical exploits), OpenZeppelin AI tools (reduce audit time by 50%).

3. AI-powered Reputation Engine (maturity level: medium-high).

To ensure Sybil resistance and reliable Activity Scoring:

- Graph Neural Network for Sybil detection — PyTorch Geometric on GyberComputer: analysis of the wallet interaction graph, detection of clusters of fake/coordinated accounts.
- Proof of Personhood — ZK cryptography (Circom/Halo2) for one-person-one-identity without disclosure of personal data.
- Anomaly Detection — ML models for detecting unusual activity patterns: farming (performing minimal tasks for Activity Score), wash trading of tasks in G-Plan, coordinated voting.
- Reputation-weighted ranking — every action = endorsement, the strength of which depends on reputation; AI forms a dynamic ranking via Soulbound tokens (SBT).

4. Federated Learning on GyberComputer (maturity level: medium-high).

Axiom A3 (data sovereignty) prohibits centralised data collection for model training. Federated Learning resolves this conflict:

```
Federated Learning Architecture:

  GyberComputer nodes:
    │
    ├─ Node 1: local data → local
    │          training → prototype/gradients
    ├─ Node 2: local data → local
    │          training → prototype/gradients
    ├─ Node N: ...
    │
    └─► Aggregator (on GyberComputer):
          – Federated Averaging / PearFL aggregation
          – Differential Privacy (PySyft, OpenDP)
          – Updated global model → back
            to nodes
          – Recording of training round on-chain (GyberNet)
          – GBR reward to participants for
            providing data/computation

  Framework: Flower (de facto FL standard, 2026)
  Protocol: PearFL (Prototype Exchange –
    exchange of lightweight prototypes instead of gradients,
    minimising communication overhead)
  Privacy: Differential Privacy + PySyft
  Incentives: GBR rewards via UnitManager
```

Federated Learning enables AiC models to be trained without violating the data sovereignty of participants: data never leaves the user's device; the model comes to the data, not the other way around. This fundamentally distinguishes AiC from centralised AI companies, which require the transfer of data to their servers.

5. cadCAD Ecosystem Simulation (maturity level: high).

To model the economic effects of proposed decisions and optimise tokenomics parameters, cadCAD (complex adaptive dynamics Computer Aided Design) is used — an agent-based modelling framework developed by BlockScience (M. Zargham):

- Modelling of token velocity, emission/burning, LP liquidity
- Stress testing: what will happen with 10× participant growth? With a market crash? With a coordinated Sybil attack?
- Parameter optimisation: quorum, voting period, staking rewards, LP Burn threshold
- A/B testing of governance parameters before a real vote

Each Economic DAO proposal is accompanied by a cadCAD simulation demonstrating the expected consequences across 10,000+ scenarios. This transforms governance from a political process into a data-driven engineering discipline - **Ethical constraints** are established democratically through Social DAO, ensuring public control over AI applications

In this way, AiC is not merely a technical project, but an **institutional innovation** — a new model for organising the development and governance of technologies that define humanity's future.

4.8.9. AiC as the Infrastructure AI Layer of the Ecosystem

The three AiC circuits described above (Internal, Research, Commercial) possess yet another, fundamental dimension: AiC is an **infrastructure layer** providing AI augmentation of all processes of the cyber-social corporation. It is not a standalone project alongside DSP, GyberNet, and G-Plan — it is a horizontal layer permeating all elements of the ecosystem.

```
Architectural position of AiC in the ecosystem:

  ┌──────────────────────────────────────────────────┐
  │        Application layer                         │
  │   DSP    G-Plan    GyberNet    Commerce projects  │
  ├──────────────────────────────────────────────────┤
  │        AiC — AI infrastructure layer             │
  │                                                   |
  │   ┌──────────────┬──────────────┬──────────────┐  │
  │   │ AI analyst   │ AI coordi-   │ AI assistant  │  │
  │   │ (E1, SES)    │ nator(PMIP)  │ (IPI, FRP)    │  │
  │   ├──────────────┼──────────────┼──────────────┤  │
  │   │ AI auditor   │ AI monitor   │ AI knowledge  │  │
  │   │ (Code DAO)   │ (Economic)   │ base(E6, cumul.)│  │
  │   └──────────────┴──────────────┴──────────────┘  │
  ├──────────────────────────────────────────────────┤
  │        Blockchain layer (BSC, GyberNet)           |
  │   Governor    UnitManager    Tokens    LP pools   |
  └──────────────────────────────────────────────────┘
```

📊 **Professional version:** Open SVG

Figure 4.5: Architectural position of AiC in the ecosystem as a horizontal layer permeating all levels. Top level — applications (DSP, G-Plan, GyberNet, Commerce projects), middle level — AiC with six types of AI agents (AI analyst for SES and E1, AI coordinator for PMIP, AI assistant for IPI and FRP, AI auditor for Code DAO, AI monitor for Economic DAO, AI knowledge base for experience accumulation E6), bottom level — blockchain (Governor, UnitManager, tokens, LP pools). Each component of the top level accesses AI services through a standardised API.

Relationship of AiC to theoretical constructs:

Theoretical element	AiC function	Result
Axiom A8 (Cognitive augmentation)	AiC — institutional realisation of A8	Democratic control over AI augmentation
PMIP: C(p) → min	AI coordinator reduces project costs	individual_cost(p | AI) < individual_cost(p)
SES: project selection	AI predictor of SR improves selection quality	Directed evolution instead of blind selection
E1: collective intelligence	AI analyst compensates for cognitive limitations	Scaling without loss of quality
E2: antifragility	AI monitor ensures preventive adaptation	Active antifragility
E6: knowledge accumulation	AI knowledge base indexes all experience	Exponential knowledge growth
FRP: fork resolution	AI mediator reduces the cost of Deliberation	FRP from costly process → routine

Principle of horizontality: each element of the ecosystem accesses AiC AI services through a standardised API. DSP uses AI for discussion summarisation and recommendations. G-Plan uses AI for optimal task assignment and deadline forecasting. GyberNet uses AI for optimisation of consensus parameters. Commerce projects receive AI code review, security audit, and analytics. At the same time, all AI services are subject to the division principle (Axiom A8): AI decides, AI informs, AI has no authority — depending on the class of the task.

Economic model of the AI layer:

```
Financing of AiC infrastructure:

  1. Basic AI services → financed from the treasury
     (public good, available to all participants)

  2. Advanced AI services → freemium model
     (basic access is free, extended access — for GBR)

  3. Commercial AI products (Circuit 3) →
     revenue returns to the ecosystem via
     distribution smart contracts

  This creates a self-sustaining cycle:
  AI layer augments the ecosystem → ecosystem grows →
  more data and resources for AI → AI is augmented →
  ecosystem is augmented further

  Formally: positive feedback loop 7
  (AI Flywheel):
  quality(AI) ↑ → productivity(CSC) ↑ →
  revenue(ecosystem) ↑ → investment(AiC) ↑ →
  quality(AI) ↑
```

In this way, AiC closes the theoretical cycle: Axiom A8 defines the principles of AI augmentation, the theoretical sections (PMIP, SES, E1–E6) describe the mechanisms, and AiC provides the institutional and infrastructural realisation, subject to community governance through the four-class DAO system.

5. DAO Taxonomy: A Four-Class Model of Decentralised Decision-Making

A distinguishing feature of the GyberExperiment governance architecture is the **four-class DAO taxonomy** — a structured classification of decentralised governance mechanisms that accounts for the different categories of collective decision-making. This taxonomy reflects the recognition that not all decisions are alike in their nature, and that different types of decisions may require different governance processes, participation criteria, and enforcement mechanisms.

5.1 Social DAO

5.1.1 Purpose

Social DAOs govern the **internal social life of the community** — decisions concerning community norms, social events, membership policies, communication standards, dispute resolution procedures, and other matters pertaining to the community as a social entity rather than as an economic or technical enterprise.

5.1.2 Scope

Social DAO decisions include, but are not limited to:

- Organisation of community events (conferences, hackathons, meetups, educational programmes) - Establishment and modification of community codes of conduct - Decisions regarding community communication channels and platforms - Membership and participation policies - Recognition and rewards for contributions to community life - Cultural and educational initiatives

5.1.3 Characteristics

- **Low economic stakes:** Social DAO decisions are generally not associated with significant financial resources (although they may include the allocation of small event budgets). - **Broad participation:** All community members may participate in Social DAO governance, reflecting the principle that social decisions should reflect the widest possible consensus. - **Simple majority:** Most Social DAO decisions may be adopted by a simple majority vote, given their relatively low stakes and reversibility. - **High frequency:** Social decisions arise frequently in the day-to-day life of the community, requiring lightweight governance processes that do not create excessive overhead.

5.1.4 Formal Governance Parameters

```` Social DAO Parameters:

Quorum: 5% of the total number of active participants (active = Activity Score ≥ 20)

Discussion period: 48 hours Voting period: 72 hours Approval threshold: >50% (simple majority)

Proposal rights: — Any participant with status ≥ Unit — Minimum staking: 100 GBR

Voting rights: — All participants with status ≥ Unit — Weight: 1 vote per participant (one-person-one-vote) — Exception: membership proposals require Activity Score ≥ 30

Execution: — Automatic via Timelock (24h delay) — Off-chain decisions: published on DSP, take effect after Timelock
````

Budget constraints: — Maximum budget of a single proposal: 500 GBR (for events) — Excess → escalation to Economic DAO ```

5.1.5 Process Model: From Initiative to Execution

The proposal life cycle in Social DAO is formalised as the following finite-state machine:

```
[Initiative]      →      [Discussion,      48h]      →      [Voting,      72h]
│                                          │ │
┌────┴────┐  │                                                │              │
│                                                          Approved    Rejected
│                                     │          │ │
[Timelock]   [Archive]  │                                                  (24h)
│                                   │ │                                [Execution]
│                                  │ │                             [Monitoring, 30
days] │                                          │ └─ Retry: no earlier than ──
[Completion] 14 days after rejection
```

Typical Social DAO precedents:

1. Organisation of a hackathon. The author proposes the format, date, and budget (≤500 GBR in prizes). Discussion: the community adjusts the parameters. Vote: simple majority. Execution: funds are allocated from the social pool, and the organiser reports via G-Plan.

2. Update to the code of conduct. The author proposes a new edition. Discussion period: extended to 7 days (by decision of moderators). Vote: simple majority. Execution: the new edition is published on DSP and all participants are notified.

3. Exclusion of a participant for violation of the code of conduct. A qualified majority (67%) is required, as fundamental rights are affected. Mandatory procedure: warning → repeated violation → exclusion proposal → vote. The excluded participant may appeal the decision through FRP.

5.2 Code DAO

5.2.1 Purpose

Code DAOs govern the **technical infrastructure of the community** — in particular, the codebase of the DSP platform and other shared technical assets. The current state of the platform code is determined by the state of the main branch of the GitHub repository, and changes to that branch require passage through the Code DAO process.

5.2.2 Scope

Code DAO decisions include:

- Merging proposed code changes (pull requests) into the main branch - Approving new modules for inclusion in the standard node configuration - Selection of technical standards and architectural patterns - Evaluation and approval of security-critical changes - Management of deprecation and removal of obsolete components - Resolution of technical disputes between competing implementations

5.2.3 Characteristics

- **Requirement of technical competence:** Voting in Code DAO may be weighted by technical reputation — participants with a confirmed track record of development contributions carry greater weight in technical decisions. This reflects the principle that technical decisions should be determined by technical expertise, while remaining transparent and contestable. - **Review process:** Proposals in Code DAO typically include a mandatory technical review period, during which qualified community members examine the proposed changes for correctness, security, performance, and architectural consistency. - **Meritocratic governance:** Although any community member may propose code changes, voting rights on technical matters are concentrated among those who have demonstrated relevant competence through their contribution history. - **Awareness of irreversibility:** Code changes in production systems may have significant consequences. Code DAO processes include mechanisms for staged deployment, testing requirements, and rollback procedures. - **Continuous AI security audit (from A8):** Every pull request affecting smart contracts (Solidity/Vyper) automatically passes through an AI audit pipeline before being admitted to a Code DAO vote. The pipeline includes:

```
AI Audit Pipeline (AiC implementation, Circuit 1):
```

Git Push → CI/CD (GitHub Actions) → AI scanning: │ │ Tools: │ ├─ Octane Security: CI/CD-native analysis, │ detection of exploitable vulnerabilities │ ├─ Sherlock AI: in-development analysis, │ targeted remediation │ ├─ AuditAgent (Nethermind): attack scenario simulation │ based on real-world audits │ ├─ Critical vulnerability → │ merge block + alert to ArchDev + Core ├─ Warning → │ comment in PR + mandatory review └─ Clear → admission to Code DAO vote

```
Deployed contracts → continuous monitoring of anomalous transaction patterns → alerts to Gnosis Safe signers
```

The AI audit does not replace human review but complements it: models detect patterns that a human may miss, while a human assesses context and business logic inaccessible to the model. Audit parameters (sensitivity thresholds, list of checked vulnerabilities) are determined by Code DAO vote.

5.2.4 Formal Governance Parameters

```
Code DAO Parameters:
```

Quorum: 10% of participants with status ≥ Dev (critical: technical decisions must be made by technically competent participants)

Technical review period: 96 hours (minimum) — For smart contracts: 7 days + AI audit — For architectural decisions: 14 days Voting period: 96 hours Approval threshold: — Routine changes (bug fixes, dependency updates): >50% — Functional changes (new features): >60% — Architectural changes: >67% + cadCAD impact analysis — Smart contracts: >75% + AI audit clear + ≥2 independent reviews

Proposal rights: — Any participant with status ≥ Unit (pull request is open to all) — Minimum staking: 500 GBR

Voting rights: — Participants with status ≥ Dev — Weight determined by technical reputation: VP_code = code_contributions × review_quality × seniority — code_contributions: number of accepted PRs in the past year — review_quality: average review score from other developers — seniority: multiplier from MemberStatus (Dev=1×, LeadDev=1.5×, ArchDev=2×, Core=3×)

Mandatory checks before admission to vote: — CI/CD pipeline passed (100% unit tests, ≥80% coverage) — Linting passed (ESLint + Prettier) — AI audit completed (for Solidity code) — ≥1 approve from a participant with status ≥ LeadDev — Does not violate existing API contracts (breaking changes require a migration plan)

Rollback procedure: — Any change may be rolled back via emergency vote of ≥3 ArchDev/Core (without full quorum) — Emergency rollback: any 2 Core within 24h — After rollback: mandatory post-mortem analysis ```

5.2.5 Process Model: From PR to Merge

The life cycle of a code change in Code DAO:

```
[Fork / Branch] │ ▼ [Development + local tests] │ ▼ [Pull Request] → [CI/CD
Pipeline] │                        │ │                  ┌────┴────┐ │
Passed    Failed │                │         │ │                ▼      [Fix → retry]
│         [AI audit] │             (Solidity only) │                │ │
┌────┴────┐ │         Clear   Alert │           │        │ │           │     [Fix/
justification] │           │       │ │          ▼       ▼ │     [Peer Review: ≥1
LeadDev+] │                │ │            ┌────┴────┐ │        Approved  Changes
Requested │           │            │ │          │      [Fix → retry] │          ▼
│     [Code DAO vote, 96h] │                  │ │            ┌────┴────┐ │
Accepted  Rejected │            │           │ │    [Timelock 24h] [Archive / FRP]
│            │ │     [Auto-merge → main] │            │ │     [Deploy pipeline]
│          │ └── [Monitoring 7 days]
```

Typical Code DAO precedents:

1. Dependency update (routine). A developer upgrades React 18→19. CI/CD passes, unit tests are green. One approve from a LeadDev. Vote: 48h, simple majority. Merge after 24h Timelock.

2. New smart contract (critical). A developer proposes LPBurnInitiator.sol. AI audit identifies 0 critical, 2 low issues (gas optimisation). Two independent reviews from ArchDev. Vote: 96h, threshold 75%. After approval: Timelock 48h. Deploy to testnet → 7 days monitoring → deploy to mainnet.

3. Architectural decision (strategic). Proposal to migrate from REST to GraphQL for GPROD API. The author provides: RFC document, proof-of-concept, performance benchmarks, migration plan. Discussion: 14 days. Vote: 96h, threshold 67%. If approved: a working group is formed in G-Plan for staged migration.

4. Emergency security patch. AI monitoring detects anomalous activity on UnitManager. Emergency pause activated (AI decision). Developer publishes a hotfix. Accelerated review: 2 ArchDev/Core within 24h. Vote: 24h (emergency mode), threshold 50%. Deploy → unpause. Post-mortem required within 7 days.

5.2.6 Open Source Governance: Distinctive Features of Code DAO

Code DAO implements a unique open-source governance model that differs both from traditional corporate governance (proprietary, closed) and from classical open source (BDFL — Benevolent Dictator For Life, as in Linux or Python).

Comparative table of code governance models:

| Parameter | Corporate | BDFL (Linux) | Meritocracy (Apache) | Code DAO (CSC) | |—|—|—|—|—| | Who decides | CTO / Tech Lead | One person (Torvalds) | Committers committee | DAO vote | | Decision criterion | Business value | Technical quality | Technical quality + consensus | VP_code × vote | | Transparency | Closed discussions | Public mailing lists | Public discussions | On-chain + DSP | | Fork right | No (proprietary) | Yes (GPL) | Yes (Apache License) | Yes (FRP protocol) | | AI audit | Internal (if any) | None | Partial | Mandatory | | Accountability | To shareholders | To community (informally) | To Foundation | On-chain, verifiable | | Incentives | Salary | Reputation, interest | Reputation | GBR + reputation + status | | Key-person risk | High | Critical | Medium | Minimal |

Code DAO reduces key-person risk, which is a fundamental problem for open-source projects. In the BDFL model (Linux, Python, Ruby), the departure of a key person creates a succession crisis. In the Code DAO model, decisions are made in a distributed manner: no single developer, however talented, holds unilateral authority over the codebase. At the same time, technical competence in decisions is preserved: the VP_code formula ensures that voting weight correlates with actual contribution to development and technical authority.

5.3 Commerce DAO

5.3.1 Purpose

Commerce DAOs implement the concept of **decentralised crowd-investment** — allowing entrepreneurs and enthusiasts within the community to propose business ideas or formal business plans for realisation using collective community resources, providing investors with the opportunity to receive a share of the resulting profit.

5.3.2 Scope

Commerce DAO decisions include:

- Evaluation and approval of proposed commercial projects - Determination of the terms for the issuance and sale of internal tokens - Determination of profit-sharing schemes between executors and investors - Approval of partnerships and client agreements - Management of project budgets and financial reporting - Decisions regarding the continuation, modification, or termination of a project based on results

5.3.3 Characteristics

- **Economic stakes:** Commerce DAO decisions involve significant financial resources and economic risk. Governance processes are correspondingly more stringent. - **Due diligence:** Proposals in Commerce DAO undergo a structured evaluation process comprising market analysis, technical feasibility assessment, financial projections, and risk evaluation. - **Investor protection:** Smart contracts governing Commerce DAO projects include mechanisms for fund custody, milestone-based staged release, transparent accounting, and (in certain circumstances) fund return. - **Results-based accountability:** Commercial projects are subject to regular performance reporting, with defined metrics and benchmarks. Sustained unsatisfactory performance may initiate governance actions, including management change or project restructuring.

5.3.4 Relationship to the IPI Model

Commerce DAOs serve as the governance layer for phases 3–5 of the project life cycle (Accumulation, Implementation, Operation) for projects with commercial objectives. They operationalise the concept of the social-investment circle for revenue-generating projects.

5.3.5 Formal Governance Parameters

```
Commerce DAO Parameters:

Quorum: 15% of the total number of active participants with status ≥ Dev

Discussion period: 7 days (includes due diligence) Voting period: 5 days Approval threshold: — Standard projects (C(p) < 10,000 GBR): >50% — Large projects (C(p) ≥ 10,000 GBR): >60% — Strategic projects (C(p) ≥ 100,000 GBR): >67% + Economic DAO approval

Proposal rights: — Participants with status ≥ Dev — Minimum staking: 1,000 GBR — Required: business plan, financial projections, risk assessment

Voting rights: — Participants with status ≥ Dev — Weight: Hybrid Voting Power VP = √(GBR_staked) × activity_multiplier — SIC(p) investors: additional weight proportional to investment

Due diligence process: — AI agent analyses the business plan: market fit, competitive analysis, financial model — Technical review from Code DAO (for projects with a technical component) — DUNA legal analysis for compliance risks — Due diligence results are published as an annex to the proposal

Investor protection mechanisms: — Escrow contract: funds are allocated by milestones — Milestone voting: SIC(p) votes for the release of each tranche — Emergency stop: Core + ArchDev may freeze a project upon critical violations — Refund mechanism: upon non-fulfilment of milestones, remaining funds are returned to investors ```
```

5.3.6 Commercial Project Life Cycle

```
Full Commerce DAO project cycle:

[Idea] ─── author publishes concept │ [Discussion, 7 days] ─── community feedback │ [Due Diligence, 14 days] │ AI analysis + technical review │ + legal review │ [Commerce DAO Vote, 5 days] │ ├── Approved │ │ │ [Formation] │ │ Formation of SIC(p) and AG(p) │ │ Issuance of project tokens │ │ LP Burn: 0.1% of budget → GBR burn │ │ │ [Accumulation] │ │ Collection of investments from SIC(p) │ │ Achievement of CC(p) ≥ C(p) │ │ │ [Implementation] │ │
```

AG(p) works | | Milestone reports every 30 days | | SIC(p) votes for each tranche | | | [Operation] | | Project generates revenue | | Profit distribution: | | 60% — SIC(p) investors | | 25% — AG(p) executors | | 10% — Governance Pool | | 5% — AiC fund | | | [Completion / Continuation] | Final report or transition | to permanent operation | └──── Rejected → [Archive] or rework ```

Examples of typical Commerce DAO projects:

- **LQD (Liquidity Deployer):** Automated deployment of liquidity pools. Budget: ~15,000 GBR. SIC: ~200 participants. AG: 5 developers + 2 analysts. Implementation period: 6 months. Revenue model: 0.1% fee on deployed liquidity.

- **SAPP (Social Application Protocol Platform):** SDK for developers of decentralised applications on top of DSP. Budget: ~50,000 GBR. SIC: ~500 participants. AG: 12 developers + 3 designers. Period: 12 months. Revenue model: application marketplace (5% commission on sales).

- **PowerSwapMeta:** DEX aggregator with meta-transactions. Budget: ~30,000 GBR. SIC: ~300 participants. AG: 8 developers. Period: 9 months. Revenue model: 0.05% of exchange volume.

5.4 Economic DAO

5.4.1 Purpose

Economic DAOs represent **a fundamentally new concept for the organisation of public financing, project management, and socio-economic interaction.** They provide for the accumulation of social, financial, and economic resources for the most effective realisation of any relevant public projects — including projects whose scale and ambitions exceed the capacity of traditional financing mechanisms.

5.4.2 Distinction from Commerce DAO

While Commerce DAOs focus on individual commercial projects with defined profit models, Economic DAOs address **systemic economic decisions** affecting the community as a whole:

- Management of the community treasury - Allocation of resources across multiple projects and initiatives - Strategic investments in infrastructure, research, and capacity building - Inter-community economic agreements and partnerships - Monetary policy of the GBR ecosystem (e.g., decisions on staking parameters, liquidity provision, token burning) - Long-term economic planning and resource allocation

5.4.3 Characteristics

- **Highest economic stakes:** Economic DAO decisions affect the economic position and long-term viability of the entire community. Governance processes are the most stringent in the taxonomy. - **Supermajority requirement:** The most important Economic DAO decisions may require approval by a qualified majority (e.g., 67% or 75%) to ensure broad consensus before disposing of community resources. - **Extended discussion periods:** Proposals in Economic DAO are subject to extended discussion periods, allowing thorough analysis and debate before voting commences. - **Expert input:** Economic DAO governance may include advisory mechanisms — expert analysis, simulation results, review of historical precedents — to inform the decision-making process, while preserving

the principle that final decisions are made by the community through voting. - **Systemic risk management:** Economic DAOs are responsible for monitoring and managing systemic risks to the community economy — excessive concentration in specific projects, liquidity risks, dependence on specific external systems, and other threats to the economic health of the ecosystem. - **AI Treasury Advisor (from A8):** Economic DAO uses an AI agent to inform economic decisions:

```` AI Treasury Advisor (AiC implementation, Circuit 1):

On-chain Analytics Agent: — Real-time monitoring of economic flows: LP positions, trading volumes, TVL, token distribution, concentration — Visualisation: integration with Dune Analytics

Risk Assessment: — Systemic risk assessment: Risk(ecosystem) = f(concentration, liquidity, correlation, external_deps) — Early warning when approaching critical thresholds

Simulation Engine (cadCAD): — Modelling the consequences of economic proposals before voting: "if 5% of treasury is burned → impact on liquidity, GBR price, TVL" — Monte Carlo simulation on GyberComputer

AI Treasury Advisor informs, does not decide: all analysis results are published as a supplement to the proposal in Economic DAO; the final decision rests with the participants ```

#### 5.4.4 Formal Governance Parameters

```` Economic DAO Parameters:

Quorum: 25% of the total number of active participants with status ≥ LeadDev

Discussion period: 14 days (mandatory, cannot be shortened) Voting period: 7 days Approval threshold: — Standard economic decisions: >67% — Changes to monetary policy: >75% — Changes to axioms or architecture: >80% + confirmation by MacroeconomicDAO

Proposal rights: — Participants with status ≥ LeadDev — Minimum staking: 10,000 GBR — Required: economic justification, cadCAD simulation (≥1000 scenarios), DUNA legal expert opinion

Voting rights: — Participants with status ≥ Dev — Weight: Hybrid Voting Power × 1.5 for statuses LeadDev and above (expertise premium)

cadCAD obligation: — Every proposal in Economic DAO is accompanied by a cadCAD simulation: (1) baseline scenario (current parameters) (2) proposed scenario (proposed changes) (3) stress scenarios (extreme conditions) — Results are published as an annex to the proposal — If the difference between baseline and proposed exceeds 20% on key metrics → mandatory extended discussion period (21 days)

Veto mechanism: — Core participants (Core status) possess the right of suspensive veto on Economic DAO decisions for 14 days — The veto is not absolute: a repeated vote with a threshold of >80% overrides the veto — The veto exists as protection against hasty decisions, not as an instrument of Core power over the community ```

#### 5.4.5 Types of Economic Decisions

Economic DAO operates in several fundamentally distinct domains, each requiring specific parameters and expertise:

**Type 1: Monetary policy (tokenomics parameters).** - Modification of the reward rate in UnitManager - Adjustment of Activity Score thresholds - Modification of the Halving schedule - Management of Buyback & Burn parameters - Quorum: elevated (30%). Threshold: 75%. cadCAD: mandatory

**Type 2: Treasury management.** - Allocation of Governance Pool funds across directions - Strategic investment decisions - Formation of a stabilisation fund - Management of DUNA Treasury (fiat portion) - Quorum: standard (25%). Threshold: 67%

**Type 3: Inter-project coordination.** - Prioritisation of ecosystem projects - Allocation of GyberComputer computational resources - Resolution of resource conflicts between Commerce DAO projects - Quorum: standard (25%). Threshold: >50% (simple majority is sufficient for coordination decisions)

**Type 4: System architecture.** - Modification of DAO class parameters (quorums, thresholds) - Addition of new DAO classes - Modification of the Hybrid Voting Power formula - Modification of the MemberStatus hierarchy - Quorum: elevated (30%). Threshold: 80%. Required: MacroeconomicDAO confirmation

**Type 5: External relations.** - Partnerships with other DAO ecosystems - Cross-chain integration decisions - Interaction with regulators via DUNA - Quorum: standard (25%). Threshold: 67%

#### 5.4.6 MacroeconomicDAO as Meta-Governance

**MacroeconomicDAO** functions as a meta-governance layer coordinating the activities of the four DAO classes. It is the institutional expression of the community's collective economic intelligence — the mechanism through which the community exercises conscious, purposeful governance of its own socio-economic system.

MacroeconomicDAO is not a centralised authority overriding the decisions of individual DAOs. Rather, it is a governance space in which inter-class issues are resolved, the allocation of resources across DAO classes is coordinated, and the general trajectory of the experiment is directed by the collective will of active participants.

**AI augmentation of DAO governance (from A8).** To counteract voter fatigue and cognitive overload as the community scales, all four DAO classes utilise AiC AI services:

- **AI proposal summariser:** For each proposal, a structured summary is generated — the essence, risks, economic consequences, and arguments for and against. Participants receive an objective overview before voting, which supplements rather than replaces discussion. - **Consequence simulator:** Before voting on economically significant proposals, the results of modelling are shown: impact on the treasury, liquidity, GyberComputer load, and SES parameters. Implemented via cadCAD on GyberComputer. - **AI discussion summariser:** Automatic summaries of discussions with the main arguments of each side — for participants who cannot follow the discussion in real time. This lowers the barrier to informed participation (A6).

### 5.5 DAO Interaction and Conflict Resolution

The four DAO classes are not isolated from one another. Decisions in one class frequently have consequences for others:

- A Code DAO decision to adopt a new technical architecture may have economic consequences (costs, performance, scalability) requiring Economic DAO consideration - A Commerce DAO project may require social coordination (community engagement, marketing) falling within the competence of Social DAO - An Economic DAO resource allocation decision may affect the priorities of Commerce DAO projects

When inter-class conflicts arise, resolution follows a structured escalation process:

1. **Informal coordination:** Representatives of the affected DAOs discuss the matter in shared communication spaces, seeking consensus. 2. **Joint proposal:** If informal coordination produces a resolution, it is formalised as a joint proposal submitted for ratification to the relevant DAOs. 3. **MacroeconomicDAO arbitration:** If informal coordination fails, the matter is escalated to MacroeconomicDAO for system-wide discussion and resolution.

```` Inter-class interaction model:

```
┌──────────────────────────────────────────────────────────────────────────┐
│        MacroeconomicDAO      │      │        (meta-governance,        arbitration)      │
└──────────────────────────────────────────────────────────────────────────┘
                                          ┬
│  ┌──────────────────────────┼──────────────────────────┐  │  │  │  ┌──────────▼──────────┐
┌──────▼──────┐  ┌──────▼──────────┐  │ Social │  │ Code │  │ Commerce │  │ DAO
│◀┤ DAO ├▶│ DAO │  │  │  │  │  │  │  │  └──────┬──────┘  └──────┬──────┘
└──────┬──────┘  │  │  │  └──────────────────────┼──────────────────────┘  │
┌──────▼──────┐  │ Economic │  │ DAO │  │ (systemic │  │ decisions) │
└──────────────┘
```

Interaction flows: Social ↔ Code: technical decisions with social consequences Social ↔ Commerce: marketing, engagement, social responsibility Code ↔ Commerce: technical implementation of commercial products Economic ↔ all: financing, resource allocation, monetary policy MacroeconomicDAO → all: arbitration, strategic direction ```

### 5.6 Formal Analysis of the Four-Class DAO System

The four-class DAO taxonomy described in sections 5.1–5.4 constitutes not an arbitrary classification but a formally grounded architecture optimised for different types of collective decisions. In this section we formalise the properties of this system and demonstrate its advantages over monolithic DAOs.

#### 5.6.1. Formal Model of a DAO Class

We define a DAO class as a tuple:

```
D = ⟨Q, V, T, P, E, B⟩
```
```

where: Q — minimum quorum (share of active participants) V — voting function T — time parameters (discussion_period, voting_period, timelock_delay) P — set of admissible proposal types E — escalation function (conditions for referring a decision upward) B — budget constraints (maximum size of a single proposal) ```

The four DAO classes form a partially ordered set with respect to the level of economic influence:

```` Social ≤ Code ≤ Commerce ≤ Economic

Formally: influence($D_1$) ≤ influence($D_2$) ⟹ Q($D_1$) ≤ Q($D_2$) — higher quorum ∧ T($D_1$) ≤ T($D_2$) — longer periods ∧ B($D_1$) ≤ B($D_2$) — higher budget thresholds ∧ approval($D_1$) ≤ approval($D_2$) — higher approval threshold ```

This monotonicity property guarantees that decisions with greater economic influence undergo more rigorous scrutiny: from a simple majority in Social DAO (50%, quorum 5%) to a qualified majority in Economic DAO (67%, quorum 25%).

#### 5.6.2. Theorem on the Reduction of Cognitive Load

**Claim.** The four-class system reduces the cognitive load on a participant compared to a monolithic DAO while maintaining or improving decision quality.

**Proof (sketch).** Let N be the total number of proposals in a given period, N_s, N_c, N_m, N_e the number of proposals in Social, Code, Commerce, and Economic DAO respectively, such that N = N_s + N_c + N_m + N_e.

In a monolithic DAO each participant must evaluate all N proposals. Cognitive load: L_mono = N × complexity_avg.

In the four-class system a participant may focus on 1–2 classes in which they are competent, delegating their vote in the rest. Average cognitive load:

```` L_classified = α × N_focused + β × N_delegated

where α ≈ 1 (full evaluation), β ≈ 0.1 (delegate check) ```

For a typical distribution (N_focused ≈ N/4, N_delegated ≈ 3N/4):

`L_classified ≈ N/4 + 0.1 × 3N/4 = N/4 + 3N/40 = 13N/40 ≈ 0.325N`

Thus, L_classified / L_mono ≈ 0.325 — cognitive load is reduced by ~67.5%.

Decision quality does not diminish in the process, because: (a) participants vote on matters in which they are competent, reducing the noise of incompetent votes; (b) delegation concentrates voting power among participants with the relevant expertise; (c) different approval thresholds and quorums are optimised for each type of decision. □

#### 5.6.3. Security Analysis: Attacks on Inter-Class Boundaries

The four-class system introduces a new class of attack absent from monolithic DAOs: **cross-class arbitrage** — exploitation of differences in governance parameters between classes.

**Example of an attack.** An attacker submits a proposal to Social DAO (low quorum 5%, simple majority) disguised as a social initiative but with economic consequences equivalent to an Economic DAO decision. For example: “organise a hackathon with a budget of 500 GBR” — formally Social DAO, but the hackathon includes “prizes” in the form of project tokens that are in effect an investment decision.

**Mitigation.** The escalation function E(D) formalises the conditions under which a proposal must be reclassified:

```
Escalation rules:

1. Budget threshold: budget(proposal) > B(D) ⟹ escalate(D_next) 2. Inter-class influence: if proposal D₁ has consequences for D₂, and influence(D₂) > influence(D₁), then escalate(D₂) 3. AI classifier: AiC analyses the text and parameters of the proposal, recommending a class with a confidence score 4. Challenge: any participant with status ≥ Dev may initiate reclassification of a proposal within the discussion_period
```

The AI Cross-DAO Coordinator (Appendix F) automatically analyses each new proposal and assigns it a recommended class. Where the declared and recommended classes diverge, the proposal is flagged as “requiring reclassification” and referred for manual review.

#### 5.6.4. Evolution of Parameters

The parameters of each DAO class (quorum, periods, thresholds) are not static. They are subject to adaptation through Economic DAO on the basis of empirical data:

```
Adaptive rules:

Rule 1 (Adaptive Quorum): If participation in votes of D over the past 30 days < 2 × Q(D), then Q(D) := max(Q(D) × 0.9, Q_min(D)) Rationale: an excessively high quorum under low turnout paralyses governance.

Rule 2 (Escalation Frequency): If >20% of proposals of D₁ are escalated to D₂ over the past 90 days, then revision of the boundary of the admissible type set P(D₁). Rationale: frequent escalation indicates suboptimal classification.

Rule 3 (Voting Period): If the average turnout in the final 20% of voting_period(D) exceeds 50% of total turnout, then voting_period(D) may be reduced. Rationale: late voters had sufficient time but did not require it.
```

All parameter adaptations pass through Economic DAO voting (the class with the highest requirements), ensuring system stability — meta-rules cannot be altered by lightweight votes.

# 6. The Applied Ecosystem: From Theory to Implementation

The theoretical constructs set forth in the preceding sections find practical expression in specific software products, protocols, and services that constitute the applied ecosystem of GyberExperiment. Each element of the ecosystem implements one or more principles of the cybersocial economy, and together they form a fully functional environment for decentralised socio-economic interaction.

## 6.1 DSP — GyberSocial Platform (Decentralised Social Platform)

DSP is the central hub of the ecosystem — a decentralised social platform that provides the infrastructure for all forms of participant interaction.

```
DSP Architecture:

  ┌──────────────────────────────────────────────┐
  │              DSP — Layers                    │
  ├──────────────────────────────────────────────┤
  │                                              │
  │  User layer:                                 │
  │    — Participant profiles (verification      │
  │      via digital signature)                  │
  │    — News feeds (aggregation                 │
  │      from project spaces)                    │
  │    — Reputation system (integration          │
  │      with UnitManager and G-Plan)            │
  │    — Direct communication (P2P encryption)   │
  │                                              │
  │  Project layer:                              │
  │    — Project spaces (for each                │
  │      IPI-model project)                      │
  │    — Discussion directories                  │
  │    — Voting systems (integration             │
  │      with Governor contracts)                │
  │    — Task tracking (G-Plan)                  │
  │                                              │
  │  Infrastructure layer:                       │
  │    — Distributed storage (IPFS)              │
  │    — Blockchain integration (BSC, Ethereum)  │
  │    — P2P network (libp2p)                    │
  │    — End-to-end encryption                   │
  │                                              │
  │  Technology stack:                           │
  │    — Frontend: React/Next.js, TypeScript     │
  │    — Backend: Node.js, Rust (core modules)   │
  │    — Protocol: ActivityPub-compatible        │
  │    — Storage: IPFS + local cache             │
  │    — Identification: DID (Decentralized      │
  │      Identifiers) + Verifiable Credentials   │
  └──────────────────────────────────────────────┘
```

**AI personalisation with data sovereignty (from A8, A3).** DSP implements intelligent content personalisation without violating user data sovereignty. Formally:

```
Let u ∈ U be a participant, D(u) their local data
(interaction history, preferences, subscriptions).

The personalisation model M(u) is trained exclusively
on the user's device (on-device inference):

  M(u) : D(u) → R(u),

where R(u) is the personalised content ranking.

Privacy invariant:
  ∀ u ∈ U : D(u) ∩ Server_Storage = ∅

No ecosystem server stores or processes
raw user data. On the server side, only
aggregated, anonymised signals are used
(federated aggregation):

  M_global = FedAvg({∇M(u₁), ∇M(u₂), ..., ∇M(uₙ)})

The global model improves through federated learning
(Flower framework), in which only gradients,
not data, are transmitted from devices.
```

In this respect DSP is fundamentally distinct from Web2 platforms, where personalisation is built on the centralised collection and monetisation of user data. In DSP, personalisation is a tool of the user, not an instrument for extracting value from the user.

DSP differs from traditional social platforms in a principled sense: data belongs to users (Axiom A3), the platform code is open (AGPL-3.0), governance is exercised through Code DAO and Social DAO, and monetisation is based on creating value for participants rather than extracting value from their data.

### 6.1.1. Identification and Authentication Architecture

The DSP identification system is built on a two-level model that balances usability with cryptographic security:

```
DSP Two-Level Authentication:

  Level 1 — Basic access (password):
    — Registration via email + password
    — Limited functionality: viewing profiles,
      reading forums, basic communication
    — Storage: bcrypt-hashed password
      in encrypted Supabase database
    — Sessions: JWT + Refresh Token (30-day rotation)
    — 2FA: optional TOTP (Google Authenticator)

  Level 2 — Full access (wallet):
    — Binding of a cryptographic wallet
      (MetaMask, WalletConnect, Ledger)
    — Unlocks: DAO voting,
      reward receipt, project creation,
      participation in Commerce DAO
    — Wallet = universal primary key
      (wallet address = primary key across all modules)
    — Session upgrade: signature challenge (EIP-4361)
    — DID: binding of a Decentralized Identifier
      (did:gyber:0x...)

  Relationship between levels:
    — User begins at Level 1
    — Binding a wallet upgrades to Level 2
    — Reverse downgrade is impossible:
      once bound, a wallet remains linked
    — Multiple wallets: a user may bind
      several wallets, but only one
      is primary (used for DAO voting)
```

**Verifiable Credentials (VC):** DSP supports the issuance and verification of Verifiable Credentials to the W3C standard for confirming participant skills, achievements, and statuses. Credentials are stored in the user's wallet (not on a server) and may be presented to any party without disclosing superfluous data (selective disclosure via ZK-proofs). Examples: "Participant completed 10 Code DAO tasks" → VC signed by G-Plan oracle; "MemberStatus: LeadDev" → VC signed by UnitManager.

### 6.1.2. Communication Protocol

DSP uses a federated communication protocol that provides decentralisation without sacrificing convenience:

```
DSP Protocol Stack:

  Public channels:
    — ActivityPub-compatible protocol
    — Federated architecture: each
      project space = a separate server
    — Cross-project communication:
      user subscribes to channels
      of projects of interest
    — Moderation: automatic (AI filters) +
      social (moderators from Social DAO)

  Private channels:
    — End-to-End encryption (Signal Protocol)
    — Group chats: MLS (Message Layer Security)
    — Keys stored on the user's device
    — Server transmits only encrypted packets

  Project spaces:
    — Dedicated forum for each IPI project
    — Binding to DAO class: Social, Code, Commerce
    — Integration with G-Plan: tasks, milestones, reports
    — Integration with Governor: proposals, votes
    — Automatic archiving of completed projects
      on IPFS

  Notifications:
    — Push notifications about DAO votes,
      G-Plan tasks, messages
    — AI prioritisation: important events
      (votes with a deadline, critical PRs)
      receive elevated priority
    — Configurable filters: by project,
      event type, DAO class
```

### 6.1.3. Modular Extension Architecture

DSP is designed as a modular platform allowing extension through standardised interfaces:

```
DSP Modular Architecture:

  Core:
    — Identification (DID, wallet)
    — Basic communication (ActivityPub)
    — Profiles and reputation
    — DAO integration (Governor, Timelock)

  Modules (extensions):
    — DSP-Marketplace: marketplace of
      decentralised applications
    — DSP-Analytics: dashboards and analytics
      (integration with Dune Analytics)
    — DSP-Education: educational platform
      (courses, certificates as VC)
    — DSP-Media: decentralised media
      (articles, video, podcasts on IPFS)
    — DSP-AI: AI assistant for ecosystem navigation
      (project recommendations,
      personalised feed)

  Extension standard (SAPP):
    — API: REST + GraphQL + WebSocket
    — Authentication: OAuth 2.0 + DID Auth
    — Publication: SAPP Registry (on-chain)
    — Monetisation: via internal tokens
      or subscriptions
    — Review: Code DAO checks security
      before publication in DSP-Marketplace
```

### 6.1.4. Comparison of DSP with Existing Platforms

| Characteristic | Facebook/ Meta | Twitter/X | Mastodon | Lens Protocol | DSP |
|---|---|---|---|---|---|
| Data ownership | Platform | Platform | Server operator | User (on-chain) | **User (DID + IPFS)** |
| Feed algorithm | Closed, optimised for engagement | Closed | Chronological | Open | **AI on-device (A3), federated learning** |
| Moderation | Centralised | Centralised | Server operators | Minimal | **Social DAO + AI filters** |
| Monetisation | Advertising (user data) | Advertising + subscription | Donations | NFT | **Value creation (PMIP, Commerce DAO)** |
| Code | Closed | Closed | Open (AGPL) | Open | **Open (AGPL-3.0)** |
| Governance | Corporate | Unilateral (CEO) | Each server separately | Lens DAO | **Code DAO + Social DAO** |
| Identification | Email/phone | Email/ phone | Email | Lens Profile (NFT) | **DID + wallet + VC** |
| Blockchain | None | None | None | Polygon | **BSC → GyberNet** |

## 6.2 GyberNet — Decentralised Network

GyberNet is a dedicated blockchain infrastructure that ensures the transparency and immutability of all ecosystem transactions. Unlike general-purpose public blockchains (Ethereum, BSC), GyberNet is designed as a specialised network optimised for the needs of the cybersocial corporation: DAO governance, participant activity verification, IPI project coordination, and enforcement of economic protocols.

### 6.2.1 Architecture and Functions

```
GyberNet Architecture Layers:

  Layer 1 — Consensus:
    — Modified BFT (Byzantine Fault Tolerant)
    — Reputation-weighted validators:
      validator_weight(v) = stake(v) × reputation_score(v)
    — Finality: 3 seconds (single-slot finality)
    — Throughput: 1000+ TPS (target)
    — Energy efficiency: PoS without PoW mining

  Layer 2 — Execution:
    — EVM-compatible virtual machine
    — Contract languages: Solidity ^0.8.20, Vyper
    — Pre-installed contracts (precompiles):
      Governor, Timelock, UnitManager, GBRVotingWrapper
    — Gas model: subsidised gas for
      DAO votes (from Governance Pool)

  Layer 3 — Storage:
    — Merkle Patricia Trie for state
    — IPFS binding for heavy data
      (documents, media, project artefacts)
    — Pruning strategy: archival data
      moved to IPFS; on-chain retains
      only hashes and metadata

  Layer 4 — Network:
    — libp2p for P2P communication
    — Gossip protocol for transaction propagation
    — DHT (Distributed Hash Table) for node discovery
    — TLS 1.3 for inter-node communication encryption

  Layer 5 — Cross-network:
    — Bridges to BSC (main GBR token network)
    — Bridges to Ethereum (DeFi integration)
    — Bridges to TON, Solana (ecosystem expansion)
    — Standard: IBC-like protocol
      (Inter-Blockchain Communication)
```

### 6.2.2 Validation Model

GyberNet uses a modified Proof-of-Stake model that integrates the ecosystem's reputation mechanism:

```
Reputation PoS (RPoS):

  Participation in validation:
    — Any participant with status ≥ Dev may
      become a validator
    — Minimum staking: 10,000 GBR
    — Activity Score ≥ 50 (confirmation of
      active participation in the ecosystem)

  Validator weight:
    validator_weight = √(staked_GBR) × (1 + AS/100)
    where AS = participant's Activity Score

    Square root of staking prevents
    domination by large holders.
    Activity Score bonuses active participants.

  Validator rewards:
    — Block reward: 10 GBR/block (decreasing
      according to the halving schedule)
    — Transaction fees: 70% to validator,
      30% to Governance Pool
    — Slashing: 5% of staking for double-signing,
      1% for missing >100 consecutive blocks

  Validator rotation:
    — Active set: up to 100 validators
    — Rotation every 1000 blocks (~50 minutes)
    — Priority: validator_weight + randomness
      (VRF — Verifiable Random Function)
```

### 6.2.3 Specialised Functions for CSC

**Decentralised Name Resolution (GNS — Gyber Name Service):** - Registration of human-readable names for participants: `alice.gyber` - Binding to wallet, DID, DSP profile, and projects - Governance through Social DAO (registration cost, rules) - Support for subdomains for projects: `lqd.commerce.gyber`

**On-chain Activity Registry:** - All activity verifications from G-Plan are recorded on-chain - Immutable history: Activity Score history for each participant - Used for the calculation of Hybrid Voting Power and rewards

**DAO Governance Hub:** - All proposals, votes, and decisions of the four DAO classes are recorded on GyberNet - Cross-DAO references: decisions of one DAO may reference decisions of another - Governance analytics: on-chain data for AI Treasury Advisor

### 6.2.4 Deployment Strategy

```
GyberNet Deployment Phases:

  Phase 0 (current): BSC as main network
    — GBR token deployed on BSC
    — UnitManager on BSC
    — Governor/Timelock on BSC
    — Advantage: ready infrastructure,
      low gas costs, PancakeSwap liquidity

  Phase 1: GyberNet Testnet
    — Deployment of test network
    — Migration of governance contracts
    — Testing of RPoS consensus
    — Parallel operation with BSC

  Phase 2: GyberNet Mainnet (Canary)
    — Launch of main network with limited
      validator set (Core + ArchDev)
    — Bridge BSC ↔ GyberNet: GBR on both networks
    — Governance migrates to GyberNet
    — UnitManager remains on BSC
      (to preserve liquidity)

  Phase 3: Full migration
    — All ecosystem contracts on GyberNet
    — BSC → bridge network for external liquidity
    — Public validation: any participant ≥ Dev
```

## 6.3 GyberComputer — Distributed Computing Network

GyberComputer is a distributed network of computing nodes provided by community members for the execution of resource-intensive tasks: training AI models, rendering, scientific computation.

```
GyberComputer Architecture:

  Participant nodes:
    — Provide computational resources
      (CPU, GPU, RAM, storage)
    — Receive reward in GBR
      proportional to provided resources
    — Verification via Proof-of-Computation

  Task scheduler:
    — Distribution of tasks across nodes
      taking into account available resources
    — Prioritisation based on DAO voting
    — Fault tolerance: duplication
      of critical computations

  Integration with AiC:
    — Provision of GPU capacity for
      model training (Circuit 1)
    — Distributed inference for
      AI services (Circuit 3)
    — Federated learning without transfer
      of raw data

  Technology stack:
    — Orchestration: Kubernetes + custom
      scheduler in Rust
    — Communication: gRPC, Protocol Buffers
    — Containerisation: Docker, WASM
    — Monitoring: Prometheus, Grafana

  AI Compute Orchestrator (from A8):
    — Predictive Scheduling: ML model
      (Reinforcement Learning, PPO/SAC)
      predicts optimal task distribution
      across nodes taking into account latency,
      capacity, and availability
    — Anomaly Detection: early detection
      of node failures via metric analysis
      (Prophet/DeepAR) before computation loss
    — Auto-scaling: predictive scaling
      based on usage patterns —
      resource allocation before peak load
    — Proof-of-Computation Verification:
      AI verifies the correctness of computations,
      critical for federated learning (AiC)
```

## 6.4 G-Plan — Task Management and Activity Verification System

G-Plan is an integrated task management system that simultaneously serves as an activity verification mechanism for participants within the UnitManager system.

```
G-Plan Functions:

  Task management:
    — Creation, assignment, tracking of tasks
      within IPI-model projects
    — Task hierarchy: epics → stories → tasks
    — Binding to project milestones
    — Cross-project dependencies

  AI Task Orchestrator (from A8):
    — Decomposition: AI agent analyses
      IPI-model project specifications
      and proposes task breakdown
      with complexity and dependency assessment
    — Assignment: recommendation of executors
      based on competency profile, current
      workload, and execution history
    — Predictive Analytics: completion deadline
      forecast, early warning
      of delay risks
    — Cross-project Dependency Resolver:
      identification of dependencies between tasks
      of different projects, order optimisation
    — All AI Task Orchestrator recommendations
      are proposals: approval of
      assignments rests with the project lead

  Activity verification:
    — Recording of completed tasks on-chain
    — Task confirmation by participants
      of higher status
    — Generation of activity reports
      for UnitManager
    — Wallet classification algorithm:
      active / inactive

  Integration with reputation system:
    — Task execution quality affects
      reputation weight
    — Reputation affects voting weight
      in Code DAO and Commerce DAO
    — Continuous verification:
      a participant who ceases activity
      loses the right to rewards

  AI task quality verification (from A8):
    — Automatic quality assessment of a completed
      task before forwarding for human confirmation:
      Q(task) = AI_Review(deliverable, spec, context)
      Q ∈ [0, 1], where Q < θ_min → return for revision,
      Q ≥ θ_min → forwarding for review by a participant
      of higher status
    — AI does not replace human confirmation
      but serves as a first filter: reduces the load
      on reviewers and ensures basic quality control
      (compliance with specification, completeness, formatting)
    — Threshold θ_min is determined by Code DAO vote
```

```
Sybil detection and farming protection (from A8):
  — Graph Neural Network analyses the wallet
    interaction graph, detecting clusters
    of coordinated/fake accounts:
    Sybil(wallet_cluster) = GNN(G_interactions)
  — Anomaly detection identifies atypical
    patterns: task wash trading, mass
    creation of identical tasks, suspicious
    confirmation chains
  — Proof of Personhood (ZK cryptography):
    proof of "one person — one account"
    without disclosure of personal data (A3)
  — Sybil analysis results are advisory:
    the final decision on blocking
    is made by Social DAO
```

## 6.5 LQD, SAPP, PowerSwapMeta, Contact — Auxiliary Ecosystem Projects

In addition to the core platform components, the GyberExperiment ecosystem includes a number of applied projects implemented or being implemented through the IPI model:

### 6.5.1. LQD (Liquidity Deployer)

LQD is an automated platform for the deployment and management of liquidity pools for ecosystem project tokens. LQD is a critical infrastructure component enabling IPI-model projects to transition from the Accumulation phase to the Implementation phase.

```
LQD Architecture:

  Components:
    — Pool Factory: smart contract for creating
      liquidity pools (GBR/project_token,
      project_token/USDT)
    — LP Strategy Engine: algorithm for determining
      optimal pool parameters
      (initial price, depth, fee tier)
    — LP Burn Module: mechanism for burning LP tokens
      as proof-of-commitment upon project initiation
    — LP Lock Manager: temporary locking of LP tokens
      to prevent rug-pull
    — Analytics Dashboard: monitoring of TVL, volumes,
      impermanent losses

  Integrations:
    — DEX: PancakeSwap V3, Uniswap V3 (via bridges)
    — Chainlink: price oracles for determining
      initial parameters
    — Commerce DAO: approval of parameters for large pools

  Deployment parameters (configurable via Commerce DAO):
    — Minimum LP Burn: 0.1% of C(p) in GBR
    — LP Lock period: 180 days (default)
    — Fee tier: 0.3% (standard), 1% (volatile tokens)
    — Initial liquidity: ≥$1,000 (in GBR equivalent)

  Economic model:
    — Fee: 0.1% of each deployment (→ Governance Pool)
    — LP incentivisation: additional GBR rewards for
      liquidity providers in the first 90 days
    — Buyback & Burn: 2% of fee revenue → purchase
      and burn of GBR

  AI integration (AiC Circuit 1):
    — Predictive pricing: AI analyses market data
      and recommends the initial token price
    — Impermanent loss forecasting: warning to LP
      of potential losses under various scenarios
    — Liquidity optimisation: automatic rebalancing
      of positions (Uniswap V3 concentrated liquidity)
```

LQD is the first Commerce DAO project to have completed the full IPI cycle: from idea (proposal for LP deployment automation) through SIC formation (~200 participants), accumulation (~15,000 GBR), implementation (6 months, AG = 5 developers + 2 analysts) to operation. Revenue model: 0.1% fee on deployed liquidity; projected annual revenue at TVL $10M = $10,000 in GBR equivalent.

### 6.5.2. SAPP (Social Application Protocol Platform)

SAPP is a standardised platform and SDK for the creation of decentralised social applications (dApps) on top of DSP. SAPP implements Axiom A4 (extensibility): any developer may create a DSP extension without permission from a central authority.

```
SAPP Stack:

  SAPP Module = {
    manifest.json:     — module metadata (name, version, author,
                         dependencies, required permissions)
    src/:              — module source code (React + TypeScript)
    api-contract.ts:   — typed API contract
                         (what the module provides to DSP)
    permissions.json:  — requested permissions
                         (user data, network, storage)
    tests/:            — automated tests
    audit.sbt:         — AI audit soulbound token (after passing)
  }

  Publication process:
    1. Developer creates a SAPP module
    2. AI Code Reviewer checks the code (vulnerabilities, compliance)
    3. Code DAO peer review (≥2 approvals from Dev+)
    4. Commerce DAO approves the business model
    5. Publication in SAPP Marketplace
    6. DSP users install the module (opt-in)

  Marketplace:
    — Catalogue of approved SAPP modules
    — Ratings and reviews (verified installations)
    — Three categories: free, freemium, paid
    — Monetisation: developer receives 85% of revenue,
      10% → Governance Pool, 5% → AiC fund
    — Internal application tokens: each SAPP
      may issue its own token via LQD

  Examples of SAPP modules:
    — GyberNotes: decentralised note-taking system
      with encryption (DDP + E2E)
    — GyberPolls: advanced polls with QV/CV mechanisms
    — GyberAnalytics: AI analytics of community activity
    — GyberBounty: bug bounty programme management
    — GyberLearn: educational platform with certificates (VC)
```

SAPP as a Commerce DAO project: budget ~50,000 GBR, SIC ~500 participants, AG = 12 developers + 3 designers, implementation period 12 months. Revenue model: application marketplace with 15% commission on sales.

### 6.5.3. PowerSwapMeta

PowerSwapMeta is a DEX aggregator with meta-transaction support that optimises token exchange within the GyberExperiment ecosystem. The core innovation: users do not pay for gas directly — the fee is paid from the GBR liquidity pool, which lowers the barrier to entry (A6, inclusivity).

```
PowerSwapMeta Architecture:

  Components:
    — Route Optimizer: algorithm for finding the optimal
      exchange route across multiple pools
      (split routes, multi-hop)
    — Meta-Transaction Relayer: EIP-2771-compatible
      relayer paying gas on behalf of the user
    — Cross-Chain Bridge: secure exchange between
      BSC ↔ Ethereum ↔ Polygon
      (lock-mint with M-of-N threshold signatures)
    — Gas Station Network: GBR pool for gas payment
      (replenished from fees)

  Meta-transactions:
    1. User signs exchange intent (off-chain)
    2. Relayer pays gas from Gas Station Pool
    3. Exchange is executed on-chain
    4. Fee (0.05% of volume) covers gas + profit

  AI Route Optimizer (AiC Circuit 3):
    — ML model predicts the optimal route
      based on current liquidity, volatility,
      gas price, and historical patterns
    — Slippage prediction: AI warns
      of potential slippage before execution
    — MEV protection: detection of sandwich attacks
      and front-running, automatic selection
      of a protected route (private mempool)

  Supported pools:
    — PancakeSwap V2/V3 (BSC)
    — Uniswap V2/V3 (Ethereum, Polygon)
    — SushiSwap (multichain)
    — Ecosystem project pools (via LQD)
```

PowerSwapMeta as a Commerce DAO project: budget ~30,000 GBR, SIC ~300 participants, AG = 8 developers, period 9 months. Revenue model: 0.05% of exchange volume; projected annual revenue at $100M volume = $50,000 in GBR equivalent.

### 6.5.4. Contact

Contact is a decentralised identity and contact management system for ecosystem participants. Contact implements Axiom A3 (data sovereignty) for the domain of digital identity: the user exercises full control over their identity and chooses what data to disclose.

```
Contact Architecture:

  DID (Decentralized Identifier):
    — Format: did:gybernaty:<wallet_address>
    — Storage: on-chain (GyberNet registry)
    — Control: only the wallet owner may
      update/revoke the DID

  Verifiable Credentials (VC):
    — MemberStatus VC: confirmation of current status
      (issued by UnitManager, signed on-chain)
    — Skill VC: confirmation of competencies
      (issued by Code DAO after peer review)
    — Project VC: confirmation of project participation
      (issued by G-Plan upon milestone completion)
    — AI Audit VC: confirmation of AI audit passage
      (issued by AiC, soulbound)

  Zero-Knowledge Proofs:
    — ZK-proof of membership: proof that
      "I am an NFC participant with status ≥ Dev"
      without disclosing wallet or name
    — ZK-proof of activity: proof that
      "my Activity Score ≥ 60"
      without disclosing the exact value
    — ZK-proof of age: proof that
      "I have been a participant for more than 6 months"
      without disclosing the registration date
    — Technology: Semaphore (ZK-SNARK group membership)

  Portable identity:
    — One DID for all ecosystem projects
    — Cross-platform: DSP, SAPP modules, GyberNet, external dApps
    — Selective disclosure: the user chooses
      which VC to present for each interaction
    — Revocation: the user may revoke
      any issued VCs at any time

  AI integration:
    — Sybil detection: GNN analyses the identity graph,
      identifying clusters of fake accounts
    — Anomaly detection: detection of suspicious
      patterns (mass registration, coordinated
      activity, wash trading)
    — Reputation scoring: AI aggregates VC, Activity Score,
      voting history to form
      a comprehensive reputation score
```

Contact as a Commerce DAO project: budget ~20,000 GBR, SIC ~250 participants, AG = 6 developers + 2 cryptographers, period 8 months. Revenue model: basic DID free, advanced VC services (enterprise verification, cross-chain bridging) — paid.

### 6.5.5. Interrelationship of Auxiliary Projects

The four auxiliary projects form an integrated infrastructure stack in which each component reinforces the others:

```
Integration matrix of auxiliary projects:

  LQD ↔ PowerSwapMeta:
    — LQD creates pools, PowerSwapMeta aggregates them
    — PowerSwapMeta routes traffic to LQD pools
    — Shared AI Route Optimizer optimises both

  SAPP ↔ Contact:
    — SAPP modules use Contact DID for authorisation
    — Contact provides VC for SAPP permissions
    — ZK-proofs ensure privacy in SAPP

  LQD ↔ Contact:
    — Contact DID identifies LP providers
    — Activity VC from LP → Activity Score increase

  SAPP ↔ PowerSwapMeta:
    — SAPP modules may embed token exchange
    — Meta-transactions lower the barrier for SAPP

  All four ↔ DSP:
    — DSP is the operating environment for all projects
    — Unified UX: users interact via DSP
    — G-Plan tracks tasks of all four projects
```

The aggregate value of these four projects exceeds the sum of their individual values — a classic manifestation of the emergent property of a complex adaptive system (section 3.6, property E2). LQD without PowerSwapMeta creates pools with limited traffic; PowerSwapMeta without LQD has no project pools to aggregate; SAPP without Contact cannot provide private authorisation; Contact without SAPP has limited application. Together they form a complete “financial-social stack” for the CyberSocium ecosystem.

## 6.6 Portfolio Validation and Ecosystem Integration

All components of the applied ecosystem are connected by a unified economic circuit:

```
Ecosystem economic circuit:

  Participant
    │
    ├─► G-Plan: completes tasks → activity verified
    │     │
    │     └─► UnitManager: receives GBR reward
    │           │
    │           └─► GBR token: used for
    │                 │
    │                 ├─► DAO voting
    │                 ├─► Staking (passive income)
    │                 ├─► Investment in projects (Commerce DAO)
    │                 └─► Access to services (GyberComputer, AiC)
    │
    ├─► DSP: participates in social life
    │     │
    │     └─► Reputation: grows with activity → increases
    │           voting weight in DAO
    │
    └─► IPI projects: invests / works
          │
          └─► Project tokens → LP with GBR → DEX
                │
                └─► Revenue → distribution via
                      smart contracts
```

This integration ensures the realisation of all eight axioms:

| Axiom | Implementation in the ecosystem |
|---|---|
| A1 (Decentralisation) | P2P architecture of DSP, distributed GyberComputer, multisig governance |
| A2 (Transparency) | On-chain transactions in GyberNet, open code (AGPL-3.0) |
| A3 (Data sovereignty) | DID identification, IPFS storage, E2E encryption |
| A4 (Extensibility) | Modular architecture, SAPP protocol, open APIs |
| A5 (Meritocratic fairness) | UnitManager, G-Plan verification, reputation system |
| A6 (Inclusivity) | PMIP (minimum participation threshold), multilingualism, cross-chain |
| A7 (Self-governance) | Four-class DAO taxonomy, Governor contracts |
| A8 (Cognitive augmentation) | AiC (three circuits), AI audit in Code DAO, AI Treasury Advisor in Economic DAO, AI proposal summarisation, federated learning on GyberComputer |

## 6.7 Deployment Architecture and Integration Testing

The GyberExperiment ecosystem consists of five software modules, each performing a specialised function. Their integration ensures the realisation of the full cycle of the cybersocial corporation.

### 6.7.1 Five Ecosystem Modules

```
GyberExperiment Modular Architecture:

   ┌──────────────────────────────────────────────────────┐
   │              DSP Client (Frontend)                   │
   │  Next.js 15, React, TypeScript                       │
   │  Role: DSP user interface                            │
   │  Interacts with: GPROD (API), CONTRACTS (Web3)       │
   └──────────────────────────────────────────────────────┘
                         │
                         │ REST/GraphQL
   ┌─────────────────────▼────────────────────────────────┐
   │                GPROD (Backend)                        │
   │  NestJS 11, Prisma 6, PostgreSQL                     │
   │  Role: blockchain middleware, API gateway,           │
   │        business logic coordination                   │
   │  Interacts with: DSP Client (API), CONTRACTS (ethers),│
   │        Supabase (data), AiC (AI services)            │
   └──────────────────────────────────────────────────────┘
                         │
                         │ ethers.js / viem
   ┌─────────────────────▼────────────────────────────────┐
   │              CONTRACTS (Blockchain)                  │
   │  Solidity ^0.8.20, OpenZeppelin, BSC                 │
   │  Role: on-chain logic — UnitManager, Governor,       │
   │        Timelock, GBRVotingWrapper, LP Burn           │
   │  Interacts with: GPROD (events/tx), DSP Client (Web3)│
   └──────────────────────────────────────────────────────┘

   ┌──────────────────────────────────────────────────────┐
   │              Supabase (Database)                     │
   │  Self-hosted PostgreSQL + Auth + Realtime            │
   │  Role: off-chain data storage — profiles,            │
   │        G-Plan tasks, project metadata,               │
   │        blockchain event cache                        │
   │  Interacts with: GPROD (Prisma ORM)                  │
   └──────────────────────────────────────────────────────┘

   ┌──────────────────────────────────────────────────────┐
   │              DDP (Phase 2)                           │
   │  IPFS + ContentRegistry + ACL + P2P messaging        │
   │  Role: decentralised content storage,                │
   │        P2P communication, media                      │
   │  Interacts with: DSP Client, GPROD (index),          │
   │        CONTRACTS (ACL on-chain)                      │
   └──────────────────────────────────────────────────────┘
```

### 6.7.2 The Principle of “Wallet = Universal Key”

A single principle connects all modules: **the wallet address is the primary key** for identifying a participant across all system components.

```
Wallet Address as Primary Key:

  DSP Client: user.walletAddress → profile, UI state
  GPROD:     User.walletAddress → API authorisation,
             blockchain queries
  CONTRACTS: msg.sender → MemberStatus, Activity Score,
             reward claims, DAO voting
  Supabase:  users.wallet_address → off-chain data,
             G-Plan tasks, project metadata
  DDP:       wallet → DID → IPFS storage,
             ACL access control

  Synchronisation:
    CONTRACTS (source of truth for on-chain data)
      → GPROD (indexes events, caches)
        → DSP Client (displays to user)
    Supabase (source of truth for off-chain data)
      → GPROD (ORM access)
        → DSP Client (displays to user)
```

### 6.7.3 Containerisation and DevOps Pipeline

Each ecosystem module is containerised using Docker and orchestrated via Docker Compose (for development and staging) and Kubernetes (for production). This ensures environment reproducibility and simplifies deployment.

```
Docker architecture (docker-compose.yml):

  services:
    dsp-client:
      build: ./DSP
      ports: ["3000:3000"]
      environment:
        - NEXT_PUBLIC_GPROD_URL=http://gprod:4000
        - NEXT_PUBLIC_CHAIN_ID=97  # BSC Testnet
        - NEXT_PUBLIC_WALLETCONNECT_PROJECT_ID=${WC_ID}
      depends_on: [gprod]
      healthcheck:
        test: ["CMD", "curl", "-f", "http://localhost:3000/api/health"]
        interval: 30s
        timeout: 10s
        retries: 3

    gprod:
      build: ./GPROD
      ports: ["4000:4000"]
      environment:
        - DATABASE_URL=postgresql://gyber:${DB_PASS}@supabase-db:5432/gybernaty
        - BSC_RPC_URL=${BSC_RPC}
        - UNIT_MANAGER_ADDRESS=${UM_ADDR}
        - GOVERNOR_ADDRESS=${GOV_ADDR}
        - JWT_SECRET=${JWT_SECRET}
      depends_on: [supabase-db, redis]

    supabase-db:
      image: supabase/postgres:15.1.1.78
      volumes: ["db-data:/var/lib/postgresql/data"]
      environment:
        - POSTGRES_PASSWORD=${DB_PASS}
        - POSTGRES_DB=gybernaty

    redis:
      image: redis:7-alpine
      volumes: ["redis-data:/data"]
      healthcheck:
        test: ["CMD", "redis-cli", "ping"]

    hardhat-node:
      build: ./CONTRACTS
      command: npx hardhat node --fork ${BSC_RPC}
      ports: ["8545:8545"]
      profiles: ["dev"]  # Development only

  volumes:
    db-data:
    redis-data:
```

**CI/CD pipeline (GitHub Actions):**

Each module has its own CI/CD pipeline, activated on push and pull request. Pipelines are unified via a monorepo strategy with path-based filters.

```
CONTRACTS pipeline (on push to contracts/**):

  1. checkout → setup-node → install dependencies
  2. compile: npx hardhat compile
  3. lint: npx solhint 'contracts/**/*.sol'
  4. test: npx hardhat test --network hardhat
  5. coverage: npx hardhat coverage → ≥90%
  6. static analysis: Slither → vulnerability report
  7. gas report: npx hardhat test --gas
  8. [on merge to main] deploy to BSC Testnet
  9. [on release tag] deploy to BSC Mainnet (manual approval)

GPROD pipeline (on push to GPROD/**):

  1. checkout → setup-node → install dependencies
  2. lint: eslint + prettier check
  3. type check: tsc --noEmit
  4. test: jest --coverage → ≥80%
  5. build: nest build
  6. docker build → push to registry
  7. [on merge to main] deploy to staging
  8. integration test: E2E tests against staging

DSP Client pipeline (on push to DSP/**):

  1. checkout → setup-node → install dependencies
  2. lint: eslint + prettier check
  3. type check: tsc --noEmit
  4. test: vitest --coverage → ≥80%
  5. build: next build
  6. lighthouse CI: performance ≥90, accessibility ≥95
  7. docker build → push to registry
  8. [on merge to main] deploy to Vercel staging
  9. E2E: Playwright against staging
```

All secrets (private keys, API keys, database passwords) are stored in GitHub Secrets and HashiCorp Vault. Private keys for smart contract deployment are held in hardware keys (Ledger), connected via Gnosis Safe for production deploys.

### 6.7.4 Integration Testing Strategy

Given the critical role of smart contracts (transaction irreversibility, financial management), integration testing follows a multi-level strategy:

```
Testing levels:

  Level 1: Unit Tests (each module separately)
    — CONTRACTS: Hardhat/Foundry tests for each
      contract (UnitManager, Governor, Timelock)
    — GPROD: Jest tests for each NestJS module
    — DSP Client: Vitest + React Testing Library
    — Coverage: ≥90% for CONTRACTS, ≥80% for GPROD/DSP Client

  Level 2: Integration Tests (pairwise integration)
    — GPROD ↔ CONTRACTS: testing on Hardhat fork
      of BSC mainnet (realistic state)
    — DSP Client ↔ GPROD: E2E tests (Playwright/Cypress)
    — GPROD ↔ Supabase: migrations, transactional integrity

  Level 3: End-to-End Scenarios
    — Full cycle: registration → task completion
      → verification → claimReward() → voting
    — Test environment: BSC Testnet + Supabase staging
    — Automation: GitHub Actions CI/CD

  Level 4: Security Testing
    — AI audit: Slither + Mythril + Octane Security
    — Formal verification: UnitManager invariants
      (balance ≥ Σ pending_rewards)
    — Fuzzing: Echidna/Medusa for edge cases
    — Penetration testing: external auditors

  Level 5: Staging / Canary Deployment
    — BSC Testnet: full replica of production environment
    — Canary: 5% of traffic on new version
    — Rollback: automatic upon detection
      of anomalies (AI monitoring)
```

### 6.7.5 Monitoring and Operational Resilience

```
Ecosystem operational monitoring:

  Infrastructure:
    — Prometheus + Grafana: metrics for all modules
    — Uptime: SLA ≥99.9% for GPROD and DSP
    — Alerts: PagerDuty/Opsgenie for Core participants

  Blockchain:
    — Dune Analytics dashboard:
      TVL, volume, active wallets, reward distributions
    — AI anomaly detection:
      anomalous transaction patterns → alert
    — Fund Monitoring: remaining UnitManager balance,
      projected depletion at current payout rate

  Application:
    — Sentry: error tracking for DSP Client and GPROD
    — Custom metrics: DAU, MAU, G-Plan tasks,
      DAO proposals, votes
    — AI agent: weekly ecosystem status report
      for Economic DAO

  Security:
    — Chainalysis monitoring: suspicious
      addresses, sanctions lists
    — Rate limiting: DDoS protection
    — WAF: Web Application Firewall for DSP Client/GPROD
    — Gnosis Safe: signature monitoring, alerts
      on unexpected transactions
```

## 6.8 Open Development Model and Community Governance

The GyberExperiment ecosystem is developed as an open-source project under the AGPL-3.0 licence (code) and CC-BY-4.0 (documentation and research). This model is not merely a technical decision but a fundamental architectural choice flowing from Axioms A1 (decentralisation) and A2 (transparency). This section describes the organisational model of open-source development, contribution policies, and quality mechanisms.

### 6.8.1. Principles of Open Development

Open-source code in the context of CyberSocium performs four functions that go beyond the traditional understanding of open source:

**1. Governance transparency.** Every participant may verify that the smart contracts deployed on BSC correspond to the published source code. This eliminates the need for trust: a participant does not need to believe that Governor works correctly — they may read

and verify the code. Formally: for each contract C_deployed there exists C_source such that bytecode(compile(C_source)) = bytecode(C_deployed), and C_source is publicly accessible.

**2. Auditability.** Open-source code may be audited by any expert without an NDA or special permission. This enables: (a) AI audit via public tools (Slither, Mythril); (b) community peer review through Code DAO; (c) professional audit by external auditors with public reports. The model of “security through transparency” is the opposite of “security through obscurity” and is consistent with Kerckhoffs’s cryptographic principle.

**3. Extensibility.** Any developer may fork the repository, modify it for their own needs, and propose changes back to upstream. This implements Axiom A4 (extensibility): the ecosystem grows not only through the core developers but also through contributors from the global community.

**4. Reproducibility.** Open code ensures the reproducibility of the scientific experiment: any researcher may deploy a complete copy of the GyberExperiment ecosystem (on a testnet) and verify the theoretical predictions of CyberSocium on their own data.

### 6.8.2. Contribution Process

Contribution to the GyberExperiment codebase is governed by Code DAO and follows a formalised process:

```
Contribution process:

  1. Task identification:
     — Participant selects a task from G-Plan (bound to DAO)
     — Or creates a GitHub issue describing a bug/feature
     — AI assistant helps classify the issue:
       type (bug, feature, refactor, docs),
       priority (critical, high, medium, low),
       complexity estimate (XS, S, M, L, XL)

  2. PR (Pull Request) formation:
     — Participant forks the repository
     — Creates a branch: feature/<name> or fix/<name>
     — Develops the solution with tests
     — Opens PR with description: what, why, how to test

  3. Automated checks (CI/CD):
     — Lint: ESLint + Prettier (TypeScript) / Solhint (Solidity)
     — Type check: tsc --noEmit / slither (Solidity)
     — Tests: unit + integration, coverage ≥ threshold
     — Build: compilation check
     — AI Code Review: automatic vulnerability and code quality
       analysis (AiC Circuit 1)

  4. Peer review (Code DAO):
     — Minimum 2 approvals from participants
       with status ≥ Dev
     — For critical modules (CONTRACTS):
       minimum 1 approval from LeadDev+
     — Review includes: correctness, security,
       style, test coverage, documentation

  5. Merge and reward:
     — Merge after approval + CI pass
     — G-Plan records the contribution
     — Activity Score updated (EIP-712 proof)
     — Contributor receives reward
       via UnitManager (by status and complexity)

  6. Deployment:
     — Auto-deploy to staging (merge to main)
     — Manual approval for production (CONTRACTS)
     — Canary deployment for DSP Client and GPROD
```

**Contribution reward model:**

Each G-Plan task has a complexity rating that determines the base reward. The Activity Multiplier modulates the reward amount based on the quality of past contributions.

| Task complexity | Base reward (GBR) | Examples |
|---|---|---|
| XS (Extra Small) | 100,000 | Typo fix, dependency update |
| S (Small) | 500,000 | Bug fix, adding a test |
| M (Medium) | 2,000,000 | New feature, module refactoring |
| L (Large) | 10,000,000 | New contract, major feature with tests |
| XL (Extra Large) | 50,000,000 | Architectural change, new module |

Final reward = base_reward × AM(contributor) × review_quality_bonus, where review_quality_bonus ∈ [1.0, 1.3] is awarded by Code DAO reviewers for exceptional code quality.

### 6.8.3. Versioning and Releases

All modules follow Semantic Versioning (SemVer 2.0.0): MAJOR.MINOR.PATCH. For smart contracts, given their irreversibility, additional constraints apply:

```
Smart contract versioning policy:

  PATCH (0.0.x):
    — Bug fixes via UUPS upgrade
    — Does not change the external interface
    — Requires: Code DAO approval (2 reviews)
    — Timelock: 24 hours

  MINOR (0.x.0):
    — Addition of new functions (backward-compatible)
    — Possible change to internal state
    — Requires: Code DAO approval + AI audit
    — Timelock: 48 hours

  MAJOR (x.0.0):
    — Breaking changes: interface change,
      data migration, new contracts
    — Requires: Code DAO + Economic DAO approval
      + external audit + AI audit
    — Timelock: 72 hours
    — Mandatory canary period: 7 days on testnet

  Each release:
    — Tagged in git: v1.2.3
    — Accompanied by CHANGELOG.md with detailed description
    — Verified on Etherscan/BscScan
    — Announced in Social DAO and DSP
```

### 6.8.4. Codebase Governance

Management of the codebase is a function of Code DAO with escalation to Economic DAO for strategic decisions. Formalisation:

```
Code DAO decision types:

  Tactical (routine):
    — Merge PR
    — Dependency updates
    — CI/CD configuration
    Process: peer review (2 approvals)
    DAO vote: not required

  Operational (significant):
    — Change of architecture decision (ADR)
    — Addition of new module
    — Technology stack change
    Process: Code DAO vote (simple majority)
    Quorum: 10%

  Strategic (critical):
    — Smart contract upgrades (MAJOR)
    — Licence change
    — Integration of a new blockchain network
    — Decisions with economic consequences >10,000 GBR
    Process: Code DAO + Economic DAO vote
    Quorum: Code DAO 10%, Economic DAO 25%
```

This three-level model ensures balance: routine operations are not slowed down by the bureaucracy of DAO voting, while strategic decisions undergo thorough community scrutiny.

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| ## 7. Comparative Analysis and Related Work |
| CyberSocial economics and GyberExperiment do not exist in a vacuum. A number of projects and theoretical frameworks address similar problems, albeit from different positions. This section situates the contribution of CyberSocium within the context of existing literature, active DAO projects, and alternative approaches to constructing decentralised socio-economic systems. The comparison is conducted along five dimensions: (1) theoretical foundations, (2) organisational architecture, (3) economic model, (4) legal integration, (5) AI integration. |
| ### 7.1 Theoretical Frameworks: Review and Positioning |
| #### 7.1.1. Commons-Based Peer Production (Benkler, 2006) |

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| Yochai Benkler, in his foundational work The Wealth of Networks, described the model of commons-based peer production (CBPP), whereby large groups of people coordinate their efforts to create public goods without market pricing or hierarchical management. Classic examples include Wikipedia, Linux, and OpenStreetMap. |
| **Connection to CyberSocium.** The PMIP model (Section 3.4) is a direct development of Benkler's ideas: the principle of minimal individual participation formalises the intuition that, given a sufficient number of participants, the individual contribution tends toward a vanishingly small quantity. The formula individual_cost(p) = C(p)/\|SIC(p)\| → 0 constitutes a mathematical axiomatisation of Benkler's observations concerning the "modularity" and "granularity" of tasks in CBPP systems. |

**Fundamental distinction.** Benkler describes CBPP as a supplement to the market economy: Linux exists within the capitalist system and depends upon it (developers receive salaries from corporations that use Linux). CyberSocium lays claim to the creation of an autonomous economic system in which participants have no need of external financing. The PMIP + SES + IPI model creates a closed cycle: idea → project → realisation → value → distribution → new ideas. Benkler also does not address the problem of formal verification of contribution: in Wikipedia, contribution is measured by the number of edits, but not by their quality. UnitManager resolves this problem through a meritocratic system of statuses (None → Unit → Dev → LeadDev → ArchDev → Core) with programmatic verification of activity.

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| #### 7.1.2. Institutional Economics of the Commons (Ostrom, 1990) |
| Elinor Ostrom, Nobel Laureate of 2009, empirically demonstrated that communities are capable of effectively managing common-pool resources without privatisation or state regulation — provided that eight institutional design principles are observed [Ostrom, 1990]. These principles were examined in detail in Section 2.2, with a mapping onto the architecture of GyberExperiment. |
| **Connection to CyberSocium.** All eight of Ostrom's principles are implemented in the CSC architecture: clearly defined boundaries (NFT membership), congruence of rules (four-class DAO taxonomy), collective choice arrangements (Governor contracts), monitoring (on-chain transparency), graduated sanctions (Activity Scoring, |

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| status degradation), conflict-resolution mechanisms (FRP), recognition of rights to organise (DUNA), nested enterprises (modular DSP architecture). | |
| **Fundamental distinction.** Ostrom studied local communities: irrigation systems in Nepal, fishing cooperatives in Turkey, forest commons in Japan. The maximum scale of her case studies was several thousand participants with geographic attachment. CyberSocium is designed for global scale with participants having no physical attachment. This requires fundamentally different mechanisms of identification (DID in place of personal acquaintance), monitoring (on-chain analytics in place of social control), and sanctions (automatic status degradation in place of social pressure). Furthermore, Ostrom did not examine digital common resources, which, unlike fish | |

stocks or pastures, are non-rivalrous: one participant's use of code does not diminish its availability to others. This radically alters the economics of resource governance and eliminates the classical "tragedy of the commons" for digital assets, but creates new problems: free-riding, Sybil attacks, governance spam.

#### 7.1.3. Platform Cooperativism (Scholz, 2016)

Trebor Scholz and Nathan Schneider, in the manifesto Ours to Hack and to Own (2016), proposed the model of "platform cooperativism" — the reform of digital platforms by transferring ownership to workers and users. Examples include Stocksy (a photographers' cooperative), Up & Go (a cleaners' cooperative), and Resonate (a musicians' cooperative).

**Connection to CyberSocium.** Platform cooperativism shares

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| with CyberSocium a critique of the extractive model of the platform economy and a commitment to democratic ownership. The DUNA structure of GyberExperiment is functionally analogous to a cooperative charter, ensuring collective ownership of infrastructure. | |
| **Fundamental distinction.** Platform cooperativism is a reform of the existing model: Stocksy remains a photo agency, simply with a different ownership structure. CyberSocium is a new paradigm: not a platform reform, but the creation of an economic system in which the “platform” is merely one of the technological layers. More significantly: Scholz’s cooperatives operate within the framework of traditional law (LLC, Cooperative Corp) and do not employ programmable economic incentives. CyberSocium implements incentives at the level of smart contracts (UnitManager, | |

GBRVotingWrapper), rendering them transparent, verifiable, and resistant to manipulation.

#### 7.1.4. Cryptoeconomics (Buterin, 2014; Zamfir, 2015)

Vitalik Buterin introduced the term “cryptoeconomics” to describe the use of economic incentives and cryptographic proofs to ensure the properties of distributed systems. Vlad Zamfir extended this approach by proposing a formal methodology for the design of cryptoeconomic protocols [Zamfir, 2015].

**Connection to CyberSocium.** Buterin’s cryptoeconomics is one of the theoretical foundations of CyberSocium (Section 2.5). The mechanisms of staking, slashing, and economic guarantees are implemented in UnitManager and the GBR tokenomics.

**Fundamental distinction.** Classical cryptoeconomics

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| focuses on the protocol level: how to ensure consensus, how to prevent double-spend, how to incentivise validators. CyberSocium extends cryptoeconomics to the socio-economic level: how to organise collective decision-making, how to finance public goods, how to ensure fair value distribution. The CyberSocium axiomatic system (A1–A8, Section 3.1) goes far beyond cryptoeconomic protocols, incorporating axioms of meritocratic fairness (A5), inclusivity (A6), and cognitive augmentation (A8) that have no analogues in the work of Buterin or Zamfir. | |
| #### 7.1.5. Token Engineering (Voshmgir, 2020; Zargham, 2020) | |
| Shermin Voshmgir (Token Economy, 2020) and Michael Zargham (BlockScience, cadCAD) developed the discipline of “token engineering” — the systematic design of token ecosystems | |

using methods from control theory, dynamical systems, and agent-based modelling.

**Connection to CyberSocium.** The cadCAD modelling methodology proposed by Zargham is applied in the simulation of GBR tokenomics (Section 6.3). Many of Voshmgir's principles — separation of governance tokens and utility tokens, velocity analysis, emission modelling — are implemented in the economic architecture of GyberExperiment.

**Fundamental distinction.** Token engineering is an engineering methodology applicable to any token project. CyberSocium is a theoretical framework defining for what purpose and why token systems are designed. Token engineering answers the question "how to design a sustainable tokenomics?"; CyberSocium answers the question "what economic system are

we building and why is it superior to existing ones?”. Furthermore, CyberSocium integrates the legal dimension (DUNA) and the AI dimension (AiC), which lie beyond the scope of the purely technical discipline of token engineering.

#### 7.1.6. Mechanism Design Theory (Hurwicz, 1960; Myerson, 1981)

Leonid Hurwicz laid the foundations of mechanism design theory, while Roger Myerson and Eric Maskin developed it into a formal discipline, awarded the Nobel Prize in 2007. Mechanism design theory studies the design of rules of interaction (mechanisms) under which rational agents with private information behave optimally for society (incentive compatibility).

**Connection to CyberSocium.** The formal mechanisms of CyberSocium — SES, PMIP, IPI, FRP — are designed with attention to the principles of incentive

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| compatibility. Activity Scoring implements a “preference revelation” mechanism: participants demonstrate their engagement through verifiable on-chain actions, rather than through declarations. |
| **Fundamental distinction.** Classical mechanism design theory presupposes rational agents with complete information about their own preferences. CyberSocium operates under conditions of bounded rationality (Simon, 1955) and evolving preferences: DAO participants do not always know what they want, and their preferences change as they learn. Axiom A8 (cognitive augmentation) is a direct response to this problem: AI agents compensate for participants’ bounded rationality by providing analytics, summarisation, and recommendations. |
| ### 7.2 Practical Projects: Detailed Comparative Analysis |

In order to position GyberExperiment among existing DAO projects, we conduct a detailed comparison along six key parameters. Projects representing different archetypes of decentralised organisations have been selected.

#### 7.2.1. MakerDAO / Sky (Ethereum, 2017)

**Description.** MakerDAO is one of the earliest and most successful DeFi protocols, managing the stablecoin DAI through a system of collateralised debt positions (CDPs). In 2024, it rebranded as Sky with the SKY/USDS tokens.

**Governance.** Token-weighted governance via the MKR/SKY token. Voting on key parameters: stability fee, collateral types, risk parameters. Executive votes are implemented through smart contract spells.

**Economics.** Dual-token model: MKR (governance + recapitalisation) and DAI (stablecoin). Revenue from stability

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| fees is distributed through MKR buyback-and-burn. | |
| **Comparison with GyberExperiment.** MakerDAO demonstrates that complex financial systems can be managed in a decentralised manner. However, MakerDAO is a single-product protocol (stablecoin), whereas GyberExperiment is a multi-project ecosystem. MakerDAO has no mechanisms for meritocratic governance: voting power is determined exclusively by the quantity of MKR held, which has led to the dominance of Rune Christensen (the founder) and several large holders. In GyberExperiment, Hybrid Voting Power combines token weight with a reputational coefficient (Activity Score), preventing plutocracy. Furthermore, MakerDAO does not integrate AI components into governance, whereas AiC is an institutional | |

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| part of the GyberExperiment architecture. |
| #### 7.2.2. Aragon (Ethereum, 2017) |
| **Description.** Aragon is a platform for creating and managing DAOs, providing modular tools: Aragon Court (decentralised arbitration), Aragon Govern (optimistic governance), Aragon Voice (voting). |
| **Governance.** ANT token for voting on protocol upgrades. Aragon Court uses ANT staking to form juries. |
| **Economics.** ANT as governance token. The value model of ANT is tied to the number of DAOs created on the platform. |
| **Comparison with GyberExperiment.** Aragon is infrastructure, not a system. Aragon provides building blocks for DAOs but does not define how DAOs should function economically. GyberExperiment defines not only technical primitives, but also the economic model (PMIP, SES), |

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| the governance architecture (four-class taxonomy), and the theoretical foundation (8 axioms). By analogy: Aragon is AutoCAD (a draughting tool); GyberExperiment is an architectural project (a building with drawings, calculations, and specifications). Aragon Court implements legal arbitration within a DAO; the FRP of CyberSocium implements economic arbitration at the level of project forks. |
| #### 7.2.3. Gitcoin (Ethereum, 2017) |
| **Description.** Gitcoin is a platform for financing public goods in Web3 through a grant system with Quadratic Funding (QF), proposed by Buterin, Hitzig, and Weyl [Buterin et al., 2019]. |
| **Governance.** GTC token for governing Gitcoin DAO. The Gitcoin Passport programme for Sybil resistance. |
| **Economics.** Matching pool + quadratic funding. Donors |

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| contribute funds; the matching pool amplifies contributions proportionally to the square root of the number of unique donors. | |
| **Comparison with GyberExperiment.** Gitcoin implements an elegant financing mechanism, but is constrained by the grant model: a project receives one-off funding and subsequently exists autonomously. GyberExperiment, through the IPI model, implements a complete lifecycle: from idea (Idea) through project (Project) to realisation (Implementation), with continuous funding, monitoring, and iteration. Furthermore, Gitcoin Passport addresses the Sybil attack problem through off-chain verification (binding to Twitter, Google, ENS), whereas GyberExperiment employs on-chain meritocratic mechanisms: a Sybil attacker may create multiple accounts but cannot falsify Activity | |

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| Score, which requires genuine verified activity. |
| #### 7.2.4. Colony (Ethereum, 2018) |
| **Description.** Colony is a protocol for the decentralised management of organisations, employing “lazy consensus” and a domain system for structuring work. |
| **Governance.** Reputation-weighted governance: voting weight is determined by reputation within a specific domain, not by token holdings. Reputation is earned by completing tasks and distributing rewards. |
| **Economics.** CLNY token + native reputation system. Reputation is non-transferable and decays over time. |
| **Comparison with GyberExperiment.** Colony is most similar to GyberExperiment in its conception of reputation-weighted governance. Colony’s domain system is analogous to GyberExperiment’s four-class DAO taxonomy — both |

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| divide governance by functional areas. However, Colony operates at the level of a single organisation: one Colony = one company/project. GyberExperiment designs an ecosystem of organisations (MacroeconomicDAO), in which multiple projects (IPI) exist and compete within a single economic space with a shared token (GBR), shared selection mechanisms (SES), and a shared legal wrapper (DUNA). Colony also does not address legal integration or AI components. |
| #### 7.2.5. Nouns DAO (Ethereum, 2021) |
| **Description.** Nouns DAO is an experimental project in which one NFT is minted daily at auction and all proceeds are directed to the DAO treasury. Each Noun NFT = one vote. |
| **Governance.** One-NFT-one-vote. Minimal quorum for proposals. Vote delegation. |

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| **Economics.** Perpetual auction → DAO treasury. At its peak, the treasury exceeded $50M. |
| **Comparison with GyberExperiment.** Nouns DAO is of interest as an experiment in perpetual funding: the daily auction creates a continuous inflow of resources to the treasury. GyberExperiment implements an analogous idea through different mechanisms: the Governance Pool with 80% of initial supply, staking rewards, and SES contributions. The fundamental distinction: Nouns DAO is a financial club without a defined mission (the treasury is spent on everything from graffiti to sports sponsorship), whereas GyberExperiment is a purposeful economic system with formalised project selection criteria (SES) and measurable performance metrics. The Nouns problem: the “fork wars” of 2023, in which a group of holders exited the DAO, taking |

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| their proportional share of the treasury. The FRP of CyberSocium formalises the fork procedure, preventing chaotic splits. |
| #### 7.2.6. DAOstack (Ethereum, 2018) |
| **Description.** DAOstack is a framework for scalable governance, based on the “holographic consensus” algorithm, in which prediction markets determine which proposals deserve the attention of the majority. |
| **Governance.** GEN token + holographic consensus: stakers predict whether a proposal will be approved and receive rewards for correct predictions. This filters proposals and reduces the burden on voters. |
| **Economics.** GEN as utility token for staking in prediction markets. |
| **Comparison with GyberExperiment.** DAOstack’s holographic consensus solves the same problem as AI summarisation in AiC GyberExperiment: |

scaling governance as the number of proposals grows. DAOstack uses a market mechanism (prediction markets); GyberExperiment uses an AI mechanism (AI analyst). DAOstack's advantage: markets are decentralised and independent of AI providers. GyberExperiment's advantage: AI summarisation is available to every participant, not only stakers, and provides deeper analysis (impact assessment, risk analysis, cross-DAO coordination).

### 7.3 Multi-Dimensional Comparative Matrix

For a systematic comparison of GyberExperiment with the projects examined above, the following dimensions are employed:

| Dimension | MakerDAO | Aragon | Gitcoin | Colony | Nouns | DAOstack | **GyberExperiment** |
|---|---|---|---|---|---|---|---|
| **Type** | DeFi protocol | DAO infrastructure | Grant platform | Organisational

protocol | Financial club | Governance framework | **CSC ecosystem** | | **Scale** | Single-product | Meta-platform | Single-function | Single organisation | Single treasury | Meta-platform | **Multi-project ecosystem** | | **Voting** | Token-weighted | Token-weighted | QF for grants, token for governance | Reputation-weighted | NFT (1=1) | Holographic consensus | **Hybrid (token + reputation + AI)** | | **Meritocracy** | No | No | Partial (QF) | Yes (reputation) | No | Partial (staking) | **Yes (UnitManager, Activity Score)** | | **Legal wrapper** | Cayman Foundation | Swiss Verein | Cayman Foundation | None | Unincorporated | None | **DUNA (Wyoming)** | | **AI integration** | No | No | Sybil scoring | No | No | No | **AiC (3 circuits)** | | **Formal theory** | No | No | QF [Buterin et al., 2019] | No | No | Holographic consensus | **8 axioms, PMIP, SES, IPI, FRP** | | **Project lifecycle** | N/A | No | Grant →

autonomy | Task-based | Ad hoc | Proposal → vote | **IPI (Idea → Project → Implementation)** | | **Fork protocol** | Emergency Shutdown | No | No | No | Ragequit (Nouns Fork) | No | **FRP (formalised)** | | **Treasury** | Stability fee → MKR burn | ANT sales | Matching pools (external) | Internal | NFT auction | GEN sales | **Governance Pool + LP Burn + SES** |

### 7.4 Thematic Comparison by Key Problems

#### 7.4.1. The Problem of Plutocracy

Token-weighted voting is the de facto standard in the DAO ecosystem, but it engenders plutocracy: power concentrates among large holders. Different projects offer different solutions:

| Project | Anti-plutocracy mechanism | Limitations | |—|—|—| | Gitcoin | Quadratic Funding (square root of number of donors) | Vulnerable to Sybil attacks without Passport | | Colony | Reputation-weighted

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| (non-transferable) \| Reputation tied to domain; manipulation via “easy” tasks \| \| Optimism \| Bicameral system (Token House + Citizens’ House) \| Centralised issuance of soulbound tokens for Citizens’ House \| \| GyberExperiment \| Hybrid Voting Power = f(GBR_staked, activity_score, member_status) \| Requires reliable Activity Scoring; AI dependency \| |
| CyberSocium offers the most comprehensive approach: not the abandonment of token weighting (which reduces economic incentives), but a correction of weight through a reputational multiplier. The formula: voting_power = sqrt(GBR_staked) × activity_multiplier × status_bonus ensures sub-quadratic scaling of token weight and a premium for activity. |
| #### 7.4.2. The Problem of Governance Scalability |
| As a DAO grows, the number of proposals exceeds participants’ cognitive capacity |

(voter fatigue), leading to decreased participation and degraded decision quality.

| Project | Scaling mechanism | Limitations | |—|—|—| | DAOstack | Holographic consensus (prediction markets) | Requires liquid market; manipulation by stakers | | Compound | Vote delegation (liquid democracy) | Professional delegates form a new elite | | Nouns | Low quorum (10%) + delegation | Low quorum reduces decision legitimacy | | GyberExperiment | Four-class taxonomy + AI summarisation + delegation | AI dependency; complexity of four-class system for new participants |

GyberExperiment's four-class DAO taxonomy is a unique solution without parallel: the division of proposals by type (SocialDAO, CodeDAO, CommerceDAO, EconomicDAO) with different voting rules, quorums, and timelines for each type. This reduces

cognitive load: a participant may focus on the type of decisions in which they are competent, delegating the remainder.

#### 7.4.3. The Problem of Legal Integration

The legal status of DAOs remains an unresolved problem. Most DAOs either ignore legal status (which creates the risk of unlimited liability for participants) or use traditional corporate forms (which contradicts the principles of decentralisation).

| Project | Legal form | Limitations | |—|—|—| | MakerDAO | Cayman Foundation | Offshore jurisdiction; limited recognition in other countries | | Aragon | Swiss Verein (Association) | Swiss Verein has operational limitations | | Lido | Cayman Foundation → DAO | Transitional period; indeterminate status | | GyberExperiment | DUNA (Wyoming, USA) | Single-state jurisdiction; requires

compliance for international participants |

DUNA (Decentralised Unincorporated Nonprofit Association) is the most advanced legal form, developed specifically for DAOs. Wyoming W.S. § 17-32 (2024) provides: limited liability for participants, recognition of on-chain voting as legally binding, and compatibility with IRS 501(c)(6). GyberExperiment is one of the first major projects to use DUNA as its legal wrapper (see Section 4.6 and Appendix E).

#### 7.4.4. The Problem of AI Integration

AI integration in DAO systems is at an early stage. Most projects do not consider AI as a systemic architectural component.

| Project | AI usage | Level of integration | |—|—|—| | MakerDAO | Risk analytics (external providers) | Ad hoc, not institutionalised | | Gitcoin | Sybil scoring (Gitcoin Passport ML)

| Single-function (anti-Sybil) | | Compound | None | — | | GyberExperiment | AiC (3 circuits: internal, research, commercial) | **Institutional** (Axiom A8, formalised AI governance) |

AiC (AI Community) of GyberExperiment is a unique institutional solution: not merely the use of AI tools, but the democratic governance of AI development through DAO mechanisms. The three AiC circuits (internal — AI for CSC operations; research — community AI research; commercial — AI products of the ecosystem) provide the full spectrum of AI integration, from the automation of routine operations to fundamental research.

### 7.5 The Unique Contribution of GyberExperiment: A Synthesis

On the basis of the analysis conducted, it is possible to identify five key dimensions along which GyberExperiment represents a unique

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| contribution not replicable by existing projects: |
| 1. **Theoretical completeness.** GyberExperiment is the only project that rests upon a formal axiomatic system (8 axioms), a set of formalised mechanisms (PMIP, SES, IPI, FRP), and mathematically proven properties (the Pareto-optimality theorem for SES, the PMIP convergence theorem). Neither MakerDAO, nor Aragon, nor Colony possesses a comparable theoretical foundation. |
| 2. **Architectural complexity.** The four-class DAO taxonomy implementing differentiated governance for different types of decisions (social, technical, commercial, economic) has no analogues in the existing DAO ecosystem. Colony divides governance by domains (functional areas), but not by types of decisions with different voting rules. |

3. **Economic model.** The integration of PMIP (principle of minimal participation), SES (evolutionary project selection), IPI (idea lifecycle), and LP Burn (liquidity burning mechanism) into a single economic system has no precedent. Existing projects implement individual components (QF in Gitcoin, reputation in Colony, auction in Nouns) but do not integrate them into a coherent economic theory.

4. **Legal integration.** The use of DUNA in combination with a two-tier treasury (DUNA Treasury for fiat, On-Chain Treasury for crypto) represents the most advanced model of DAO legal integration. The separation of concerns between the legal and technical levels ensures compliance without sacrificing decentralisation.

5. **AI integration as a systemic principle.** Axiom A8 (cognitive augmentation) introduces AI integration at the level of the axiomatic

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| system, rather than as an ad hoc tool. AiC institutionalises the governance of AI development through DAO mechanisms. None of the projects examined includes AI at the level of architectural principles. |
| **Synthetic claim.** GyberExperiment represents the first attempt to construct a complete socio-economic system on the basis of blockchain technologies, encompassing all levels — from formal theory (axioms, theorems) through technical architecture (smart contracts, protocols) to legal integration (DUNA) and the institutional governance of AI (AiC). This distinguishes the project from existing DAOs, which, as a rule, operate on one or two of the above-mentioned levels. |
| ### 7.6 Why the Synthesis Had Not Been Achieved Earlier: An Analysis of Barriers |

If the need for a complete decentralised socio-economic system is so evident, why did none of the projects examined attain the completeness of CyberSocium? This question is not rhetorical — its answer sheds light on the necessary conditions that coincided only by 2024–2025, and explains why the synthesis became possible precisely now.

#### 7.6.1. Technological Barrier: Infrastructure Immaturity

The first DAOs (TheDAO, 2016) appeared on Ethereum, which at the time processed approximately 15 transactions per second, with gas costs on the order of $10–50 for complex operations. CyberSocium's four-class DAO system generates an order of magnitude more transactions than a monolithic DAO: each class requires its own voting processes, cross-class proposals

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| generate additional on-chain operations, and Activity Score is updated periodically for each participant. With approximately 1,000 participants and four DAO classes, the system would generate approximately 4,000 votes/month × approximately 3 transactions/vote = approximately 12,000 transactions/month for governance alone — which was economically infeasible on Ethereum prior to 2021 (the emergence of L2) and practically inadvisable prior to BSC, Polygon, and analogous high-performance networks. | |
| Key technological preconditions that became available by 2024: | |
| - **EVM-compatible L1/L2 with low gas** (BSC, Polygon, Arbitrum): gas < $0.01 per transaction, making the four-class DAO architecture economically viable; - **OpenZeppelin Governor** (2021): standardised contract for on-chain | |

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| governance, reducing development cost from months to days; - **UUPS Proxy pattern** (2021): the ability to upgrade contracts without state migration, critically important for iterative calibration of parameters; - **Account Abstraction** (ERC-4337, 2023): reduction of the cognitive entry barrier for non-crypto users; - **IPFS + Filecoin** (mature by 2022): decentralised storage, necessary for DDP; - **Federated Learning frameworks** (PySyft, Flower, 2020+): infrastructure for implementing A3 (data sovereignty) without loss of AI model quality; - **cadCAD** (BlockScience, 2018+): a modelling tool enabling simulation of cryptoeconomic systems prior to deployment. | |
| Prior to the convergence of all these components, designing a system of the CyberSocium level was technically feasible, but economically inadvisable and practically risky. | |

| #### 7.6.2. Theoretical Barrier: Disciplinary Silos |
| --- |
| The problem addressed by CyberSocium lies at the intersection of at least six disciplines: economic theory (mechanism design, tokenomics), computer science (distributed systems, cryptography), law (corporate law, DAO regulation), sociology (organisation theory, collective action), philosophy (ethics, epistemology), and AI/ML (LLM, GNN, Federated Learning). Traditional academic and industrial structures are organised along disciplinary lines: economists design tokenomics without understanding gas costs; engineers write smart contracts without understanding mechanism design; lawyers analyse compliance without understanding cryptography. |
| This disciplinary silo explains the characteristic limitations of each project examined. MakerDAO was |

created by financial engineers — a brilliant DeFi architecture, but without a theory of collective action. Aragon was created by developers — a modular DAO platform, but without an economic model for project financing. Colony was created by managers — a task and role system, but without formal axiomatics. Gitcoin was created by crypto activists — innovative financing mechanisms (QF), but without legal integration.

CyberSocium became possible thanks to the conscious decision to design the system as a transdisciplinary synthesis from the outset, rather than as an extension of one of the existing disciplines. The formalisation as CSC = ⟨N, R, P, G, M, E⟩ is not merely a notation, but a statement that all six components must be designed simultaneously, taking into account their mutual dependencies. The axiomatic system A1–A8 serves as the interface between disciplines: each axiom has an

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| interpretation in each of the six disciplines, ensuring the coherence of the entire design. |
| #### 7.6.3. Regulatory Barrier: Absence of a Legal Form |
| Prior to 2024, no legal form existed that adequately described a DAO as a legal entity. Projects used surrogates: BVI Foundation Company (MakerDAO), Cayman Foundation (Uniswap), Swiss Verein (Aragon), or simply remained without legal entity status (the majority of DAOs). Each surrogate created a fundamental contradiction: the legal form presupposed centralised governance (board of directors, registered agent), whereas actual governance was conducted through DAO voting. This gap generated legal uncertainty, made banking relationships impossible, and created tax compliance problems for participants. |

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| Wyoming DUNA (Decentralised Unincorporated Nonprofit Association, W.S. § 17-32, entered into force 1 July 2024) for the first time provided a legal form in which: - DAO participants are legally recognised as members of the association; - Governance is conducted through smart contracts and on-chain voting (Article 17-32-107); - There is no requirement for a centralised board of directors; - Limited liability for participants (Article 17-32-104); - Tax status 501(c)(6) is available (nonprofit). |
| Without DUNA or an analogous form, building a complete socio-economic system (and not merely a DeFi protocol) was legally impossible: a system claiming to coordinate real economic activity (labour relations, goods, services) must possess legal capacity. The emergence of DUNA in 2024 removed the last of the three fundamental barriers. |

#### 7.6.4. Cultural Barrier: From “Code Is Law” to Institutional Design

The early DAOs (2016–2020) proceeded from the philosophy of “code is law” — the conviction that a smart contract fully replaces legal norms and social institutions. TheDAO hack (2016), the loss of funds due to the vulnerability in Parity Multisig (2017), governance attacks on various DeFi protocols — a series of catastrophic failures gradually brought the community to the understanding that code cannot be the sole regulator. What are required are: (a) formal theory to justify architectural decisions; (b) a legal wrapper for interaction with the external world; (c) social norms and processes for resolving situations not anticipated by the code.

This cultural evolution — from crypto-anarchist maximalism to mature institutional design — took nearly

a decade. CyberSocium is a product of this evolution: the project consciously combines cryptographic trustlessness (smart contracts ensure security without trust in specific agents) with institutional design (DUNA, FRP, and MacroeconomicDAO ensure adaptability and conflict resolution).

**Section conclusion.** A synthesis of the CyberSocium level became possible at the intersection of four conditions: (1) technological maturity of EVM-compatible blockchains and AI/ML tools; (2) a transdisciplinary approach consciously overcoming disciplinary silos; (3) legal innovation (DUNA, 2024); (4) cultural maturation of the blockchain community from “code is law” to institutional design. Not one of these conditions was fulfilled prior to 2024, which explains why existing projects — for

all their merits — remained partial solutions.

# 8. Discussion: Implications, Limitations, and Open Questions

## 8.1 Theoretical Implications

The model of CyberSocial economics presented in this document carries several important theoretical implications, touching on fundamental questions in economic theory, organisation theory, political philosophy, and computer science.

### 8.1.1. Implications for Economic Theory

CyberSocium proposes a model of economic organisation in which value is not concentrated but distributed through the PMIP mechanism. The formula individual_cost(p) = C(p) / |SIC(p)| → 0 as |SIC(p)| grows demonstrates that collective action is capable of overcoming the constraints of individual capital without recourse to centralised institutions — banks, state funds, or venture capital. This extends the models of Ostrom (commons governance) and Benkler (commons-based peer production) into the domain of full-fledged economic governance.

A deeper implication of PMIP is that it blurs the traditional distinction between microeconomics and macroeconomics. In classical economic theory there is a clear division: microeconomics describes the behaviour of individual agents and firms; macroeconomics describes the aggregate indicators of the economy as a whole. In the CSC model this division loses its meaning: every individual act of participation (a micro-decision to join a SIC) is immediately translated into a macro-effect (reduction of individual_cost for all participants). The PMIP mechanism creates a direct, formalised link between individual action and systemic outcome — a link that in the traditional economy is mediated by a multitude of institutions (banks, exchanges, state regulators) and is accompanied by significant information losses.

For the theory of the firm, CyberSocium proposes a rethinking of the boundaries of the firm. Ronald Coase (1937) explained the existence of the firm through the minimisation of transaction costs: the firm arises where internal coordination is cheaper than market coordination [8]. Oliver Williamson (1985) developed this theory by demonstrating the role of opportunism and asset specificity [9]. CSC proposes a third model that is reducible neither to the firm (hierarchical coordination) nor to the market (price coordination): protocolised coordination through smart contracts, in which transaction costs are

minimised not by centralisation but by automation. The boundaries of CSC are defined not by organisational structure but by the perimeter of the protocol — all those interacting through shared contracts are “inside” CSC.

For the theory of public goods, the implications are no less substantial. The classical problem of public goods (Samuelson, 1954) holds that rational agents are not inclined to voluntarily finance goods they can use for free (the free-rider problem). Olson (1965) showed that collective action is impossible without coercion or selective incentives [29]. PMIP in combination with SES offers a third path: not coercion and not external incentives, but structural design in which participation becomes a rational strategy. When individual_cost → 0, and access to the results of a project is conditioned on participation (through the project token), the free-rider problem is solved architecturally rather than institutionally.

### 8.1.2. Implications for Organisation Theory

The four-class DAO taxonomy proposes a new approach to structuring organisational governance, in which different types of decisions are processed by different, purpose-optimised processes. This contrasts both with the monolithic corporate hierarchy and with undifferentiated DAOs, where all decisions pass through a single mechanism.

Classical organisation theory (Mintzberg, 1979; Burns and Stalker, 1961) distinguishes several basic organisational forms: machine bureaucracy, professional bureaucracy, simple structure, divisional form, adhocracy. CSC fits none of these categories. The closest analogy is Mintzberg’s adhocracy — a flexible, project-oriented structure — but CSC surpasses it in several dimensions: (a) formalised coordination through smart contracts in place of informal mechanisms; (b) democratic governance of parameters in place of managerial control; (c) transparent and verifiable history of all decisions; (d) built-in economic incentives in place of employment contracts.

The implication for corporate governance theory (Berle & Means, 1932 [91]) lies in the elimination of the fundamental problem of the separation of ownership and control. In the traditional corporation, shareholders (owners) delegate management to managers, which creates the agency problem — a misalignment of interests. In CSC this problem is resolved structurally: each participant is simultaneously a “shareholder” (GBR holder), a “manager” (voting in the DAO), and a “worker” (contributor to a G-Plan). The Hybrid Voting Power formula VP = √(GBR_staked) × ActivityMultiplier × StatusBonus ensures that voting power depends not only on capital but also on real contribution, eliminating both the plutocracy of pure token-weighted voting and the inefficiency of pure democracy (one person = one vote).

### 8.1.3. Implications for Game Theory and Mechanism Design

The SES (socio-economic selection) model proposes an evolutionary mechanism for project selection that requires no centralised arbiter, yet avoids the tragedy of the commons problem through formalised incentives (GBR burn, reputation, staking).

From the perspective of mechanism design theory, GyberExperiment constitutes a large-scale institutional experiment (mechanism design experiment) — an attempt to realise in practice what mechanism design theory describes abstractly: the design of rules of the game under which rational agents, acting in their own interests, achieve a socially optimal outcome. The key distinction from classical mechanism design is that the rules of GyberExperiment are not fixed — they are themselves subject to voting and evolution. This is a "meta-mechanism" — a mechanism for designing mechanisms.

Formally, if a mechanism M = (S, f) is defined by a strategy space S and an outcome function f, then MacroeconomicDAO constitutes a meta-mechanism M∗ = (S, f), where S∗ is the set of possible mechanisms M, and f* is the function determining which mechanism M is selected on the basis of DAO voting. This creates a recursive structure: participants vote not only on outcomes, but on the rules by which they vote — up to and including changes to the voting rules themselves (changes to quorum, voting period, Hybrid Voting Power formula). The stability of such a system is an open theoretical question (Section 8.5.4).

### 8.1.4. Implications for Philosophy and Social Theory

CyberSocium implicitly proposes a new ontology of economic relations. In classical political economy, the subjects of the economy are individuals, firms, and states. In the CSC model a new type of subject appears — the CyberSocial corporation, which is neither a firm (no hierarchy and no employment contracts), nor a state (no territory and no monopoly on violence), nor a market (there is explicit governance and collective decisions), nor a commune (there are formalised economic incentives and tokenomics).

This resonates with Michel Foucault's concept of "heterotopia" — a space that functions according to different rules from the surrounding society, yet actually exists (as distinct from utopia). CSC is a heterotopia in the economic space: it functions within the existing economic system (it uses fiat money through DUNA, tokens on BSC, servers in traditional data centres) but is subject to different rules of coordination and distribution.

## 8.2 Practical Implications

### 8.2.1. Implications for the Blockchain Industry

GyberExperiment demonstrates that DAOs can move beyond the governance of a single protocol or financial product and function as full-fledged economic systems with diversified projects, differentiated governance, and legal legitimacy.

The current state of the DAO ecosystem is characterised by several patterns that GyberExperiment seeks to overcome:

**Mono-functionality.** Most successful DAOs manage a single product or protocol: MakerDAO manages the stablecoin DAI, Uniswap DAO manages the DEX protocol, Aave DAO manages the lending protocol. GyberExperiment proposes a model of a multi-product DAO, in which a single ecosystem coordinates multiple projects of varying nature — from social (DSP) to technical (GyberNet) and commercial (Commerce DAO projects).

**Plutocratic governance.** According to DeepDAO data (2024), in the majority of large DAOs more than 80% of voting power is concentrated among fewer than 1% of participants [92]. GyberExperiment addresses this problem through Hybrid Voting Power, in which the square root of stake reduces the weight of large holders, and Activity Multiplier and Status Bonus ensure that real contribution carries weight.

**Regulatory grey zone.** The overwhelming majority of DAOs operate without a legal entity, which creates risks for participants (unlimited personal liability) and barriers to interaction with the traditional economy (inability to enter into contracts, open bank accounts, hire employees). DUNA resolves this problem by providing a legal wrapper that preserves the principles of decentralisation.

**Voter apathy.** Average turnout in DAO votes amounts to 3–5% of the total number of token holders (Chainalysis, 2024 [93]). GyberExperiment proposes a comprehensive solution: AI summarisation of proposals, tiered governance (minor decisions are automated), reputation incentives for participation, and liquid democracy for delegation.

### 8.2.2. Implications for Policy and Regulation

The use of DUNA as a legal wrapper offers a model for the regulatory recognition of decentralised organisations that preserves the principles of decentralisation while ensuring legal accountability.

Specific implications for regulators:

- **The DUNA model as precedent.** The Wyoming Decentralised Unincorporated Nonprofit Association Act (W.S. § 17-32, 2024) is the world's first specialised law for DAOs. Successful operation of GyberExperiment under DUNA will create a precedent for other jurisdictions and may stimulate the adoption of analogous legislation.
- **Tax model for DAO participants.** The formalisation of relations between DUNA and participants (compensation for services, 1099 in the United States) creates a transparent tax model applicable to other DAOs.
- **Howey test for utility tokens.** The architecture of GBR (distribution for work rather than sale to investors; utility functions of governance and staking; absence of "promoters" — value depends on collective efforts) offers a utility token model that minimises the risk of classification as a security.
- **Cross-jurisdictional compatibility.** The interaction of DUNA with European regulation (MiCA — Markets in Crypto-Assets Regulation [95]), Singaporean regulation (Payment Services Act [96]), and Swiss regulation (DLT Act [97]) forms a map of regulatory compatibility for global DAOs.

### 8.2.3. Implications for the AI Industry

AiC offers an alternative model of AI development organisation, based on a community rather than a corporate structure, with democratic governance of research priorities and ethical constraints.

The current AI industry is characterised by extreme concentration: OpenAI, Anthropic, Google DeepMind, and Meta AI control the development of the most powerful models. This creates several problems that AiC seeks to address:

- **Concentration of control over AI.** Decisions about the capabilities, constraints, and ethical boundaries of AI models are made by a narrow circle of corporate executives. AiC proposes a model in which AI development priorities are determined by the community through Code DAO.
- **Opacity of data and training process.** Commercial AI models are trained on proprietary data; the training process is opaque. AiC uses open-source models (AGPL) with a transparent training process on GyberComputer.
- **Dependence on API providers.** Most organisations use AI through cloud provider APIs, which creates dependency and privacy risks. GyberComputer + Federated Learning provides computational autonomy.

### 8.2.4. Implications for Education and Universities

The GyberExperiment model proposes a radically new model for organising academic research and educational processes. The traditional academic system is characterised by several problems that CyberSocium addresses directly.

**The problem of grant funding.** Modern science depends on grant agencies (NSF, ERC, RFBR), which act as centralised arbiters determining which research receives funding. The grant review process is opaque, subjective, and prone to conservatism: revolutionary ideas that do not fit the mainstream are systematically underfunded (Nicholson & Ioannidis, 2012). The IPI model of CyberSocium offers an alternative: a research project receives funding proportional to the number of interested participants (|SIC|), not to a subjective expert committee assessment. This does not eliminate expert evaluation (Code DAO conducts peer review), but supplements it with a democratic mechanism for priority determination.

**The problem of publication pressure.** The "publish or perish" principle stimulates quantity of publications at the expense of quality, leading to a replication crisis and journal inflation. In the CyberSocium model, where contribution is measured through Activity Score and G-Plan, a researcher is rewarded for real contribution (code, data, reproducible results), not for number of publications. Verifiable Credentials issued by Code DAO provide verifiable confirmation of contribution without dependence on h-index metrics.

**The model of the decentralised university.** CyberSocium enables one to envision a "decentralised university" — an educational and research organisation functioning as a CSC: students and faculty are participants in NFC; courses and research are organised as IPI projects; quality is assessed through peer review (Code DAO) and social relevance (SES); degrees and certificates are Verifiable Credentials on the blockchain. The SAPP module GyberLearn (Section 6.5.2) is the first step towards realising this model.

### 8.2.5. Implications for Developing Countries

The PMIP model has particular significance for countries with limited access to traditional financing. In developing countries, entrepreneurs and inventors face a double barrier: the absence of local capital and the impossibility of attracting international financing due to institutional constraints (corruption, instability, absence of legal infrastructure).

CyberSocium offers a bypass of these barriers: a participant from Ghana or Bangladesh may: (a) join NFC by confirming activity through G-Plan, without requiring a bank account or credit history; (b) obtain Unit → Dev status through verified work, without requiring formal education or references; (c) initiate a project through Commerce DAO, attracting participants from the global NFC; (d) receive rewards in GBR, convertible through a DEX, without dependence on local banking infrastructure.

Formally: PMIP eliminates the geographic bias of traditional venture capital (85% of VC investments are concentrated in 3 countries: USA, China, United Kingdom). In CSC the sole criterion for project financing is social relevance SR(p) ≥ 1, determined by the global |SIC(p)|, not by the geographic location of the initiator.

**Case study (hypothetical): Project "CleanWater".** An engineer from Kenya develops a water purification system applicable in rural areas. In the traditional economy: requirement of $50,000 for prototyping → approach to local banks (refusal due to absence of collateral) → approach to international grant agencies (competition with thousands of applications, wait of 6–12 months) → possible rejection. In the CSC model: publication of the idea on DSP → formation of SIC (15,000 participants from NFC are interested in accessible water) → individual_cost = $50,000 / 15,000 = $3.33 per participant → project is socially relevant (SR = 1) → Commerce DAO approves → AG is formed (engineer + 8 developers + 3 water treatment experts from NFC) → AiC optimises design → prototype → testing → scaling. Full cycle: 3–6 months instead of 12–24 in the traditional model.

### 8.2.6. Implications for the Cooperative Movement

CyberSocium may be regarded as the digital evolution of the cooperative movement — a tradition tracing back to the Rochdale Pioneers (1844). Trebor Scholz (2016) introduced the term "platform cooperativism" to describe digital platforms owned and managed by their users [74]. Nathan Schneider (2018) developed this idea by proposing the concept of "exit to community" — the process of transferring control of a platform from founders to users [75].

GyberExperiment realises both of these principles: the platform (DSP) belongs to participants through DAO governance, and progressive decentralisation formalises the "exit to community" process. The distinction from traditional cooperatives: (a) global scale without geographic constraints; (b) formalised governance through smart contracts rather than meetings; (c) crypto-economic incentives in place of share contributions; (d) AI augmentation of participants' cognitive capacities.

## 8.3 Limitations and Challenges

We acknowledge the following limitations and challenges. For each limitation, the degree of mitigation provided by AI integration (Axiom A8, Section 5.6) is indicated:

1. **The governance scalability problem:** as the community grows, the number of proposals and votes may exceed participants' cognitive capacity, leading to voter fatigue and degraded decision quality. Vote delegation (liquid democracy) and AI summarisation of proposals are partial solutions. > **AI mitigation (substantial).** The AI analyst (5.6) generates personalised summaries of proposals, impact analysis,

and recommendations for each participant, reducing cognitive load from O(n) to O(1). The AI Cross-DAO Coordinator automatically routes proposals, eliminating information noise. Formally: cognitive_load(participant | AI) ≈ const for any number of proposals. The governance scalability problem is transformed from a fundamental constraint into an engineering challenge.

2. **The bootstrapping problem:** the PMIP model is most effective when |SIC| is large, but the initial stages of a project inevitably proceed with a small number of participants. Mechanisms for attracting early participants (elevated statuses in UnitManager, early access to tokens) are a compromise. > **AI mitigation (moderate).** AI agents compensate for small |SIC| in the early stages by performing coordination functions that in a mature system are distributed among participants: the AI assistant manages projects, the AI code reviewer ensures code quality, the AI analyst conducts due diligence. Effectively: effective_capacity(SIC | AI) > |SIC| even when |SIC| is small. However, the problem of attracting initial human participants remains.

3. **Regulatory uncertainty:** despite DUNA as a legal wrapper, the regulatory landscape for DAOs continues to evolve, and legislative changes may require adaptation of the organisational structure.

4. **Technical risks:** dependence on smart contracts introduces the risks of software errors with potentially irreversible financial consequences. Auditing, formal verification, and multi-level Timelock mechanisms reduce but do not eliminate these risks. > **AI mitigation (substantial).** The AI Code Reviewer (5.6.2) provides multi-level automated checking: static analysis, vulnerability detection (integration with Octane Security, Sherlock, AuditAgent from Nethermind), formal verification of invariants. The AI auditor blocks merges upon detecting critical vulnerabilities (class "AI decides" per A8). At the same time, AI introduces its own class of risks — AI model errors, hallucinations, adversarial attacks on AI systems — which require additional protective mechanisms (see point 6 below).

5. **The plutocracy problem:** despite mechanisms for activity verification and reputational weighting, token voting remains substantially weighted by capital, creating a risk of domination by large holders. Quadratic voting and conviction voting are being studied as possible mitigation mechanisms.

6. **AI dependency risks (new limitation).** Deep AI integration (Axiom A8) introduces its own class of risks: dependence on AI providers, adversarial attacks on AI models, risk of "bias amplification" (automation of bias), potential atrophy of collective judgement under excessive AI delegation. The division principle (A8)

and soulbound tokens for AI auditing are the primary protective mechanisms, but the long-term effects of AI dependency in decentralised systems are an open research question.

7. **The problem of verifying AI decisions (new limitation).** When an AI agent acts in the class "AI decides" (automatic blocking of vulnerable code, emergency stop upon anomaly detection), the community must have the ability to verify the correctness of these decisions. For complex AI models (LLM, deep learning), full explainability of decisions remains an unsolved problem (the black box problem). On-chain logging and DAO voting on challenging AI decisions are partial solutions.

8. **Key-person risk (bootstrap-phase limitation).** The current phase of the experiment is characterised by elevated key-person risk — dependence on a narrow group of initiators controlling the Governance Pool, GitHub repositories, and DUNA registration. This risk is a deliberate property of the bootstrap phase and is reduced as the transition to MacroeconomicDAO proceeds: transfer of funds to a stabilisation fund (smart contract), expansion of the circle of participants with ArchDev/Core rights, diversification of Gnosis Safe signers.

## 8.4 Threat Model and Security Analysis

A systematic analysis of attack vectors and their mitigation mechanisms is a necessary condition for transitioning from the experimental to the production phase. This section formalises the threat model for the GyberExperiment architecture, following the STRIDE methodology (Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege) and supplementing it with threats specific to decentralised systems.

### 8.4.1. Threat Classification

Threats are classified at three levels: protocol (attacks on smart contracts and blockchain), governance (attacks on DAO mechanisms and voting), and economic (attacks on tokenomics and financial flows).

**Protocol threats:**

| Attack vector | Description | Mitigation | Status |
|---|---|---|---|
| Flash loan attack on governance | Attacker takes a flash loan, acquires GBR, votes for a malicious proposal, and repays the loan within a single transaction | Snapshot-at-proposal-time in Governor contract: voting weight is fixed at the time of proposal creation, not at voting time. Voting delay between creation and start of voting excludes the use of borrowed tokens | Implemented in Governor contract |
| Smart contract vulnerabilities | Exploitation of vulnerabilities (reentrancy, overflow, logic errors) | Multi-level audit: (1) static analysis (Slither, Mythril); (2) formal verification of invariants (Certora Prover); (3) AI audit pipeline (Octane Security, Sherlock AI); (4) Timelock + Gnosis Safe as additional barriers; (5) bug bounty programme | AI audit — target architecture; Timelock and Safe — implemented |
| Oracle manipulation | Manipulation of price oracles to exploit contracts dependent on external data | TWAP (Time-Weighted Average Price) instead of spot price. Multiple data sources (Chainlink, Band Protocol). AI detection of anomalous price deviations | Target architecture |
| MEV (Maximal Extractable Value) | Sandwich attacks, front-running on user transactions | Use of MEV-protected RPC (Flashbots Protect), commit-reveal schemes for critical transactions, batch processing of votes | Target architecture |

| Attack vector | Description | Mitigation | Status |
|---|---|---|---|
| Social engineering on signers | Compromise of Gnosis Safe signer keys | Emergency multisig: pause only, not withdraw. M-of-N threshold scheme. Scheduled signer rotation. Hardware keys (Ledger/Trezor) mandatory | Implemented |

**Governance threats:**

| Attack vector | Description | Mitigation | Status |
|---|---|---|---|
| Governance capture | Accumulation of a significant volume of GBR to control voting | Supermajority for critical decisions. Timelock delay (48h minimum). Quadratic voting reduces weight of large holders. Reputational multipliers. AI monitoring of anomalous token accumulation | Partially implemented |
| Sybil attacks on G-Plan | Creation of fictitious accounts to receive rewards | GNN Sybil detection: interaction graph analysis. Proof of Personhood. Activity confirmation by higher-status participants. Activity Scoring (0–100) with minimum threshold | Manual verification — implemented; AI detection — target |
| Voter apathy | Low turnout (<5%) gives a small group disproportionate influence | Vote delegation (liquid democracy). AI summarisation of proposals reduces cognitive load. Reputational incentives for voting participation. Tiered governance: minor decisions automated; voting only on significant matters | Target architecture |
| Collusion | Secret collusion between participants to extract rent | Transparency of on-chain data. AI detection of suspicious voting patterns. Mandatory cooling period for repeat proposals by | Target architecture |

| Attack vector | Description | Mitigation | Status |
|---|---|---|---|
| | | the same author. Public discussion prior to voting | |

**Economic threats:**

| Attack vector | Description | Mitigation | Status |
|---|---|---|---|
| Death spiral (GBR) | Price decline → participant exodus → project contraction → further decline | Fixed supply. Buyback & burn from project revenues. x5 multiplier incentivises project completion. Two-tier DUNA treasury limits fiat losses. Stabilisation fund (Phase II) | Partial — requires empirical verification |
| Fund exhaustion (UnitManager) | Depletion of the 60% supply allocated for rewards | Adaptive rewards: Fund Monitoring contract tracks remaining balance and automatically adjusts payout size. Activity Scoring (multiplier 50–150%) prevents free-riding. Halving mechanism upon reaching thresholds (50%, 25%, 10% of remaining fund) | Target architecture |
| Liquidity fragmentation | Multiple token/GBR pools create illiquid markets | Liquidity consolidation through LP aggregator. Minimum pool size requirements for listing. Pools created only for projects that have passed the Implementation stage | Target architecture |
| Regulatory risk | Qualification of GBR as security (Howey test) | DUNA as nonprofit form: revenue is reinvested, not distributed. Buyback & burn — liquidity management mechanism, not | Target architecture; legal opinion — in progress |

| Attack vector | Description | Mitigation | Status |
|---|---|---|---|
| | | dividend. Absence of ICO/IEO: tokens distributed through UnitManager for activity. Legal opinion from specialist firm | |

### 8.4.2. Howey Analysis: GBR Token as Utility

To determine whether GBR constitutes an “investment contract” (security), the four-part Howey test (SEC v. W.J. Howey Co., 1946) is applied:

1. **Investment of money** — GBR is not sold to investors; tokens are distributed through UnitManager as a reward for verified activity. A participant does not “invest” money but receives tokens for labour.
2. **In a common enterprise** — GyberExperiment is a common enterprise (common enterprise); this criterion is fulfilled.
3. **With an expectation of profit** — Participants receive GBR as compensation for work, not as an investment with an expectation of profit. An increase in the value of GBR is not guaranteed and is not the primary motivation for participation.
4. **Solely from the efforts of others** — The value of GBR depends on the collective efforts of all participants. Each participant makes their own contribution; there are no “promoters” upon whose efforts profit depends.

**Conclusion:** GBR conforms to the utility token model (analogous to ETH), rather than to a security. However, this qualification requires confirmation by a specialised legal opinion and may depend on jurisdiction.

### 8.4.3. Tax Analysis for Participants

Participants in GyberExperiment who receive rewards in GBR through UnitManager may be subject to different tax regimes depending on jurisdiction:

- **United States:** GBR received for work qualifies as ordinary income at the time of receipt. The tax base is the fair market value of GBR at the time of receipt. Upon subsequent sale — capital gains/losses.
- **EU (general approach):** Analogously: income at the time of receipt + capital gains upon sale. Specific rules vary by country.

- **Through DUNA:** Participants receiving compensation as contractors of DUNA receive standard tax documents (1099 in the United States).

DUNA is not an employer of participants. All rewards are paid as compensation for services, not as employment income.

### 8.4.4. Mitigation Plan: Prioritisation

```
Priority 1 (Critical — prior to launch):
  — Smart contract audit (UnitManager, Governor, Timelock)
  — Gnosis Safe deployment with M-of-N signers
  — Howey analysis by a law firm
  — Snapshot-at-proposal-time in Governor

Priority 2 (High — first 6 months):
  — AI audit pipeline (Slither + Mythril + Certora)
  — GNN Sybil detection for G-Plan
  — Fund Monitoring contract for UnitManager
  — Bug bounty programme

Priority 3 (Medium — first year):
  — Quadratic voting
  — AI monitoring of governance anomalies
  — Stabilisation fund
  — cadCAD simulation of tokenomics

Priority 4 (Low — Phase II+):
  — Formal verification of contracts
  — MEV protection
  — Cross-chain security
  — Federated learning for Sybil detection
```

## 8.5 Open Research Questions

The theory of CyberSocium generates a series of open questions, each of which represents an independent direction for academic research.

**8.5.1. Formal verification of macroeconomic properties.** Is it possible to formally prove that the GyberExperiment system possesses certain desirable macroeconomic properties (stability, equitable distribution, antifragility)? Existing approaches (cadCAD, TokenSPICE, Gauntlet) allow the modelling of specific scenarios but do not provide formal guarantees. The application of model checking and theorem proving to economic systems is a promising but underdeveloped direction.

**8.5.2. Optimal DAO parameters.** What should the optimal parameters of quorum, voting periods, and approval thresholds be for each DAO class? Empirical data from existing DAOs (MakerDAO: quorum approximately 10%, Compound: approximately 4%, Uniswap:

approximately 4%) shows significant variation. Required: (a) collection of empirical data from GyberExperiment; (b) agent-based modelling (ABM) for parameter assessment; (c) adaptive mechanisms for automatic parameter self-correction.

**8.5.3. Inter-system interaction of CSCs.** How can multiple CyberSocial corporations interact with one another? What protocols are required for cross-CSC coordination? Does a "meta-SES" emerge — evolutionary selection between CSCs? And how does a CSC interact with traditional economic institutions (banks, exchanges, state regulators)?

**8.5.4. Long-term evolution.** How will the system evolve on the scale of decades? What emergent properties may arise from the interaction of hundreds of DAOs? Is there an attractor for the system — a stationary state towards which it tends? What are the conditions for its existence and stability?

**8.5.5. AI and the quality of democratic decisions.** How does the integration of AI systems into governance processes (AI summarisation, AI analysis of proposals, AI monitoring) affect the quality and democratic character of decisions? Is there a risk of "AI laundering" — a situation in which decisions are de facto made by AI, even though humans formally vote? How can the atrophy of collective judgement be measured and prevented?

**8.5.6. Scalability of PMIP.** The PMIP universal implementability theorem is mathematically sound, but ignores transaction costs, coordination losses, and the problem of attention (attention economy). At what N does the model break down? What is the empirical upper bound of PMIP efficiency?

**8.5.7. The initial bootstrap problem.** How can the "cold start" be overcome — the initial period when N is small and PMIP does not provide substantial resources? What is the minimum critical mass for CSC viability? Empirical data from the current experiment should provide an answer.

**8.5.8. Formalisation of SES as an evolutionary process.** Is it possible to formalise SES in the terms of evolutionary game theory (Maynard Smith, 1982)? Are there evolutionarily stable strategies (ESS) for agents in NFC? What is the convergence rate of SES towards an optimal distribution of resources?

**8.5.9. Legal compatibility of CSC.** Can a CSC function within the existing legal systems of various jurisdictions? DUNA resolves this problem for Wyoming, but how does a CSC interact with the legal systems of the EU (MiCA), Singapore (PSA), and Switzerland (DLT Act)?

**8.5.10. Measurement of social relevance.** SR(p) in the current model is binary (0 or 1). Is it possible to develop a continuous metric of social relevance that accounts not only for the number of SIC participants, but also for the intensity of their interest, their expertise, and the public significance of the problem that the project addresses?

## 8.6 GyberExperiment as a Controlled Socio-Economic Experiment

GyberExperiment is not simply a startup or a technological platform. It is a **controlled socio-economic experiment** — a conscious attempt to test the theoretical predictions of CyberSocium on a real system with real participants and real economic stakes. This section formalises the experimental methodology, its limitations, and its connection with the traditions of experimental economics.

### 8.6.1. Methodological Tradition: From Laboratory Economics to Field Experiments

Experimental economics as a discipline was founded by Vernon Smith (Nobel Prize 2002) and Daniel Kahneman (Nobel Prize 2002). Smith demonstrated that laboratory experiments allow economic theories to be tested under controlled conditions; Kahneman showed that real behaviour systematically deviates from the predictions of rational choice theory. Esther Duflo and Abhijit Banerjee (Nobel Prize 2019) developed the approach of randomised controlled trials (RCT) for assessing the effectiveness of economic interventions “in the field.”

GyberExperiment occupies an intermediate position between laboratory experiment and field study. It possesses properties of a controlled experiment: the system parameters (quorums, thresholds, reward formulas) are defined by the designer and can be modified; results are measurable (on-chain data provides full observability). But it also possesses properties of field research: participants are real people with real motivations; economic stakes are real (GBR has market value); the environment is not fully controlled (external shocks, regulatory changes, technical failures).

### 8.6.2. Experimental Design

We formalise GyberExperiment as a series of nested experiments, each testing a specific hypothesis of CyberSocium theory:

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Experimental design GyberExperiment:

  Experiment E1: Verification of PMIP
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    Hypothesis H1: individual_cost(i,p) = O(1/|SIC(p)|)
    Observable variables: C(p), |SIC(p)|, individual_cost(i,p)
    Method: regression analysis of the dependence of individual_cost
      on |SIC| for all completed projects
    Counterfactual: comparison with analogous projects
      on traditional crowdfunding platforms
    Minimum sample: ≥30 completed projects
    Expected time: Phase II (2025-2026)

  Experiment E2: Hybrid VP vs. Token-Weighted Voting
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    Hypothesis H2: Gini(VP_hybrid) < 0.7 × Gini(VP_token)
    Observable variables: VP distribution, voting
      outcomes, participation
    Method: quasi-experiment — parallel votes
      in two modes on the same cohort
    Counterfactual: voting on the same proposals
      with VP = GBR_staked (without square root)
    Minimum sample: ≥50 participants, ≥20 votes
    Expected time: Phase I-II (2025)

  Experiment E3: SES and evolutionary project selection
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    Hypothesis H3: projects with f(p) > f̄ attract more
      resources (participants, GBR) in the next period
    Observable variables: f(p) (fitness), x(p) (resource
      share), Δx(p)/Δt (rate of change)
    Method: panel analysis of all projects
      with monthly observations
    Counterfactual: random allocation of resources
      (Monte Carlo simulation)
    Minimum sample: ≥20 active projects
    Expected time: Phase II-III (2026-2027)

  Experiment E4: Activity Score and motivation
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    Hypothesis H4: introduction of Activity Score increases
      average participant activity by ≥30%
    Observable variables: on-chain activity,
      contribution rate, retention
    Method: before/after comparison + diff-in-diff
      (with a control group of participants unaffected
       by Activity Score)
    Counterfactual: period prior to introduction of Activity Score
    Minimum sample: ≥100 participants
    Expected time: Phase I (2025)

  Experiment E5: AI augmentation of governance
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    Hypothesis H5: AI summariser increases voter
      participation by ≥20% and informed decision-making by ≥30%
    Observable variables: voter turnout, time to
      review proposal, decision quality (expert assessment)
    Method: A/B test (group with AI summariser vs.
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      group without)
    Counterfactual: voting on identical proposals
      without AI summarisation
    Minimum sample: ≥50 participants in each group
    Expected time: Phase II (2026)
```

### 8.6.3. Advantages of the On-Chain Experimental Environment

The blockchain as an experimental platform possesses unique properties unavailable in traditional laboratory or field experimental economics:

**Full observability.** Every transaction, vote, reward claim, LP Burn, and IPI transition is recorded in an immutable ledger. This eliminates measurement error, characteristic of survey data and self-reported metrics. A researcher can reconstruct the complete interaction history of any participant to the second.

**Verifiability.** Any researcher (not only the GyberExperiment team) can reproduce the analysis using public on-chain data. This implements the principle of scientific reproducibility (reproducibility) at the architectural level — not as a voluntary commitment of the researcher, but as a technical property of the system.

**Impossibility of retroactive falsification.** Data recorded on-chain cannot be altered after the fact. This eliminates p-hacking, selective reporting, and other forms of scientific misconduct identified in the “reproducibility crisis” of the social sciences (Ioannidis, 2005; Open Science Collaboration, 2015).

**Real stakes.** Unlike laboratory experiments with symbolic rewards ($5–50), participants in GyberExperiment operate with real assets (GBR with market value). This ensures ecological validity — the experimental results reflect real behaviour, not laboratory artefacts.

### 8.6.4. Limitations of the Experimental Methodology

We acknowledge fundamental limitations of GyberExperiment as an experiment:

**Selection bias.** Participants in GyberExperiment are a self-selected group: people who have consciously chosen to participate in a cryptoeconomic system. They are not a representative sample of the general population. Consequently, the results of the experiment cannot be automatically extrapolated to an arbitrary group of people. Mitigation: as NFC grows and becomes geographically diversified (KPI: Diversity Index), selection bias decreases but is not eliminated.

**Absence of a control group.** A true randomised experiment requires a control group not subjected to the "treatment" (CyberSocium). This is practically impossible: one cannot randomly assign people to "decentralised" and "centralised" groups with identical tasks. Mitigation: quasi-experimental methods (difference-in-differences, regression discontinuity, instrumental variables) allow an approximation of causal identification.

**Hawthorne effect.** Participants who know they are participating in an experiment may alter their behaviour (the Hawthorne effect). In GyberExperiment this effect is attenuated: (a) on-chain data records real actions (not self-reports); (b) as participants become accustomed to the system, the "novelty effect" diminishes; (c) financial incentives (GBR rewards) ensure that rational behaviour dominates over the desire to "make an impression."

**Confounding variables.** External factors (the crypto market, regulatory changes, technical failures) affect results in parallel with the variables of interest. Mitigation: cadCAD modelling allows the separation of endogenous dynamics from exogenous shocks; time series of on-chain data allow the use of event study methodology for the identification of causal effects.

**Ethical considerations.** GyberExperiment involves real people with real economic stakes. This creates ethical obligations: informed consent (participants know that the system is experimental), minimisation of risks (contract audits, Fund Monitoring, Timelock), and the right to exit (a participant may withdraw their stake and leave the system at any time). DUNA as a legal wrapper provides limited liability for participants, reducing personal economic risk.

### 8.6.5. Open Data and Academic Collaboration

CyberSocium adheres to the principle of **Open Science** — all experimental data (with the exception of private off-chain data of participants, protected under A3) is published in the public domain:

- On-chain data is accessible via BSCScan/block explorer;
- Aggregated metrics (KPIs from Section 9.5) are published daily on DSP;
- cadCAD models and simulations are available on GitHub (AGPL-3.0);
- Experimental protocols (E1–E5) and pre-registrations are published prior to the commencement of the experiment;
- Results of analyses are submitted to peer-reviewed journals (Journal of Institutional Economics, Cryptoeconomic Systems, Journal of Political Economy).

This openness reflects Axiom A2 (transparency): the scientific value of GyberExperiment is determined not only by its practical results, but also by its contribution to the common knowledge of decentralised socio-economic systems. Even if the specific configuration of

GyberExperiment proves to be suboptimal, the experimental data will enable other researchers and projects to learn from its experience — in full accordance with the epistemological position of Section 1.6.

## 9. Development Roadmap: From Experiment to Ecosystem

The implementation of CyberSocium follows the principle of **progressive decentralization** — beginning with a minimally viable prototype under the coordination of a narrow group of initiators and progressively transferring control to the community as it grows and institutional mechanisms mature. This approach is driven by two factors: first, the bootstrapping problem, wherein PMIP does not yet provide sufficient resources; second, the necessity of empirically calibrating parameters (quorums, thresholds, rewards) prior to their codification in autonomous smart contracts.

Each phase is defined by: (a) priority objectives, (b) key performance indicators (KPIs), (c) transition criteria for the subsequent phase, and (d) risks and mitigations.

### 9.1 Phase I: Foundation (2024–2025)

Phase I constitutes a period of “managed genesis”: the creation of the legal, technical, and social infrastructure necessary for ecosystem viability. The paradox of the initial phase

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| lies in the fact that a decentralized system inevitably begins with a centralized act — just as Bitcoin was launched by a single individual (Nakamoto) and Ethereum by a Foundation. CyberSocium acknowledges this paradox and formalizes it through the mechanism of Progressive Decentralization: every action taken by the initiators creates infrastructure that renders their own influence progressively less necessary. | |
| Key dependencies of Phase I: DUNA → bank account → fiat operations → payment of auditors and infrastructure; GBR token → PancakeSwap LP → initial liquidity → economic participation; DSP (MVP) → first projects → formation of SIC → empirical data for PMIP calibration; UnitManager v1 → first rewards → participant incentives → community growth. These dependencies define the critical path and task prioritization. | |
| Special attention is devoted to security auditing: prior to deploying significant funds into smart contracts, they must undergo a minimum of two independent audits — an AI audit (Slither + Mythril | |

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| + Octane within the AiC pipeline) and a manual audit by a specialized firm. This is not merely best practice, but a consequence of Axiom A2 (transparency): the security of the system must be verifiable by any participant. | |
| ``` Priorities: — DUNA registration (Wyoming): Filing → BOI report → Operating Agreement → Bank account → EIN — Deployment of Governor + Timelock + Gnosis Safe: Multisig 3-of-5 for Core participants Timelock delay: 48 hours — DSP launch (MVP): — Participant profiles (wallet = identifier) — Project spaces — Basic communication (forums, chats) — GitHub integration for Code DAO — UnitManager v1: reward accrual for initial participants based on a fixed schedule — G-Plan v1: basic task management with manual activity verification — Formation of the community core: Core (≥1), ArchDev (≥3), LeadDev (≥5), Dev (≥15) — Publication of CyberSocium Foundation v2.0 as the foundational white paper — GBR token launch on BSC: BEP-20, PancakeSwap LP, initial liquidity — Smart contract audit: minimum 2 independent auditors (AI audit + manual) | |

Key Performance Indicators (KPIs): — ≥50 active participants (Activity Score ≥ 20) — ≥5 projects at the Discussion/Formation stage — DUNA registered and operational — DSP functional with ≥100 registered users — GBR listed on PancakeSwap with ≥$50K TVL — ≥3 publications / presentations in relevant communities — UnitManager: ≥100 successful claimReward() transactions

Phase II Transition Criteria: — All Phase I KPIs met — Contract audit completed without critical vulnerabilities — DUNA operational: bank account, BOI, EIN — Community has demonstrated sustained organic growth (≥10% participant growth per month for ≥3 months)

Risks: — Key-person risk: dependence on 1–3 initiators Mitigation: documentation of all processes, cross-training — Regulatory risk: uncertainty regarding GBR classification Mitigation: Howey analysis by a legal firm — Low engagement: <50 participants Mitigation: early participant programs (enhanced rewards) ```

### 9.2 Phase II: Growth (2025–2026)

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| Phase II marks the transition from a minimally viable product to a fully functional ecosystem. Whereas Phase I creates infrastructure, Phase II tests the viability hypothesis: is the ecosystem capable of attracting a critical mass of participants (≥500) and generating genuine economic value (≥3 revenue-generating projects)? |
| The central event is the launch of the GyberNet testnet — the first step toward sovereign blockchain infrastructure. The RPoS consensus (Reputational Proof-of-Stake) undergoes live testing: 10+ validators, thousands of transactions, real DAO votes. Testnet data are critically important for calibrating mainnet parameters (Phase III). |
| Simultaneously, the first Commerce DAO projects take shape — projects that have completed the full IPI cycle (Idea → Discussion → Formation → Implementation) and are generating real revenue. This constitutes a test of the key mechanism: LP Burn as proof-of-commitment, milestone voting, and revenue distribution (CSC 10%, AG Reward). Each Commerce DAO project is a micro- |

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| experiment verifying a distinct aspect of CyberSocium theory. | |
| AiC moves beyond the prototype stage: the AI audit pipeline enters production, and the AI proposal summarizer constitutes the first implementation of Axiom A8 in governance. Empirical data on the influence of AI on decision quality form the basis for RQ8 (Section 10.4). | |
| International community expansion (DSP localization into a minimum of 3 languages) tests the scalability of CyberSocium beyond a single linguistic community — a critically important condition for the global NFC underlying PMIP. | |
| ``` Priorities: — DSP v2: fully functional platform with IPFS storage and end-to-end encryption — ActivityPub compatibility for federation — DID (Decentralized Identifiers) for identification — Verifiable Credentials for skill attestation — GyberNet testnet: test network with RPoS consensus and 10+ validators — First Commerce DAO projects: LQD, SAPP, PowerSwapMeta — Due diligence process for each project — Escrow contracts with milestone voting — | |

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| AiC: formation of research groups, commencement of infrastructure work (Circuit 1) — AI audit pipeline: Slither + Mythril + Octane — AI proposal summarizer (prototype) — Reputation system v1: integration of G-Plan + UnitManager + DAO voting — Automatic on-chain Activity Score — Hybrid Voting Power formula in GBRVotingWrapper — International community expansion: DSP localization (EN, ES, ZH at minimum) partnerships with university laboratories — Fund Monitoring contract: tracking UnitManager remaining balance, automatic reward adjustment — Bug bounty program via Immunefi | |
| Key Performance Indicators (KPIs): — ≥500 active participants (Activity Score ≥ 20) — ≥20 projects at various IPI stages — ≥3 projects generating revenue (Operation stage) — GyberNet testnet operating stably (uptime >99%) — GBR TVL on PancakeSwap ≥$500K — ≥10 GyberNet testnet validators — AiC Circuit 1: AI audit pipeline in production — ≥1 academic publication on CyberSocium — DUNA annual report filed on time — ≥3 Commerce DAO projects approved and funded | |

Phase III Transition Criteria: — All Phase II KPIs met — GyberNet testnet: 3+ months of stable operation — Commerce DAO: at least 1 project at the Operation stage — Hybrid Voting Power tested across ≥50 votes — AI audit: ≥100 PRs analyzed automatically — Community: ≥500 participants, of whom ≥50 Dev+

Risks: — Slow growth: <500 participants Mitigation: partnerships with DAO ecosystems, grant programs for developers — Commerce DAO projects fail to generate revenue Mitigation: due diligence, milestone voting, cadCAD modeling prior to launch — GyberNet testnet is unstable Mitigation: extend the testing period, engage external consensus auditors ```

### 9.3 Phase III: Scaling (2026–2028)

Phase III marks the transition of GyberExperiment from an experimental platform to a fully operational ecosystem. The key event is the launch of GyberNet mainnet — a dedicated blockchain network based on Reputational Proof-of-Stake (RPoS), which liberates the ecosystem from dependence on BSC and

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| establishes sovereignty over transactional infrastructure in accordance with Axiom A1 (Decentralization). In parallel, GyberComputer is launched — a distributed computing network critically required for the realization of AiC (Circuit 3) and providing the ecosystem with its own computational resources. | |
| At this phase, the four-class DAO taxonomy must demonstrate its viability at scale: Social DAO governs a community of thousands of participants; Code DAO coordinates the development of a distributed codebase; Commerce DAO finances and curates dozens of projects; Economic DAO manages tokenomics under real market dynamics. The empirical data of this phase enable verification of hypotheses H1–H3 from the research program (Section 2.6.3). | |
| A principally important aspect is Progressive Decentralization: the transfer of control from a narrow group of initiators to the community. By the end of Phase III, Core participants must not control more than 50% of voting power, and all critical functions (UnitManager, Fund Monitoring) must | |

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| operate autonomously, without manual intervention. This constitutes the transition from “managed decentralization” (bootstrap) to “self-governed decentralization” (target state). | |
| ``` Priorities: — GyberNet mainnet: full-scale network with cross-chain bridges (BSC, Ethereum) — RPoS consensus with ≥50 validators — Single-slot finality (3 seconds) — Subsidized gas for DAO votes — GyberComputer v1: distributed computing network — Kubernetes + custom scheduler (Rust) — Proof-of-Computation verification — GPU nodes for AI training (AiC) — Federated Learning infrastructure — AiC: first trained models, launch of AI services (Circuit 3) — GNN Sybil detection in production — cadCAD simulation for each Economic DAO proposal — AI Governance Agents: summarizer + simulator — Cross-chain integration: BSC ↔ GyberNet ↔ Ethereum bridges (IBC-like) Research into TON and Solana bridges — Liquid democracy: vote delegation within the four-class DAO structure — Partial delegation (per DAO class) — Automatic re-delegation upon inactivity — | |

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| Formalization of inter-community protocols: — Inter-CSC communication standard — Pilot partnership with 2–3 external DAOs — Storage decentralization: — Full migration to IPFS/Filecoin — Incentivized storage on GyberComputer — Progressive Decentralization: — Transfer of Gnosis Safe from Core to an expanded set of signers (ArchDev+) — Autonomous contracts: UnitManager, Fund Monitoring require no manual intervention — Automatic on-chain status degradation |
| Key Performance Indicators (KPIs): — ≥5,000 active participants — ≥100 projects at various IPI stages — ≥10 projects generating revenue — GyberComputer: ≥100 nodes, ≥50 GPU nodes — AiC: ≥3 AI models in production — GyberNet mainnet: uptime >99.9%, ≥50 validators — GBR total TVL (all chains) ≥$5M — ≥5 academic publications on CyberSocium — ≥2 inter-community partnerships with external DAOs — Governance participation rate ≥15% (of total active participants) |
| Phase IV Transition Criteria: — GyberNet mainnet stable for ≥12 months — |

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| Progressive decentralization: Core participants do not control >50% voting power — AiC: all three circuits operational — Ecosystem is self-sustaining: project revenues cover DUNA operating costs — Community has demonstrated capacity for self-governance without initiator intervention ``` |
| ### 9.4 Phase IV: Maturity (2028+) |
| Phase IV is not so much a conclusion as a transition to a steady state. GyberExperiment ceases to be an “experiment” and becomes a functioning system — a cybersocial corporation in the full sense of Definition 3.2.2: CSC = ⟨N, R, P, G, M, E⟩, wherein each component of the six-tuple functions autonomously and is verifiable on-chain. |
| The central objective of Phase IV is to demonstrate that CyberSocium is not merely viable, but surpasses traditional institutions on key metrics: efficiency of resource allocation (Pareto-optimality, measured through cadCAD), fairness (Gini index for GBR distribution < 0.5, key-person risk ≤ 5%), and resilience (resistance to external shocks). These metrics provide a falsifiable |

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| test of the theoretical claims advanced in Sections 3–4. | |
| Inter-community economics (inter-CSC protocols) move to the foreground: if CyberSocium proves viable, other communities will inevitably create their own CSCs, and the question of their interaction will become central. Meta-SES (RQ9, Section 10.4) — evolutionary selection among CSCs — is the theoretical framework for this interaction: CSCs with more efficient institutional designs attract more participants, creating selection pressure at the institutional level. This is Darwinian evolution for organizations — but with the possibility of deliberate mutation design (through FRP and DAO voting). | |
| Full realization of Axiom A8 (Cognitive Augmentation) means that AI ceases to be an auxiliary tool and becomes an integral participant in governance: AI agents summarize proposals, simulate consequences, optimize resource allocation, and detect Sybil attacks — all while preserving the principle of division (AI recommends, humans decide). Federated Learning provides on-device inference for all DSP | |

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| participants, eliminating dependence on centralized AI providers. |
| ``` Priorities: — Full autonomy: all critical functions governed by smart contracts and DAO voting without manual override — Gnosis Safe: ≥9-of-15 signers — Timelock: all significant changes pass through Governor + Timelock — UnitManager: fully autonomous — Inter-community economics: protocols for interaction among cybersocial corporations — Inter-CSC bridge: resource exchange — Meta-SES: evolutionary selection among CSCs — Cross-CSC DAO: joint proposals — AI governance: full realization of A8 — AI agents for all four DAO classes — AI orchestration of GyberComputer — AI predictor of SR for SES — AI coordinator of PMIP — Federated Learning: on-device inference for all DSP participants — AI ethics governance through Social DAO — Full realization of the eight axioms (A1–A8) at the scale of a global ecosystem — Each axiom verifiable on-chain — Annual axiom compliance audit — Influence on regulatory frameworks: — Promotion of DUNA-like legislation in the EU, Singapore, Switzerland — |

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| | Participation in FATF, MiCA working groups — Development of model legislation for jurisdictions recognizing DAOs — Academic recognition: — Publication of results in peer-reviewed journals (Journal of Institutional Economics, Cryptoeconomic Systems, etc.) — Collaborative research programs with universities — Open datasets for researchers in DAO governance |
| | Maturity Metrics: — ≥50,000 active participants — ≥1,000 projects at various stages — GyberComputer: ≥1,000 nodes — AiC: ≥10 AI models, ≥3 commercial AI products — GBR total TVL ≥$50M — DUNA recognized as a model structure in a minimum of 3 jurisdictions — ≥20 academic publications — Governance participation rate ≥25% — Gini index for GBR distribution <0.5 (moderate inequality) — Key-person risk: ≤5% (no single participant controls >5% voting power) |
| | Vision: MacroeconomicDAO demonstrates that a decentralized, self-governed, transparent economic community is capable not only of competing with traditional economic institutions, but |

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| of surpassing them in efficiency, fairness, and resilience. |
| CyberSocium ceases to be an experiment and becomes a validated model — not the only one, but one of the viable alternatives available to humanity for organizing global economic interaction. ``` |
| ### 9.5 Success Metrics: The Ecosystem KPI System |
| A roadmap without measurable success criteria is nothing more than a list of aspirations. The present section defines the system of key performance indicators (KPIs) that enables objective assessment of GyberExperiment ecosystem progress and well-founded decision-making regarding phase transitions. KPIs are organized along four dimensions: social (participants and community), economic (tokenomics and finance), technical (infrastructure and security), and governance (management and decentralization). |
| #### 9.5.1. Social KPIs: Community Growth and Quality |
| Quantitative community growth is a necessary but insufficient condition for success. CyberSocium, |

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| unlike traditional platforms, optimizes not for “number of registered accounts” (a metric easily susceptible to Sybil attacks and statistical manipulation), but for quality of participation, measured through activity and contribution. |
| ``` Social KPIs: |
| NFC (Number of Full Citizens): Definition: participants with MemberStatus ≥ Unit(1) and Activity Score ≥ θ_min over the preceding 30 days Phase I target: NFC ≥ 50 Phase II target: NFC ≥ 500 Phase III target: NFC ≥ 5,000 Phase IV target: NFC ≥ 50,000 |
| DAU/MAU ratio (Daily Active / Monthly Active Users): Definition: ratio of daily active wallets to monthly active wallets on DSP Target: ≥ 0.30 (benchmark: Discord ~0.25, Telegram ~0.35) Significance: durability of engagement, not “register and leave” |
| MemberStatus Distribution: Target: pyramidal structure Core: 1–3 \| ArchDev: 5–15 \| LeadDev: 15–50 \| Dev: 50–200 \| Unit: 200+ \| None: unlimited Red flag: more than 80% of participants at the Unit level (= insufficient vertical mobility) Red flag: inverted pyramid (= status inflation) |

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| Retention Rate: Definition: percentage of participants with Activity Score > 0 90 days after first transaction Target: ≥ 40% (benchmark: DeFi protocols ~15–25%) Significance: co-metric for measuring the quality of onboarding and the value of participation |
| SIC Formation Rate: Definition: number of new SIC(p) created per 30 days Target: ≥ 3 (Phase I), ≥ 15 (Phase II), ≥ 50 (Phase III) Significance: SIC = the atomic unit of collective action; without SIC — no PMIP effect |
| Diversity Index: Definition: Herfindahl-Hirschman Index (HHI) by participant jurisdictions Target: HHI < 0.25 (no dominance by a single jurisdiction) Significance: realization of A6 (inclusivity); geographic concentration = risk factor ``` |
| #### 9.5.2. Economic KPIs: Tokenomics Sustainability |
| Economic metrics measure the sustainability of the incentive system and the ecosystem's capacity to generate and distribute value. |
| ``` Economic KPIs: |
| Fund Solvency Ratio: Definition: UnitManager.fund_balance / projected_rewards_12m |

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| Target: ≥ 2.0 (twofold reserve against obligations) Red flag: < 1.0 (= obligations exceed available funds; trigger for Halving) Monitoring: daily, automatically via Fund Monitor | |
| GBR Velocity: Definition: transaction_volume / circulating_supply Target: 0.5–3.0 (benchmark: ETH ~1.5–2.5) Red flag: < 0.1 (= token is not being used, a “dead” asset) Red flag: > 10 (= purely speculative use) | |
| LP Burn Commitment Rate: Definition: number of LP Burn events / number of Commerce DAO proposals Target: ≥ 50% of proposals with LP Burn (proof of commitment) Significance: LP Burn = an irreversible signal of seriousness of project intent | |
| PMIP Efficiency: Definition: individual_cost(i,p) = C(p) / \|SIC(p)\| for realized projects Theoretical bound: O(1/n) Practical target: individual_cost < $10 for projects with C(p) ≤ $50,000 at \|SIC(p)\| ≥ 5000 Significance: the primary indicator of PMIP operability; if individual_cost does not decrease as N grows — the theory is not confirmed | |
| Burn/Mint Ratio: Definition: GBR_burned / GBR_minted for a given period Phase I | |

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| target: 0.1–0.3 (supply growth for bootstrapping) Phase III+ target: ≥ 0.8 (approaching a deflationary model) Significance: tokenomics sustainability requires a balanced emission/burn cycle ``` |
| #### 9.5.3. Technical KPIs: Reliability and Security |
| ``` Technical KPIs: |
| Smart Contract Security Score: Definition: (1 - critical_vulnerabilities / total_audited_functions) Target: ≥ 0.999 (zero critical vulnerabilities in production) Monitoring: AI audit (Slither + Mythril) at each deployment + manual audit every 6 months |
| System Uptime: Definition: percentage of time DSP + GPROD are available Target: ≥ 99.5% (permissible downtime ~44 hours/year) Monitoring: health checks every 30 seconds |
| Transaction Finality Time: BSC (Phase I–II): < 5 seconds GyberNet (Phase III+): < 2 seconds (single-slot finality) |
| Test Coverage: Definition: line coverage for each module Target: CONTRACTS ≥ 95%, GPROD ≥ 80%, DSP Client ≥ 70% Significance: a baseline requirement for safe iterative development |

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| Bug Bounty Payout Rate: Definition: total bug bounty payouts / TVL Target: 0.1–1% of TVL allocated to bug bounty pool Significance: economic incentive for white-hat researchers ``` |
| #### 9.5.4. Governance KPIs: Decentralization and Participation |
| Governance KPIs are the most specific to CyberSocium, as governance is precisely what distinguishes a CSC from an ordinary DeFi protocol. |
| ``` Governance KPIs: |
| Voter Participation Rate: Definition: unique_voters / eligible_voters for each DAO class Target: ≥ 30% (benchmark: on-chain governance ~5–15% for the majority of DAOs) Significance: low participation = de facto oligarchy; CyberSocium sets an ambitious target of 30%+ |
| Gini Coefficient of Voting Power: Definition: Gini coefficient for VP distribution after Hybrid formula Target: Gini(VP) < 0.60 (benchmark: token-weighted DAO ~0.85–0.95) Significance: the primary indicator of Hybrid VP effectiveness; if Gini(VP) ≈ Gini(token_holdings) → the formula is not working |

| |
|---|
| Proposal Diversity Index: Definition: Shannon entropy of proposals distribution across DAO classes per quarter Target: H > 1.5 (out of a maximum of $\log_2(4) \approx 2.0$) Red flag: H < 0.5 (= one class monopolizes the governance agenda) |
| Nakamoto Coefficient: Definition: minimum number of participants capable of controlling 51% of VP Phase I target: ≥ 3 (= Gnosis Safe multisig) Phase II target: ≥ 10 Phase III+ target: ≥ 50 Significance: the primary metric of decentralization; growth of Nakamoto coefficient = progress of Progressive Decentralization |
| Cross-Class Proposal Success Rate: Definition: percentage of cross-class proposals approved by all affected DAO classes Target: 40–70% Red flag: < 20% (= inter-class coordination is not functioning) Red flag: > 90% (= rubber-stamping, absence of genuine deliberation) |
| Decision Time (average time to decision): Target Social DAO: < 72 hours Target Code DAO: < 48 hours Target Commerce DAO: < 120 hours Target Economic DAO: < 168 hours |

| |
|---|
| Significance: balance between speed and quality of deliberation ``` |
| #### 9.5.5. Aggregate Metric: Ecosystem Health Score |
| For operational monitoring, it is appropriate to aggregate the key KPIs into a single composite indicator — the **Ecosystem Health Score (EHS)** — computed as the weighted mean of normalized indicators: |
| ``` EHS = $w_1$ × norm(NFC) + $w_2$ × norm(Fund_Solvency) + $w_3$ × norm(Security_Score) + $w_4$ × norm(Voter_Participation) + $w_5$ × norm(Retention) + $w_6$ × norm(SIC_Formation) |
| where: norm(x) = min(x / target(x), 1.0) — normalization to [0, 1] $w_1$... $w_6$ — weights determined by Economic DAO (defaults: $w_1$ = 0.20, $w_2$ = 0.20, $w_3$ = 0.15, $w_4$ = 0.20, $w_5$ = 0.15, $w_6$ = 0.10) |
| Interpretation: EHS ≥ 0.80: Healthy ecosystem (green zone) 0.50 ≤ EHS < 0.80: Requires attention (yellow zone) EHS < 0.50: Critical condition (red zone) Trigger: EHS < 0.50 for three consecutive days → automatic creation of an Emergency Proposal in MacroeconomicDAO ``` |

EHS is published on DSP in real time (updated every 24 hours) and is accessible via the GPROD API. The transparency of EHS realizes Axiom A2: every participant can view the state of the ecosystem and verify the calculation.

# 10. Conclusion

## 10.1 Summary: From Problem to Solution

The present document has set forth the theoretical foundations and architecture of cybersocial economics — a new interdisciplinary field studying the regularities of the transition of the global socio-economic mechanism toward decentralized, self-governed, cybersocial forms of organization.

We have demonstrated that:

1. **The problem is real and systemic:** concentration of economic power, appropriation of data, opacity of governance, and inefficiency in resource allocation are not isolated deficiencies but structural properties of centralized economic institutions ill-suited to the reality of the era of supercommunication. The gap between the technological possibilities of coordination (billions of connected people) and the institutional reality (decisions made by boards of dozens) defines the central contradiction of our epoch.

2. **A decentralized alternative is formalizable:** through the concepts of the cybersocial corporation (CSC), the Principle of Minimal Individual Participation (PMIP), socio-economic selection (SES), the Idea-Project-Implementation model (IPI), and the Fork Resolution Protocol (FRP), we have provided a rigorous description of the mechanisms of decentralized economic interaction. Each of these mechanisms is formalized mathematically, possesses demonstrable properties, and is linked to a specific technical implementation.

3. **Governance can be simultaneously decentralized and efficient:** the four-class DAO taxonomy (Social, Code, Commerce, Economic) with the coordinating role of MacroeconomicDAO demonstrates that different types of decisions can be processed by specialized, optimized mechanisms without recourse to a centralized hierarchy. Hybrid Voting Power ($VP = \sqrt{GBR_staked} \times ActivityMultiplier \times StatusBonus$) resolves the plutocracy problem inherent in the majority of existing DAOs.

4. **Practical implementation is feasible:** GyberExperiment, its applied ecosystem (DSP, GyberNet, GyberComputer, G-Plan, AiC), and its legal wrapper (DUNA, Wyoming) constitute a concrete, functioning implementation of the theoretical constructs described. Five modules (DSP Client, GPROD, CONTRACTS, Supabase, DDP) provide a full stack — from user interface to blockchain layer.

5. **AI is a precondition for realizing the full potential of CyberSocium:** Axiom A8 (Cognitive Augmentation) and its institutional realization through AiC transform the fundamental limitations of direct democracy — cognitive overload and coordination costs — from intractable problems into engineering challenges. Without AI — an elegant theory with practical scalability limits. With AI — a functioning system at planetary scale.

6. **The mathematical corollary of PMIP — "Millions are Billions" — is not a utopian slogan but a formal proposition:** given a sufficiently large number of NFC participants, the threshold cost of social relevance is overcome for a project of any cost. This means that a cybersocial corporation is, in principle, capable of implementing projects of any scale — without centralization of capital or power.

7. **The DUNA legal wrapper provides a bridge between the decentralized and traditional worlds:** the Wyoming Decentralized Unincorporated Nonprofit Association (W.S. § 17-32, 2024) provides a legal framework within which a DAO can interact with banks, regulators, and counterparties while preserving the internal logic of decentralization. The two-tier treasury (on-chain + DUNA treasury) minimizes regulatory and financial risks.

## 10.2 Architecture of Contributions: What This Work Proposes That Is Novel

For clarity of positioning, we enumerate the original contributions of the present document, distinguishing it from the existing literature:

**Theoretical contributions:**

| Contribution | Novelty | Related works |
|---|---|---|
| 8 CyberSocium axioms (A1–A8) | First formal axiomatic system for CSC design; A8 (Cognitive Augmentation) is an original axiom with no analogues in the literature | Ostrom (8 design principles, 1990), Buterin (cryptoeconomics, 2014) |
| PMIP (Principle of Minimal Individual Participation) | Formal model with proven properties (universal realizability, scale inversion, antifragility) | Benkler (commons-based peer production, 2006), Weyl & Posner (Radical Markets, 2018) |
| SES (Socio-Economic Selection) | Formalization as an evolutionary game with proven Pareto-optimality and evolutionary stability of cooperation | Maynard Smith (ESS, 1982), Page (diversity theorem, 2007) |
| IPI (Idea-Project-Implementation) | Formalization of the project lifecycle as a finite automaton linked to DAO votes | Agile methodology, Stage-Gate (Cooper, 1990) |
| FRP (Fork Resolution Protocol) | Formal conflict resolution protocol for decentralized systems via on-chain arbitration | Kleros (decentralized arbitration), Aragon Court |
| Four-class DAO taxonomy | First DAO typology based on functional differentiation of decision types | Aragon (general-purpose DAO), Colony (task-based) |
| CAS model of CyberSocium | Formal description of CSC as a complex adaptive system with identified attractors and feedback loops | Holland (CAS, 1992), Kauffman (NK model, 1993) |

**Architectural contributions:**

| Contribution | Novelty |
|---|---|
| Hybrid Voting Power | Combination of quadratic voting, reputation, and status in a single formula |
| LP Burn as proof-of-commitment | Liquidity burning rather than staking as a project initiation mechanism |
| Two-tier DUNA treasury | On-chain + off-chain finances under unified DAO governance |
| AiC as institutional AI | AI integrated into the DAO structure with a formalized principle of division |
| MemberStatus hierarchy (6 levels) | Formalized meritocratic hierarchy with on-chain management |
| Reputational PoS (RPoS) | Modified consensus accounting for ecosystem reputation |

## 10.3 Limitations of This Work

We acknowledge the following limitations of the present document:

1. **Empirical verification is absent.** All theoretical models (PMIP, SES, CAS) remain mathematical constructions that have not undergone empirical testing against real data. GyberExperiment is designed as a site for such verification, but results will not be available before the completion of Phase II (2026).

2. **Parameters are not calibrated.** The specific numerical parameters (quorums, voting thresholds, reward sizes, slashing conditions) have been chosen on the basis of theoretical considerations and analysis of existing DAOs, but have not been optimized for the specific GyberExperiment ecosystem. Calibration through cadCAD modeling and empirical data is a Phase II objective.

3. **Scalability has not been proven.** The PMIP universal realizability theorem is mathematically correct but ignores transaction costs, coordination losses, and the attention problem. The effective upper bound on CSC scalability is an open question.

4. **Bootstrapping dependence.** The current phase is characterized by significant key-person risk and dependence on a narrow group of initiators. This is a deliberate compromise of the bootstrap phase, but it creates fragility that contradicts the stated principle of decentralization.

5. **Regulatory uncertainty.** DUNA is a relatively new legal form (the statute was enacted in 2024), without substantial case law or precedents. Its resilience under conditions of regulatory change has not been tested.

## 10.4 Directions for Future Research

The theory of CyberSocium opens several promising directions for academic research:

**Short-term (1–2 years):**

RQ1: Empirical verification of PMIP. The central question: is the theoretical O(1/n) convergence property of individual contribution confirmed by real-world data? Methodology: collection of data on all SIC formations in GyberExperiment Phase I; statistical analysis of the dependence of individual_cost(i) on |SIC(p)|; comparison of the observed distribution with the theoretical one. Expected outcomes: (a) confirmation or refutation of O(1/n) convergence; (b) identification of practical deviations from the theoretical model (coordination costs, free-riding, heterogeneity bias); (c) calibration of parameters for the cadCAD model. Publication target: peer-reviewed journal (Journal of Institutional Economics or Cryptoeconomic Systems).

RQ2: cadCAD modeling of GBR tokenomics. Construction of an agent-based model of GBR tokenomics using the cadCAD framework (BlockScience). Scenario modeling: (a) organic growth (10% per month), (b) stagnation (0%), (c) crisis (−20% participants), (d) speculative shock (5× increase in GBR price). For each scenario: analysis of the resilience of incentive mechanisms (Activity Score, MemberStatus degradation, LP Burn); identification of critical thresholds; recommendations for parameter calibration. Methodological contribution: an open cadCAD framework for modeling four-class DAO systems, available for use by other projects.

RQ3: Hybrid Voting Power vs. token-weighted voting. Quasi-experimental design: parallel conduct of votes on identical proposals in two modes (Hybrid VP and token-weighted) on a cohort of ≥50 participants. Metrics: (a) Gini index of voting power distribution; (b) correlation between stake size and vote outcome; (c) subjective participant satisfaction (survey); (d) Pareto efficiency of outcomes (expert assessment). Hypothesis H1: Hybrid VP reduces Gini voting power by ≥30% compared to token-weighted voting at comparable decision quality.

RQ4: Effectiveness of AI audit of smart contracts. Quantitative assessment of the AI audit pipeline (Slither + Mythril + Octane): percentage of detected vulnerabilities (recall), percentage of false positives (precision), detection time vs. manual audit. Benchmark: use of known vulnerabilities from the Immunefi bug bounty database and SWC Registry. Practical question: at what recall/precision level can AI audit replace (rather than supplement) manual audit?

**Medium-term (3–5 years):**

RQ5: Formal verification of CSC through model checking. Application of formal verification tools (TLA+, Alloy, SPIN) to verify macroscopic properties of CSC: (a) absence of deadlock in the four-class DAO system; (b) guarantee of FRP termination within a finite number of steps; (c) impossibility of accumulating >50% voting power by a single agent under a functioning Activity Score; (d) preservation of fund solvency under given emission/burn parameters. Methodological novelty: the first application of model checking to the full architecture of a DAO (rather than only to individual smart contracts). Expected outcome: formal safety and liveness guarantees for key CSC subsystems.

RQ6: A continuous social relevance metric SR(p). The current PMIP model uses a binary SR(p) ∈ {0, 1}. Research: development of a continuous metric SR(p) ∈ [0, 1] incorporating: (a) the size of SIC(p) as a function of total NFC; (b) intensity of participation (mean contribution); (c) temporal stability (retention); (d) qualitative factors (expert assessments). Theoretical question: are the key PMIP properties (universal realizability, scale inversion) preserved under continuous SR? Practical question: does continuous SR allow for more efficient resource allocation among projects?

RQ7: Cross-CSC interaction protocols. Formalization of protocols for interaction among independent cybersocial corporations. Three levels: (a) informational (exchange of project and participant metadata); (b) economic (token exchange, cross-CSC investments); (c) governance (joint proposals, inter-system voting). Formal requirements: interoperability without violation of autonomy; preservation of A1 (decentralization) during interaction; conflict resolution mechanisms at the meta-level. Analogues: IBC protocol (Cosmos), cross-chain bridges, but at the governance level, not only the transactional level.

RQ8: The influence of AI on the quality of democratic decisions. Experimental research: measurement of the influence of an AI summarizer and AI simulator (AiC Circuit 1) on the quality of decisions in MacroeconomicDAO. Methodology: A/B testing on two cohorts (with and without AI support). Metrics: (a) time to decision; (b) voter informedness (post-vote test); (c) decision quality (expert assessment after 6 months); (d) participation rate. Ethical question: does the AI summarizer create “filter bubbles” in governance? How can “adversarial diversity” be ensured in AI aggregation of arguments?

**Long-term (5+ years):**

RQ9: Meta-SES — evolutionary selection among CSCs. Formalization of an “ecosystem of ecosystems”: multiple interacting CSCs compete for participants, resources, and attention. Meta-SES as a second-order evolutionary game: CSCs with more efficient institutional designs attract more participants, while less efficient ones adapt or dissolve. Theoretical questions: (a) does a Nash equilibrium exist in meta-SES? (b) does meta-SES converge to

an optimal institutional design? (c) what is the role of “institutional diversity” (analogous to biodiversity)? Connection to the literature: institutional Darwinism (Hodgson, 2002), cultural evolution (Boyd & Richerson, 1985).

RQ10: Emergent properties of large-scale DAO ecosystems. Agent-based modeling of an ecosystem comprising hundreds of interacting DAOs (four-class). Questions: (a) do stable patterns of specialization emerge among DAOs? (b) do “DAO clusters” form analogous to Porter’s industrial clusters? (c) what is the critical mass of DAOs required for a self-sustaining ecosystem to emerge? (d) is a “tragedy of the commons” possible at the meta-level (among DAOs) and how can it be prevented? Methodology: ABM (Mesa / NetLogo) calibrated against GyberExperiment data.

RQ11: CSC vs. traditional institutions: a comparative analysis. The first rigorous empirical comparison of CSC and traditional organizational effectiveness. Metrics: (a) transaction costs (Williamson framework); (b) Pareto efficiency of resource allocation; (c) innovation rate (new projects/year per participant); (d) participant satisfaction (survey); (e) resilience to shocks (observed response to external crises). Design: matched comparison — CSC projects vs. comparable startups/non-profit organizations of similar size. Limitation: GyberExperiment at Phase III stage (≥5,000 participants) is required for statistical significance.

RQ12: The philosophy of CSC as a socio-economic subject. Fundamental question: is CSC a “subject” in the philosophical sense? Does CSC possess a moral status distinct from the sum of the moral statuses of its participants? Is CSC, as a unified entity, responsible for decisions made through DAO voting? Connection to the debates: corporate moral responsibility (French, 1984; List & Pettit, 2011), agency of AI systems, collective intentionality (Searle, 1995). Practical question: how does the legal person of DUNA relate to the “moral person” of CSC? Can CSC be a “bearer of rights” in a sense analogous to human rights?

## 10.5 Invitation to Participate

CyberSocium is not a finished solution. It is an experimental platform, a research program, and a working hypothesis — one in which AI is not an external supplement but an integral component that amplifies every theoretical mechanism, from PMIP to SES, from feedback loops to the four-class DAO system.

We invite:

- **Researchers** — for empirical verification of theoretical models, development of new formal results, and critical analysis of the proposed architecture
- **Developers** — for implementing the architecture in code, auditing smart contracts, and developing DSP, GyberNet, and GyberComputer

- **Economists** — for calibrating tokenomics parameters, modeling scenarios, and analyzing the economic properties of the system
- **Legal scholars** — for developing the DUNA legal framework, analyzing regulatory compatibility across jurisdictions, and drafting contract templates
- **AI engineers** — for implementing AiC, developing Sybil detection models, AI audit, and Federated Learning pipelines
- **Designers and UX researchers** — for creating interfaces that lower the cognitive barrier to participation in DAO governance
- **All who are dissatisfied with the status quo** — for participation in ecosystem projects, testing governance mechanisms, and forming a new type of socio-economic relations

To join, use the following channels: - **GitHub:** `github.com/gybernaty` — code, documentation, issues - **DSP:** Decentralised Social Platform — direct engagement with the community - **GBR Token:** `0xa970cae9fa1d7cca913b7c19df45bf33d55384a9` (BSC) — economic participation

> "The exclusive goal and opportunity of a society is the most effective unification of its participants in the interests of the society itself."

This principle is not an ideal to be attained in the future. It is an engineering specification that can be realized now. MacroeconomicDAO is the instrument of this realization. And CyberSocium is the theoretical foundation upon which this realization is built.

We stand at the beginning of the path. The theoretical apparatus has been set forth, the architecture has been designed, and the first components have been implemented. What lies ahead is the most challenging part: empirical verification, parameter calibration, attracting a critical mass of participants, and demonstrating that a decentralized, self-governed, AI-augmented economic system is capable not merely of existing, but of surpassing traditional institutions in efficiency, fairness, and resilience.

If this succeeds — GyberExperiment will become not simply a successful project, but a proof of existence for a new class of socio-economic systems. If not — the experimental data will enrich theory and reveal which of our assumptions were mistaken. In both cases, knowledge expands, and the next attempt will be better informed. This is the essence of the scientific method, and this is the essence of GyberExperiment.

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# 12. Appendices

## Appendix A. Glossary

### A.1. Core CyberSocium Concepts

| Term | Definition | Section |
|---|---|---|
| **CyberSocium** | A new interdisciplinary field studying the patterns of transition of socioeconomic systems toward decentralized, self-governing, cyber-social forms of organization | 1.1 |
| **Cyber-social corporation (CSC)** | A decentralized, autonomous, self-governing economic unit of a new type, defined as CSC = ⟨N, R, P, G, M, E⟩ (participants, resources, protocols, governance, market, environment) | 3.2 |
| **Cyber-social economy** | An economic system in which coordination is performed by protocols owned by the participants themselves, rather than by hierarchical intermediaries | 1.1, 3.3 |
| **GyberExperiment** | The first concrete realization of CyberSocium theory; includes an ecosystem of projects (DSP, GyberNet, GyberComputer, AiC) and a legal wrapper (DUNA) | 4.1 |

## A.2. Theoretical Mechanisms

| Term | Definition | Section |
|---|---|---|
| **PMIP** | Principle of Minimal Individual Participation: individual_cost(p) = C(p) / \|SIC(p)\| → 0 as \|SIC(p)\| grows. The mechanism for distributing the cost of a project across all SIC participants | 3.4 |
| **SES** | Socioeconomic Selection — an evolutionary mechanism for selecting projects through market and social feedback, formalized as an evolutionary game | 3.5 |
| **IPI** | Idea-Project-Implementation model — a formalized project lifecycle: Idea → Project → Implementation → Operation → Legacy | 3.4.2 |
| **FRP** | Fork Resolution Protocol — a mechanism for resolving disagreements through three phases: discussion → synthesis → separation (fork) | 3.4.5 |
| **SIC** | Social Investment Circle — the set of participants N_SIC ⊆ N_FC interested in the realization of a specific project; forms the financial and resource base of the project | 3.4.1 |
| **AG** | Active Group — a subset of SIC (AG ⊂ SIC) directly participating in project implementation (developers, managers, researchers) | 3.4.1 |
| **NFC** | Full Circle of participants in the cyber-social corporation — the entire set of registered participants in the ecosystem | 3.4.1 |

| Term | Definition | Section |
|---|---|---|
| **SR(p)** | Social relevance of project p — a threshold function: SR(p) = 1 if \|SIC(p)\| ≥ threshold(C(p)), otherwise SR(p) = 0 | 3.4.3 |
| **CAS** | Complex Adaptive System — a system composed of multiple interacting agents, exhibiting emergent properties (self-organization, adaptation, evolution) | 3.6 |

## A.3. Architectural Components

| Term | Definition | Section |
|---|---|---|
| **MacroeconomicDAO** | The meta-governance level, coordinating four classes of DAO (Social, Code, Commerce, Economic) through Cross-DAO coordination mechanisms | 4.6 |
| **Social DAO** | A class of DAO for managing social processes: community rules, moderation, event organization, membership. Quorum 5%, 72-hour voting period | 4.6.3, 5.1 |
| **Code DAO** | A class of DAO for managing the technical stack: PR merges, architectural decisions, quality standards. Quorum 10%, 96-hour voting period | 4.6.3, 5.2 |
| **Commerce DAO** | A class of DAO for commercial projects: due diligence, financing, milestone tracking, profit distribution. Quorum 15%, 120-hour voting period | 4.6.3, 5.3 |
| **Economic DAO** | A class of DAO for strategic economic decisions: tokenomics parameters, treasury allocation, cross-chain strategy. Quorum 25%, 168-hour voting period | 4.6.3, 5.4 |
| **Gbr (GBR)** | GyberCommunityToken — the primary economic instrument of the ecosystem. BEP-20 on BSC, supply 10B, address: `0xa970cae9fa1d7cca913b7c19df45bf33d55384a9` | 4.2 |
| **UnitManager** | A smart contract on BSC (UUPS + Ownable + ReentrancyGuard + Pausable + EIP-712), managing participant registration, statuses, and reward distribution | 4.2.3, App D.1 |
| **MemberStatus** | Enumeration of participant statuses: None(0) → Unit(1) → Dev(2) → LeadDev(3) → ArchDev(4) → Core(5). Determines rights, obligations, and reward amounts | 4.2.3 |
| **Hybrid Voting Power** | A combined voting formula: VP = √(GBR_staked) × ActivityMultiplier × StatusBonus. The square root reduces the weight of large holders (anti-plutocracy) | 4.6.5 |
| **Activity Score** | A numerical evaluation of participant activity (0-100), computed from G-Plan data, voting participation, and project contributions | 4.6.7 |
| **G-Plan** | | 6.4 |

| Term | Definition | Section |
|---|---|---|
| | The task management and activity verification system: planning, assignment, tracking, confirmation. Data are used for Activity Score and rewards | |
| **DSP** | Decentralised Social Platform — the operating system of CSC: an environment for the IPI cycle, SIC formation, DAO governance, and everyday interaction. Implementation: DSP Client (Next.js 15) + GPROD (NestJS 11) | 4.3, 6.1 |
| **GyberNet** | The specialized blockchain of CSC. Phases I–II: functions are executed via BSC. Phase III+: proprietary blockchain with RPoS consensus | 4.4, 6.2 |
| **GyberComputer** | The community's distributed computing network: hosting DSP, training/inference AI, IPFS pinning. Participants stake GBR and provide resources (CPU/GPU/storage) | 4.5, 6.3 |
| **AiC** | Artificial Intelligence Community — the AI department of the Gybernaty ecosystem. Three circuits: internal (AI for the ecosystem), research (R&D), commercial (AI-as-a-service) | 4.8 |
| **DUNA** | Decentralized Unincorporated Nonprofit Association — the legal wrapper under Wyoming law (W.S. § 17-32, 2024). Provides legal personhood to CSC | 4.7 |

## A.4. Technical Terms

| Term | Definition |
| --- | --- |
| **DSP Client** | The client-side component of DSP — the ecosystem's frontend. Next.js 15, React 19, Server Components |
| **GPROD** | Gyber Production Backend — blockchain middleware. NestJS 11, Prisma 6, PostgreSQL |
| **DDP** | Distributed Data Protocol — the distributed data storage protocol (IPFS + Content Registry + ACL). Phase 2 |
| **LP Burn** | A liquidity pool burn mechanism for initiating a project: the initiator burns LP tokens (GBR/USDT), demonstrating proof-of-commitment |
| **Buyback & Burn** | A deflationary mechanism: 2% of Commerce DAO project revenue is directed toward purchasing and burning GBR |
| **RPoS** | Reputational Proof-of-Stake — GyberNet's modified consensus: validator_weight = √(staked_GBR) × (1 + ActivityScore/100) |
| **SAPP** | Socio-App Protocol Package — the DSP extensions standard: code module + manifest + API contract |
| **DID** | Decentralized Identifier — a decentralized participant identifier: `did:gybernaty:<wallet_address>` |
| **VC** | Verifiable Credentials — verifiable attestations (status, skills, participation), signed by the issuing party's digital signature |
| **EIP-712** | An Ethereum standard for typed structured data signatures; used for activity proofs and off-chain votes |
| **UUPS** | Universal Upgradeable Proxy Standard — a pattern for upgradeable smart contracts (EIP-1822) |
| **Gnosis Safe** | A multisig wallet for treasury management: M-of-N threshold signature scheme |
| **Timelock** | |

| Term | Definition |
|---|---|
| | An execution-delay contract: after a DAO decision is adopted, its execution is deferred by 24–48 hours for review |
| **cadCAD** | Complex Adaptive Dynamics Computer-Aided Design — the BlockScience framework for modeling complex systems |
| **GNN** | Graph Neural Network — a neural network for graph analysis; used for Sybil detection |
| **Federated Learning** | Distributed training of ML models: each node trains locally and transmits only gradient updates |
| **Soulbound Token (SBT)** | A non-transferable token bound to a wallet. Used for AI-audit attestations and reputation records |

## A.5. Economic and Legal Terms

| Term | Definition |
|---|---|
| **TVL** | Total Value Locked — the aggregate value of assets locked in the ecosystem's smart contracts |
| **Howey Test** | The SEC's four-part test for determining whether an asset constitutes an "investment contract" (security) |
| **MiCA** | Markets in Crypto-Assets Regulation — the European Union's regulation of crypto-assets |
| **PSA** | Payment Services Act — Singapore's payment services law regulating tokens |
| **DLT Act** | Distributed Ledger Technology Act — Switzerland's law on distributed ledgers |
| **AGPL-3.0** | Affero General Public License — a license guaranteeing code openness, including network use |
| **Quadratic voting** | A mechanism in which the cost of N votes = $N^2$. Reduces the weight of large holders |
| **Liquid Democracy** | Delegable voting: a participant may delegate their vote to another participant on a specific issue |
| **Conviction Voting** | A mechanism in which a vote's weight increases over time: the longer a participant supports a proposal, the greater their weight |

## Appendix B. Smart Contract Addresses

```
Network: Binance Smart Chain (BSC)

  Gbr Token (BEP-20):
    0xa970cae9fa1d7cca913b7c19df45bf33d55384a9

  UnitManager:
    [address will be published after v2 deployment]

  Governor:
    [address will be published after deployment]

  Timelock:
    [address will be published after deployment]

  Gnosis Safe (Treasury):
    [address will be published after deployment]

Status: contracts are under active development.
All addresses will be verified and published
on BSCScan with open source code.
```

## Appendix C. The Eight Axioms of CyberSocium — Summary

```
A1 — DECENTRALIZATION
  No single node, agent, or group can
  exercise control over the system as a whole.
  Implementation: P2P architecture, multisig,
  distributed storage.

A2 — TRANSPARENCY
  All economic transactions and governance
  decisions are recorded on the blockchain and available
  for verification by any participant.
  Implementation: on-chain governance, AGPL-3.0.

A3 — DATA SOVEREIGNTY
  The user is the sole owner of
  all rights to generated content, metadata,
  and any other types of data.
  Implementation: DID, IPFS, E2E encryption,
  zero-knowledge proofs.

A4 — EXTENSIBILITY
  The system is designed for expansion
  without permission from a central authority.
  Implementation: modular architecture, SAPP,
  open APIs, permissionless participation.

A5 — MERITOCRATIC FAIRNESS
  Reward is proportional to verified
  contribution, not to capital, status, or connections.
  Implementation: UnitManager, G-Plan, reputation
  system, verification of activity.

A6 — INCLUSIVITY
  The entry threshold is minimized to ensure
  the broadest possible participation.
  Implementation: PMIP, multilingual support,
  cross-chain compatibility, free
  basic access.

A7 — SELF-GOVERNANCE
  All rules of the system are established
  and modified by the participants themselves
  through formalized voting mechanisms.
  Implementation: four-class DAO taxonomy,
  Governor contracts, Timelock.

A8 — COGNITIVE AUGMENTATION
  The system employs artificial intelligence
  as a tool for augmenting the collective
  cognitive capacities of participants,
  while preserving human control
  over strategic decisions.
  Implementation: AiC (three circuits), federated
  learning, AI audit, AI summarization,
  human-in-the-loop for all critical decisions.
```

## Appendix D. Smart Contract Interface Specification

This appendix describes the key smart contract interfaces of the GyberExperiment architecture. All contracts are written in Solidity ^0.8.20 using the OpenZeppelin library and deployed on BNB Smart Chain (BSC).

### D.1. UnitManager (GybernatyUnitManager)

Canonical contract: `CONTRACTS/unit-manager-dao/UnitManager.sol`

```
// SPDX-License-Identifier: GPL-3.0
pragma solidity ^0.8.20;

// Patterns: UUPS Proxy + Ownable + ReentrancyGuard + Pausable + EIP-712

enum MemberStatus {
    None,       // 0 — not registered
    Unit,       // 1 — basic participant
    Dev,        // 2 — developer
    LeadDev,    // 3 — lead developer
    ArchDev,    // 4 — architect
    Core        // 5 — core (added in v2)
}

interface IUnitManager {
    // ═══ Registration and statuses ═══

    /// @notice Register a new participant with initial Unit status
    /// @param wallet Participant's wallet address
    function registerMember(address wallet) external;

    /// @notice Promote a participant's status
    /// @param wallet Participant's address
    /// @param newStatus New status
    /// @dev Requires two-level approval:
    ///      confirmation by participants 1 and 2 levels above
    function promoteStatus(
        address wallet,
        MemberStatus newStatus
    ) external;

    /// @notice Demote status (degradation for inactivity)
    /// @param wallet Participant's address
    /// @dev Called automatically when Activity Score < threshold
    function degradeStatus(address wallet) external;

    // ═══ Rewards ═══

    /// @notice Claim reward for the current period
    /// @dev Checks: status, Activity Score (via G-Plan),
    ///      time since last payout (≥ 30 days)
    function claimReward() external;

    /// @notice Claim a project completion bonus (×5)
    /// @param projectId Identifier of the completed project
    /// @dev Checks: project in Operation status,
    ///      msg.sender ∈ AG(project)
    function claimProjectBonus(uint256 projectId) external;

    // ═══ Activity Scoring ═══

    /// @notice Update a participant's Activity Score
    /// @param wallet Participant's address
    /// @param score New Activity Score (0-100)
    /// @param proof EIP-712 signature from the verifier
    /// @dev Verifier = G-Plan oracle or signer
    function updateActivityScore(
        address wallet,
```

```
        uint8 score,
        bytes calldata proof
    ) external;

    // ═══ State reads ═══

    function getMemberStatus(address wallet)
        external view returns (MemberStatus);
    function getActivityScore(address wallet)
        external view returns (uint8);
    function getRewardAmount(MemberStatus status)
        external view returns (uint256);
    function getLastClaimTime(address wallet)
        external view returns (uint256);

    // ═══ Events ═══

    event MemberRegistered(address indexed wallet, MemberStatus status);
    event StatusChanged(address indexed wallet,
        MemberStatus oldStatus, MemberStatus newStatus);
    event RewardClaimed(address indexed wallet,
        uint256 amount, bool isProjectBonus);
    event ActivityScoreUpdated(address indexed wallet, uint8 score);
}
```

**Reward table:**

| Status | Reward/period | Project Bonus (×5) |
|---|---|---|
| Unit | 10,000,000 GBR | 50,000,000 GBR |
| Dev | 100,000,000 GBR | 500,000,000 GBR |
| LeadDev | 1,000,000,000 GBR | 5,000,000,000 GBR |
| ArchDev | 10,000,000,000 GBR | 50,000,000,000 GBR |
| Core | Determined by governance | — |

## D.2. GBRVotingWrapper (Hybrid Voting Power)

```
interface IGBRVotingWrapper {
    /// @notice Compute Hybrid Voting Power
    /// @param account Address of the voter
    /// @return votingPower Voting power
    /// @dev VP = sqrt(GBR_staked) × ActivityMultiplier × StatusBonus
    function getVotingPower(address account)
        external view returns (uint256 votingPower);

    /// @notice Delegate voting power
    /// @param delegatee Address of the delegate
    function delegate(address delegatee) external;

    /// @notice Stake GBR to participate in governance
    /// @param amount Amount of GBR to stake
    function stake(uint256 amount) external;

    /// @notice Withdraw GBR from stake
    /// @param amount Amount of GBR to withdraw
    /// @dev Timelock: 7 days after withdrawal request
    function unstake(uint256 amount) external;

    // ═══ Activity Multiplier ═══
    // score ∈ [0, 30)  → AM = 0.5  (50%)
    // score ∈ [30, 60) → AM = 0.75 (75%)
    // score ∈ [60, 80) → AM = 1.0  (100%)
    // score ∈ [80, 100] → AM = 1.5  (150%)

    // ═══ Status Bonus ═══
    // Unit: 1.0, Dev: 1.1, LeadDev: 1.2,
    // ArchDev: 1.3, Core: 1.5
}
```

### D.3. LPBurnInitiator

```
interface ILPBurnInitiator {
    /// @notice Burn LP tokens to initiate a project
    /// @param lpToken Address of the LP token (CAKE-LP GBR-BNB
    ///                or CAKE-LP GBR-USDT)
    /// @param amount Amount of LP tokens to burn
    /// @param proposalType Type of proposal being initiated
    /// @return initiationId Unique initiation ID
    /// @dev LP tokens are sent to 0x000...dead (irreversible)
    ///      Minimum threshold: minBurnValue (in USD)
    function burnForInitiation(
        address lpToken,
        uint256 amount,
        ProposalType proposalType
    ) external returns (uint256 initiationId);

    /// @notice Check the minimum burn threshold
    function estimateBurnValue(
        address lpToken,
        uint256 amount
    ) external view returns (uint256);

    enum ProposalType {
        Research,       // Research project
        Product,        // Product development
        CriticalVote    // Critical vote
    }

    event LPBurned(address indexed burner,
        uint256 amount, uint256 usdValue,
        ProposalType proposalType, uint256 initiationId);
}
```

### D.4. Governor and Timelock

Standard OpenZeppelin contracts with customization:

```
Governor: OpenZeppelin Governor + GovernorSettings
  + GovernorCountingSimple + GovernorVotesQuorumFraction
  + GovernorTimelockControl

Customization:
  — getVotes() overridden to use
    IGBRVotingWrapper instead of standard ERC20Votes
  — proposalThreshold() depends on DAO class
  — quorum() depends on DAO class and proposal type

Timelock: OpenZeppelin TimelockController
  — minDelay: 24-72 hours (depends on DAO class)
  — proposers: Governor contract
  — executors: Governor contract + any (auto-execution)
  — admin: Gnosis Safe (pause only)
```

# Appendix E. DUNA Legal Framework: Key Provisions

This appendix contains excerpts from the key provisions of the Association Agreement of Gybernaty DUNA, registered in the state of Wyoming (USA) in accordance with W.S. § 17-32 (Decentralized Unincorporated Nonprofit Associations Act, 2024).

## E.1. Association Structure

**Legal form:** Decentralized Unincorporated Nonprofit Association (DUNA), state of Wyoming.

**Jurisdiction:** Wyoming W.S. § 17-32 (enacted in 2024 specifically for DAOs).

**Key legal properties:** - Limited liability of participants for the obligations of the association - Exclusion of fiduciary duties of members (Notice of Restrictions on Duties) - Recognition of on-chain voting as a legally binding decision-making mechanism - The public identifier of the governance smart contract (Governor contract address) is specified at registration - Compatibility with IRS 501(c)(6) for nonprofit status

## E.2. Roles and Powers

```
Organizer:
  Role: nominal party / registered agent
  Powers: filing documents at registration
  Restrictions: after registration — no management
    rights or access to assets

Administrator (Ministerial Agent):
  Role: external legal/accounting firm
  Powers:
    — Opening and maintaining fiat bank accounts
    — Signing tax returns
      (Form 8832, 990-N) and liaising with IRS
    — Receiving official notices
    — Executing fiat payments
      approved by vote
    — Maintaining bookkeeping and compliance
  Restrictions:
    — MAY NOT initiate expenditures
      without on-chain voting
    — MAY NOT access the crypto treasury
    — MAY NOT block approved decisions
    — May be instantly replaced
      by a Social DAO vote
  Oversight:
    — AI monitor continuously verifies actions
    — Monthly reporting to the community

Token Holders (Gybernaty Community):
  Powers:
    — Initiating and voting on all proposals
    — Controlling the crypto treasury via Governor + Timelock
    — Appointing and replacing the Administrator
    — Modifying governance parameters
  Restrictions:
    — >25% control → BOI Report (FinCEN)
    — Not formal directors of DUNA
```

## E.3. Key Articles of the Association Agreement

**Article 1. Purpose and Mission.** Gybernaty DUNA is established to develop and support the ecosystem of decentralized socioeconomic interaction (CyberSocium): financing open-source development, supporting research, providing infrastructure, and coordinating community activities.

**Article 3. Membership.** A member of DUNA is any holder of the GBR token with an active Activity Score. Membership is determined by on-chain state and requires no formal application.

**Article 5. Governance.** All decisions are made through on-chain voting in the Governor contract. The outcome of a vote is legally binding for DUNA, the Administrator, and all participants.

**Article 7. Treasury.** Two-level system: crypto treasury (Timelock) and fiat treasury (Treasury Fiat Account + Operating Fiat Account).

**Article 9. Buyback & Burn.** Buyback & Burn is a protocol liquidity management operation (asset management operation) and is NOT a profit distribution. GBR tokens do NOT confer rights to dividends.

**Article 11. Dissolution.** Assets are not distributed among participants. They are transferred to another nonprofit organization with a similar mission. A qualified majority (≥66%) in the Economic DAO with a quorum of ≥30% is required.

**Article 13. Document Priority.** In case of discrepancy between the Agreement and smart contracts, the Agreement takes legal precedence. The code is brought into conformity through the Code DAO.

**Article 15. Intellectual Property.** Code: AGPL-3.0. Brand “Gybernaty”: owned by DUNA. Research: CC-BY-4.0.

## E.4. Two-Level Treasury: Architecture and Control

One of the key architectural decisions of DUNA is the two-level treasury system, which ensures the separation of on-chain and off-chain financial flows while maintaining unified governance through the DAO.

```
Two-Level Treasury:

  Level 1 — Crypto Treasury (On-Chain Treasury):
    Control: Governor + Timelock smart contracts
    Assets: GBR, stablecoins (USDT, USDC), LP tokens
    Location: Gnosis Safe multisig on BSC
    Budget: ~60% of Governance Pool (6B GBR)
    Operations:
      — Reward payouts via UnitManager
      — Staking rewards
      — Buyback & Burn
      — Financing Code DAO and Commerce DAO projects
    Auditability: fully on-chain; any participant
      can verify balances and transaction history
    Signers: M-of-N (current configuration: 3-of-5)

  Level 2 — Fiat Treasury (Off-Chain Treasury):
    Control: Administrator (external firm),
      acts only upon approval by on-chain vote
    Assets: USD, EUR (bank accounts)
    Location: Bank account in Wyoming
    Budget: formed through crypto→fiat conversion
    Operations:
      — Payment of legal and accounting services
      — Payment of hosting and infrastructure
      — Regulatory payments (filing fees, taxes)
      — Payment of external auditors (smart contracts)
      — Payment of contractor services (per 1099)
    Auditability: monthly Administrator report
      to the community + AI monitoring of discrepancies

  Fiat Expenditure Protocol:
    1. Proposal in Economic DAO: "Spend $X on Y"
    2. Vote: qualified majority (67%),
       quorum 25%
    3. On-chain approval → Timelock (48h)
    4. Administrator receives notification + proof of approval
    5. Administrator executes the fiat payment
    6. Administrator provides confirmation (receipt)
    7. AI monitor verifies compliance:
       amount, recipient, purpose
```

The principal advantage of the two-level architecture is that the crypto treasury is fully controlled by smart contracts and requires no trust in physical persons, while the fiat treasury minimizes trust in the Administrator through procedural constraints and AI monitoring. No single individual has simultaneous access to both treasury levels — this ensures the principle of separation of concerns.

## E.5. Compliance and Taxation

**DUNA's tax status.** Gybernaty DUNA files for 501(c)(6) status (business league) with the IRS. This status provides exemption from federal income tax, provided that revenues are reinvested in the activities of the association and not distributed among participants. Buyback & Burn is classified as an asset management operation, not as a distribution.

**DUNA obligations:** - Annual filing of Form 990-N (e-Postcard) or Form 990-EZ with the IRS - Beneficial Ownership Information (BOI) Report with FinCEN if participants with >25% control are present - Annual Report with the state of Wyoming - Maintenance of bookkeeping records (GAAP or Cash Basis) through the Administrator - AML/KYC: not required for on-chain operations; required for fiat operations through a bank

**International participants.** Participants from outside the United States are subject to the tax laws of their own jurisdiction. GBR rewards are classified as income at fair market value at the time of receipt. DUNA does not withhold taxes for non-US participants; responsibility for tax reporting rests with the participant. For participants from EU jurisdictions, consultation with a tax advisor is recommended, given variations in the application of MiCA (Markets in Crypto-Assets Regulation) across countries.

## E.6. Comparison of DUNA with Alternative Legal Forms

| Parameter | DUNA (Wyoming) | Cayman Foundation | Swiss Verein | Marshall Islands DAO LLC | BVI Ltd |
|---|---|---|---|---|---|
| Jurisdiction | USA (Wyoming) | Cayman Islands | Switzerland | Marshall Islands | British Virgin Islands |
| DAO recognition | Yes (W.S. § 17-32) | No (common law) | No (common law) | Yes (DAO Act 2022) | No |
| On-chain voting | Legally binding | Undefined | Undefined | Legally binding | Undefined |
| Limited liability | Yes | Yes | Partial | Yes | Yes |
| Nonprofit compatibility | 501(c)(6) | No | Limited | No | No |
| Bank accounts | Available (Wyoming banks) | Difficult | Available | Extremely difficult | Difficult |
| Taxation | 0% federal (under 501c6) | 0% (tax haven) | Cantonal tax | 0% | 0% |
| Reputational risks | Low (US jurisdiction) | High (offshore) | Low | High (association with shell companies) | High |
| Registration cost | ~$200 | ~$5,000–15,000 | ~$3,000–10,000 | ~$5,000 | ~$3,000–8,000 |
| Annual maintenance | ~$500 | ~$5,000–15,000 | ~$2,000–5,000 | ~$3,000–5,000 | ~$3,000–5,000 |

DUNA represents the optimal choice for GyberExperiment across a combination of factors: legal recognition of DAO mechanisms, compatibility with nonprofit status, accessibility of banking services, low cost, and minimal reputational risks. The primary limitation is single-state jurisdiction, which requires additional work to ensure cross-jurisdictional compatibility.

# Appendix F. Technical Specification of AI Integration

## F.1. AI Agent Architecture

```
Multi-agent AiC system (Circuit 1):

  Orchestrator: CrewAI / Microsoft AutoGen
  Models: open-source LLMs on GyberComputer
    — Primary: Llama 3 (70B) or Mixtral (8x22B)
    — Fallback: Llama 3 (8B) for edge tasks
    — Specialized: fine-tuned models for
      specific tasks (audit, moderation)

  Agents:

  1. AI Proposal Summarizer
     Input: proposal text + context
     Output: {essence, risks, economic_consequences,
       arguments_for, arguments_against}
     Deployment: GyberComputer
     Trigger: new proposal in Governor contract

  2. AI Code Reviewer
     Input: PR diff + context
     Output: {vulnerabilities, warnings,
       code_quality, recommendations}
     Integration: GitHub Actions CI/CD
     Tools: Octane Security, Sherlock AI,
       AuditAgent (Nethermind), Slither, Mythril

  3. AI Activity Verifier
     Input: on-chain data + G-Plan + GitHub
     Output: Activity Score (0-100) +
       EIP-712 signature for UnitManager
     Frequency: daily update

  4. AI Treasury Advisor
     Input: on-chain analytics + market
     Output: risk + simulations + recommendations
     Tools: cadCAD, Dune Analytics,
       Chainlink price feeds

  5. AI Moderator (DSP)
     Input: message + Social DAO rules
     Output: {ok, review, reject} + explanation
     Deployment: edge (on-device) for
       privacy + GyberComputer for
       complex cases

  6. AI Cross-DAO Coordinator
     Input: multi-DAO proposal
     Output: routing + impact_assessment
```

## F.2. Federated Learning

```
Framework: Flower 1.x (flower.ai)
Aggregation protocol: FedAvg + PearFL (adaptive)

Communication:
  — gRPC between nodes and aggregator
  — TLS 1.3 for encryption
  — Differential Privacy: ε = 1.0, δ = 10⁻⁵

Incentivization:
  — GBR reward per training round
  — On-chain recording of rounds in GyberNet
  — Slashing for poisoning attacks

Tasks:
  — LLM fine-tuning (LoRA/QLoRA)
  — GNN training for Sybil detection
  — Recommendation models for DSP
  — CV for content moderation

Node hardware requirements:
  Minimum: 16GB RAM, GPU 8GB VRAM (RTX 3060+)
  Recommended: 32GB RAM, GPU 24GB VRAM (RTX 4090)
  Cluster: A100/H100 (financed by Economic DAO)
```

## F.3. Sybil Detection: GNN

```
Graph:
  Nodes: wallets (BSC addresses)
  Edges: transactions, project participation,
    votes, DSP interactions
  Features: Activity Score, MemberStatus,
    registration time, transaction volume

Model:
  Architecture: GraphSAGE + attention
  Framework: PyTorch Geometric
  Training: semi-supervised
  Inference: daily on GyberComputer

Metrics:
  Precision ≥ 0.95 (minimizing false positives)
  Recall ≥ 0.85

Integration:
  Sybil flag → AM = 0 → neutralization of voting power
  Challenge via Social DAO (human-in-the-loop)
```

### F.4. cadCAD Ecosystem Model

```
Agents:
  – Participants: N ∈ [100, 1,000,000]
  – Projects: M ∈ [1, 10,000] with IPI cycle
  – AI agents: K ∈ [1, 100]

State space:
  – GBR_supply(t), GBR_price(t), TVL(t)
  – N_active(t), M_projects(t)
  – treasury_balance(t)

Governed parameters:
  – staking_reward_rate, burn_rate
  – quorum_threshold, voting_period
  – lp_burn_minimum, activity_threshold

Stress scenarios:
  – 10× participant growth over 6 months
  – GBR_price crash → 0.1× over 1 week
  – Coordinated Sybil attack (1,000 accounts)
  – Exit of 30% of Core/ArchDev participants
  – Regulatory ban in a major jurisdiction

Health metric:
  Health(t) = w₁·liquidity + w₂·participation
    + w₃·project_success_rate + w₄·treasury_runway
```

### F.5. AI Governance: AI Resource Management Protocol

Governance of AI components in a decentralized system raises unique challenges absent from traditional organizations: who decides which data are used for training? who controls fine-tuning? how is an audit trail ensured for AI decisions? This section specifies the AI Governance protocol, implementing the principle “AI is subordinate to the DAO, not the reverse.”

```
AI Governance Protocol:

  Level 1: Model Configuration
  ────────────────────────────────────
    Decides: Code DAO (simple majority)
    Scope: choice of base model (Llama/Mixtral/etc.),
      framework versions (Flower, PyTorch),
      inference parameters (temperature, top_p)
    Timelock: 24 hours
    Rationale: technical decisions, reversible,
      low risk impact

  Level 2: Training Policy
  ────────────────────────────────────
    Decides: Social DAO + Code DAO (cross-class;
      both classes must approve)
    Scope: which data are permissible for training,
      Differential Privacy parameters (ε, δ),
      fairness constraints, content policies
    Timelock: 48 hours
    Rationale: affects participant privacy
      (A3) and social norms (A6)

  Level 3: Deployment
  ────────────────────────────────────
    Decides: MacroeconomicDAO (qualified
      majority 67%)
    Scope: deployment of a new AI agent,
      removal of an existing agent, modification of AI
      access rights to on-chain operations
    Timelock: 72 hours
    Rationale: AI with on-chain rights can
      execute financial operations; maximum
      caution is required

  Level 4: Kill Switch
  ────────────────────────────────────
    Decides: Emergency multisig (Core + ArchDev, 3-of-5)
    Scope: immediate shutdown of any AI agent
    Timelock: 0 (instant execution)
    Post-hoc: Emergency Proposal in Social DAO
      within 24 hours for ratification
    Rationale: protection against compromised AI, hallucination
      with financial consequences, data breach
```

Every consequential action of an AI agent (Activity Score assignment, Sybil flag, treasury recommendation) is recorded in an on-chain audit log through the specialized AiCAuditRegistry contract:

```
// AiCAuditRegistry pseudocode
interface IAiCAuditRegistry {
    event AiAction(
        uint256 indexed actionId,
        address indexed aiAgent,
        bytes32 actionType,     // SCORE_UPDATE, SYBIL_FLAG, etc.
        bytes32 inputHash,      // SHA-256 hash of input data
        bytes32 outputHash,     // SHA-256 hash of output
        bytes32 modelHash,      // hash of model weights
        uint256 timestamp
    );

    // Any participant can challenge an AI action
    function challengeAction(
        uint256 actionId,
        string calldata reason
    ) external;

    // A challenge routes the action to Social DAO
    // for human review
}
```

This mechanism ensures transparent AI accountability: every AI decision has a cryptographically verifiable audit trail (which model, on which data, what result), and every decision can be challenged by a participant through the DAO. AI in CyberSocium is not a “black box” but a verifiable instrument subordinate to collective governance.

### F.6. AI Model Update Protocol

Updating models in production is one of the highest-risk operations in AI systems. In centralized organizations a model is updated by decision of the ML team; in CyberSocium — by decision of the DAO, with safety guarantees.

```
Model Update Protocol (MU):

  Stage 1: Proposal (Code DAO)
  ────────────────────────────────
    Initiator: any Dev+ participant
    Content: {model_hash, training_report,
      benchmark_results, diff_analysis}
    Requirement: benchmark on standard dataset
      ≥ baseline (current model) on all metrics

  Stage 2: Shadow Testing (7 days)
  ────────────────────────────────
    New model runs in parallel with the current one
    Results of both are recorded, but only
    the current model affects the system
    Metrics: accuracy, latency, fairness_score,
      resource_consumption
    Automatic rejection if:
      — accuracy < baseline - 2%
      — latency > baseline × 1.5
      — fairness_score < baseline - 5%

  Stage 3: Canary Deployment (7 days)
  ────────────────────────────────
    5% of requests processed by the new model
    Monitoring: anomaly detection on outputs
    Automatic rollback if:
      — anomaly rate > 3%
      — user complaints > threshold
      — resource consumption > budget × 1.2

  Stage 4: Gradual Rollout (14 days)
  ────────────────────────────────
    5% → 25% → 50% → 100%
    At each stage: 48 hours of monitoring
    DAO vote for the 50% → 100% transition

  Stage 5: Decommission Old Model (7 days)
  ────────────────────────────────
    Old model remains available as fallback
    after 7 days — archiving
    Full cycle: ~5 weeks
```

This protocol ensures a balance between innovation (regular model updates) and stability (multi-layered protection against regressions). The key difference from industry practice: the decision on final deployment (50% → 100%) is made by the DAO, not the ML team.

### F.7. Ethical Constraints on AI in the CyberSocium Ecosystem

The AiC AI components operate under a set of formal constraints derived from the axioms of CyberSocium:

**Constraint 1: Prohibition on autonomous financial operations.** No AI agent may independently initiate a GBR transfer, a modification of staking parameters, or an LP Burn. AI may recommend (Treasury Advisor), but execution always requires DAO voting or multisig confirmation. Rationale: A7 (self-governance) means governance by people, not by algorithms.

**Constraint 2: Transparency of training data.** For each model deployed in AiC, the following is published: (a) a description of the training dataset (without disclosure of private data); (b) the training method (architecture, hyperparameters, number of epochs); (c) fairness metrics across protected groups (jurisdiction, MemberStatus, duration of participation). Rationale: A2 (transparency) extends to AI components.

**Constraint 3: Right to Human Review.** Any participant affected by an AI agent decision (Activity Score, Sybil flag, content moderation) has the right to have the decision reviewed by a human. A review request is submitted to the Social DAO and considered within 72 hours. The AI decision is suspended until the review is completed. Rationale: A5 (meritocratic fairness) requires the ability to correct errors; A6 (inclusivity) prohibits discrimination on the basis of algorithmic bias.

**Constraint 4: Prohibition on profiling for influence.** AiC AI models may not be used to create personalized “nudge” mechanisms directed at altering participants’ voting behavior. The AI Summarizer provides the same information to all participants; personalization is permitted only for UX (language, display preferences), not for the content of governance materials. Rationale: A1 (decentralization) is incompatible with algorithmic manipulation of governance; A3 (data sovereignty) prohibits the use of behavioral data for influence.

**Constraint 5: Graceful degradation.** The system is designed to function without AI components — with degraded performance but without loss of functionality. If all AI agents are unavailable: Activity Score is computed by a simplified formula (on-chain metrics, without AI verification); Sybil detection switches to manual mode (Social DAO monitoring); DAO proposals are submitted without AI summarization (participants read the full text). Rationale: A4 (extensibility) means that AI is an extension, not a dependency; the system must be robust to the failure of any component.

## Appendix G. Mathematical Notation and Symbols

The following is a complete list of the mathematical notations used throughout the document, provided for the reader’s convenience.

## G.1. Sets and Elements

| Symbol | Meaning | Section |
|---|---|---|
| N | Set of all CSC participants | 3.2, 3.6 |
| N_FC (NFC) | Full Circle of participants in the cyber-social corporation | 3.4.1 |
| N_SIC (SIC) | Social Investment Circle of project p | 3.4.1 |
| AG | Active Group (AG ⊂ SIC ⊆ NFC) | 3.4.1 |
| P | Set of all projects | 3.4.2 |
| R | Set of CSC resources | 3.2 |
| G | Set of governance protocols | 3.2 |
| Ω | Space of organizational forms ⟨D, T, A, S, C⟩ | 3.3 |

## G.2. Functions and Operators

| Symbol | Definition | Section |
|---|---|---|
| C(p) | Cost of project p | 3.4.1 |
| individual_cost(p) | C(p) / \|SIC(p)\| — individual cost of participation | 3.4.3 |
| SR(p) | Social relevance of project p ∈ {0, 1} | 3.4.3 |
| EV(p) | Expected economic value of project p | 3.4.3 |
| f(p) | Fitness function of project p in SES | 3.5.5 |
| VP(i) | Voting Power of participant i = √(GBR_staked_i) × AM_i × SB_i | 4.6.5 |
| AS(i) | Activity Score of participant i ∈ [0, 100] | 4.6.7 |
| AM(i) | Activity Multiplier: 0.5 + (AS(i) / 100) ∈ [0.5, 1.5] | 4.6.5 |
| SB(i) | Status Bonus: depends on MemberStatus(i) | 4.6.5 |
| validator_weight(v) | √(staked_GBR(v)) × (1 + AS(v)/100) — validator weight in GyberNet | 4.4.3, 6.2.2 |
| Health(t) | Ecosystem health metric at time t | App F.4 |
| x(p_i, t) | Share of resources allocated to project p_i at time t (replicator dynamics) | 3.5.6 |
| dx(p_i)/dt | Rate of change of resource share = x(p_i) × [f(p_i) - f̄] | 3.5.6 |

## G.3. Parameters and Constants

| Symbol | Meaning | Current value |
|---|---|---|
| GBR_total_supply | Total GBR token supply | 10,000,000,000 (10B) |
| quorum_social | Quorum for Social DAO | 5% |
| quorum_code | Quorum for Code DAO | 10% |
| quorum_commerce | Quorum for Commerce DAO | 15% |
| quorum_economic | Quorum for Economic DAO | 25% |
| voting_period_social | Voting period for Social DAO | 72h |
| voting_period_code | Voting period for Code DAO | 96h |
| voting_period_commerce | Voting period for Commerce DAO | 120h |
| voting_period_economic | Voting period for Economic DAO | 168h |
| timelock_delay | Execution delay after a decision is adopted | 24–48h |
| burn_rate | Share of project revenue directed to buyback & burn | 2% |
| lp_burn_threshold | Minimum LP amount for project initiation | $1,000 in GBR/USDT LP |
| min_stake_validator | Minimum stake for a GyberNet validator | 100,000 GBR |
| $w_1$, $w_2$, $w_3$ | Weights in the SES fitness function | 0.4, 0.4, 0.2 (calibratable) |
| $\delta$ | Phase transition threshold in space $\Omega$ | > 0 (determined empirically) |

## Appendix H. Index of Diagrams

| Figure | Title | File | Section |
|---|---|---|---|
| 3.1 | Comparison of organizational forms (Corporation / DAO / CSC) | `diagrams/conceptual/organizational_forms_comparison.svg` | 3.2 |
| 3.2 | Feedback loops in CyberSocium | `diagrams/conceptual/cybersocium_feedback_loops.svg` | 3.6.3 |
| 3.3 | Space of organizational forms Ω | `diagrams/conceptual/organizational_space.svg` | 3.3 |
| 4.1 | Full architecture of the GyberExperiment ecosystem | `diagrams/architecture/system_architecture.svg` | 4.1 |
| 4.2 | Mapping of theory → implementation correspondences | `diagrams/architecture/theory_implementation_map.svg` | 4.1 |
| 4.3 | GBR emission schedule | `diagrams/architecture/gbr_emission_schedule.svg` | 4.2 |
| 4.4 | Decision-making process in MacroeconomicDAO | `diagrams/process/dao_decision_flow.svg` | 4.6.2 |
| 4.5 | GyberNet architecture (5 layers) | `diagrams/architecture/gybernet_architecture.svg` | 4.4.3, 6.2.1 |
| 5.1 | Four-class DAO taxonomy | `diagrams/conceptual/dao_taxonomy.svg` | 5 |
| 6.1 | Five ecosystem modules (DSP Client, GPROD, CONTRACTS, Supabase, DDP) | `diagrams/architecture/five_modules.svg` | 6.7.1 |

## Appendix I. Timeline of Key Events

| Date | Event | Significance |
|---|---|---|
| 2023 Q3 | Beginning of CyberSocium conceptual development | Formulation of axioms A1–A7 |
| 2023 Q4 | Deployment of the GBR token on BSC | First on-chain component of the ecosystem |
| 2024 Q1 | Development of UnitManager v1 | Smart contract for statuses and rewards |
| 2024 Q2 | Adoption of the Wyoming DUNA Act | Legal basis for the legal wrapper |
| 2024 Q3 | Registration of Gybernaty DUNA | Legal personhood of CSC |
| 2024 Q4 | Formulation of axiom A8 (Cognitive Augmentation) | Integration of AI into the theoretical foundation |
| 2025 Q1 | CyberSocium Foundation Document v1.0 | First publication of the theoretical foundations |
| 2025 Q2 | Start of DSP Client development (Next.js 15) | Ecosystem frontend |
| 2025 Q3 | Launch of GPROD (NestJS 11) | Backend / blockchain middleware |
| 2025 Q4 | UnitManager v2 with MemberStatus (6 levels) | Updated contract with complete hierarchy |
| 2026 Q1 | CyberSocium Foundation Document v2.0 | The present document: complete theoretical specification |
| 2026 Q2 | (planned) DSP v2 with IPFS and E2E encryption | Fully functional social platform |
| 2026 Q3 | (planned) GyberNet testnet | Test network with RPoS consensus |
| 2026 Q4 | (planned) AiC Circuit 1: AI audit pipeline | First production AI component |
| 2027 | (planned) First Commerce DAO projects in Operation | Economic self-sufficiency of the ecosystem |
| 2028 | (planned) GyberNet mainnet | CSC's proprietary blockchain |

## Appendix J. Map of Theoretical Dependencies

This appendix visualizes the logical structure of CyberSocium theory — the dependencies between axioms, mechanisms, theorems, and technical components. This map enables the reader to: (a) trace the logical chain from each axiom to a specific technical implementation; (b) identify which components are required for each theoretical result; (c) assess the impact of each individual decision on the system as a whole.

### J.1. Axioms → Mechanisms → Technical Components

```
Dependency graph (simplified):

  AXIOMS                  MECHANISMS                TECHNOLOGY
  ═══════                 ══════════                ══════════

  A1 (Decentralization) —> PMIP                   —> UnitManager
                        —> Four-class DAO          —> Governor × 4
                        —> Progressive Decentr.    —> Timelock + Gnosis Safe

  A2 (Transparency)     —> Open-source (AGPL)     —> GitHub repos
                        —> On-chain audit trail    —> BSC/GyberNet explorer
                        —> Public cadCAD           —> cadCAD simulations

  A3 (Data             —> Federated Learning      —> Flower + PySyft
      Sovereignty)     —> Client-side encryption  —> DDP (IPFS + ACL)
                        —> DID/VC                  —> W3C standards

  A4 (Extensibility)   —> UUPS Proxy pattern      —> Upgradeable contracts
                        —> SAPP modules            —> DSP plugin system
                        —> Modular degradation     —> Microservices arch.

  A5 (Merit. fairness) —> Hybrid VP               —> GBRVotingWrapper
                        —> MemberStatus hierarchy  —> UnitManager.setStatus()
                        —> Activity Score          —> G-Plan + AI Verifier

  A6 (Inclusivity)     —> PMIP (cost → 0)         —> Low gas (BSC)
                        —> Low barrier entry       —> Wallet = PK
                        —> Accessibility (WCAG)    —> DSP Client UX

  A7 (Self-governance) —> MacroeconomicDAO         —> Cross-DAO Coordinator
                        —> FRP                      —> Fork Protocol contracts
                        —> SES                      —> Replicator dynamics

  A8 (Cognitive        —> AiC (3 circuits)         —> AI agent pipeline
      augmentation)    —> AI Summarizer             —> LLM (Llama/Mixtral)
                        —> GNN Sybil detector       —> PyTorch Geometric
                        —> AI audit                 —> Slither + Mythril
```

## J.2. Theorems and Their Dependencies

```
Proof structure:

  Theorem 1 (Universal Realizability of PMIP)
    Depends on: definition of PMIP, NFC, SIC, SR(p)
    Used in: justification of A1, A6; comparative analysis (7.5)
    Technical implementation: UnitManager reward distribution

  Theorem 2 (Scale Inversion)
    Depends on: Theorem 1, definition of individual_cost
    Used in: the "millions = billions" argument (3.4.3)
    Technical implementation: SIC formation incentives

  Theorem 3 (Antifragility of PMIP)
    Depends on: Theorem 1, definition of NFC, external shocks
    Used in: justification of CSC resilience (3.6.4)
    Technical implementation: Fund Monitoring + Halving

  Theorem 4 (O(n) coordination complexity with AI)
    Depends on: A8 (cognitive augmentation), definition of AiC
    Used in: justification of DAO scalability (4.8)
    Technical implementation: AI Summarizer + AI Cross-DAO Coordinator

  Theorem 5 (Reduction of cognitive load)
    Depends on: four-class taxonomy, definition of L(t)
    Used in: justification of DAO differentiation (5.6.2)
    Technical implementation: Governor × 4 with different parameters

  Theorem 6 (Incentive compatibility of SES)
    Depends on: definition of M_SES, Activity Score, Hybrid VP
    Used in: justification of CSC mechanism design (3.5.7)
    Technical implementation: GBRVotingWrapper + UnitManager

  Theorem 7 (Stability of multi-population SES)
    Depends on: Theorem 6, four-class taxonomy
    Used in: justification of the ecological model (3.5.8)
    Technical implementation: Four DAO treasuries + MacroeconomicDAO
```

## J.3. Critical Implementation Path

Based on the dependency graph, the critical path is identified — the minimal sequence of implementations required for the MVP to function:

```
MVP Critical Path:

  1. GBR Token (BEP-20 on BSC)               — tokenomics foundation
       ↓
  2. UnitManager v2 (MemberStatus 6)          — statuses + rewards
       ↓
  3. GBRVotingWrapper (Hybrid VP)             — voting
       ↓
  4. Governor × 4 + Timelock                  — four DAO classes
       ↓
  5. DSP Client (wallet + governance UI)      — user interface
       ↓
  6. GPROD (blockchain middleware)            — link between UI and blockchain
       ↓
  7. G-Plan v1 (task management)             — Activity Score
       ↓
  8. DUNA registration (Wyoming)             — legal status

  All components (1)-(8) are blocking: without any one of them
  the MVP does not function. The order is determined by technical dependencies.
  Expected time: Phase I (2024-2025).
```

This dependency map is a working tool for developers and researchers within the ecosystem. It will be updated as the theory and implementation evolve — every change in the axiomatic system or architecture is traceable through the dependency graph to all affected components.

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